

JEE-ADVANCED QUESTION BANK

PART - II (XII TOPIC)

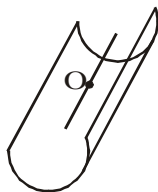
PHYSICS

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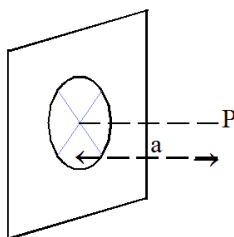
ELECTROSTATICS

EXERCISE (S-2)

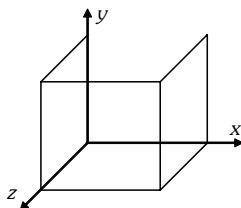
1. A thin long strip whose cross-section is a semicircle carries a uniform surface charge of density σ on its inner surface. Find the electric field at a point O located midway on its axis.



2. An infinitely long non-conducting plane of charge density σ has circular aperture of certain radius carved out from it. The electric field at a point which is at a distance 'a' from the centre of the aperture is $\sigma/2\sqrt{2}\epsilon_0$. Find the radius of aperture.

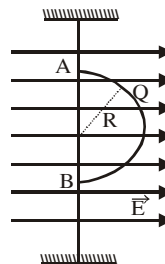


3. A disc of radius R is kept such that its axis coincide with the x-axis and its centre is at (d,0,0). The thickness of disc is t and it carries a uniform volume charge density ρ . The external electric field in the space is given by $\vec{E} = K\vec{r}$ where $K = \text{Constant}$ and $\vec{r}$ is position vector of any point in space with respect to the origin of the coordinate system. Find the electric force on the disc.
4. An electric field given by $\vec{E} = 4\hat{i} - 3(y^2 + 2)\hat{j}$ pierces Gaussian's cube of side 1m placed at origin such that its three sides represents x, y and z axes. The net charge enclosed within the cube is given by $-n\epsilon_0$. Find the value of n.

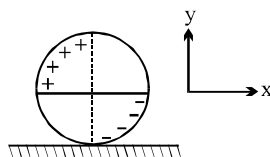


5. The charge $Q = \pi C$ is distributed on a thin semicircular ring of radius $R = 2\text{m}$.

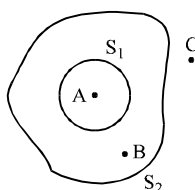
There is a uniform electrostatic field $|\vec{E}| = 2\text{N/C}$ directed horizontally. The semicircular ring can rotate freely about a fixed vertical axis AB. Initially the ring is in static equilibrium as shown in figure. If we want to rotate it about the fixed axis by 90° then minimum work required on the ring is $x\text{J}$. Find the value of x .



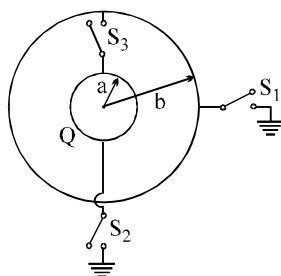
6. Four point charges $+8\text{ }\mu\text{C}$, $-1\text{ }\mu\text{C}$, $-1\text{ }\mu\text{C}$ and $+8\text{ }\mu\text{C}$, are fixed at the points, $-\sqrt{\frac{27}{2}}\text{ m}$, $-\sqrt{\frac{3}{2}}\text{ m}$, $+\sqrt{\frac{3}{2}}\text{ m}$ and $+\sqrt{\frac{27}{2}}\text{ m}$ respectively on the y -axis. A particle of mass $6 \times 10^{-4}\text{ kg}$ and of charge $+0.1\text{ }\mu\text{C}$ moves along the $-x$ direction. Its speed at $x = +\infty$ is v_0 . Find the least value of v_0 for which the particle will cross the origin. Find also the kinetic energy of the particle at the origin. Assume that space is gravity free. (Given : $1/(4\pi\epsilon_0) = 9 \times 10^9\text{ Nm}^2/\text{C}^2$)
7. A nonconducting ring of mass m and radius R is charged as shown. The charged density i.e. charge per unit length is λ . It is then placed on a rough nonconducting horizontal surface plane. At time $t=0$, a uniform electric field $\vec{E} = E_0\hat{i}$ is switched on and the ring start rolling without sliding. Determine the friction force (magnitude and direction) acting on the ring, when it starts moving.



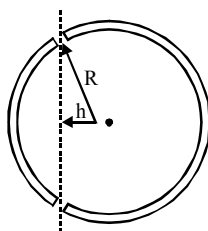
8. Small identical balls with equal charges are fixed at vertices of regular 2008-gon with side a . At a certain instant, one of the balls is released & a sufficiently long time interval later, the ball adjacent to the first released ball is freed. The kinetic energies of the released balls are found to differ by K at a sufficiently long distance from the polygon. Determine the charge q of each ball.
9. S_1 and S_2 are two conducting surfaces. Between S_1 and S_2 and inside S_1 is air. S_1 is spherical with A its centre. S_1 has total charge Q . S_2 is uncharged. Find (if possible) :
- Charges induced on inner and outer surface of S_2 .
 - Total electric field at A, B.
 - Electric field at B due to induced charges on S_2 .
 - Electric field at C due to induced charge on inner surface of S_2 .
 - Electric field produced by induced charge on outer surface of S_2 inside the body of S_2 .
 - Can you find electric field at C easily ?
- (take the required distances from A). Which charge will produce electric field here.



10. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' ($b > a$). Initially switches S_1 , S_2 and S_3 are open and sphere 'A' carries a charge Q . First the switch ' S_1 ' is closed to connect the shell B with the ground and then opened. Now the switch ' S_2 ' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch ' S_3 ' is closed to connect the spheres together. Find the heat (in joule) which is produced after closing the switch S_3 . [Consider $b = 4$ cm, $a = 2$ cm and $Q = 80 \mu\text{C}$]



11. Consider a metal sphere, of radius R that is cut in two along a plane whose minimum distance from the sphere's centre is h . Sphere is uniformly charged by a total electric charge Q . What force is necessary to hold the two parts of the sphere together ?



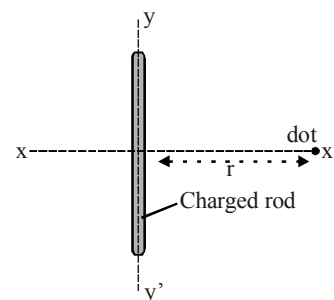
12. Two fixed charges $-2Q$ and Q are located at the points with co-ordinates $(-3a, 0)$ and $(3a, 0)$ respectively in the x - y plane.
- Show that all the points in the x - y plane where the electric potential due to the two charges is zero lie on a circle. Find its radius and the location of its centre.
 - Give the expression for the potential $V_{(x)}$ at a general point on the x -axis and sketch the function $V_{(x)}$ on the whole x -axis.
 - If a particle of charge $+q$ starts from rest at the centre of the circle, show by a short qualitative argument that the particle eventually crosses the circle. Find its speed when it does so.

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A square of side b centred at the origin with sides parallel to axes of x and y has surface charge density $\sigma(x, y) = \sigma_0 xy$ within its boundaries. Total charge on the square is
 (A) 0 (B) $\sigma_0 b^2$ (C) $2\sigma_0 b^2$ (D) $4\sigma_0 b^2$

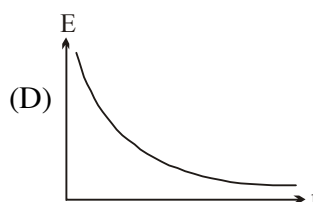
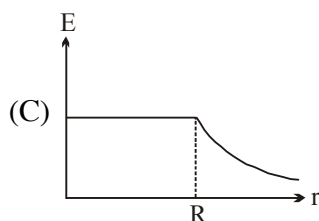
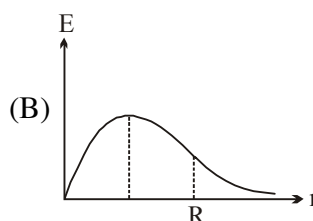
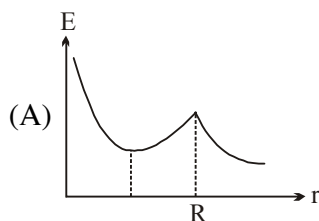
2. A uniformly charged rod is kept on y -axis with centre at origin, as shown. Which of the following actions will increase the electric field strength at the position of the dot?



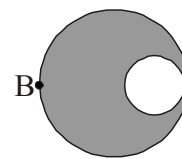
- (A) make the rod longer without changing the charge
 (B) make the rod shorter without changing the charge
 (C) make the rod shorter without changing the linear charge density
 (D) rotate the rod about yy'
3. **Statement-1 :** A positive point charge initially at rest in a uniform electric field starts moving along electric lines of forces. (Neglect all other forces except electric forces)

Statement-2 : Electric lines of force represents path of charged particle which is released from rest in it.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
4. A spherical insulator of radius R is charged uniformly with a charge Q throughout its volume and contains a point charge $\frac{Q}{16}$ located at its centre. Which of the following graphs best represent qualitatively, the variation of electric field intensity E with distance r from the centre.

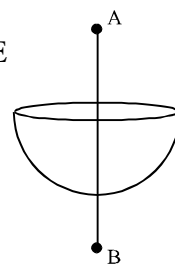


5. A positively charged sphere of radius r_0 carries a volume charge density ρ_E (Figure). A spherical cavity of radius $r_0/2$ is then scooped out and left empty, as shown. What is the direction and magnitude of the electric field at point B ?



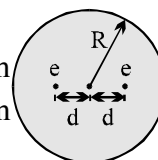
- (A) $\frac{17\rho r_0}{54\epsilon_0}$ left (B) $\frac{\rho r_0}{6\epsilon_0}$ left (C) $\frac{17\rho r_0}{54\epsilon_0}$ right (D) $\frac{\rho r_0}{6\epsilon_0}$ right

6. The diagram shows a uniformly charged hemisphere of radius R . It has volume charge density ρ . If the electric field at a point $2R$ distance above its center is E then what is the electric field at the point which is $2R$ below its center?



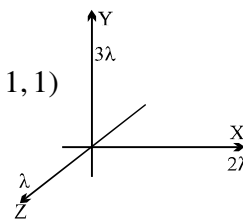
- (A) $\rho R/6\epsilon_0 + E$
(B) $\rho R/12\epsilon_0 - E$
(C) $-\rho R/6\epsilon_0 + E$
(D) $\rho R/24\epsilon_0 + E$

7. Using Thomson's model of the atom, consider an atom consisting of two electrons, each of charge $-e$, embedded in a sphere of charge $+2e$ and radius R . In equilibrium each electron is at distance d from the centre of the atom. What is equilibrium separation between electrons?



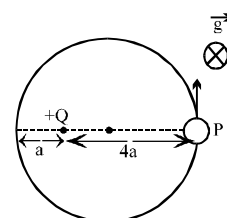
- (A) R (B) $R/2$ (C) $R/3$ (D) $R/4$

8. The diagram shows three infinitely long uniform line charges placed on the X, Y and Z axis. The work done in moving a unit positive charge from $(1, 1, 1)$ to $(0, 1, 1)$ is equal to :-



- (A) $(\lambda \ln 2) / 2\pi\epsilon_0$ (B) $(\lambda \ln 2) / \pi\epsilon_0$
(C) $(3\lambda \ln 2) / 2\pi\epsilon_0$ (D) None

9. The diagram shows a small bead of mass m carrying charge q . The bead can freely move on the smooth fixed ring placed on a smooth horizontal plane. In the same plane a charge $+Q$ has also been fixed as shown. The potential at the point P due to $+Q$ is V . The velocity with which the bead should be projected from the point P so that it can complete a circle should be greater than

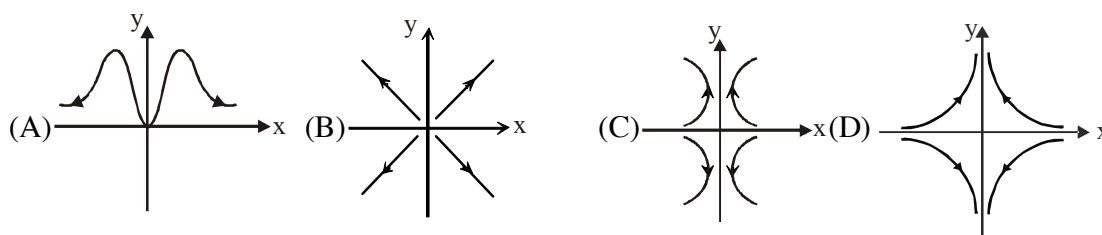


- (A) $\sqrt{\frac{6qV}{m}}$ (B) $\sqrt{\frac{qV}{m}}$ (C) $\sqrt{\frac{3qV}{m}}$ (D) none

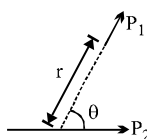
10. A charged particle of charge Q is held fixed and another charged particle of mass m and charge q (of the same sign) is released from a distance r . The impulse of the force exerted by the external agent on the fixed charge by the time distance between Q and q becomes $2r$ is

- (A) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mr}}$ (B) $\sqrt{\frac{Qqm}{4\pi\epsilon_0 r}}$ (C) $\sqrt{\frac{Qqm}{\pi\epsilon_0 r}}$ (D) $\sqrt{\frac{Qqm}{2\pi\epsilon_0 r}}$

11. In a certain region of space, the potential field depends on x and y coordinates as $V = (x^2 - y^2)$. The corresponding electric field lines in x - y plane are correctly represented by :

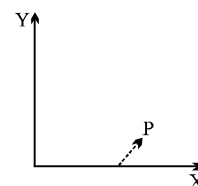


12. Two short electric dipoles are placed as shown. The energy of electric interaction between these dipoles will be :-



- (A) $\frac{2kP_1P_2 \cos \theta}{r^3}$ (B) $\frac{-2kP_1P_2 \cos \theta}{r^3}$ (C) $\frac{-2kP_1P_2 \sin \theta}{r^3}$ (D) $\frac{-4kP_1P_2 \cos \theta}{r^3}$

13. A small electric dipole $\vec{P}$ is placed on the X axis at the point $(1, 0)$. The dipole vector forms an angle of 30° with the X axis. Consider a non uniform electric field to have been applied in the region given by the vector $\vec{E} = x^2\hat{i} + y^2\hat{j}$. What is the force acting on the dipole?



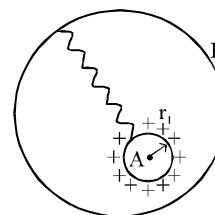
- (A) $2\vec{P} \cos 30^\circ (\hat{i} + 2\hat{j})$ (B) $2\vec{P} \cos 30^\circ (\hat{i})$ (C) $2\vec{P} \cos 30^\circ (2\hat{j})$ (D) None
14. A metal sphere A of radius r_1 charged to a potential ϕ_1 is enveloped by a thin walled conducting spherical shell B of radius r_2 . Then ϕ_2 of the sphere A after it is connected by a thin wire to the shell B will be :-

(A) $\phi_1 \frac{r_1}{r_2}$

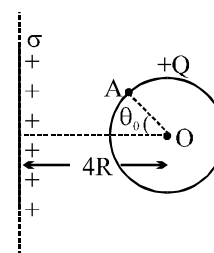
(B) $\phi_1 \left(\frac{r_2}{r_1} \right)$

(C) $\phi_1 \left(1 - \frac{r_1}{r_2} \right)$

(D) $\phi_1 \left(\frac{r_1 r_2}{r_1 + r_2} \right)$



15. A conducting sphere of radius R and charge Q is placed near a uniformly charged nonconducting infinitely large thin plate having surface charge density σ . Then find the potential at point A (on the surface of sphere) due to charge on sphere (here $K = \frac{1}{4\pi\epsilon_0}$, $\theta_0 = \frac{\pi}{3}$)



(A) $K \frac{Q}{R} - \frac{\sigma}{4\epsilon_0} R$

(B) $K \frac{Q}{R} - \frac{\sigma R}{\epsilon_0}$

(C) $K \frac{Q}{R}$

(D) none of these

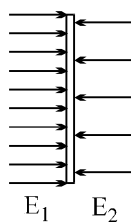
16. The intensity of an electric field depends only on the coordinates x, y and z as follows :

$$\vec{E} = a \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} \text{ unit}$$

The electrostatic energy stored between two imaginary concentric spherical shells of radii R and $2R$ with centre at origin is :-

- (A) $\frac{4\pi\epsilon_0 a^2}{R}$ (B) $\frac{2\pi\epsilon_0 a^2}{R}$ (C) $\frac{\pi\epsilon_0 a^2}{R}$ (D) $\frac{\pi\epsilon_0 a^2}{2R}$

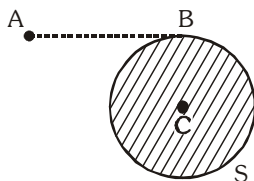
17. A charged large metal sheet is placed into uniform electric field, perpendicularly to the electric field lines. After placing the sheet into the field, the electric field on the left side of the sheet is $E_1 = 5 \times 10^5 \text{ V/m}$ and on the right it is $E_2 = 3 \times 10^5 \text{ V/m}$. The sheet experiences a net electric force of 0.08 N . Find the area of one face of the sheet. Assume external field to remain constant after introducing the large sheet. Use $\left(\frac{1}{4\pi\epsilon_0}\right) = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$



- (A) $3.6 \pi \times 10^{-2} \text{ m}^2$ (B) $0.9 \pi \times 10^{-2} \text{ m}^2$ (C) $1.8 \pi \times 10^{-2} \text{ m}^2$ (D) none

MULTIPLE CORRECT TYPE QUESTIONS

18. S is a solid neutral conducting sphere. A point charge q of $1 \times 10^{-6} \text{ C}$ is placed at point A . C is the centre of sphere and AB is a tangent. $BC = 3 \text{ m}$ and $AB = 4 \text{ m}$.

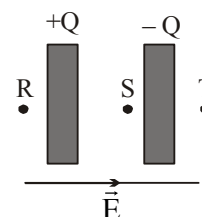


- (A) The electric potential of the conductor is 1.8 kV .
 (B) The electric potential of the conductor is 2.25 kV .
 (C) The electric potential at B due to induced charges on the sphere is -0.45 kV .
 (D) The electric potential at B due to induced charges on the sphere is 0.45 kV .

19. A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges $+Q$ and $-Q$ respectively. A third plate C, identical to A and B, with charge $+Q$, is now introduced midway between A and B, parallel to them. Which of the following is/are **CORRECT**?

- (A) The charge on the inner face of B is now $-\frac{3Q}{2}$
 (B) There is no change in the potential difference between A and B.
 (C) The potential difference between A and C is one-third of the potential difference between B and C.
 (D) Plate C is at a potential higher than plate B

20. Two large thin conducting plates with small gap in between are placed in a uniform electric field E (perpendicular to the plates). Area of each plate is A and charges $+Q$ and $-Q$ are given to these plates as shown in the figure. If points R, S and T as shown in the figure are three points in space, then the



- (A) field at point R is E (B) field at point S is E
 (C) field at point T is $\left(E + \frac{Q}{\epsilon_0 A}\right)$ (D) field at point S is $\left(E + \frac{Q}{A\epsilon_0}\right)$

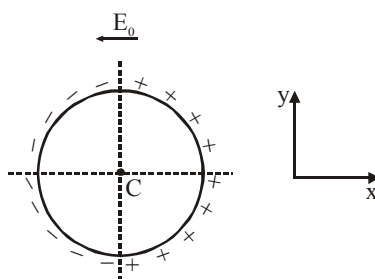
21. Equipotential surfaces :-

- (A) are closer in regions of large electric fields compared to regions of lower electric fields.
 (B) will be more crowded near sharp edges of a conductor.
 (C) will be more crowded near regions of large charge densities.
 (D) will always be equally spaced.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 22 and 23

A uniform ring of mass m and radius R can rotate freely about an axis passing through centre C and perpendicular to plane of paper. Half of ring is positively charged and other half is negatively charged. Uniform electric field E_0 is switched on along $-ve$ x -axis (axis are shown in figure) [Magnitude of charge density λ]



22. The dipole moment of ring is :-

- (A) $2\lambda R^2$ (B) $4\lambda R^2$ (C) $2\pi\lambda R^2$ (D) $4\pi\lambda R^2$

23. If ring is slightly disturb from given position, find the angular speed of ring when it rotate by $\pi/2$.

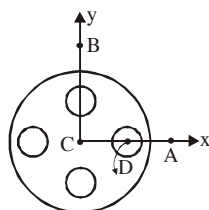
- (A) $2\sqrt{\frac{\lambda E_0}{m}}$ (B) $\sqrt{\frac{\lambda E_0}{m}}$ (C) $\sqrt{\frac{8\lambda E_0}{m}}$ (D) None

MATRIX MATCH TYPE QUESTION

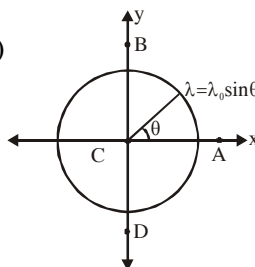
24. Column-II shows some charge distributions and column-I has some statements about electric field at four points A, B, C, D. Match column-I with column-II.

Column-I

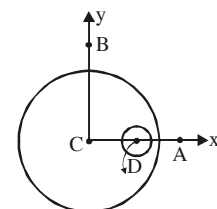
(A) $\vec{E}_A$ has x component only (P)



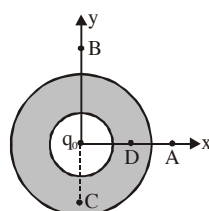
(B) $\vec{E}_B$ has y component only (Q)



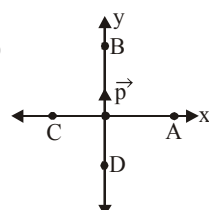
(C) $\vec{E}_C$ has y component only (R)



(D) $|\vec{E}_D|$ is zero (S)



(T)



Column-II

A solid non conducting sphere of radius R of volumetric charge density ρ with four symmetrical spherical cavities. All the five sphere's centre lie in same plane.

A very small circular filament lying in xy-plane. All points lie in same plane. A, B and D are at large distance compared to radius of circle.

A charged spherical conductor with an empty cavity in it.

A hollow thick spherical neutral conductor with a concentric cavity. Charge q_0 is placed inside at centre of cavity.

A small electric dipole $\vec{p}$ is placed at origin. A, B, C and D are four points at large distance from origin.

EXERCISE-JA

(Not to be discussed in class)

ALL ANSWERS OF THE QUESTIONS **JEE-2007** ONWARDS ARE AS GIVEN ON **IIT-JEE** WEBSITE.

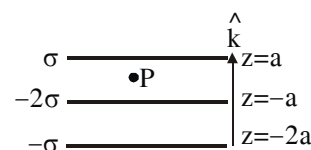
1. Three large parallel plates have uniform surface charge densities as shown in the figure. What is the electric field at P ? [IIT-JEE 2005 (Scr)]

(A) $-\frac{4\sigma}{\epsilon_0} \hat{k}$

(B) $\frac{4\sigma}{\epsilon_0} \hat{k}$

(C) $-\frac{2\sigma}{\epsilon_0} \hat{k}$

(D) $\frac{2\sigma}{\epsilon_0} \hat{k}$



2. A conducting liquid bubble of radius a and thickness t ($t \ll a$) is charged to potential V . If the bubble collapses to a droplet, find the potential on the droplet. [IIT-JEE 2005]

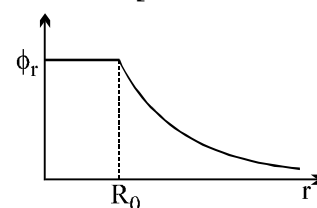
3. The electrostatic potential (ϕ_r) of a spherical symmetric system, kept at origin, is shown in the adjacent

figure, and given as $\phi_r = \frac{q}{4\pi\epsilon_0 r}$ ($r \geq R_0$); $\phi_r = \frac{q}{4\pi\epsilon_0 R_0}$ ($r \leq R_0$)

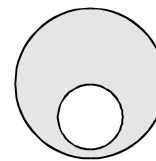
[IIT-JEE 2006]

Which of the following option(s) is/are correct?

- (A) For spherical region $r \leq R_0$, total electrostatic energy stored is zero.
 (B) Within $r = 2R_0$, total charge is q .
 (C) There will be no charge anywhere except at $r = R_0$.
 (D) Electric field is discontinuous at $r = R_0$.
4. A long, hollow conducting cylinder is kept coaxially inside another long, hollow conducting cylinder of larger radius. Both the cylinders are initially electrically neutral. [IIT-JEE 2007]
 (A) A potential difference appears between the two cylinders when a charge density is given to the inner cylinder
 (B) A potential difference appears between the two cylinders when a charge density is given to the outer cylinder
 (C) No potential difference appears between the two cylinders when a uniform line charge is kept along the axis of the cylinder
 (D) No potential difference appears between the two cylinders when same charge density is given to both the cylinders
5. Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. The net charge on the sphere is then, [IIT-JEE 2007]
 (A) negative and distributed uniformly over the surface of the sphere
 (B) negative and appears only at the point on the sphere closest to the point charge
 (C) negative and distributed non-uniformly over the entire surface of the sphere
 (D) zero



6. A spherical portion has been removed from a solid sphere having a charge distributed uniformly in its volume as shown in the figure. The electric field inside the emptied space is [IIT-JEE 2007]
- (A) zero everywhere (B) nonzero and uniform
(C) nonuniform (D) zero only at its center



7. Positive and negative point charges of equal magnitude are kept at $(0, 0, \frac{a}{2})$ & $(0, 0, -\frac{a}{2})$, respectively.

The work done by the electric field when another positive point charge is moved from $(-a, 0, 0)$ to $(0, a, 0)$ is [IIT-JEE 2007]

- (A) positive (B) negative
(C) zero (D) depends on the path connecting the initial and final positions
8. Consider a system of three charges $\frac{q}{3}$, $\frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C, respectively, as shown in the figure. Take O to be the centre of the circle of radius R and angle CAB = 60°

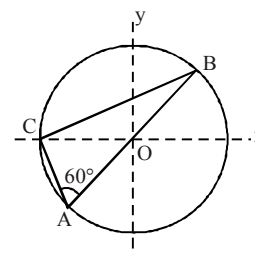
[IIT-JEE 2008]

(A) The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ directed along the negative x-axis

(B) The potential energy of the system is zero

(C) The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$

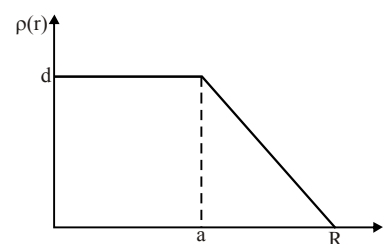
(D) The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$



Paragraph for Question No. 9 to 11

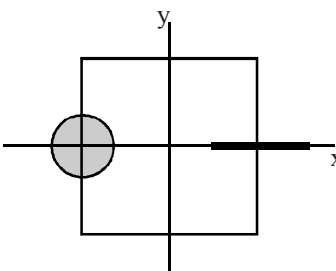
The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R . The charge density $\rho(r)$ [charge per unit volume] is dependent only on the radial distance r from the centre of the nucleus as shown in figure. The electric field is only along the radial direction.

[IIT-JEE 2008]



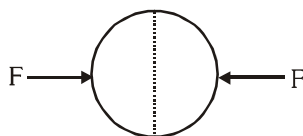
9. The electric field at $r = R$ is :
- (A) independent of a (B) directly proportional to a
(C) directly proportional to a^2 (D) inversely proportional to a
10. For $a = 0$, the value of d (maximum value of ρ as shown in the figure) is :
- (A) $\frac{3Ze}{4\pi R^3}$ (B) $\frac{3Ze}{\pi R^3}$ (C) $\frac{4Ze}{3\pi R^3}$ (D) $\frac{Ze}{3\pi R^3}$
11. The electric field within the nucleus is generally observed to be linearly dependent on r . This implies
- (A) $a = 0$ (B) $a = \frac{R}{2}$ (C) $a = R$ (D) $a = \frac{2R}{3}$

12. A disk of radius $a/4$ having a uniformly distributed charge $6C$ is placed in the x - y plane with its centre at $(-a/2, 0, 0)$. A rod of length a carrying a uniformly distributed charge $8C$ is placed on the x -axis from $x = a/4$ to $x = 5a/4$. Two point charges $-7C$ and $3C$ are placed at $(a/4, -a/4, 0)$ and $(-3a/4, 3a/4, 0)$, respectively. Consider a cubical surface formed by six surfaces $x = \pm a/2, y = \pm a/2, z = \pm a/2$. The electric flux through this cubical surface is



[IIT-JEE 2009]

- (A) $\frac{-2C}{\epsilon_0}$ (B) $\frac{2C}{\epsilon_0}$ (C) $\frac{10C}{\epsilon_0}$ (D) $\frac{12C}{\epsilon_0}$
13. Three concentric metallic spherical shells of radii $R, 2R, 3R$, are given charges Q_1, Q_2, Q_3 , respectively. It is found that the surface charge densities on the outer surfaces of the shells are equal. Then, the ratio of the charges given to the shells, $Q_1 : Q_2 : Q_3$, is [IIT-JEE 2009]
 (A) $1 : 2 : 3$ (B) $1 : 3 : 5$ (C) $1 : 4 : 9$ (D) $1 : 8 : 18$
14. Under the influence of the Coulomb field of charge $+Q$, a charge $-q$ is moving around it in an elliptical orbit. Find out the correct statement(s). [IIT-JEE 2009]
 (A) The angular momentum of the charge $-q$ is constant
 (B) The linear momentum of the charge $-q$ is constant
 (C) The angular velocity of the charge $-q$ is constant
 (D) The linear speed of the charge $-q$ is constant
15. A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = Kr^a$, where K and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, find the value of a . [IIT-JEE 2009]
16. A uniformly charged thin spherical shell of radius R carries uniform surface charge density of σ per unit area. It is made of two hemispherical shells, held together by pressing them with force F (see figure). F is proportional to [IIT-JEE 2010]

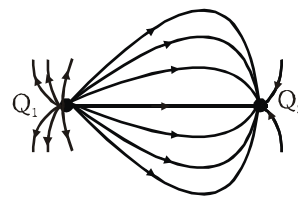


- (A) $\frac{1}{\epsilon_0} \sigma^2 R^2$ (B) $\frac{1}{\epsilon_0} \sigma^2 R$ (C) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R}$ (D) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R^2}$
17. A tiny spherical oil drop carrying a net charge q is balanced in still air with a vertical uniform electric field of strength $\frac{81\pi}{7} \times 10^5 \text{ V m}^{-1}$. When the field is switched off, the drop is observed to fall with terminal velocity $2 \times 10^{-3} \text{ m/s}$. Given $g = 9.8 \text{ m/s}^2$, viscosity of the air $= 1.8 \times 10^{-5} \text{ N-s m}^{-2}$ and the density of oil $= 900 \text{ kg m}^{-3}$, the magnitude of q is [IIT-JEE 2010]
 (A) $1.6 \times 10^{-19} \text{ C}$ (B) $3.2 \times 10^{-19} \text{ C}$ (C) $4.8 \times 10^{-19} \text{ C}$ (D) $8.0 \times 10^{-19} \text{ C}$

18. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on the x -axis are shown in the figure. These lines suggest that

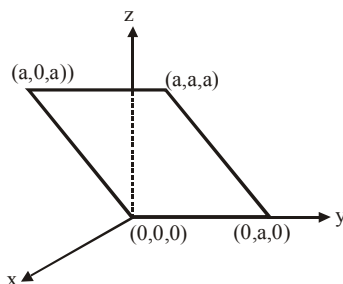
[IIT-JEE 2010]

- (A) $|Q_1| > |Q_2|$
(B) $|Q_1| < |Q_2|$
(C) at a finite distance to the left of Q_1 the electric field is zero
(D) at a finite distance to the right of Q_2 the electric field is zero



19. Consider an electric field $\vec{E} = E_0 \hat{x}$, where E_0 is a constant. The flux through the shaded area (as shown in the figure) due to this field is

[IIT-JEE 2011]



- (A) $2E_0 a^2$ (B) $\sqrt{2}E_0 a^2$ (C) $E_0 a^2$ (D) $\frac{E_0 a^2}{\sqrt{2}}$

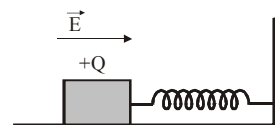
20. A spherical metal shell A of radius R_A and a solid metal sphere B of radius $R_B (< R_A)$ are kept far apart and each is given charge $+Q$. Now they are connected by a thin metal wire. Then [IIT-JEE 2011]

- (A) $E_A^{\text{inside}} = 0$ (B) $Q_A > Q_B$ (C) $\frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$ (D) $E_A^{\text{on surface}} < E_B^{\text{on surface}}$

21. A wooden block performs SHM on a frictionless surface with frequency, ν_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be

[IIT-JEE 2011]

- (A) of the same frequency and with shifted mean position
(B) of the same frequency and with the same mean position
(C) of changed frequency and with shifted mean position
(D) of changed frequency and with the same mean position



22. Which of the following statement(s) is/are correct? [IIT-JEE 2011]

- (A) If the electric field due to a point charge varies as $r^{-2.5}$ instead of r^{-2} , then the Gauss law will still be valid.
(B) The Gauss law can be used to calculate the field distribution around an electric dipole.
(C) If the electric field between two point charges is zero somewhere, then the sign of the two charges is the same.
(D) The work done by the external force in moving a unit positive charge from point A at potential V_A to point B at potential V_B is $(V_B - V_A)$

23. Four point charges, each of $+q$, are rigidly fixed at the four corners of a square planar soap film of side ' α '. The surface tension of the soap film is γ . The system of charges and planar film are in equilibrium,

and $a = k \left[\frac{q^2}{\gamma} \right]^{1/N}$, where ' k ' is a constant. Then N is

[IIT-JEE 2011]

Paragraph for Question Nos. 24 and 25

A dense collection of equal number of electrons and positive ions is called neutral plasma. Certain solids containing fixed positive ions surrounded by free electrons can be treated as neutral plasma. Let ' N ' be the number density of free electrons, each of mass ' m '. When the electrons are subjected to an electric field, they are displaced relatively away from the heavy positive ions. If the electric field becomes zero, the electrons begin to oscillate about the positive ions with a natural angular frequency ' ω_p ', which is called the plasma frequency. To sustain the oscillations, a time varying electric field needs to be applied that has an angular frequency ω , where a part of the energy is absorbed and a part of it is reflected. As ω approaches ω_p , all the free electrons are set to resonance together and all the energy is reflected. This is the explanation of high reflectivity of metals. [IIT-JEE 2011]

24. Taking the electronic charge as ' e ' and the permittivity as ' ϵ_0 ', use dimensional analysis to determine the correct expression for ω_p .

(A) $\sqrt{\frac{Ne}{m\epsilon_0}}$

(B) $\sqrt{\frac{m\epsilon_0}{Ne}}$

(C) $\sqrt{\frac{Ne^2}{m\epsilon_0}}$

(D) $\sqrt{\frac{m\epsilon_0}{Ne^2}}$

25. Estimate the wavelength at which plasma reflection will occur for a metal having the density of electrons $N \approx 4 \times 10^{27} \text{ m}^{-3}$. Take $\epsilon_0 \approx 10^{-11}$ and $m \approx 10^{-30}$, where these quantities are in proper SI units

(A) 800 nm

(B) 600 nm

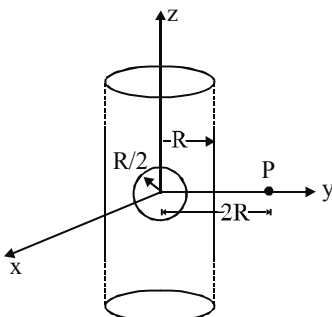
(C) 300 nm

(D) 200 nm

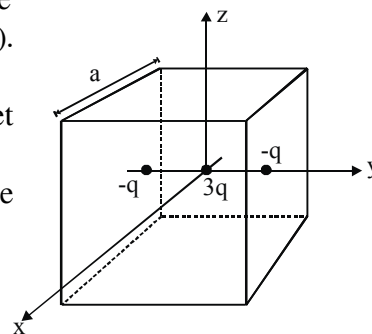
26. An infinitely long solid cylinder of radius R has a uniform volume charge density ρ . It has a spherical cavity of radius $R/2$ with its centre on the axis of the cylinder, as shown in the figure. The magnitude of the electric field at the point P , which is at a distance $2R$ from the axis of the cylinder, is given by

the expression $\frac{23\rho R}{16k\epsilon_0}$. The value of k is :-

[IIT-JEE 2012]



27. A cubical region of side a has its centre at the origin. It encloses three fixed point charges, $-q$ at $(0, -a/4, 0)$, $+3q$ at $(0, 0, 0)$ and $-q$ at $(0, +a/4, 0)$. Choose the correct option(s). [IIT-JEE 2012]

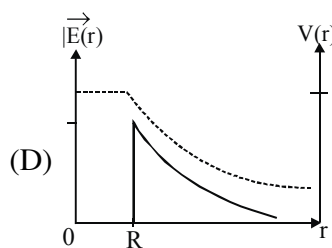
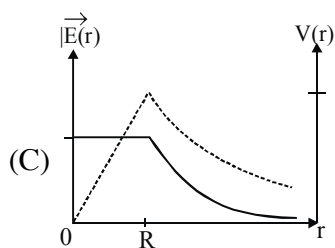
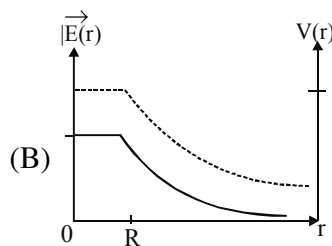
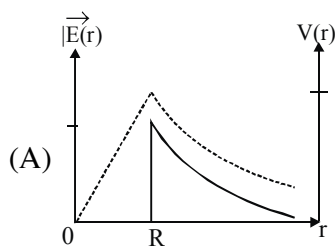


- (A) The net electric flux crossing the plane $x = +a/2$ is equal to the net electric flux crossing the plane $x = -a/2$
 (B) The net electric flux crossing the plane $y = +a/2$ is more than the net electric flux crossing the plane $y = -a/2$.
 (C) The net electric flux crossing the entire region is $\frac{q}{\epsilon_0}$
 (D) The net electric flux crossing the plane $z = +a/2$ is equal to the net electric flux crossing the plane $z = -a/2$.

28. Two large vertical and parallel metal plates having a separation of 1 cm are connected to a DC voltage source of potential difference X . A proton is released at rest midway between the two plates. It is found to move at 45° to the vertical JUST after release. Then X is nearly [IIT-JEE 2012]

- (A) 1×10^{-5} V (B) 1×10^{-7} V (C) 1×10^{-9} V (D) 1×10^{-10} V

29. Consider a thin spherical shell of radius R with its centre at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential $V(r)$ with the distance r from the centre, is best represented by which graph? [IIT-JEE 2012]

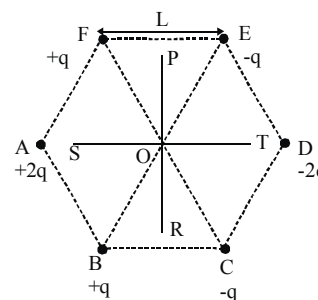


30. Six point charges are kept at the vertices of a regular hexagon of side L

and centre O , as shown in figure. Given that $K = \frac{1}{4\pi\epsilon_0} \frac{q}{L^2}$, which of

the following statement(s) is(are) correct? [IIT-JEE 2012]

- (A) The electric field at O is $6K$ along OD .
 (B) The potential at O is zero.
 (C) The potential at all points on the line PR is same.
 (D) The potential at all points on the line ST is same.



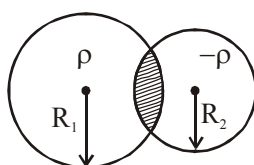
31. Two non-conducting solid spheres of radii R and $2R$, having uniform volume charge densities ρ_1 and ρ_2 respectively, touch each other. The net electric field at a distance $2R$ from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero. The ratio $\frac{\rho_1}{\rho_2}$ can be

[JEE-Advance-2013]

- (A) -4 (B) $-\frac{32}{25}$ (C) $\frac{32}{25}$ (D) 4

32. Two non-conducting spheres of radii R_1 and R_2 and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed such that they partially overlap, as shown in the figure. At all points in the overlapping region :-

[JEE-Advance-2013]



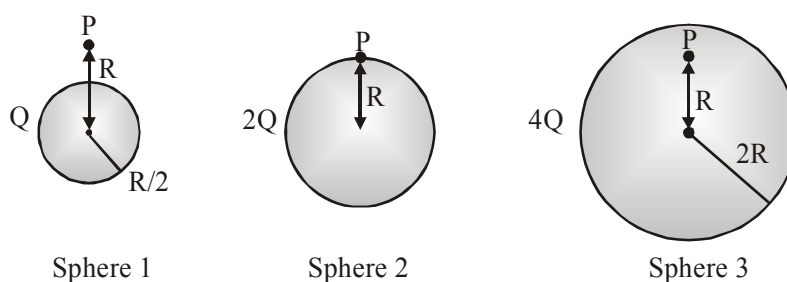
- (A) the electrostatic field is zero
 (B) the electrostatic potential is constant
 (C) the electrostatic field is constant in magnitude
 (D) the electrostatic field has same direction
33. Let $E_1(r)$, $E_2(r)$ and $E_3(r)$ be the respective electric fields at a distance r from a point charge Q , an infinitely long wire with constant linear charge density λ , and an infinite plane with uniform surface charge density σ . If $E_1(r_0) = E_2(r_0) = E_3(r_0)$ at a given distance r_0 , then :-

[JEE-Advance-2014]

- (A) $Q = 4\sigma\pi r_0^2$ (B) $r_0 = \frac{\lambda}{2\pi\sigma}$
 (C) $E_1(r_0/2) = 2E_2(r_0/2)$ (D) $E_2(r_0/2) = 4E_3(r_0/2)$

34. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then

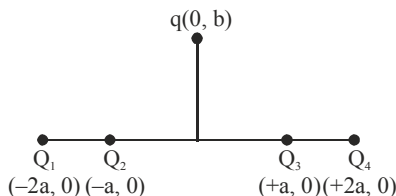
[JEE-Advance-2014]



- (A) $E_1 > E_2 > E_3$ (B) $E_3 > E_1 > E_2$ (C) $E_2 > E_1 > E_3$ (D) $E_3 > E_2 > E_1$

35. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.

[JEE-Advance-2014]



List-I

- (P) Q_1, Q_2, Q_3, Q_4 all positive
(Q) Q_1, Q_2 positive ; Q_3, Q_4 negative
(R) Q_1, Q_4 positive ; Q_2, Q_3 negative
(S) Q_1, Q_3 positive ; Q_2, Q_4 negative

List-II

- (1) $+x$
(2) $-x$
(3) $+y$
(4) $-y$

Code :

(A) P-3, Q-1, R-4, S-2 (B) P-4, Q-2, R-3, S-1 (C) P-3, Q-1, R-2, S-4 (D) P-4, Q-2, R-1, S-3

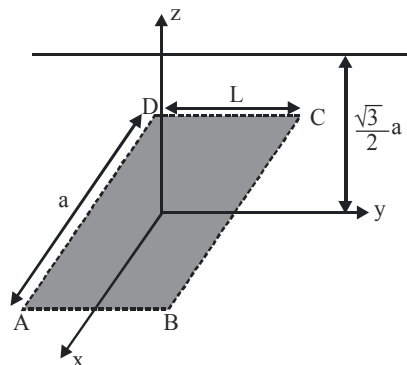
36. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at

$z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric

field through the rectangular surface ABCD lying in the x-y

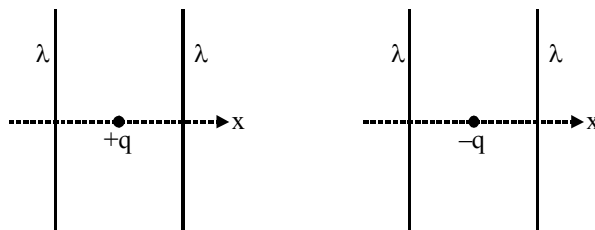
plane with its centre at the origin is $\frac{\lambda L}{n\epsilon_0}$ (ϵ_0 = permittivity of

free space) then the value of n is. [JEE-Advance-2015]



37. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x direction only. If they are given a small displacement about their equilibrium positions, then the correct statement(s) is (are) :

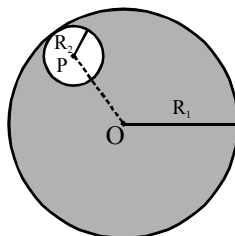
[JEE-Advance-2015]



- (A) Both charges execute simple harmonic motion
(B) Both charges will continue moving in the direction of their displacement
(C) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement.
(D) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement.

38. Consider a uniform spherical distribution of radius R_1 centred at the origin O. In this distribution, a spherical cavity of radius R_2 , centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is(are) :

[JEE-Advance-2015]

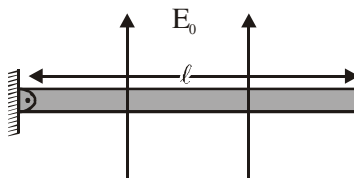


- (A) $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$
(B) $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$
(C) $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$
(D) $\vec{E}$ is uniform and both its magnitude and direction depend on $\vec{a}$

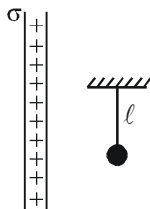
REVIEW TEST

(Not to be discussed in class)

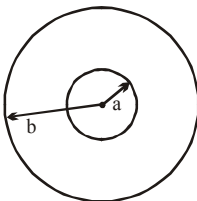
1. A thin insulating uniformly charged (linearly charged density λ) rod is hinged about one of its ends. It can rotate in vertical plane. If rod is in equilibrium by applying vertical electric field E as shown in figure. Find the value of E (in N/C). (Given that mass of rod 2 kg, $\lambda = 10 \text{ C/m}$, $\ell = 1 \text{ m}$, $g = 10 \text{ m/s}^2$)



2. A simple pendulum of length ℓ and bob mass m is hanging in front of a large nonconducting sheet of surface charge density σ . If suddenly a charge $+q$ is given to the bob in the position shown in figure. Find the maximum angle through which the string is deflected from vertical.



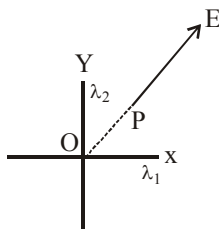
3. Outer cylinder of the coaxial nonconductor of radius 'b' is given a positive potential V relative to the inner cylinder of radius 'a' as shown in the figure (charge distribution is uniform). A charge q (mass = m) is set free with negligible velocity at the surface of the inner cylinder. Find the velocity (in m/s), when it hits the outer cylinder. [consider $V = 10$, $q = -20$, $m = 1$ all in S.I. Units]



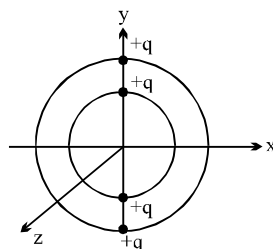
4. A positive charge Q is uniformly distributed throughout the volume of a nonconducting sphere of radius R . A point mass having charge $+q$ and mass m is fired towards the centre of the sphere with velocity v from a point at distance r ($r > R$) from the centre of the sphere. Find the minimum velocity v so that it can penetrate $R/2$ distance of the sphere. Neglect any resistance other than electric interaction. Charge on the small mass remains constant throughout the motion.
5. A charge $+Q$ is fixed at the origin of the coordinate system while a small electric dipole of dipole-moment $\vec{p}$ pointing away from the charge along the x -axis is set free from a point far away from the origin.
 - (a) calculate the K.E. of the dipole when it reaches to a point $(d, 0)$ **[IIT-JEE 2003]**
 - (b) calculate the force on the charge $+Q$ at this moment.

6. Two mutually perpendicular infinite wires along x-axis and y-axis carry charge densities λ_1 and λ_2 .

The electric line of force at P is along the line $y = \frac{1}{\sqrt{3}}x$, where P is also a point lying on the same line then find λ_2 / λ_1 .

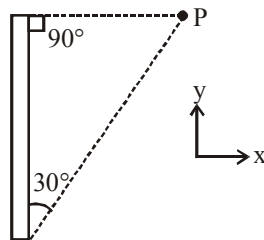


7. The electric potential in a region is given by $V(x, y, z) = ax^2 + ay^2 + abz^2$. 'a' is a positive constant of appropriate dimensions and b, a positive constant such that V is volts when x, y, z are in m. Let $b = 2$. The work done by the electric field when a point charge $+4\mu\text{C}$ moves from the point $(0, 0, 0.1\text{m})$ to the origin is $50\mu\text{J}$. The radius of the circle of the equipotential curve corresponding to $V = 6250$ volts and $z = \sqrt{2}\text{m}$ is αm . Fill α^2 in OMR sheet.
8. Two concentric rings of radii r and $2r$ are placed with centre at origin. Two charges $+q$ each are fixed at the diametrically opposite points of the rings as shown in figure. Smaller ring is now rotated by an angle 90° about Z-axis then it is again rotated by 90° about Y-axis. Find the work done by electrostatic forces in each step. If finally larger ring is rotated by 90° about X-axis, find the total work required to perform all three steps.



9. A non-conducting disc of radius a and uniform positive surface charge density σ is placed on the ground, with its axis vertical. A particle of mass m & positive charge q is dropped, along the axis of the disc, from a height H with zero initial velocity. The particle has $\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma}$.
- (a) Find the value of H if the particle just reaches the disc.
- (b) Sketch the potential energy of the particle as a function of its height and find its equilibrium position.

10. The direction (θ) of $\vec{E}$ at point P due to uniformly charged finite rod will be :-



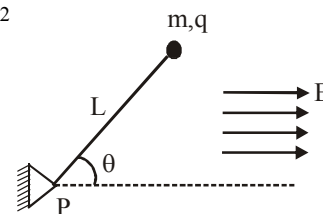
- (A) at angle 30° from x-axis
(B) 45° from x - axis
(C) 60° from x-axis
(D) none of these
11. The maximum electric field at a point on the axis a uniformly charged ring is E_0 . At how many points on the axis will the magnitude of electric field be $E_0/2$:-
(A) 1
(B) 2
(C) 3
(D) 4
12. A particle of mass m and charge q is attached to a light rod of length L . The rod can rotate freely in the plane of paper about the other end, which is hinged at P. The entire assembly lies in a uniform electric field E also acting in the plane of paper as shown. The rod is released from rest when it makes an angle θ with the electric field direction. Determine the speed of the particle when the rod is parallel to the electric field.

(A) $\left(\frac{2qEL(1 - \cos \theta)}{m}\right)^{1/2}$

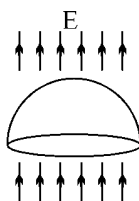
(B) $\left(\frac{2qEL(1 - \sin \theta)}{m}\right)^{1/2}$

(C) $\left(\frac{qEL(1 - \cos \theta)}{2m}\right)^{1/2}$

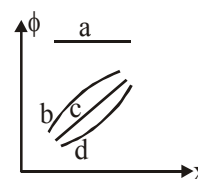
(D) $\left(\frac{2qEL \cos \theta}{m}\right)^{1/2}$



13. A hemispherical surface (half of a spherical surface) of radius R is located in a uniform electric field E that is parallel to the axis of the hemisphere. What is the magnitude of the electric flux through the hemisphere surface?



- (A) 0
(B) $4\pi R^2 E/3$
(C) $2\pi R^2 E$
(D) $\pi R^2 E$
14. A line of charge extends along a X-axis whose linear charge density varies directly as x . Imagine a spherical volume with its centre located on X-axis and is moving gradually along it. Which of the graphs shown in figure correspond to the flux ϕ with the x coordinate of the centre of the volume?
- (A) a
(B) b
(C) c
(D) d



15. Potential energy of a system comprising of point charges is U_1 . When a charge q is added in the system without disturbing other charges, the potential energy becomes U_2 . The potential of the point where the charge q is placed in the system is

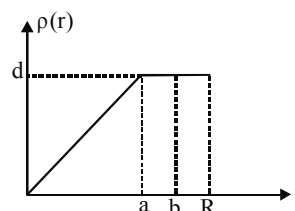
(A) $\frac{U_2 - U_1}{q}$ (B) $\frac{U_1 - U_2}{q}$ (C) $\frac{U_1 + U_2}{2q}$ (D) $\frac{U_2 - U_1}{2q}$

16. Two fixed charges A and B of $5 \mu\text{C}$ each are separated by a distance of 6m . C is the mid point of the line joining A and B. A charge 'Q' of $-5 \mu\text{C}$ is shot perpendicular to the line joining A and B through C with a kinetic energy of 0.06J . The charge 'Q' comes to rest at a point D. The distance CD is:-

(A) 3 m (B) $\sqrt{3}\text{ m}$ (C) $3\sqrt{3}\text{ m}$ (D) 4 m

17. The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R . The charge density $\rho(r)$ [charge per unit volume] is dependent on the radial distance r from the centre of the nucleus as shown in figure. Select correct alternative/s.

- (A) Electric field at $r = R$ is independent of b
 (B) Electric potential at $r = R$ is proportional to b
 (C) Electric field at $r = R$ is proportional to a
 (D) Electric potential at $r = R$ is proportional to a



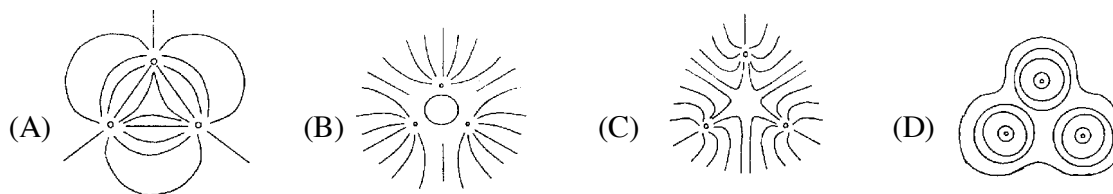
18. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?

(A) R (B) $R/2$ (C) $R/3$ (D) $2R$

19. Equipotential at a great distance from a collection of charges whose total sum is not zero are approximately

- (A) spheres. (B) planes. (C) paraboloids (D) ellipsoids.

20. Three positive charges of equal value q are placed at the vertices of an equilateral triangle. The resulting lines of force should be sketched as in [JEE 2001 (Scr)]



21. Consider the following conclusions regarding the components of an electric field at a certain point in space given by

$$E_x = -Ky$$

$$E_y = Kx$$

$$E_z = 0$$

(I) The field is conservative.

(II) The field is non-conservative.

(III) The lines of force are straight lines

(IV) The lines of force are circles.

Of these conclusions

(A) II and IV are valid

(B) I and III are valid

(C) I and IV are valid

(D) II and III are valid

22. Point P lies on the axis of a dipole. If the dipole is rotated by 90° anticlockwise, the electric field vector $\vec{E}$ at P will rotate by

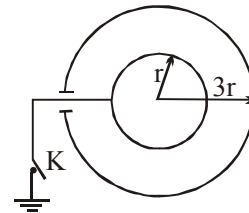
- (A) 90° clockwise (B) 180° clockwise (C) 90° anticlockwise (D) none

23. n small drops of same size are charged to V volts each. If they coalesce to form a single large drop, then its potential will be :-

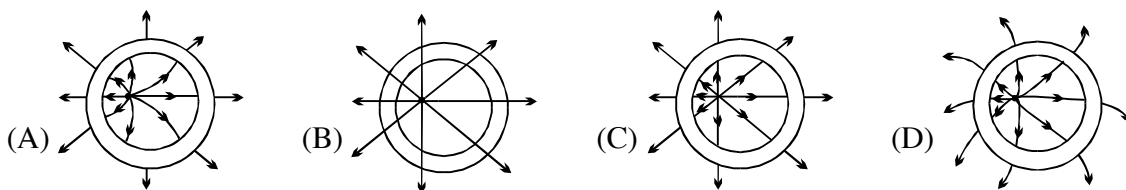
(A) V/n (B) Vn (C) $Vn^{1/3}$ (D) $Vn^{2/3}$

24. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q and inner shell is neutral. The amount of charge which flows from inner shell to the earth after the key K is closed, is equal to :-

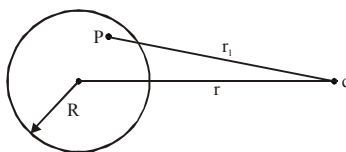
(A) $-q/3$ (B) $q/3$
(C) $3q$ (D) $-3q$



25. A point charge q is placed at a point inside a hollow conducting sphere. Which of the following electric force pattern is correct ? [IIT-JEE'2003 (scr)]



26. A point charge q is placed at a distance r from center of a conducting neutral sphere of radius R ($r > R$). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by



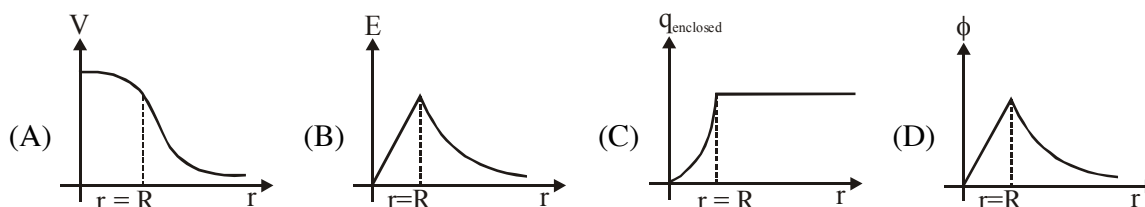
(A) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (B) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (C) $\frac{Kq}{r} - \frac{Kq}{r_1}$ (D) $-\frac{Kq}{r_1} + \frac{Kq}{R}$

27. Two non-conducting hemispherical surfaces, which are having uniform charge density σ are placed on smooth horizontal surface as shown in figure. Assuming springs are ideal, calculate compression in each spring if both the hemispherical surface are just touching each other.



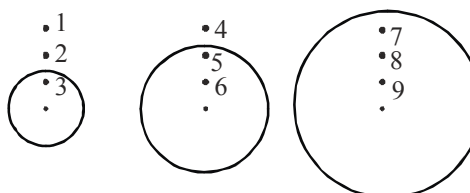
(A) $\frac{\sigma^2 R^2}{2\epsilon_0 k}$ (B) R (C) $\frac{\sigma^2 \pi R^2}{2\epsilon_0 k}$ (D) None of these

28. Graph shows the variation of potential (V), magnitude of electric field (E), charge enclosed (q_{enclosed}) within the concentric sphere and net flux (ϕ) through a concentric spherical Gaussian surface as a function of distance (r) from centre of a uniformly positive charged solid sphere of radius R . Choose the **CORRECT** option(s):-



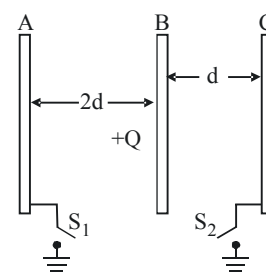
29. Figure shows three spherical shells in separate situations, with each shell having the same uniform positive net charge. Points 1, 4 and 7 are at the same radial distances from the centre of the their respective shells; so are points 2, 5 and 8; and so are points 3, 6 and 9. With the electric potential taken equals to zero at an infinite distance, choose correct statement.

- (A) Point 3 has highest potential
 (B) point 1, 4 and 7 are at same potential
 (C) Point 9 has lowest potential
 (D) point 5 and 8 are at same potential

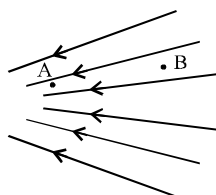


30. Three identical, parallel conducting plates A, B and C are placed as shown. Switches S_1 and S_2 are open, and can connect A and C to earth when closed. $+Q$ charge is given to B.

- (A) If S_1 is closed with S_2 open, a charge of amount Q will pass through S_1
 (B) If S_2 is closed with S_1 open, a charge of amount Q will pass through S_2
 (C) If S_1 and S_2 are closed together, a charge of amount $Q/3$ will pass through S_1 , and a charge of amount $2Q/3$ will pass through S_2 .
 (D) All the above statements are incorrect

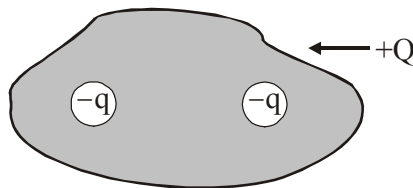


31. Which of the following is true for the figure showing electric lines of force? (E is electrical field, V is potential)



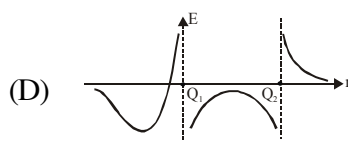
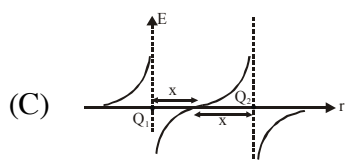
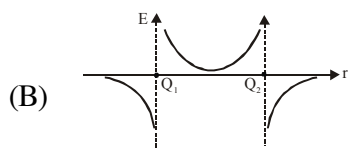
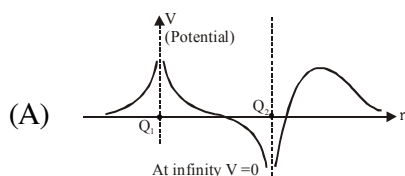
- (A) $E_A > E_B$ (B) $E_B > E_A$ (C) $V_A > V_B$ (D) $V_B > V_A$
32. Consider a uniform electric field in the $\hat{z}$ direction. The potential is a constant
- (A) in all space. (B) for any x for a given z .
 (C) for any y for a given z . (D) on the x - y plane for a given z .

33. A hollow closed conductor of irregular shape is given some charge. Which of the following statements are correct ?
 (A) The entire charge will appear on its outer surface.
 (B) All points on the conductor will have the same potential.
 (C) All points on its surface will have the same charge density.
 (D) All points just outside it will have the same electric intensity.
34. A conducting body is given charge Q and charge $-q$ has been placed in each of the cavity, which of the following statements is/are true?



- (A) If $Q = 2q$, then conducting body will be at zero potential.
 (B) If an external electric field is applied then the charge distribution on the outer surface of conductor would change.
 (C) The potential of any point inside the cavity is less than that of conducting body.
 (D) None of these
35. As shown in column-I their are graphs of electric field (E) and potential (V) along the line joining charges Q_1 and Q_2 are drawn against distance (r) on x -axis for charges Q_1 and Q_2 . Take potential at infinity equal to zero. [Take direction of E in rightward direction as positive]

Column-I



Column-II

(P) $Q_1 < 0, Q_2 > 0$

(Q) $Q_1 > 0, Q_2 < 0$

(R) $|Q_1| > |Q_2|$

(S) $Q_1 < 0, Q_2 < 0$

(T) $|Q_1| = |Q_2|$

36. A spherical metallic conductor has a spherical cavity. A positive charge is placed inside the cavity at its centre. Another positive charge is placed outside it. The conductor is initially electrically neutral.

Column-I**(Cause)**

- (A) If outside charge is shifted to other position
 (B) If inside charge is shifted to other position within cavity
 (C) If magnitude of charge inside cavity is increased
 (D) If conductor is earthed

Column-II**(Effect)**

- (P) distribution of charge on inner surface of cavity changes
 (Q) distribution of charge on outer surface of conductor changes
 (R) electric potential at centre of conductor due to charges present on outer surface of conductor changes
 (S) force on the charge placed inside cavity changes

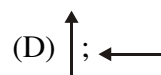
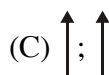
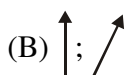
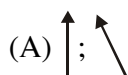
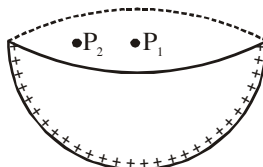
37. A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R . The force of gravity acts downwards. The equilibrium of the particle will be stable

- (A) for all values of H (B) only if $H > \frac{R}{\sqrt{2}}$ (C) only if $H < \frac{R}{\sqrt{2}}$ (D) only if $H = \frac{R}{\sqrt{2}}$

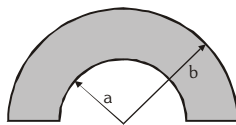
38. A system consists of uniformly charged sphere of radius R and a surrounding medium filled by a charge with the volume density $\rho = \frac{\alpha}{r}$, where α is a positive constant and r is the distance from the centre of the sphere. The charge of the sphere for which electric field intensity E outside the sphere is independent of r is :-

- (A) $\frac{\alpha}{2\epsilon_0}$ (B) $\frac{2}{\alpha\epsilon_0}$ (C) $2\pi\alpha R^2$ (D) αR^2

39. Consider a uniformly charged hemispherical shell shown below. Indicate the directions (not magnitude) of the electric field at the central point P_1 and an off-centre point P_2 on the drumhead of the shell.

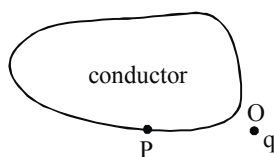


40. A non conducting semicircular disc (as shown in figure) has a uniform surface charge density σ . The electric potential at the centre of the disc :-



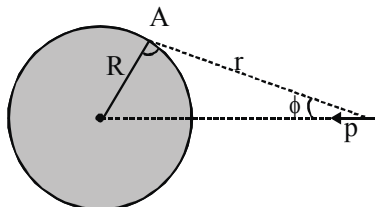
- (A) $\frac{\sigma}{2\pi\epsilon_0} \frac{\ln(b/a)}{(b-a)}$ (B) $\frac{\sigma(b-a)}{2\epsilon_0}$ (C) $\frac{\sigma(b-a)}{4\epsilon_0}$ (D) $\frac{\sigma(b-a)}{4\pi\epsilon_0}$

41. The density of charge at P on the conductor is σ . The resultant electric field near P will [$\hat{n}$ = unit normal vector at point P. $\vec{r}$ = vector along OP] where $K = 1/4 \pi\epsilon$



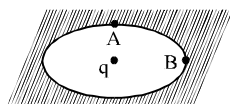
- (A) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n} + \left(\frac{Kq\vec{r}}{r^3}\right)$ (B) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n} + \left(\frac{Kq\vec{r}}{r^3}\right)$ (C) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n}$ (D) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$

42. A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is :-



- (A) $\frac{kp \cos \phi}{r^2}$ (B) $\frac{kp \cos^2 \phi}{r^2}$
(C) 0 (D) $\frac{2kp \cos^2 \phi}{r^2}$

43. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the center of the cavity. The points A & B are on the cavity surface as shown in the figure. Then :



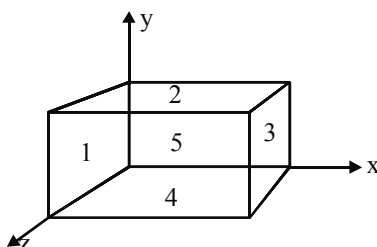
- (A) electric field near A in the cavity = electric field near B in the cavity
(B) charge density at A = charge density at B
(C) potential at A = potential at B
(D) total electric field flux through the surface of the cavity is q/ϵ_0 .

44. Four identical particles each having mass m and charge q are placed at the vertices of a square of side ℓ . All the particles are free to move without any friction and released simultaneously from rest. Then
- (A) At all instants, the particles remain at vertices of square whose edge length is changing
 (B) The configuration is changing (not remaining square) as the time passes

(C) The speed of the particles when one of the particles get displaced by $\frac{\ell}{\sqrt{2}}$ is $\sqrt{\frac{q^2}{8\pi\epsilon_0 m \ell} \left(2 + \frac{1}{\sqrt{2}}\right)}$

(D) Speed of the particles can not be found

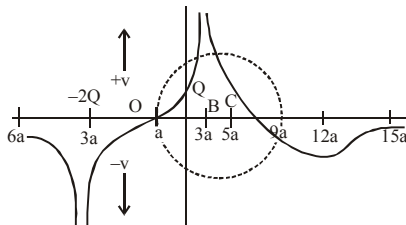
45. In a region of space, the electric field $\vec{E} = E_0 x \hat{i} + E_0 y \hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



- (A) the total electric flux through the faces 1 and 3 is $E_0 a^3$
 (B) the charge inside the cubical volume is $2\epsilon_0 E_0 a^3$
 (C) the total electric flux through the faces 2 and 4 is $2E_0 a^3$
 (D) the charge inside the cubical volume is $\epsilon_0 E_0 a^3$

ANSWER KEY

EXERCISE (S-2)

1. Ans. $\frac{\sigma}{\pi \epsilon_0}$ 2. Ans. a 3. Ans. $\pi K d t p R^2$
4. Ans. 3 5. Ans. 8
6. Ans. $v_0 = 3 \text{ m/s}$; K.E. at the origin = $(27 - 10\sqrt{6}) \times 10^{-4} \text{ J}$ approx. $2.5 \times 10^{-4} \text{ J}$
7. Ans. $\lambda R E_0 \hat{i}$ 8. Ans. $\sqrt{4\pi\epsilon_0 K a}$
9. Ans. (i) -Q, +Q (ii) $E_A = 0$, can't be found (iii) can't be found (iv) can't be found (v) 0 (vi) No, induced charge on outer surface]
10. Ans. 180 11. Ans. $F = \frac{Q^2}{32\pi\epsilon_0 R^4} (R^2 - h^2)$
12. Ans. (a) radius = $4a$ & center $(5a, 0)$
- (b) $V = \frac{1}{4\pi\epsilon_0} \left[\frac{-2Q}{|x+3a|} + \frac{Q}{|x-3a|} \right]$
- (c) speed $v = \sqrt{\frac{1}{4\pi\epsilon_0} \times \frac{qQ}{2ma}}$
- 

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. Ans. (A) 2. Ans. (B) 3. Ans. (C) 4. Ans. (A) 5. Ans. (A) 6. Ans. (B)
7. Ans. (A) 8. Ans. (B) 9. Ans. (A) 10. Ans. (B) 11. Ans. (D) 12. Ans. (B)
13. Ans. (B) 14. Ans. (A) 15. Ans. (A) 16. Ans. (C) 17. Ans. (A)

MULTIPLE CORRECT TYPE QUESTIONS

18. Ans. (A, C) 19. Ans. (A, B, C, D) 20. Ans. (A, D) 21. Ans. (A, B, C)

COMPREHENSION TYPE QUESTIONS

22. Ans. (B) 23. Ans. (C)

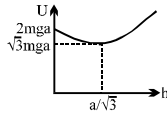
MATRIX MATCH TYPE QUESTION

24. Ans. (A) P, R, S (B) P, Q, R, S, T (C) Q, T (D) R, S

EXERCISE-(JA)

1. Ans. (C) 2. Ans. $V' = \left(\frac{a}{3t}\right)^{1/3} V$ 3. Ans. (A,B,C,D) 4. Ans. (A) 5. Ans. (D)
6. Ans. (B) 7. Ans. (C) 8. Ans. (C) 9. Ans. (A) 10. Ans. (B)
11. Ans. (C) 12. Ans. (A) 13. Ans. (B) 14. Ans. (A) 15. Ans. 2
16. Ans. (A) 17. Ans. (D) 18. Ans. (A,D) 19. Ans. (C)
20. Ans. (A,B,C,D) 21. Ans. (A) 22. Ans. (C,D) 23. Ans. 3
24. Ans. (C) 25. Ans. (B) 26. Ans. 6 27. Ans. (A,C,D)
28. Ans. (C) 29. Ans. (D) 30. Ans. (A, B, C) 31. Ans. (B, D) 32. Ans. (C, D)
33. Ans. (C) 34. Ans. (C) 35. Ans. (A) 36. Ans. 6 37. Ans. (C)
38. Ans. (D)

REVIEW TEST

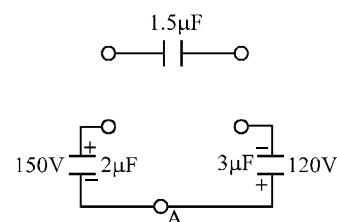
1. Ans. 2 2. Ans. $2 \tan^{-1} \left(\frac{\sigma q}{2\epsilon_0 mg} \right)$ 3. Ans. 20
4. Ans. $\left[\frac{2KQq}{mR} \left(\frac{r-R}{r} + \frac{3}{8} \right) \right]^{1/2}$
5. Ans. (a) $K.E = \frac{P}{4\pi\epsilon_0} \frac{Q}{d^2}$, (b) $\frac{QP}{2\pi\epsilon_0 d^3}$ along positive x-axis
6. Ans. 3 7. Ans. 6
8. Ans. $W_{\text{first step}} = \left(\frac{8}{3} - \frac{4}{\sqrt{5}} \right) \frac{Kq^2}{r}$, $W_{\text{second step}} = 0$, $W_{\text{total}} = 0$
9. Ans. (a) $H = \frac{4a}{3}$, (b) $U = mg \left[2\sqrt{h^2 + a^2} - h \right]$ equilibrium at $h = \frac{a}{\sqrt{3}}$, 
10. Ans. (A)
11. Ans. (D) 12. Ans. (A) 13. Ans. (D) 14. Ans. (C) 15. Ans. (A)
16. Ans. (D) 17. Ans. (A) 18. Ans. (C) 19. Ans. (A) 20. Ans. (C)
21. Ans. (A) 22. Ans. (A) 23. Ans. (D) 24. Ans. (B) 25. Ans. (A)
26. Ans. (C) 27. Ans. (C) 28. Ans. (A,B,C) 29. Ans. (A,C) 30. Ans. (A,B,C)
31. Ans. (A, D) 32. Ans. (B,C,D) 33. Ans. (A, B) 34. Ans. (A,B,C)
35. Ans. (A) $\rightarrow$ (Q, R) ; (B) $\rightarrow$ (Q,T) ; (C) $\rightarrow$ (S,T) (D) $\rightarrow$ (P)
36. Ans. A $\rightarrow$ Q; B $\rightarrow$ P, S; C $\rightarrow$ P, Q, R; D $\rightarrow$ Q, R
37. Ans. (B) 38. Ans. (C) 39. Ans. (C) 40. Ans. (C) 41. Ans. (C)

CAPACITANCE

EXERCISE (O-2)

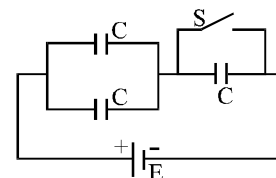
1. Two capacitors of $2\ \mu\text{F}$ and $3\ \mu\text{F}$ are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity $1.5\ \mu\text{F}$ falls to the free ends of the wire. Then

- (A) charge on the $1.5\ \mu\text{F}$ capacitor is $180\ \mu\text{C}$
(B) charge on the $2\ \mu\text{F}$ capacitor is $120\ \mu\text{C}$
(C) positive charge flows through A from right to left.
(D) positive charge flows through A from left to right.



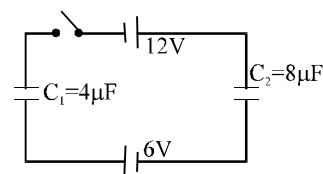
2. In the circuit shown, each capacitor has a capacitance C . The emf of the cell is E . If the switch S is closed

- (A) positive charge will flow out of the positive terminal of the cell
(B) positive charge will enter the positive terminal of the cell
(C) the amount of charge flowing through the cell will be CE .
(D) the amount of charge flowing through the cell will be $\frac{4}{3} CE$.



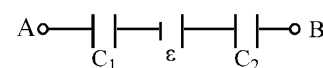
3. In the circuit shown initially C_1 & C_2 are uncharged. After closing the switch

- (A) The charge on C_2 is greater than that on C_1
(B) The charge on C_1 and C_2 are the same
(C) The potential drops across C_1 and C_2 are the same
(D) The potential drops across C_2 is greater than that across C_1



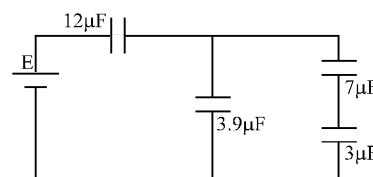
4. A circuit shown in the figure consists of a battery of emf 10 V and two capacitance C_1 and C_2 of capacitances $1.0\ \mu\text{F}$ and $2.0\ \mu\text{F}$ respectively. The potential difference $V_A - V_B$ is 5V

- (A) charge on capacitor C_1 is equal to charge on capacitor C_2
(B) Voltage across capacitor C_1 is 5V.
(C) Voltage across capacitor C_2 is 10 V
(D) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2 .

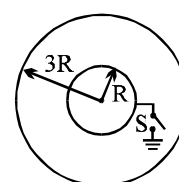


5. Four capacitors and a battery are connected as shown. The potential drop across the $7\ \mu\text{F}$ capacitor is 6 V. Then the :

- (A) potential difference across the $3\ \mu\text{F}$ capacitor is 10 V
(B) charge on the $3\ \mu\text{F}$ capacitor is $42\ \mu\text{C}$
(C) e.m.f. of the battery is 30 V
(D) potential difference across the $12\ \mu\text{F}$ capacitor is 10 V.



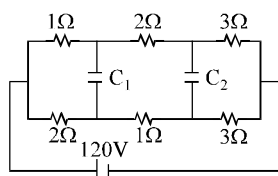
6. The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k . The capacitor is connected to a cell of emf E , and the slab is taken out
- (A) charge $CE(k - 1)$ flows through the cell
 (B) energy $E^2C(k - 1)$ is absorbed by the cell.
 (C) the energy stored in the capacitor is reduced by $E^2C(k - 1)$
 (D) the external agent has to do $\frac{1}{2}E^2C(k - 1)$ amount of work to take the slab out.
7. A parallel plate air-core capacitor is connected across a source of constant potential difference. When a dielectric plate is introduced between the two plates then :
- (A) some charge from the capacitor will flow back into the source.
 (B) some extra charge from the source will flow back into the capacitor.
 (C) the electric field intensity between the two plate does not change.
 (D) the electric field intensity between the two plates will decrease.
8. A parallel plate capacitor of plate area A and plate separation d is charged to potential difference V and then the battery is disconnected. A slab of dielectric constant K is then inserted between the plates of the capacitor so as to fill the space between the plates. If Q , E and W denote respectively, the magnitude of charge on each plate, the electric field between the plates (after the slab is inserted) and the work done on the system, in question, in the process of inserting the slab, then
- (A) $Q = \frac{\epsilon_0 AV}{d}$ (B) $Q = \frac{\epsilon_0 KAV}{d}$
 (C) $E = \frac{V}{Kd}$ (D) $W = -\frac{\epsilon_0 AV^2}{2d} \left(1 - \frac{1}{K}\right)$
9. A parallel plate capacitor has a parallel slab of copper inserted between and parallel to the two plates, without touching the plates. The capacity of the capacitor after the introduction of the copper sheet is :
- (A) minimum when the copper slab touches one of the plates.
 (B) maximum when the copper slab touches one of the plates.
 (C) invariant for all positions of the slab between the plates.
 (D) greater than that before introducing the slab.
10. Two thin conducting shells of radii R and $3R$ are shown in the figure. The outer shell carries a charge $+Q$ and the inner shell is neutral. The inner shell is earthed with the help of a switch S .
- (A) With the switch S open, the potential of the inner sphere is equal to that of the outer.
 (B) When the switch S is closed, the potential of the inner sphere becomes zero.
 (C) With the switch S closed, the charge attained by the inner sphere is $-Q/3$.
 (D) By closing the switch the capacitance of the system increases.



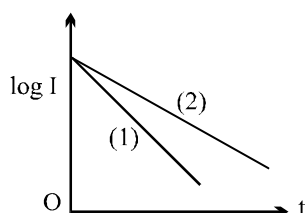
11. A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges $+Q$ and $-Q$ respectively. A third plate C, identical to A and B, with charge $+Q$, is now introduced midway between A and B, parallel to them. Which of the following are correct?

- (A) The charge on the inner face of B is now $-\frac{3Q}{2}$
 (B) There is no change in the potential difference between A and B.
 (C) The potential difference between A and C is one-third of the potential difference between B and C.
 (D) The charge on the inner face of A is now $Q/2$.

12. In the circuit shown in figure $C_1 = C_2 = 2\mu\text{F}$. Then charge stored in

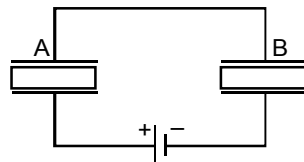


- (A) capacitor C_1 is zero
 (B) capacitor C_2 is zero
 (C) both capacitor is zero
 (D) capacitor C_1 is $40\mu\text{C}$
13. A capacitor of capacity C is charged to a steady potential difference V and connected in series with an open key and a pure resistor 'R'. At time $t = 0$, the key is closed. If $I =$ current at time t , a plot of $\log I$ against ' t ' is as shown in (1) in the graph. Later one of the parameters i.e. V , R or C is changed keeping the other two constant, and the graph (2) is recorded. Then

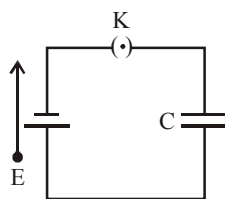


- (A) C is reduced
 (B) C is increased
 (C) R is reduced
 (D) R is increased
14. A dielectric slab fills the space between the plates of a parallel-plate capacitor. The magnitude of the bound charge on the slab is 75% of the magnitude of the free charge on the plates. The capacitance of the capacitor is $480\mu\text{F}$ with the slab inserted. The maximum charge that can be stored on the capacitor is $240\epsilon_0 L^2 E_{\text{max}}$, where E_{max} is the breakdown field. Choose the **CORRECT** statement(s):-
- (A) The dielectric constant for the dielectric slab is 4
 (B) Without the dielectric, the capacitance of the capacitor would be $360\mu\text{F}$.
 (C) The plate area is $60 L^2$
 (D) If the dielectric slab is having the same area as the capacitor plate but the width half that of the capacitor, the capacitance would be $192\mu\text{F}$.

15. Figure shows two identical capacitors A & B with identical dielectric inserted between them and are connected to a battery. Now the slab of capacitor B is pulled out with battery connected :

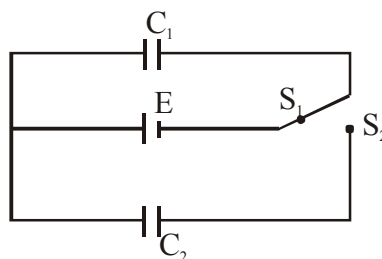


- (A) During the process current flows counter clockwise in the circuit.
 (B) Finally charge on capacitor B will be less than that on capacitor A.
 (C) Electric field in capacitor A reduces in magnitude.
 (D) During the process, internal energy of the battery increases.
16. A parallel plate capacitor is connected to a battery as shown in Fig. 2.5. Consider two situations:
 A: Key K is kept closed and plates of capacitors are moved apart using insulating handle.
 B: Key K is opened and plates of capacitors are moved apart using insulating handle.



Choose the correct option(s).

- (A) In A : Q remains same but C changes.
 (B) In B : V remains same but C changes.
 (C) In A : V remains same and hence Q changes.
 (D) In B : Q remains same and hence V changes.
17. As shown in circuit C_2 is initially uncharged. Now switch is shifted from position S_1 to S_2 then

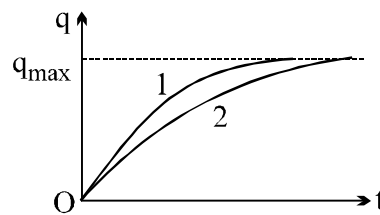


- (A) Charge distribution will not take place
 (B) Charge & energy distribution will take place but total energy & charge of system (circuit) will remain constant
 (C) Charge will decrease
 (D) Circuit will release some energy

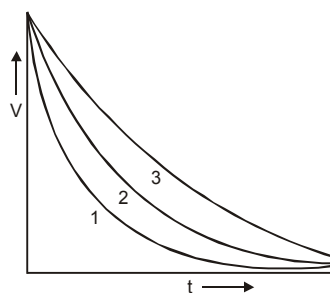
- 18.** A parallel plate capacitor is charged from a cell and then isolated from it. The separation between the plate is now increased
- (A) the force of attraction between the plates will decrease
 - (B) The field in the region between the plates will not change
 - (C) The energy stored in the capacitor will increase
 - (D) The potential difference between the plates will decreases.

Question No.19 to 20 (2 questions)

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.



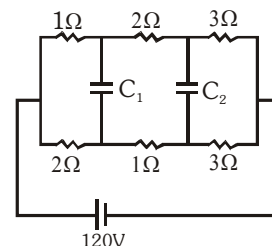
19. Choose the correct statement(s) related to the two circuits.
(A) Both the capacitors are charged to the same charge.
(B) The emf's of cells in both the circuit are equal.
(C) The emf's of the cells may be different.
(D) The emf E_1 is more than E_2
20. Identify the correct statement(s) related to the R_1, R_2, C_1 and C_2 of the two RC circuits.
(A) $R_1 > R_2$ if $E_1 = E_2$
(B) $C_1 < C_2$ if $E_1 = E_2$
(C) $R_1 C_1 > R_2 C_2$
(D) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$
21. A parallel plate capacitor is connected to a battery. The quantities charge, voltage, electric field and energy associated with the capacitor are given by Q_0, V_0, E_0 and U_0 respectively. A dielectric slab is introduced between plates of capacitor but battery is still in connection. The corresponding quantities now given by Q, V, E and U related to previous ones are
(A) $Q > Q_0$ (B) $V > V_0$ (C) $E > E_0$ (D) $U < U_0$
22. Three identical capacitors A, B and C are charged to the same potential and then made to discharge through three resistances R_A, R_B and R_C , where $R_A > R_B > R_C$. Their potential differences (V) are plotted against time t , giving the curves 1, 2 and 3. The relations between A, B, C and 1, 2, 3 is/are-



- (A) $1 \rightarrow A$ (B) $2 \rightarrow B$ (C) $1 \rightarrow C$ (D) $3 \rightarrow A$

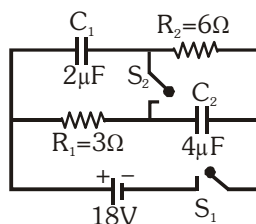
23. In the circuit shown in figure $C_1 = C_2 = 2\mu\text{F}$. Then charge stored in

- (A) capacitor C_1 is zero
(B) capacitor C_2 is zero
(C) both capacitor is zero
(D) capacitor C_1 is $40\mu\text{C}$

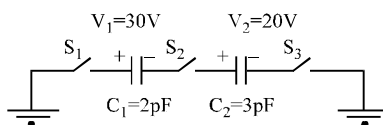


Question No. 24 to 26 (3 questions)

In the circuit shown initially the switches are open and capacitors are uncharged. Switches S_1 and S_2 are closed simultaneously at $t = 0$.



24. Charge on capacitor C_2 is
(A) $12\mu\text{C}$ (B) $24\mu\text{C}$ (C) $48\mu\text{C}$ (D) None of these
25. Now switch S_2 is opened after a long time interval then charge flow through the S_1 is
(A) $12\mu\text{C}$ (B) $24\mu\text{C}$ (C) $36\mu\text{C}$ (D) $48\mu\text{C}$
26. In above question the amount of heat dissipated in resistors
(A) $136\mu\text{J}$ (B) $272\mu\text{J}$ (C) $68\mu\text{J}$ (D) None of these
27. For the circuit shown, which of the following statements is true ? [IIT-JEE 1999]

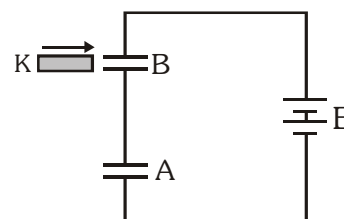


- (A) with S_1 closed, $V_1 = 15\text{ V}$, $V_2 = 20\text{ V}$
(B) with S_3 closed, $V_1 = V_2 = 25\text{ V}$
(C) with S_1 & S_2 closed, $V_1 = V_2 = 0$
(D) with S_1 & S_2 closed, $V_1 = 30\text{ V}$, $V_2 = 20\text{ V}$
28. Two identical capacitors, have the same capacitance C . One of them is charged to potential V_1 and the other to V_2 . The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is [IIT-JEE 2002 (Scr)]

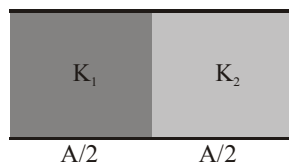
- (A) $\frac{1}{4}C(V_1^2 - V_2^2)$ (B) $\frac{1}{4}C(V_1^2 + V_2^2)$
(C) $\frac{1}{4}C(V_1 - V_2)^2$ (D) $\frac{1}{4}C(V_1 + V_2)^2$

Question No. 29 to 33 (4 questions)

Two identical capacitor A and B, each of capacitance 'C' are connected in series. The combination connected to a battery of emf E. A dielectric slab of dielectric constant K is introduced between the plates of capacitor B to cover entire space between the plates.



29. After introduction of dielectric slab in B, the ratio of capacitance of A and B is
(A) 1:1 (B) 1:K (C) K:1 (D) 1: $\sqrt{K}$
30. After introduction of dielectric slab in B, the ratio of potential differences across A and B will be
(A) 1:1 (B) 1:K (C) K:1 (D) 1: $\sqrt{K}$
31. The ratio of potential differences across A before and after the introduction of dielectric slab in B will be
(A) 1:1 (B) 1:K (C) (K+1):2 (D) K+1:2K
32. The ratio of potential difference across B before and after the introduction of dielectric slab in B will be
(A) 1:1 (B) K:1 (C) (K+1):2 (D) K+1:2K
33. The ratio of energy stored in capacitors A and B after the introduction of dielectric slab in B is
(A) 1:1 (B) 1:K (C) K:1 (D) (K+1)²:K²
34. Inside a parallel plate capacitor of capacitance C (without dielectrics) dielectrics are filled which have dielectric constant k_1 and k_2 such that $k_1 > k_2$. The capacitor is now connected to a battery of voltage V then :-



(A) New capacitance of the capacitor will be $\left(\frac{k_1 + k_2}{2} \right) C$

- (B) Field in dielectric 1 will be more than the field in dielectric 2
(C) Surface charge density on the capacitor plates will be uniform
(D) Field in both the dielectric regions will be the same.

35. Column-I

- (A) Plates of an isolated, charged, parallel plate, air core capacitor are slowly pulled apart.
(B) A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor to completely fill the space between plates.
(C) Plates of a parallel plate capacitor connected across a battery are slowly pulled apart.

Column-II

- (P) Electric energy stored inside capacitor increases in the process.
(Q) Force between the two plates of the capacitor remain unchanged.
(R) Electric field in the region between plates remain unchanged.

- (D) A dielectric slab is slowly inserted inside a parallel plate capacitor connected across a battery to completely fill the space between plates.

- (S) Total electric energy stored inside capacitor decreases in the process.

- (T) Electric field in the region decreases.

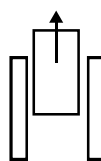
36. Column-I shows some parameters after completion of event and Column-II shows the arrangements to which these parameters belong.

Column-I

- (A) Work done by external agent is negative

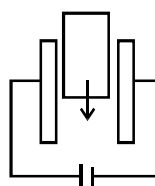
Column-II

- (P) Dielectric is being removed slowly from a charged capacitor



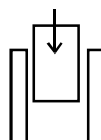
- (B) Electric field between plates increases

- (Q) Dielectric being introduced slowly while battery is connected



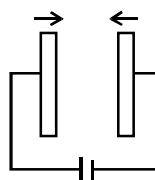
- (C) Energy of capacitor decreases

- (R) Dielectric is being introduced slowly in a charged capacitor

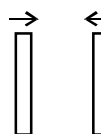


- (D) Force between plates remains constant

- (S) Plates of capacitor are being slowly moved closer



- (T) Plates of capacitor are being moved slowly in a charged capacitor

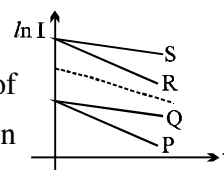


EXERCISE (J-A)

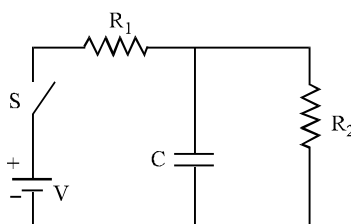
(Not to be discussed in class)

ALL ANSWERS OF THE QUESTIONS **JEE-2007** ONWARDS ARE AS GIVEN ON **IIT-JEE** WEBSITE.

1. In an RC circuit while charging, the graph of $\ln I$ versus time is as shown by the dotted line in the adjoining diagram where I is the current. When the value of the resistance is doubled, which of the solid curves best represents the variation of $\ln I$ versus time? [IIT-JEE' 2004 (Scr)]



- (A) P (B) Q (C) R (D) S
2. In the given circuit, the switch S is closed at time $t = 0$. The charge Q on the capacitor at any instant t is given by $Q(t) = Q_0 (1 - e^{-\alpha t})$. Find the value of Q_0 and α in terms of given parameters shown in the circuit. [IIT-JEE 2005]

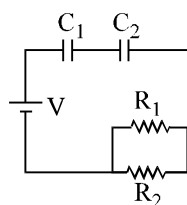


3. An uncharged capacitor of capacitance $4\mu\text{F}$, a battery of emf 12 volt and a resistor of $2.5\text{ M}\Omega$ are connected in series. The time after which $v_c = 3v_R$ is (take $\ln 2 = 0.693$) [IIT-JEE' 2005 (Scr)]

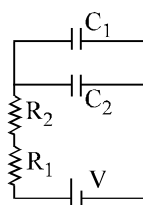
(A) 6.93 sec. (B) 13.86 sec. (C) 20.52 sec. (D) none of these

4. Given : $R_1 = 1\Omega$, $R_2 = 2\Omega$, $C_1 = 2\mu\text{F}$, $C_2 = 4\mu\text{F}$ [IIT-JEE 2006]

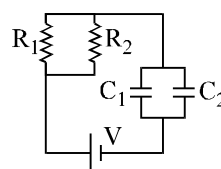
The time constants (in μs) for the circuits I, II, III are respectively



(I)



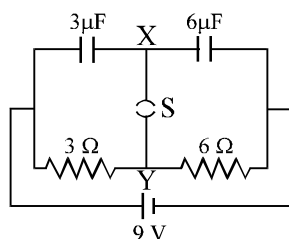
(II)



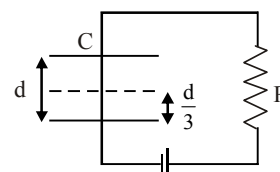
(III)

(A) 18, 8/9, 4 (B) 18, 4, 8/9 (C) 4, 8/9, 18 (D) 8/9, 18, 4

5. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is [IIT-JEE 2007]

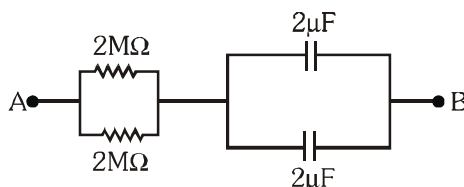


- (A) 0 (B) $54 \mu\text{C}$ (C) $27 \mu\text{C}$ (D) $81 \mu\text{C}$
6. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$. The level of liquid is $\frac{d}{3}$ initially. Suppose the liquid level decreases at a constant speed V ,

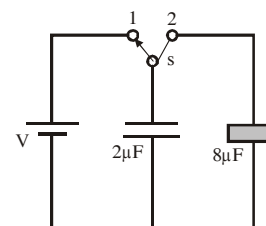


the time constant as a function of time t is [IIT-JEE 2008]

- (A) $\frac{6\epsilon_0 R}{5d + 3Vt}$ (B) $\frac{(15d + 9Vt)\epsilon_0 R}{2d^2 - 3dVt - 9V^2t^2}$ (C) $\frac{6\epsilon_0 R}{5d - 3Vt}$ (D) $\frac{(15d - 9Vt)\epsilon_0 R}{2d^2 + 3dVt - 9V^2t^2}$
7. **Statement-1** : For practical, the earth is used as a reference at zero potential in electrical circuits.
and
Statement-2 : The electrical potential of a sphere of radius R with charge Q uniformly distributed on the surface is given by $\frac{Q}{4\pi\epsilon_0 R}$. [IIT-JEE 2008]
- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
 (B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
8. At time $t=0$, a battery of 10V is connected across points A and B in the given circuit. If the capacitors have no charge initially, at what time (in seconds) does the voltage across them become 4V ? [Take : $\ln 5 = 1.6$, $\ln 3 = 1.1$] [IIT-JEE 2010]

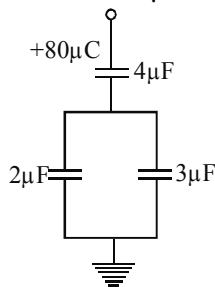


9. A $2 \mu\text{F}$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is [IIT-JEE 2011]

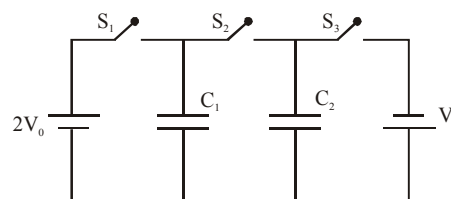


- (A) 0% (B) 20%
 (C) 75% (D) 80%

10. In the given circuit, a charge of $+80\ \mu\text{C}$ is given to the upper plate of the $4\ \mu\text{F}$ capacitor. Then in the steady state, the charge on the upper plate of the $3\ \mu\text{F}$ capacitor is [IIT-JEE 2012]

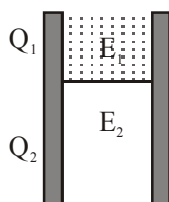


- (A) $+32\ \mu\text{C}$ (B) $+40\ \mu\text{C}$ (C) $+48\ \mu\text{C}$ (D) $+80\ \mu\text{C}$
11. In the circuit shown in the figure, there are two parallel plate capacitors each of the capacitance C . The switch S_1 is pressed first to fully charge the capacitor C_1 and then released. The switch S_2 is then pressed to charge the capacitor C_2 . After some time, S_2 is released and then S_3 is pressed, After some time,



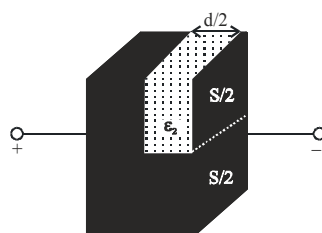
[JEE-Advance 2013]

- (A) the charge on the upper plate of C_1 is $2CV_0$ (B) the charge on the upper plate of C_1 is CV_0
(C) the charge on the upper plate of C_2 is 0. (D) the charge on the upper plate of C_2 is $-CV_0$
12. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that covers $1/3$ of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_1 . When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2 . The electric field in the dielectric is E_1 and that in the other portion is E_2 . Choose the correct option/options, ignoring edge effects.



[JEE-Advance 2014]

- (A) $\frac{E_1}{E_2} = 1$ (B) $\frac{E_1}{E_2} = \frac{1}{K}$ (C) $\frac{Q_1}{Q_2} = \frac{3}{K}$ (D) $\frac{C}{C_1} = \frac{2+K}{K}$
13. A parallel plate capacitor having plates of area S and plate separation d , has capacitance C_1 in air. When two dielectrics of different relative permittivities ($\epsilon_1 = 2$ and $\epsilon_2 = 4$) are introduced between the two plates as shown in the figure, the capacitance becomes C_2 . The ratio $\frac{C_2}{C_1}$ is :-



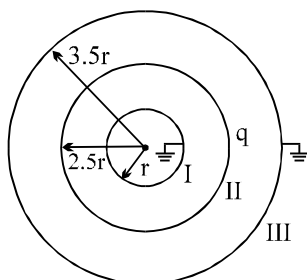
[JEE-Advanced 2015]

- (A) $\frac{6}{5}$ (B) $\frac{5}{3}$ (C) $\frac{7}{5}$ (D) $\frac{7}{3}$

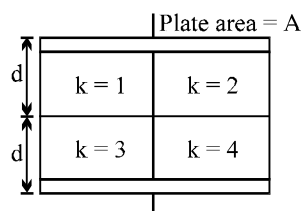
REVIEW TEST

(Not to be discussed in class)

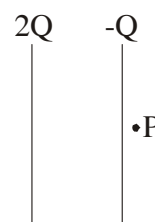
1. Figure shows three concentric conducting spherical shells with inner and outer shells earthed and the middle shell is given a charge q . Find the electrostatic energy of the system stored in the region I and II.



2. Find the capacitance of the system shown in figure.



3. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is 'C'. P is a point outside the capacitor and close to the plate of charge $-Q$. The distance between the plates is 'd' then which statement is wrong



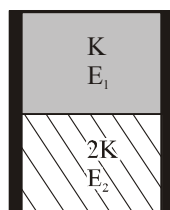
(A) A point charge at point 'P' will experience electric force due to capacitor

(B) The potential difference between the plates will be $\frac{3Q}{2C}$

(C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$

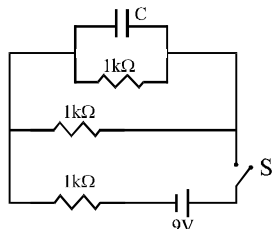
(D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$

4. A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and $2K$ are now introduced in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then

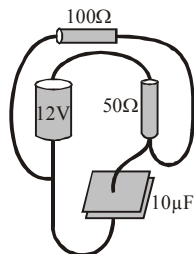


- (A) $E_1 = 2E_2$
 (B) Electrostatic potential energy of upper half is less than that of lower half
 (C) Induced charges on both slabs are same
 (D) Charge distribution on the plates remains same after insertion of dielectric

5. A capacitor $C = 100 \mu\text{F}$ is connected to three resistor each of resistance $1 \text{ k}\Omega$ and a battery of emf 9V . The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant



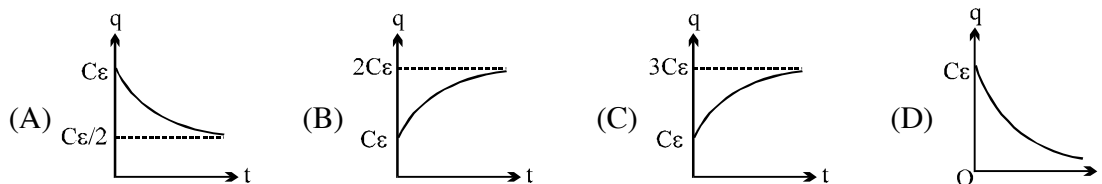
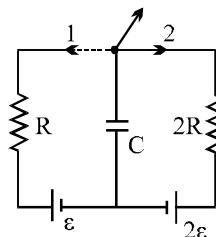
- (A) 33 ms (B) 5 ms (C) 3.3 ms (D) 50 ms
6. In the circuit shown in figure.



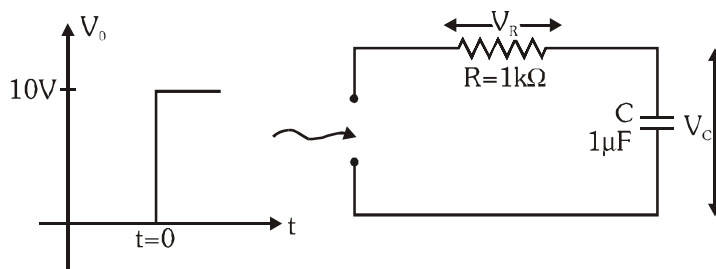
- (A) In steady state, there is no current in the 100Ω resistor
 (B) In steady state, the current in 100Ω resistor is 0.08 A
 (C) In steady state, there is no current in the 50Ω resistor
 (D) In steady state, the current in 50Ω resistor is 0.04 A
7. A capacitor of capacitance C is charged by a battery whose internal resistance is R . The time after which potential difference across resistor becomes n times to that across capacitor is

- (A) $RC \ln\left(\frac{1+n}{n}\right)$ (B) $RC \ln\left(\frac{n}{1+n}\right)$
 (C) $RC \ln\left(\frac{n}{n-1}\right)$ (D) $RC \ln(1+n)$

8. In the circuit shown, the switch is shifted from position $1 \rightarrow 2$ at $t = 0$. The switch was initially in position 1 since a long time. The graph between charge on capacitor C and time ' t ' is



9. A step voltage V_0 is applied to a series combination of R and C as shown in the figure. Then,



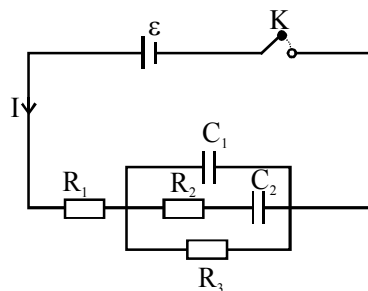
- (A) After sufficiently long time $V_R = 0$ (B) As time passes V_R decreases as $\left(\frac{1}{t}\right)$
 (C) After 1 ms, $V_C = 6.3$ volt (approximately) (D) Initially current through R is 10 mA
10. Mark the **CORRECT** statement(s) regarding the current I through the battery in the circuit shown in figure.

(A) Immediately after the key K is closed, $I = \frac{\varepsilon}{R_1}$

(B) Immediately after the key K is closed, $I = \frac{\varepsilon}{R_1 + R_3}$

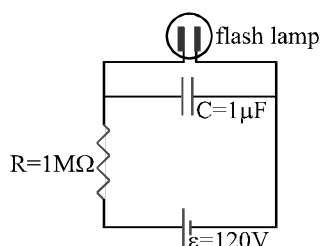
(C) Long time after key K is closed, $I = \frac{\varepsilon}{R_1 + R_3}$

(D) Long time after key K is closed, $I = \frac{\varepsilon}{R_1 + R_2}$



Question No. 11 to 14 (4 questions)

A highway emergency flasher uses a 120 volt battery, a $1 \text{ M}\Omega$ resistor, a $1 \mu\text{F}$ capacitor and a neon flash lamp in the circuit shown in the figure. The flash lamp has a resistance more than $10^{10} \Omega$ when the voltage across it is less than 110V. Above 110 V, the neon gas ionizes, the lamp's resistance drops to 10Ω , and the capacitor discharges completely. Until the capacitor voltage reaches the breakdown voltage $V_b = 110 \text{ V}$, the large resistance of the flash lamp ensures that it draws a negligible current.



The capacitor charges as if the lamp were absent. At V_b , however, the lamp resistance quickly becomes negligible, and the capacitor discharges through the lamp as if the battery and the series resistor were absent. The time between the flashes is the time for the capacitor to charge to V_b . The flash duration is roughly the time for the capacitor to discharge through the lamp, or about 3 time constant of the capacitor–lamp circuit. The flash energy is the stored energy in the capacitor at 110 volt.

11. The flash interval is found by solving for the time when the capacitor voltage is $V_b = 110$ V.

$V_b = \varepsilon(1 - e^{-t/CR})$, $\ln 12 = 2.5$. Flash interval is

- (A) 2 s (B) 2/5 s (C) 5/2 s (D) 1 s

12. Time constant (τ_c) of the capacitor–lamp circuit is–

- (A) 20 μ s (B) 15 μ s (C) 30 μ s (D) 10 μ s

13. Flash duration is

- (A) 10 μ s (B) 20 μ s (C) 30 μ s (D) 5 μ s

14. The energy in the flash is

- (A) 6.1 mJ (B) 6.1 J (C) 3 mJ (D) 12.2 mJ

ANSWER KEY

EXERCISE (O-2)

1. Ans. (A, B, C) 2. Ans. (A, D) 3. Ans. (B) 4. Ans. (A, D) 5. Ans. (B, C, D)
 6. Ans. (A, B, D) 7. Ans. (B, C) 8. Ans. (A, C, D) 9. Ans. (C, D) 10. Ans. (A, B, C, D)
 11. Ans. (A, B, C, D) 12. Ans. (B, D) 13. Ans. (B) 14. Ans. (A, C, D)
 15. Ans. (A, C, D) 16. Ans. (C, D) 17. Ans. (D) 18. Ans. (B, C) 19. Ans. (A, C)
 20. Ans. (D) 21. Ans. (A) 22. Ans. (B, C, D) 23. Ans. (B, D) 24. Ans. (C)
 25. Ans. (D) 26. Ans. (A) 27. Ans. (D) 28. Ans. (C) 29. Ans. (B) 30. Ans. (C)
 31. Ans. (D) 32. Ans. (C) 33. Ans. (C) 34. Ans. (A, D)
 35. Ans. (A) $\rightarrow$ (PQR); (B) $\rightarrow$ (QST); (C) $\rightarrow$ (ST); (D) $\rightarrow$ (PR)
 36. Ans. (A) $\rightarrow$ (Q, R, S, T); (B) $\rightarrow$ (P, S); (C) $\rightarrow$ (R, T); (D) $\rightarrow$ (P, R, T)

EXERCISE (J-A)

1. Ans. (B) 2. Ans. $Q_0 = \frac{CVR_2}{R_1 + R_2}$ and $\alpha = \frac{R_1 + R_2}{CR_1R_2}$ 3. Ans. (B)
 4. Ans. (D) 5. Ans. (C) 6. Ans. (A) 7. Ans. (A)
 8. Ans. (2) 9. Ans. (D) 10. Ans. (C) 11. Ans. (B, D)
 12. Ans. (A, D) 13. Ans. (D)

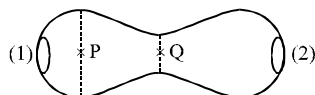
REVIEW TEST

1. Ans. $U_I = \frac{3kq_1^2}{10r}$ where $q_1 = -\frac{4q}{25}$; $U_{II} = 2K(q + q_1)^2/35r$
 2. Ans. $\frac{25 \epsilon_0 A}{24 d}$ 3. Ans. (D) 4. Ans. (B) 5. Ans. (D)
 6. Ans. (B) 7. Ans. (A) 8. Ans. (B) 9. Ans. (A, C, D)
 10. Ans. (A, C) 11. Ans. (C) 12. Ans. (D) 13. Ans. (C)
 14. Ans. (A)

Current Electricity

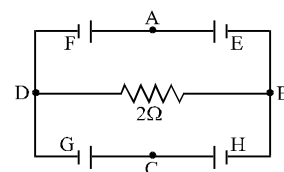
EXERCISE (O-2)

1. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



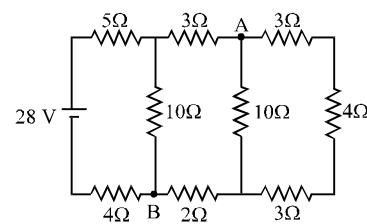
- (A) the current at the cross-section P equals the current at the cross-section Q
 (B) the electric field intensity at P is less than that at Q.
 (C) the rate of heat generated per unit time at Q is greater than that at P
 (D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.
2. Which of the following quantities do not change when an ohmic resistor connected to a battery is heated due to the current?
 (A) drift speed (B) resistivity (C) resistance (D) number of free electrons
3. A current passes through an ohmic conductor of nonuniform cross section. Which of the following quantities are independent of the cross-section?
 (A) the charge crossing in a given time interval.
 (B) drift speed
 (C) current density
 (D) free-electron density

4. In the circuit shown E, F, G and H are cells of e.m.f. 2V, 1V, 3V and 1V respectively and their internal resistances are 2Ω , 1Ω , 3Ω and 1Ω respectively.

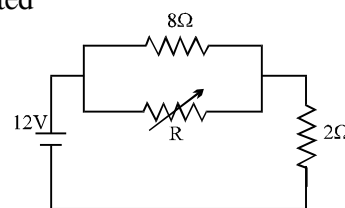
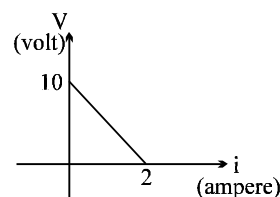


- (A) $V_D - V_B = -2/13$ V
 (B) $V_D - V_B = 2/13$ V
 (C) $V_G = 21/13$ V = potential difference across G.
 (D) $V_H = 19/13$ V = potential difference across H.

5. Consider the circuit shown in the figure
 (A) the current in the 5Ω resistor is 2 A
 (B) the current in the 5Ω resistor is 1 A
 (C) the potential difference $V_A - V_B$ is 10 V
 (D) the potential difference $V_A - V_B$ is 5 V



6. A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r .
- (A) Potential difference across points A and B must be more than E .
 (B) A must be at higher potential than B
 (C) In battery, current flows from positive terminal to the negative terminal
 (D) No current flows through battery
7. A simple circuit contains an ideal battery and a resistance R . If a second resistor is placed in parallel with the first,
- (A) the potential across R will decrease
 (B) the current through R will decreased
 (C) the current delivered by the battery will increase
 (D) the power dissipated by R will increased.
8. The equivalent resistance of a group of resistances is R . If another resistance is connected in parallel to the group, its new equivalent becomes R_1 & if it is connected in series to the group, its new equivalent becomes R_2 we have :
- (A) $R_1 > R$ (B) $R_1 < R$ (C) $R_2 > R$ (D) $R_2 < R$
9. Two identical fuses are rated at 10A. If they are joined
- (A) in parallel, the combination acts as a fuse of rating 20A
 (B) in parallel, the combination acts as a fuse of rating 5A
 (C) in series, the combination acts as a fuse of rating 10A.
 (D) in series, the combination acts as a fuse of rating 20A.
10. A battery of emf E and internal resistance r is connected across a resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative(s).
- (A) internal resistance of battery is 5Ω
 (B) emf of the battery is 20V
 (C) maximum current which can be taken from the battery is 4A
 (D) $V-i$ graph can never be a straight line as shown in figure.
11. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition
- (A) $R = 0$
 (B) $R = 8\Omega$
 (C) power dissipated in the 2Ω resistor is 72 W.
 (D) power dissipated in the 2Ω resistor is 8 W.



12. A galvanometer may be converted into ammeter or voltmeter. In which of the following cases the resistance of the device will be the largest ? (Assume maximum range of galvanometer = 1 mA)

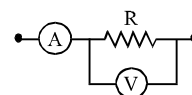
(A) an ammeter of range 10A (B) a voltmeter of range 5 V
(C) an ammeter of range 5 A (D) a voltmeter of range 10 V.

13. Mark out the correct options.

(A) An ammeter should have small resistance. (B) An ammeter should have large resistance.
(C) A voltmeter should have small resistance. (D) A voltmeter should have large resistance.

14. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :

(A) 5Ω
(B) less than 5Ω
(C) greater than 5Ω
(D) between 4Ω & 5Ω



15. A micrometer has a resistance of 100Ω and a full scale range of $50\mu\text{A}$. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).

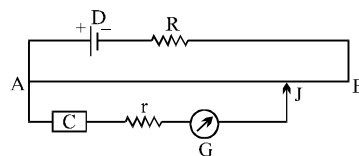
(A) 50 V range with $10\text{ k}\Omega$ resistance in series. (B) 10 V range with $200\text{ k}\Omega$ resistance in series.
(C) 5 mA range with 1Ω resistance in parallel. (D) 10 mA range with $1\text{ k}\Omega$ resistance in parallel.

16. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20V and its internal resistance is 5Ω . There is a resistance box in series with the battery and the potentiometer wire, whose resistance can be varied from 120Ω to 170Ω . Resistance of the potentiometer wire is 75Ω . The following potential differences can be measured using this potentiometer.

(A) 5V (B) 6V (C) 7V (D) 8V

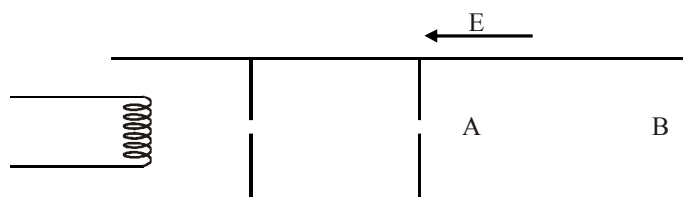
17. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0 . C is a cell of internal resistance r .

The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this ?



(A) $r > R_0$ (B) $R > R_0$
(C) emf of C > emf of D (D) The negative terminal of C is connected to A.

18. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electron move from A to B :

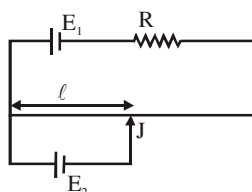
**Column-I**

- (A) Speed of an electron
 (B) Number of free electrons per unit volume
 (C) Current density
 (D) Electric potential

Column-II

- (P) Increases
 (Q) Decreases
 (R) Remains same
 (S) any of the above is possible

19. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ .

**Column I**

- (A) If E_1 is increased
 (B) If R is increased
 (C) If E_2 is increased

Column II

- (P) ℓ should increase
 (Q) ℓ should decrease
 (R) ℓ should remain the same to again get the null point

Paragraph

Important aspect of fuse wire and battery

Electric fuse is a protective device used in series with an electric circuit or an electric appliance to save it from damage due to overheating produced by strong current in the circuit or appliance. Fuse wire is generally made from an alloy of lead and tin which has high resistance and low melting point. It is connected in series in an electric installation. If a circuit gets accidentally short-circuited, a large current flows, then fuse wire melts away which causes a break in the circuit.

The power through fuse (P) is equal to heat energy lost per unit area per unit time (H) (neglecting

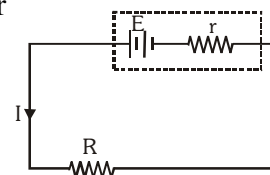
heat losses from ends of the wire). $P = I^2 R = H \times 2\pi r \ell$ $\left[R = \frac{\rho \ell}{\pi r^2} \right]$ r and ℓ are the length

and radius of fuse wire. A battery is described by its Emf (E) and internal resistance r .) Efficiency of battery is defined as the ratio of the output power and the input power

$$\eta = \frac{\text{output power}}{\text{input power}} \times 100 \% \quad \text{but } I = \frac{E}{R+r}$$

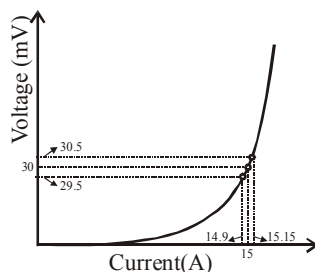
$$\text{input power} = EI, \text{ output power} = EI - I^2 r$$

$$\text{then } \eta = \left(\frac{EI - I^2 r}{EI} \right) \times 100 = \left(1 - \frac{Ir}{E} \right) \times 100 = 1 - \left(\frac{E}{R+r} \right) \left(\frac{r}{E} \right) \times 100 \quad \eta = \left(\frac{R}{R+r} \right) \times 100$$



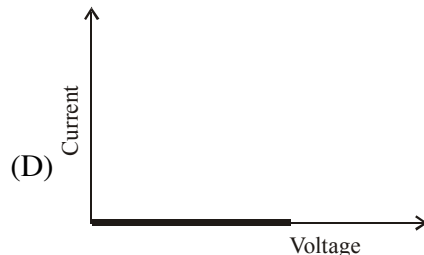
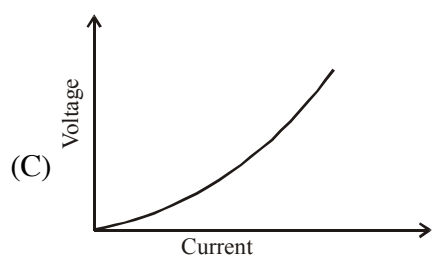
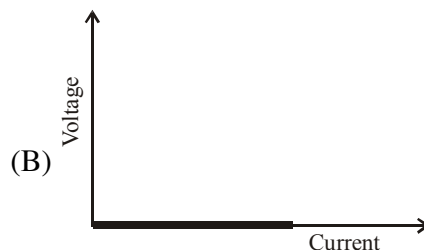
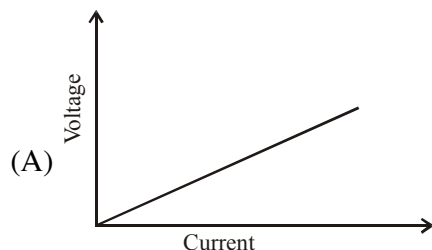
We know that output power of a source is maximum when the external resistance is equal to internal resistance, i.e., $R = r$.

20. Two fuse wire of same material are having length ratio 1 : 2 and radius ratio 4 : 1. Then respective ratio of their current rating will be—
(A) 8 : 1 (B) 2 : 1 (C) 1 : 8 (D) 4 : 1
21. The maximum power rating of a 20.0Ω fuse wire is 2.0 KW, then this fuse wire can be connected safely to a D.C. source (negligible internal resistance) of—
(A) 300 volt (B) 190 volt (C) 250 volt (D) 220 volt
22. Efficiency of a battery (non-ideal) when delivering maximum power is—
(A) 100 % (B) 50 % (C) 90 % (D) 40 %
23. Calculate the resistance of the material whose V-I graph is shown at 15 A.

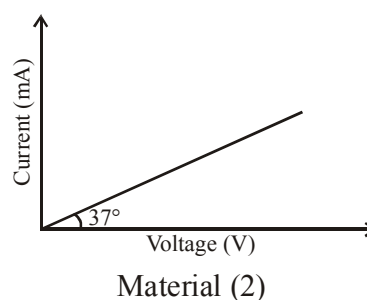
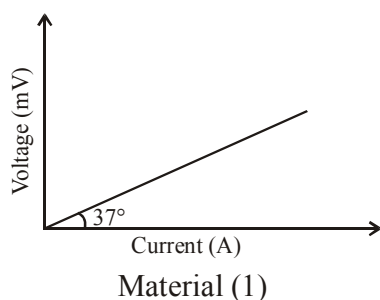


- (A) 4 Ω (B) 4 m Ω (C) 4 k Ω (D) 250 Ω

24. Which of the following V-I characteristic is most correct for a super-conductor (Superconductors are substance which offer no resistance to the flow of current) ?



25. For two different materials V-I characteristic is drawn. What is the ratio of resistances of these two materials ?



- (A) 1 (B) 10^{-6} (C) $\frac{16}{9} \times 10^{-6}$ (D) $\frac{9}{16} \times 10^{-6}$

26. Temperature dependence of resistivity $\rho(T)$ of semiconductors, insulators and metals is significantly based on the following factors:

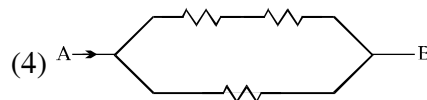
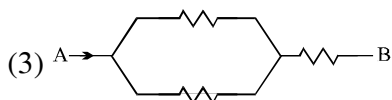
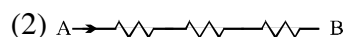
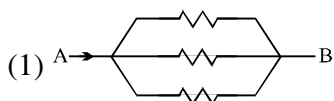
- (A) number of charge carriers can change with temperature T .
- (B) time interval between two successive collisions can depend on T .
- (C) length of material can be a function of T .
- (D) mass of carriers is a function of T .

EXERCISE - JA

(Not to be discussed in class)

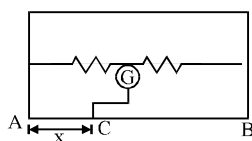
ALL ANSWERS OF THE QUESTIONS **JEE-2007** ONWARDS ARE AS GIVEN ON **IIT-JEE** WEBSITE.

1. Arrange the order of power dissipated in the given circuits, if the same current is passing through all circuits and each resistor is 'r' [IIT-JEE' 2003 (Scr)]



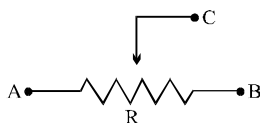
(A) $P_2 > P_3 > P_4 > P_1$ (B) $P_3 > P_2 > P_4 > P_1$ (C) $P_4 > P_3 > P_2 > P_1$ (D) $P_1 > P_2 > P_3 > P_4$

2. In the given circuit, no current is passing through the galvanometer. If the cross-sectional diameter of AB is doubled then for null point of galvanometer the value of AC would [IIT-JEE' 2003 (Scr)]

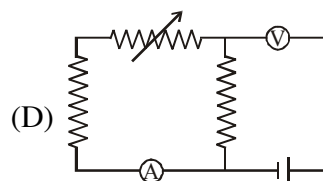
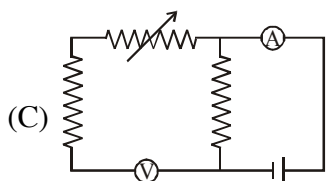
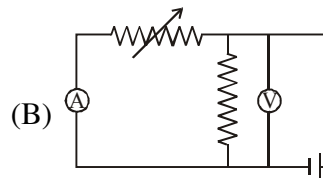
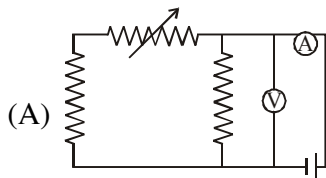


(A) x (B) $x/2$ (C) $2x$ (D) None

3. How a battery is to be connected so that shown rheostat will behave like a potential divider? Also indicate the points about which output can be taken. [IIT-JEE' 2003]

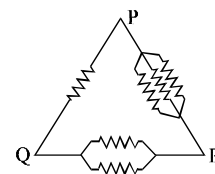


4. Which of the following set-up can be used to verify Ohm's law ? [IIT-JEE' 2003]



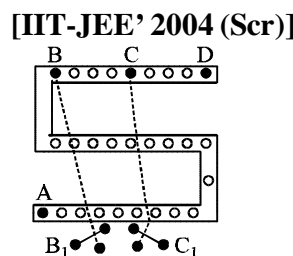
5. Six equal resistances are connected between points P, Q and R as shown in the figure. Then the net resistance will be maximum between :-
[IIT-JEE' 2004 (Scr)]

- (A) P and Q
 (B) Q and R
 (C) P and R
 (D) any two points



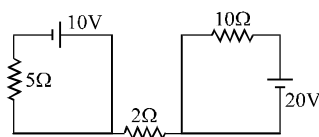
6. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between :-
[IIT-JEE' 2004 (Scr)]

- (A) B and C
 (B) C and D
 (C) A and D
 (D) B₁ and C₁



7. Draw the circuit for experimental verification of Ohm's law using a source of variable D.C. voltage, a main resistance of $100\ \Omega$, two galvanometers and two resistances of values $10^6\ \Omega$ and $10^{-3}\ \Omega$ respectively. Clearly show the positions of the voltmeter and the ammeter.
[IIT-JEE' 2004]

8. In the figure shown the current through $2\ \Omega$ resistor is
[IIT-JEE 2005 (Scr)]



- (A) 2 A (B) 0 A (C) 4 A (D) 6 A

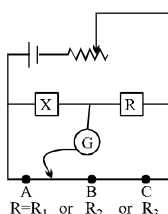
9. Which of the following groups do not have same dimensions
[IIT JEE' 2005 (Scr)]

- (A) Young's modulus, pressure, stress
 (B) work, heat, energy
 (C) electromotive force, potential difference, voltage
 (D) electric dipole, electric flux, electric field

10. A galvanometer has resistance $100\ \Omega$ and it requires current $100\ \mu\text{A}$ for full scale deflection. A resistor $0.1\ \Omega$ is connected to make it an ammeter. The smallest current required in the circuit to produce the full scale deflection is
[IIT-JEE 2005 (Scr)]

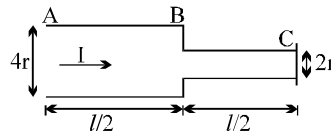
- (A) 1000.1 mA (B) 1.1 mA (C) 10.1 mA (D) 100.1 mA

11. An unknown resistance X is to be determined using resistances R_1 , R_2 or R_3 . Their corresponding null points are A, B and C. Find which of the above will give the most accurate reading and why?
[IIT-JEE 2005]



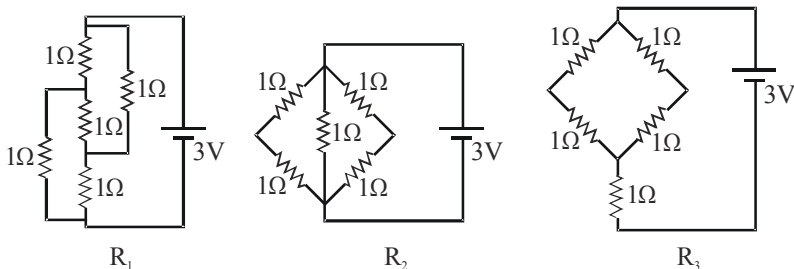
12. A rigid container with thermally insulated walls contains a coil of resistance $100\ \Omega$, carrying current 1 A. Change in internal energy after 5 min will be [IIT-JEE 2005]
(A) zero (B) 10 kJ (C) 20 kJ (D) 30 kJ

13. Consider a cylindrical element as shown in the figure. Current flowing through the element is I and resistivity of material of the cylinder is ρ . Choose the correct option out the following. [IIT-JEE 2006]
(A) Power loss in second half is four times the power loss in first half.
(B) Voltage drop in first half is twice of voltage drop in second half.
(C) Current density in both halves are equal.
(D) Electric field in both halves is equal.



14. A resistance of $2\ \Omega$ is connected across one gap of a metre-bridge (the length of the wire is 100 cm) and an unknown resistance, greater than $2\ \Omega$, is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any corrections, the unknown resistance is [IIT-JEE 2007]
(A) $3\ \Omega$ (B) $4\ \Omega$ (C) $5\ \Omega$ (D) $6\ \Omega$

15. Figure shows three resistor configurations R_1 , R_2 and R_3 connected to 3V battery. If the power dissipated by the configuration R_1 , R_2 and R_3 is P_1 , P_2 and P_3 respectively, then [IIT-JEE 2008]



- (A) $P_1 > P_2 > P_3$ (B) $P_1 > P_3 > P_2$ (C) $P_2 > P_1 > P_3$ (D) $P_3 > P_2 > P_1$

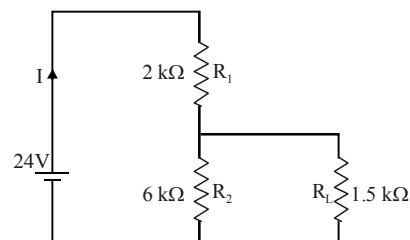
16. **Statement-1 :** In a Meter Bridge experiment, null point for an unknown resistance is measured. Now, the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance.
and [IIT-JEE 2008]

Statement-2 : Resistance of a metal increases with increase in temperature.

- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

17. For the circuit shown in the figure [IIT-JEE 2009]

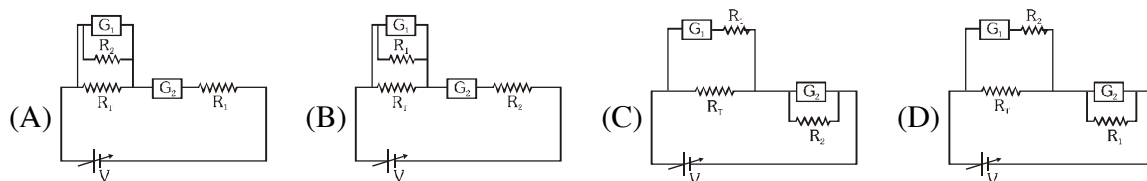
- (A) the current I through the battery is 7.5 mA
(B) the potential difference across R_L is 18V
(C) ratio of powers dissipated in R_1 and R_2 is 3
(D) if R_1 and R_2 are interchanged, magnitude of the power dissipated in R_L will decrease by a factor of 9



18. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistance R_{100} , R_{60} and R_{40} , respectively, the relation between these resistances is [IIT-JEE 2010]

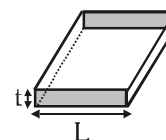
(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (B) $R_{100} = R_{40} + R_{60}$ (C) $R_{100} > R_{60} > R_{40}$ (D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

19. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is :- [IIT-JEE 2010]



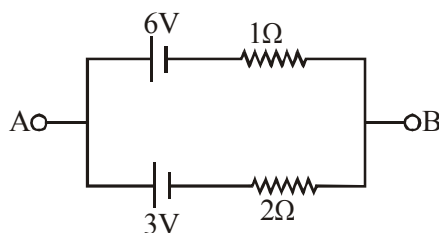
20. Consider a thin square sheet of side L and thickness t , made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is [IIT-JEE 2010]

- (A) directly proportional to L
 (B) directly proportional to t
 (C) independent of L
 (D) independent of t

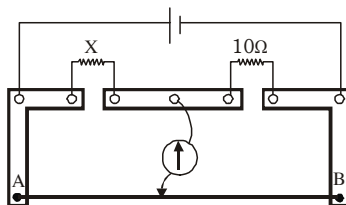


21. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R , the rate of heat produced in R is J_1 . When the same batteries are connected in parallel across R , the rate is J_2 . If $J_1 = 2.25 J_2$ then the value of R in Ω is [IIT-JEE 2010]

22. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volts is [IIT-JEE 2011]



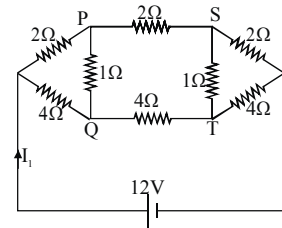
23. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10Ω resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is [IIT-JEE 2011]



- (A) 10.2 ohm (B) 10.6 ohm (C) 10.8 ohm (D) 11.1 ohm

24. For the resistance network shown in the figure, choose the correct option(s). [IIT-JEE 2012]

- (A) the current through PQ is zero
(B) $I_1 = 3A$
(C) The potential at S is less than that at Q
(D) $I_2 = 2A$

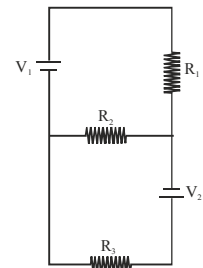


25. Heater of an electric kettle is made of a wire of length L and diameter d . It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter $2d$. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 K ? [JEE Advanced 2014]

- (A) 4 if wires are in parallel (B) 2 if wires are in series
(C) 1 if wires are in series (D) 0.5 if wires are in parallel

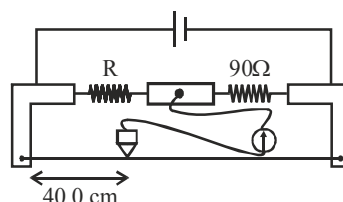
26. Two ideal batteries of emf V_1 and V_2 and three resistances R_1 , R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if :- [JEE Advanced 2014]

- (A) $V_1 = V_2$ and $R_1 = R_2 = R_3$
(B) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$
(C) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$
(D) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$



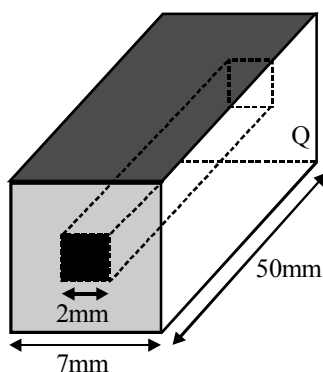
27. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a $4990\ \Omega$ resistance, it can be converted into a voltmeter of range 0 - 30 V. If connected to a $\frac{2n}{249}\ \Omega$ resistance, it becomes an ammeter of range 0 - 1.5 A. The value of n is :- [JEE Advanced 2014]

28. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of $90\ \Omega$, as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is :- [JEE Advanced 2014]

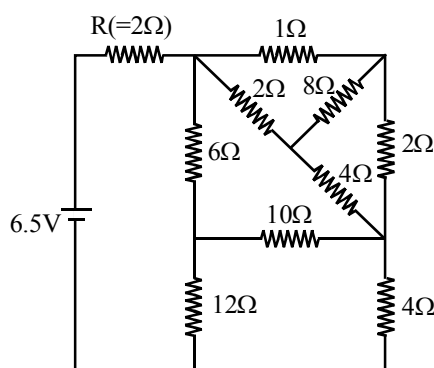


- (A) $60 \pm 0.15\ \Omega$ (B) $135 \pm 0.56\ \Omega$ (C) $60 \pm 0.25\ \Omega$ (D) $135 \pm 0.23\ \Omega$

29. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8} \Omega \text{ m}$ and $1.0 \times 10^{-7} \Omega \text{ m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is : **[JEE Advanced 2015]**



- (A) $\frac{2475}{64} \mu\Omega$ (B) $\frac{1875}{64} \mu\Omega$ (C) $\frac{1875}{49} \mu\Omega$ (D) $\frac{2475}{132} \mu\Omega$
30. In the following circuit, the current through the resistor $R (= 2\Omega)$ is I Amperes. The value of I is **[JEE Advanced 2015]**

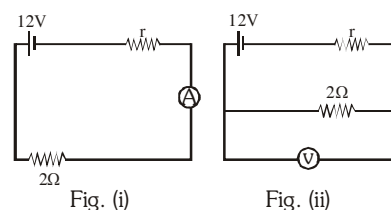


REVIEW TEST

(Not to be discussed in class)

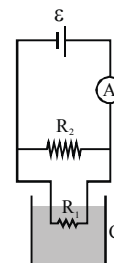
1. An accumulator of emf 2 Volt and negligible internal resistance is connected across a uniform wire of length 10m and resistance 30Ω . The appropriate terminals of a cell of emf 1.5 Volt and internal resistance 1Ω is connected to one end of the wire, and the other terminal of the cell is connected through a sensitive galvanometer to a slider on the wire. What length of the wire will be required to produce zero deflection of the galvanometer? How will the balancing change (a) when a coil of resistance 5Ω is placed in series with the accumulator, (b) the cell of 1.5 volt is shunted with 5Ω resistor?

2. A galvanometer (coil resistance 99Ω) is converted into a ammeter using a shunt of 1Ω and connected as shown in the figure (i). The ammeter reads 3A. The same galvanometer is converted into a voltmeter by connected a resistance of 101Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be $4/5$ of the full scale reading. Find

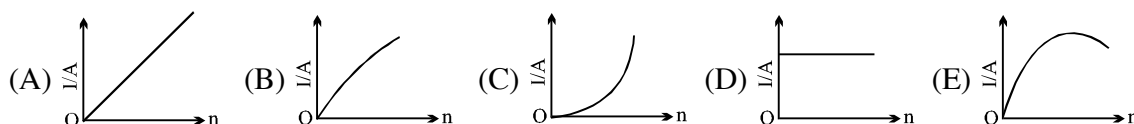


- (i) internal resistance r of the cell
- (ii) range of the ammeter and voltmeter
- (iii) full scale deflection current of the galvanometer

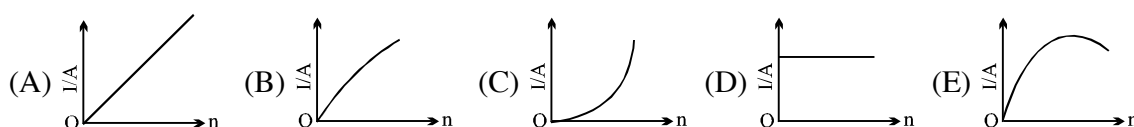
3. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature ($^{\circ}\text{C}$) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flows through the coil and the ammeter shows 3A? The resistance $R_2 = 30\Omega$. [Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = $4200 \text{ J/kg}^{\circ}\text{C}$]



4. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n ?



5. In **previous problem**, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n ?

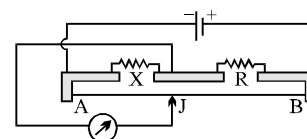


6. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t . N number of similar cells is now connected in series with a wire of the same material and cross section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . The value of N is :

(A) 4 (B) 6 (C) 8 (D) 9

7. The figure shows a metre-bridge circuit, with $AB = 100$ cm, $X = 12\Omega$ and $R = 18\Omega$, and the jockey J in the position of balance. If R is now made 8Ω , through what distance will J have to be moved to obtain balance?

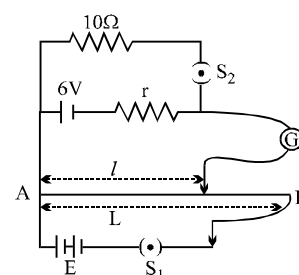
(A) 10 cm (B) 20 cm (C) 30 cm



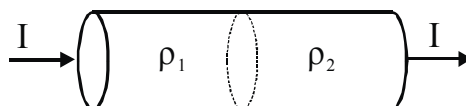
(D) 40 cm

8. In the arrangement shown in figure when the switch S_2 is open, the galvanometer shows no deflection for $l = L/2$. When the switch S_2 is closed, the galvanometer shows no deflection for $l = 5L/12$. The internal resistance (r) of 6 V cell, and the emf E of the other battery are respectively

(A) 3Ω , 8V (B) 2Ω , 12V
(C) 2Ω , 24V (D) 3Ω , 12V

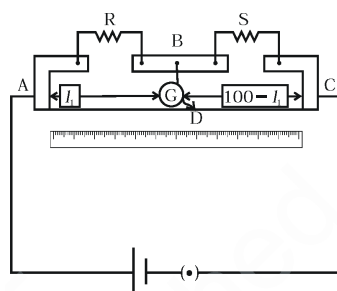


9. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



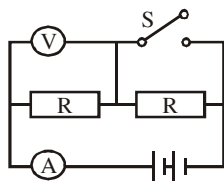
(A) zero (B) $\frac{(\rho_1 - \rho_2) I \epsilon_0}{2}$ (C) $\epsilon_0 I |\rho_1 - \rho_2|$ (D) $\epsilon_0 I |\rho_1 + \rho_2|$

10. In a meter bridge the point D is a neutral point.



- (A) The meter bridge can have no other neutral point for this set of resistances.
(B) When the jockey contacts a point on meter wire left of D, current flows to B from the wire through galvanometer.
(C) When the jockey contacts a point on the meter wire to the right of D, current flows from B to the wire through galvanometer.
(D) When R is increased, the neutral point shifts to left.

11. The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed ?



Column I

- (A) Ammeter reading
(B) Voltmeter reading
(C) Equivalent resistance of circuit
(D) Power dissipated across R in right branch

Column II

- (P) Increases
(Q) Decreases
(R) Does not change
(S) Becomes zero

Paragraph

Ram and Shyam purchased two electric tea kettles 'A' and 'B' of same size, same thickness and same volume of 0.4 litre. They studied the specification of kettles as under.

Kettle – A

Specific heat capacity = 1680 J/kg–K
mass = 200 gm
cost = 400 Rs.

Kettle – B

Specific heat capacity = 2520 J/kg–K
mass = 400 gm
cost = 400 Rs.

When kettle 'A' is switched on with constant source, the tea begins to boil in 6 minute. When kettle 'B' is switched on with the same source separately then tea begins to boil in 8 minute.

The efficiency of kettle is defined as = $\frac{\text{energy used for liquid heating}}{\text{total energy supplied}}$.

They made discussion on specification and efficiency of kettles and subsequently prepared a list of questions to draw the conclusions. Some of them are as under – (Assume specific heat of tea liquid as 4200 J/Kg–K and density as 1000 kg/m³)

12. Efficiency of kettle 'A' is–
(A) 63.34 % (B) 83.33 % (C) 93.34 % (D) 73.34 %
13. Efficiency of kettle 'B' is–
(A) 82.5 % (B) 72.5 % (C) 92.5 % (D) 62.5 %
14. Ratio of electricity consumed charges for one time boiling of tea in kettle A and kettle B.
(A) 3 : 5 (B) 2 : 3 (C) 3 : 4 (D) 1 : 1
15. The resistance of coil of kettles A and B are 'R_A' and 'R_B' respectively, then we can say–
(A) R_A > R_B (B) R_A = R_B
(C) R_A < R_B (D) can not be ascertained by above data.
16. If both the kettles are joined with the same source in series one after the other. Then boiling starts in kettle 'A' and kettle 'B' after–
(A) 4 times of their original time (B) equal to their original time
(C) 2 times of their original time (D) can not be ascertained by above data

ANSWER KEY

EXERCISE (O-2)

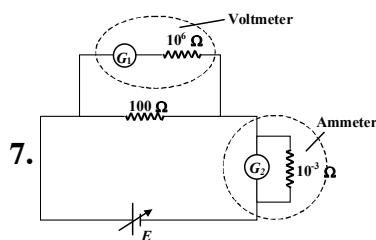
1. (A, B, C, D) 2. (D) 3. (A, D) 4. (A, C, D) 5. (A) 6. (A, B, C)
7. (C) 8. (B, C) 9. (A, C) 10. (A) 11. (A, C) 12. (D) 13. (A, D)
14. (C) 15. (B, C) 16. (A, B, C) 17. (A)
18. (A) $\rightarrow$ (P) ; (B) $\rightarrow$ (Q) ; (C) $\rightarrow$ (R) ; (D) $\rightarrow$ (P) 19. (A) $\rightarrow$ (Q) ; (B) $\rightarrow$ (P) ; (C) $\rightarrow$ (P)
20. (A) 21. (B) 22. (B) 23. (B) 24. (B) 25. (D) 26. (A, B)

EXERCISE (JA)

1. (A) 2. (A)

3. Battery should be connected across A and B. Out put can be taken across the terminals A and C or B and C

4. (A) 5. (A) 6. (C)



8. (B) 9. (D) 10. (D) 11. This is true for $r_1 = r_2$; So R_2 given most accurate value
12. (D) 13. (A) 14. (A) 15. (C) 16. (D) 17. (A, D) 18. (D) 19. (C) 20. (C)
21. 4 22. 5 23. (B) 24. (A, B, C, D) 25. (B, D) 26. (A, B, D)
27. 5 28. (C) 29. (B) 30. 1

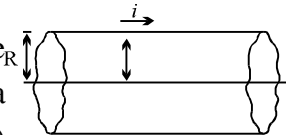
REVIEW TEST

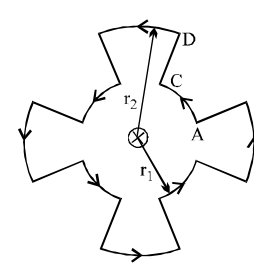
1. 7.5 m, 8.75 m, 6.25 m 2. (i) 1.01Ω (ii) 0-5 A, 0-10 V, (ii) 0.05 A
3. (025) 4. (D) 5. (A) 6. (B) 7. (B) 8. (B) 9. (C)
10. (A, C) 11. (A) $\rightarrow$ (P) ; (B) $\rightarrow$ (P) ; (C) $\rightarrow$ (Q) ; (D) $\rightarrow$ (Q, S) 12. (B)
13. (D) 14. (C) 15. (B) 16. (A)

MAGNETIC EFFECT OF CURRENT

EXERCISE (S-2)

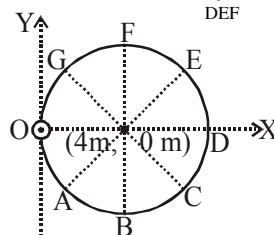
- A cylindrical conductor of radius R carries a current along its length. The current density J , however, it is not uniform over the cross section of the conductor but is a function of the radius according to $J = br$, where b is a constant. Find an expression for the magnetic field B . (a) at $r_1 < R$ & (b) at distance $r_2 > R$, measured from the axis.


- Two coils each of 100 turns are held such that one lies in the vertical plane with their centres coinciding. The radius of the vertical coil is 20 cm and that of the horizontal coil is 30 cm. How would you neutralize the magnetic field of the earth at their common centre? What is the current to be passed through each coil? Horizontal component of earth's magnetic induction = 3.49×10^{-5} T and angle of dip = 30° .
- A current of 10A flows around a closed path in a circuit which is in the horizontal plane as shown in the figure. The circuit consists of eight alternating arcs of radii $r_1 = 0.08$ m and $r_2 = 0.12$ m. Each arc subtends the same angle at the centre.

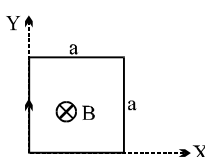


(a) Find the magnetic field produced by this circuit at the centre.
 (b) An infinitely long straight wire carrying a current of 10A is passing through the centre of the above circuit vertically with the direction of the current being into the plane of the circuit. What is the force acting on the wire at the centre due to the current in the circuit? What is the force acting on the arc AC and the straight segment CD due to the current at the centre? [JEE 2001]
- Electric charge q is uniformly distributed over a rod of length l . The rod is placed parallel to a long wire carrying a current i . The separation between the rod and the wire is a . Find the force needed to move the rod along its length with a uniform velocity v .
- A proton beam passes without deviation through a region of space where there are uniform transverse mutually perpendicular electric and magnetic field with E and B . Then the beam strikes a grounded target. Find the force imparted by the beam on the target if the beam current is equal to I .
- An infinite uniform current carrying wire is kept along z -axis, carrying current I_0 in the direction of the positive z -axis. OABCDEFG represents a circle (where all the points are equally spaced), whose centre at

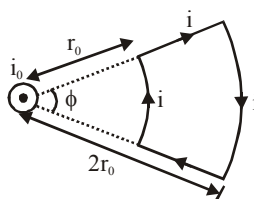
point $(4\text{m}, 0\text{m})$ and radius 4 m as shown in the figure. If $\int_{DEF} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 I_0}{k}$ in S.I. unit, then find the value of k .



7. A rod of length ℓ and total charge 'q' which is uniformly distributed is rotating with angular velocity ω about an axis passing through the centre of rod and perpendicular to rod. Find the magnitude of magnetic dipole moment (in Amp. m^2) of rod. (Take : $q = 4\text{C}$, $\omega = 6\text{ rad/s}$ and $\ell = 2\text{m}$.)
8. 3 infinitely long thin wires each carrying current i in the same direction, are in the x-y plane of a gravity free space. The central wire is along the y-axis while the other two are along $x = \pm d$.
 (i) Find the locus of the points for which the magnetic field B is zero.
 (ii) If the central wire is displaced along the z-direction by a small amount & released, show that it will execute simple harmonic motion. If the linear density of the wires is λ , find the frequency of oscillation.
9. A rectangular loop of wire is oriented with the left corner at the origin, one edge along X-axis and the other edge along Y-axis as shown in the figure. A magnetic field is into the page and has a magnitude that is given by $\beta = \alpha y$ where α is constant. Find the total magnetic force on the loop if it carries current i .



10. A closed loop carrying a current i is placed so that its plane is perpendicular to the long straight conductor which carries a current i_0 as shown in the figure. The torque acting on the current loop is $\alpha \mu_0 \text{ N-m}$. Then find the value of α . (Given $i = 2\text{ A}$, $i_0 = 2\text{ A}$, $r_0 = 2\pi\text{ m}$, $\phi = 60^\circ$)



11. In a moving coil galvanometer, torque on the coil can be expressed as $\tau = ki$, where i is current through the wire and k is constant. The rectangular coil of the galvanometer having numbers of turns N , area A and moment of inertia I is placed in magnetic field B . Find
 (a) k in terms of given parameters N , I , A and B .
 (b) the torsional constant of the spring, if a current i_0 produces a deflection of $\pi/2$ in the coil in reaching equilibrium position.
 (c) the maximum angle through which coil is deflected, if charge Q is passed through the coil almost instantaneously. (Ignore the damping in mechanical oscillations)

[JEE 2005]

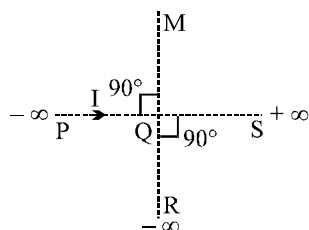
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

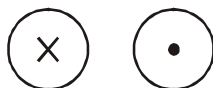
1. A ring like metallic conductor of resistance R and radius a , carries a constant current I . The ratio of the angular momentum L of the conduction electrons (about the axis of the ring) and the magnetic field B at the centre of the ring satisfy [where e and m represent the magnitudes of the electronic charge and mass]

(A) $\frac{B}{L} \propto \frac{e^2}{m}$ (B) $\frac{B}{L} \propto e.m$ (C) $\frac{B}{L} \propto \frac{e}{m}$ (D) $\frac{B}{L} \propto \frac{m}{e}$

2. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field due to this current at the point M is H_1 . Now, another infinitely long straight conductor QS is connected at Q so that the current in PQ remaining unchanged. The magnetic field at M is now H_2 . The ratio H_1/H_2 is given by :-



- (A) $1/2$ (B) 1 (C) $2/3$ (D) 2
3. Equal antiparallel currents are directed in two parallel wires so that one is out of the page and the other is into the page as shown. Compare the magnitude of the magnetic field B_2 at any arbitrary point equidistant from the wires to the magnitude of the field B_1 at that point from one wire alone :-

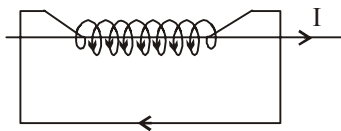


- (A) $B_2 > B_1$ for all equidistant points
(B) $B_2 < B_1$ for all equidistant points
(C) $B_2 > B_1$ for closer equidistant points only
(D) $B_2 < B_1$ for closer equidistant points only
4. A long straight wire along the z -axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is :-

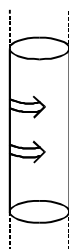
[JEE 2002 (screening)]

(A) $\frac{\mu_0 I}{2\pi} \frac{(y\hat{i} - x\hat{j})}{(x^2 + y^2)}$ (B) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} + y\hat{j})}{(x^2 + y^2)}$ (C) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{j} - y\hat{i})}{(x^2 + y^2)}$ (D) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} - y\hat{j})}{(x^2 + y^2)}$

5. In the diagram shown, a wire carries current I . What is the value of the $\oint \vec{B} \cdot d\vec{s}$ (as in Ampere's law) on the helical loop shown in the figure? The integration is done in the sense shown. The loop has N turns and part of helical loop on which arrows are drawn is outside the plane of paper.

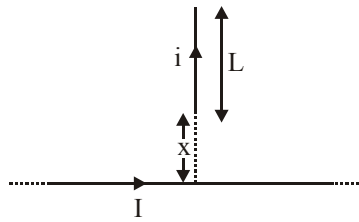


- (A) $-\mu_0(NI)$ (B) $\mu_0(I)$ (C) $\mu_0(NI)$ (D) Zero
6. A hollow cylinder having infinite length and carrying uniform current per unit length λ along the circumference as shown. Magnetic field inside the cylinder is :-



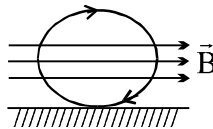
- (A) $\frac{\mu_0 \lambda}{2}$ (B) $\mu_0 \lambda$ (C) $2\mu_0 \lambda$ (D) none
7. An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+x$ direction and a magnetic field along the $+z$ direction, then [JEE 2000 (Scr)]
 (A) positive ions deflect towards $+y$ direction and negative ions towards $-y$ direction
 (B) all ions deflect towards $+y$ direction.
 (C) all ions deflect towards $-y$ direction
 (D) positive ions deflect towards $-y$ direction and negative ions towards $+y$ direction.
8. A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} - 3\hat{j})$. The value of x_0 is:
 (A) $\frac{13}{2} \frac{\alpha E_0}{B_0}$ (B) $\frac{16 \alpha B_0}{E_0}$ (C) $\frac{25}{2 \alpha E_0}$ (D) $\frac{5 \alpha}{2 B_0}$
9. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity $[u\hat{i} - v\hat{j}]$. Uniform electric and magnetic fields exist in the region along the $+y$ direction, of magnitude E and B . The particle will definitely return to the origin once if
 (A) $[vB/2\pi E]$ is an integer (B) $(u^2 + v^2)^{1/2} [B/\pi E]$ is an integer
 (C) $[vB/\pi E]$ is an integer (D) $[uB/\pi E]$ is an integer
10. An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}$ N. If the electron is moving with a velocity $\vec{V}_2 = 2\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = +2\hat{i}$ N. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is
 (A) zero (B) $2\hat{k}$ N (C) $-2\hat{k}$ N (D) information is insufficient

11. The magnetic force between wires as shown in figure is :-



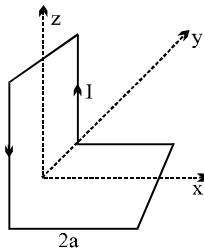
- (A) $\frac{\mu_0 i I^2}{2\pi} \ln\left(\frac{x+\ell}{2x}\right)$ (B) $\frac{\mu_0 i I^2}{2\pi} \ln\left(\frac{2x+\ell}{2x}\right)$ (C) $\frac{\mu_0 i I}{2\pi} \ln\left(\frac{x+\ell}{x}\right)$ (D) None of these

12. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current $i = 4\text{A}$. A horizontal magnetic field $B = 10\text{T}$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of the ring will be :-



- (A) $40\pi \text{ rad/s}^2$ (B) $20\pi \text{ rad/s}^2$ (C) $5\pi \text{ rad/s}^2$ (D) $15\pi \text{ rad/s}^2$

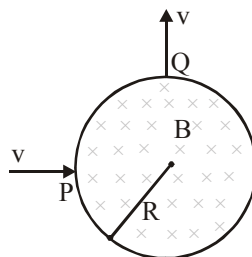
13. A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction :-
[JEE, 2001 (Scr)]



- (A) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{j} + \hat{k} + \hat{i})$ (C) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ (D) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$

MULTIPLE CORRECT TYPE QUESTIONS

14. A particle of charge ' q ' and mass ' m ' enters normally (at point P) in a region of magnetic field with speed v . It comes out normally from Q after time T as shown in figure. The magnetic field B is present only in the region of radius R and is uniform. Initial and final velocities are along radial direction and they are perpendicular to each other. For this to happen, which of the following expression(s) is/are correct :-



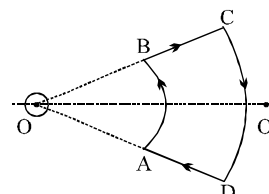
- (A) $B = \frac{mv}{qR}$ (B) $T = \frac{\pi R}{2v}$ (C) $T = \frac{\pi m}{2qB}$ (D) None of these

EXERCISE-JA

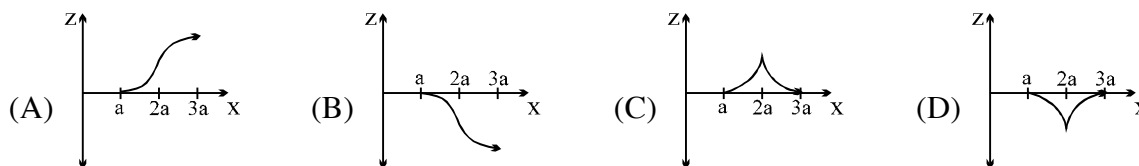
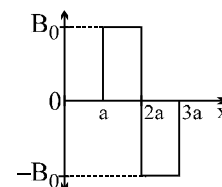
(Not to be discussed in class)

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

1. An infinite current carrying wire passes through point O and is perpendicular to the plane containing a current carrying loop ABCD as shown in the figure. Choose the correct option (s). [JEE 2006]



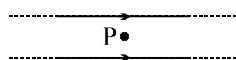
- (A) Net force on the loop is zero.
 (B) Net torque on the loop is zero.
 (C) As seen from O, the loop rotates clockwise.
 (D) As seen from O, the loop rotates anticlockwise.
2. A magnetic field $\vec{B} = B_0 \hat{j}$ exists in the region $a < x < 2a$ and $\vec{B} = -B_0 \hat{j}$ in the region $2a < x < 3a$, where B_0 is a positive constant. A positive point charge moving with a velocity $\vec{v} = v_0 \hat{i}$, where v_0 is a positive constant, enters the magnetic field at $x = a$. The trajectory of the charge in this region can be like, [JEE 2007]



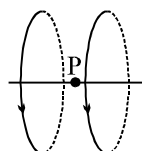
3. Two wires each carrying a steady current I are shown in four configurations in **Column I**. Some of the resulting effects are described in **Column II**. Match the statements in **Column I** with the statements in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS. [JEE 2007]

Column I

- (A) Point P is situated midway between the wires



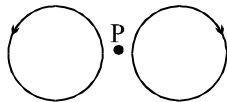
- (B) Point P is situated at the mid-point of the line joining the centers of the circular wires, which have same radii.



Column II

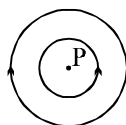
- (P) The magnetic fields (B) at P due to the currents in the wires are in the same direction.
 (Q) The magnetic fields (B) at P due to the currents in the wires are in opposite directions

- (C) Point P is situated at the mid-point of the line joining the centers of the circular wires, which have same radii.



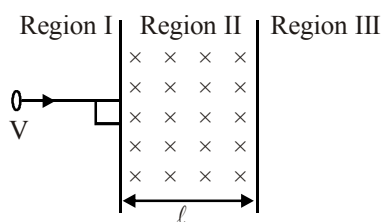
- (R) There is no magnetic field at P

- (D) Point P is situated at the common center of the wires



- (S) The wires repel each other

4. A particle of mass m and charge q , moving with velocity V enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is ℓ . Choose the correct choice(s) [JEE 2008]



- (A) The particle enters Region III only if its velocity $V > \frac{q\ell B}{m}$

- (B) The particle enters Region III only if its velocity $V < \frac{q\ell B}{m}$

- (C) Path length of the particle in Region II is maximum when velocity $V = \frac{q\ell B}{m}$

- (D) Time spent in Region II is same for any velocity V as long as the particle returns to Region I

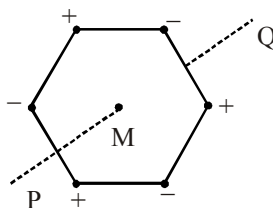
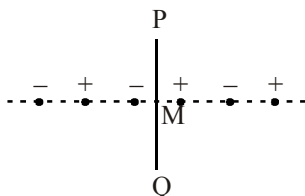
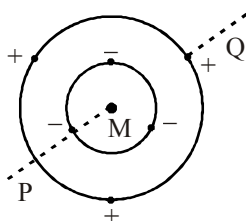
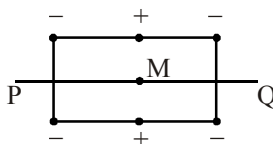
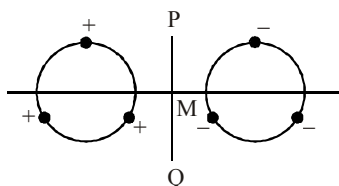
5. **Statement-1** : The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.

and

Statement-2 : Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized. [JEE 2008]

- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

6. Six point charges, each of the same magnitude q , are arranged in different manners as shown in **Column II**. In each case, a point M and a line PQ passing through M are shown. Let E be the electric field and V be the electric potential at M (potential at infinity is zero) due to the given charge distribution when it is at rest. Now, the whole system is set into rotation with a constant angular velocity about the line PQ . Let B be the magnetic field at M and μ be the magnetic moment of the system in this condition. Assume each rotating charge to be equivalent to a steady current. [JEE-2009]

Column-I(A) $E = 0$ (B) $V \neq 0$ (C) $B = 0$ (D) $\mu \neq 0$ **Column-II**(p) Charges are at the corners of a regular hexagon. M is the centre of the hexagon. PQ is perpendicular to the plane of the hexagon.(q) Charges are on a line perpendicular to PQ at equal intervals. M is the mid-point between the two innermost charges.(r) Charges are placed on two coplanar circles of radius a and $2a$. M is at the centre of the circle. PQ is perpendicular to the plane of the circles.(s) Charges are placed at the corners of a rectangle of sides a and $2a$ and at the mid points of the longer sides. M is at the centre of the rectangle. PQ is parallel to the longer sides.(t) Charges are placed on two coplanar, identical insulating rings at equal intervals. M is the mid-point between the centres of the rings. PQ is perpendicular to the line joining the centres and coplanar to the rings.

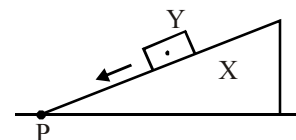
7. **Column II** shows five systems in which two objects are labelled as X and Y. Also in each case a point P is shown. **Column I** gives some statements about X and/or Y. Match these statements to the appropriate system(s) from **Column II**. [JEE-2009]

Column-I

Column-II

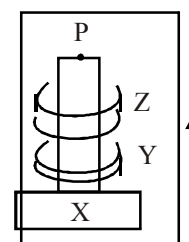
(A) The force exerted by X on Y has a magnitude Mg .

(p) Block Y of mass M left on a fixed inclined plane X, slides on it with a constant velocity.



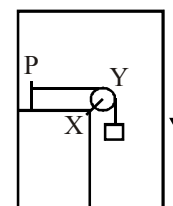
(B) The gravitational potential energy of X is continuously increasing

(q) Two ring magnets Y and Z, each of mass M, are kept in frictionless vertical plastic stand so that they repel each other. Y rests on the base X and Z hangs in air in equilibrium. P is the topmost point of the stand on the common axis of the two rings. The whole system is in a lift that is going up with a constant velocity.



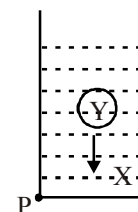
(C) Mechanical energy of the system X + Y is continuously decreasing

(r) A pulley Y of mass m_0 is fixed to a table through a clamp X. A block of mass M hangs from a string that goes over the pulley and is fixed at point P of the table. The whole system is kept in a lift that is going down with a constant velocity.

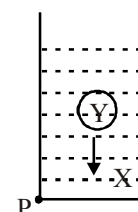


(D) The torque of the weight of Y about point P is zero.

(s) A sphere Y of mass M is put in a nonviscous liquid X kept in a container at rest. The sphere is released and it moves down in the liquid.

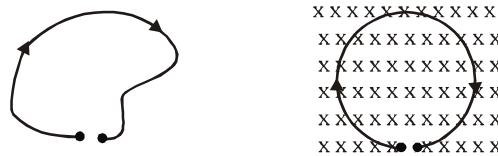


(t) A sphere Y of mass M is falling with its terminal velocity in a viscous liquid X kept in a container.



8. A steady current I goes through a wire loop PQR having shape of a right angle triangle with $PQ = 3x$, $PR = 4x$ and $QR = 5x$. If the magnitude of the magnetic field at P due to this loop is $k \left(\frac{\mu_0 I}{48\pi x} \right)$, find the value of k . [2009, 6M]

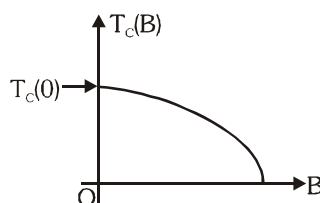
9. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is : [2010, 3M]



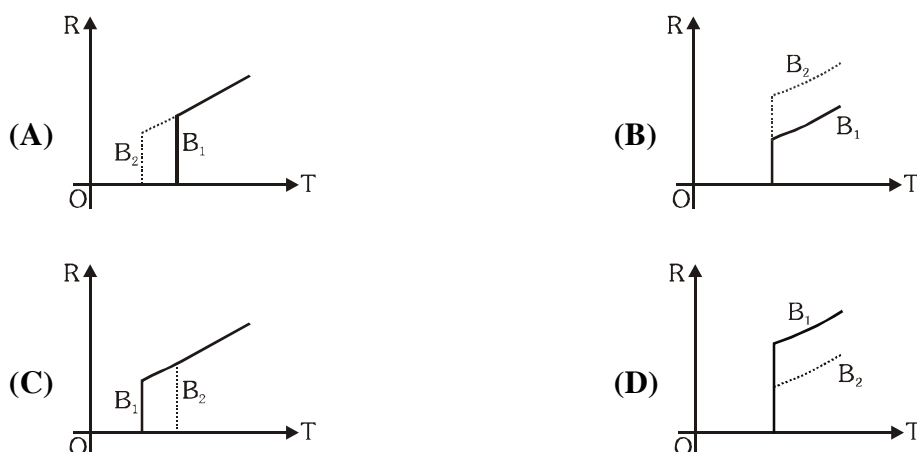
- (A) IBL (B) $\frac{IBL}{\pi}$ (C) $\frac{IBL}{2\pi}$ (D) $\frac{IBL}{4\pi}$

Paragraph for Question no. 10 and 11

Electrical resistance of certain materials, known as superconductors, changes abruptly from a nonzero value to zero as their temperature is lowered below a critical temperature $T_c(0)$. An interesting property of superconductors is that their critical temperature becomes smaller than $T_c(0)$ if they are placed in a magnetic field, i.e., the critical temperature $T_c(B)$ is a function of the magnetic field strength B . The dependence of $T_c(B)$ on B is shown in the figure. [JEE2010]



10. In the graphs below, the resistance R of a superconductor is shown as a function of its temperature T for two different magnetic fields B_1 (solid line) and B_2 (dashed line). If B_2 is larger than B_1 , which of the following graphs shows the correct variation of R with T in these fields?



11. A superconductor has $T_c(0) = 100$ K. When a magnetic field of 7.5 Tesla is applied, its T_c decreases to 75 K. For this material one can definitely say that when
- (A) $B=5$ Tesla, $T_c(B) = 80$ K (B) $B=5$ Tesla, $75 \text{ K} < T_c(B) < 100$ K
 (C) $B=10$ Tesla, $75 \text{ K} < T_c(B) < 100$ K (D) $B=10$ Tesla, $T_c(B) = 70$ K

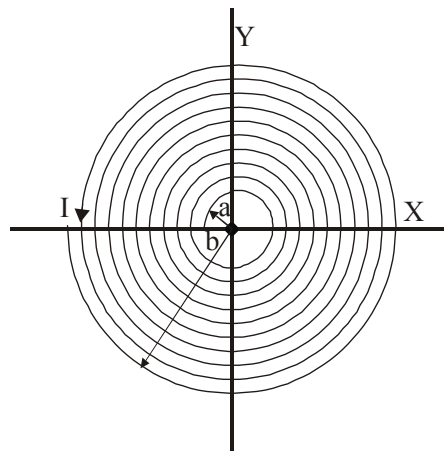
12. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi-infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statement(s) is/are true?

[IIT-JEE 2011]

- (A) they will never come out of the magnetic field region
(B) they will come out travelling along parallel paths
(C) they will come out of the same time
(D) they will come out at different times

13. A long insulated copper wire is closely wound as a spiral of 'N' turns. The spiral has inner radius 'a' and outer radius 'b'. The spiral lies in the X-Y plane and a steady current 'I' flows through the wire. The Z-component of the magnetic field at the center of the spiral is [IIT-JEE 2011]

- (A) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b}{a}\right)$
(B) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b+a}{b-a}\right)$
(C) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b}{a}\right)$
(D) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b+a}{b-a}\right)$



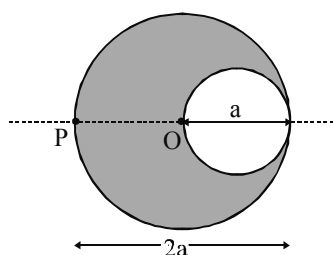
14. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $\vec{E} = E_0 \hat{j}$ and $\vec{B} = B_0 \hat{j}$. At time $t=0$, this charge has velocity $\vec{v}$ in the x-y plane, making an angle θ with the x-axis. Which of the following option (s) is (are) correct for time $t > 0$? [IIT-JEE 2012]

- (A) If $\theta = 0^\circ$, the charge moves in a circular path in the x-z plane.
(B) If $\theta = 0^\circ$, the charge undergoes helical motion with constant pitch along the y-axis.
(C) If $\theta = 10^\circ$, the charge undergoes helical motion with its pitch increasing with time, along the y-axis
(D) If $\theta = 90^\circ$, the charge undergoes linear but accelerated motion along the y-axis

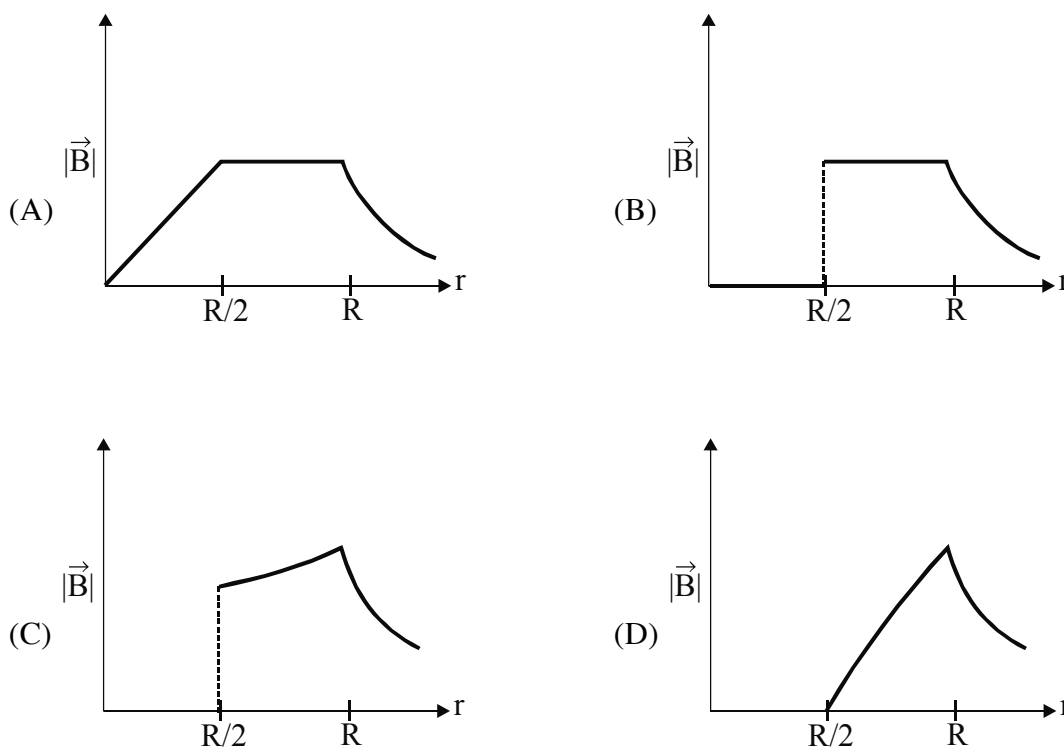
15. A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If

the magnitude of the magnetic field at the point P is given by $\frac{N}{12} \mu_0 a J$, then the value of N is :

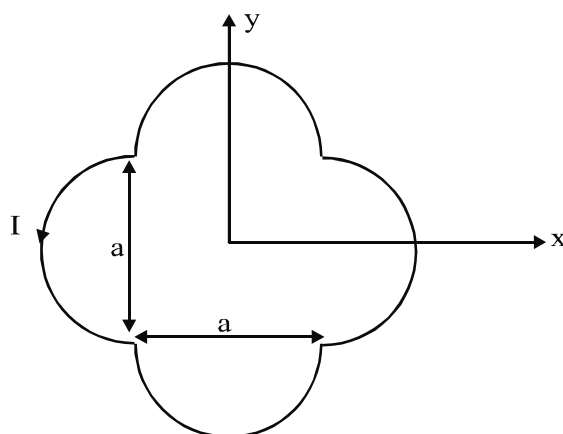
[IIT-JEE 2012]



16. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by
[IIT-JEE 2012]



17. A loop carrying current 'I' lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is
[JEE 2012]



- (A) $a^2 I \hat{k}$ (B) $\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$ (C) $-\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$ (D) $(2\pi + 1) a^2 I \hat{k}$

18. A particle of mass M and positive charge Q , moving with a constant velocity $\vec{u}_1 = 4\hat{i} \text{ ms}^{-1}$, enters a region of uniform static magnetic field normal to the x - y plane. The region of the magnetic field extends from $x = 0$ to $x = L$ from all values of y . After passing through this region, the particle emerges on the other side after 10 milliseconds with a velocity $\vec{u}_2 = 2(\sqrt{3}\hat{i} + \hat{j}) \text{ ms}^{-1}$. The correct statement(s) is (are) :-

[JEE-Advanced-2013]

- (A) The direction of the magnetic field is $-z$ direction.
 (B) The direction of the magnetic field is $+z$ direction.
 (C) The magnitude of the magnetic field $\frac{50\pi M}{3Q}$ units.
 (D) The magnitude of the magnetic field is $\frac{100\pi M}{3Q}$ units.

19. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R . This cylinder is placed coaxially inside an infinite solenoid of radius $2R$. The solenoid has n turns per unit length and carries a steady current I . Consider a point P at a distance r from the common axis. The correct statement(s) is (are) :-

[JEE-Advanced-2013]

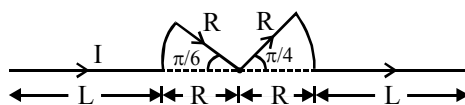
- (A) In the region $0 < r < R$, the magnetic field is non-zero
 (B) In the region $R < r < 2R$, the magnetic field is along the common axis
 (C) In the region $R < r < 2R$, the magnetic field is tangential to the circle of radius r , centred on the axis
 (D) In the region $r > 2R$, the magnetic field is non-zero

20. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1 . In contrast, if the currents I in the two wires have directions opposite to each other, the radius of curvature of the path is R_2 . If $\frac{X_0}{X_1} = 3$, and value of $\frac{R_1}{R_2}$ is:-

[JEE-Advanced-2014]

21. A conductor (shown in the figure) carrying constant current I is kept in the x - y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are)

[JEE-Advanced-2015]



- (A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$
 (B) If $\vec{B}$ is along $\hat{x}$, $F = 0$
 (C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$
 (D) If $\vec{B}$ is along $\hat{z}$, $F = 0$

REVIEW TEST

(Not to be discussed in class)

1. Two moving coil meters. M_1 and M_2 have the following particulars :

(NCERT)

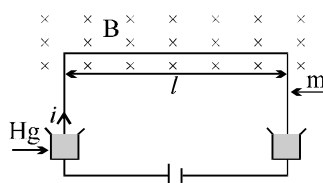
$$R_1 = 10 \, \Omega, N_1 = 30,$$

$$A_1 = 3.6 \times 10^{-3} \, \text{m}^2, B_1 = 0.25 \, \text{T}$$

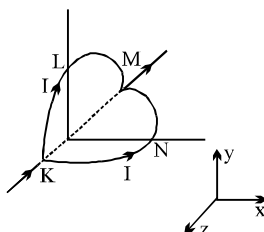
$$R_2 = 14 \, \Omega, N_2 = 42, A_2 = 1.8 \times 10^{-3} \, \text{m}^2, B_2 = 0.50 \, \text{T}$$

(The spring constants are identical for the two meters). Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1 .

2. A U-shaped wire of mass m and length l is immersed with its two ends in mercury (see figure). The wire is in a homogeneous field of magnetic induction $\mathbf{B}$. If a charge, that is, a current pulse $q = \int i dt$, is sent through the wire, the wire will jump up. Calculate, the height h that the wire reaches, the size of the charge or current pulse, assuming that the time of the current pulse is very small in comparison with the time of flight. Make use of the fact that impulse of force equals $\int \mathbf{F} dt$, which equals $m\mathbf{v}$. Evaluate q for $B = 0.1 \, \text{Wb/m}^2$, $m = 10 \, \text{gm}$, $l = 20 \, \text{cm}$ & $h = 3 \, \text{meters}$. [$g = 10 \, \text{m/s}^2$]



3. A particle of mass $1 \times 10^{-26} \, \text{kg}$ and charge $+1.6 \times 10^{-19} \, \text{C}$ travelling with a velocity $1.28 \times 10^6 \, \text{m/s}$ in the $+x$ direction enters a region in which a uniform electric field $\mathbf{E}$ and a uniform magnetic field of induction $\mathbf{B}$ are present such that $E_x = E_y = 0$, $E_z = -102.4 \, \text{kV/m}$ and $B_x = B_z = 0$, $B_y = 8 \times 10^{-2} \, \text{weber/m}^2$. The particle enters this region at the origin at time $t = 0$. Determine the location (x , y and z coordinates) of the particle at $t = 5 \times 10^{-6} \, \text{s}$. If the electric field is switched off at this instant (with the magnetic field still present), what will be the position of the particle at $t = 7.45 \times 10^{-6} \, \text{s}$?
4. A circular loop of radius R is bent along a diameter and given a shape as shown in the figure. One of the semicircles (KNM) lies in the $x-z$ plane and the other one (KLM) in the $y-z$ plane with their centers at the origin. Current I is flowing through each of the semicircles as shown in figure.



(i) A particle of charge q is released at the origin with a velocity $\vec{v} = -v_0 \hat{i}$. Find the instantaneous force $\vec{f}$ on the particle. Assume that space is gravity free.

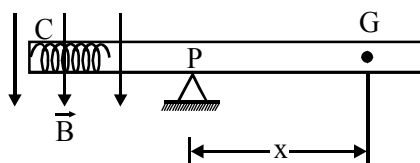
(ii) If an external uniform magnetic field $B \hat{j}$ is applied, determine the forces F_1 and F_2 on the semicircles KLM and KNM due to this field and the net force F on the loop. [JEE 2000]

5. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E \hat{i}$ and a magnetic field $B \hat{k}$ enters in I quadrant of a coordinate system at a point $(0, a)$ with initial velocity $v \hat{i}$ and leaves the quadrant at a point $(2a, 0)$ with velocity $-2v \hat{j}$. Find

- Magnitude of electric field
- Rate of work done by the electric field at point $(0, a)$
- Rate of work done by both the fields at $(2a, 0)$.

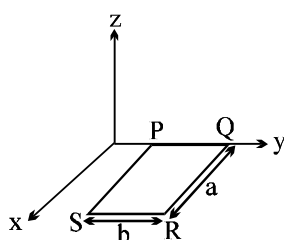
6. There exists a uniform magnetic and electric field of magnitude 1 T and 1 V/m respectively along positive y-axis. A charged particle of mass 1 kg and charge 1 C moving with velocity 1 m/sec along x-axis is at origin at $t = 0$. If the coordinates of particle at time π seconds is given as (X, Y, Z) in meter, then find the value of $(XY + YZ + ZX) \times \frac{5}{\pi^2}$.

7. A uniform wooden bar of mass $\frac{\pi}{100}$ kg and radius of cross-section 10cm carries a light coil C of 100 turns. The bar is smoothly pivoted at P. If the coil carries a current 2A and subjected to an external magnetic field 10^{-2} T, the bar remains in equilibrium. Find the distance x (in cm) of the C.M. of the rod from the pivot.

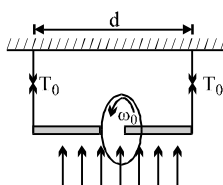


8. A rectangular loop PQRS made from a uniform wire has length a , width b and mass m . It is free to rotate about the arm PQ, which remains hinged along a horizontal line taken as the y-axis (see figure). Take the vertically upward direction as the z-axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})B_0$ exists in the region. The loop is held in the x-y plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium. [JEE 2002]

- What is the direction of the current I in PQ?
- Find the magnetic force on the arm RS.
- Find the expression for I in terms of B_0 , a , b and m .

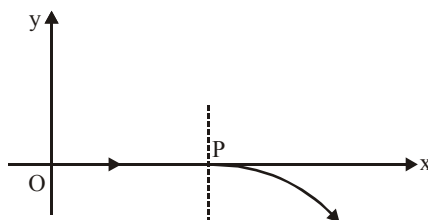


9. A wheel of radius R having charge Q , uniformly distributed on the rim of the wheel is free to rotate about a light horizontal rod. The rod is suspended by light inextensible strings and a magnetic field B is applied as shown in the figure. The initial tensions in the strings are T_0 . If the breaking tension of the strings are $\frac{3T_0}{2}$, find the maximum angular velocity ω_0 with which the wheel can be rotate. [JEE 2003]



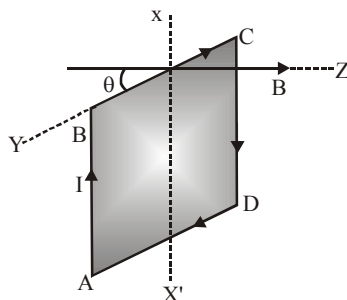
10. A charged particle enters a non-uniform uni-directional field such that initial velocity is parallel to magnetic field, then the radius of curvature of its path is (in standard notation):
 (A) mV/qB (B) 0 (C) ∞ (D) qB/mV
11. An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero
 (A) positive z- axis (B) negative z-axis (C) positive y-axis (D) negative y-axis
12. For a positively charged particle moving in a $x-y$ plane initially along the x -axis, there is a sudden change in its path due to the presence of electric and/or magnetic field beyond P. The curved path is shown in the $x-y$ plane and is found to be non-circular. Which one of the following combinations is possible?

[JEE 2004]

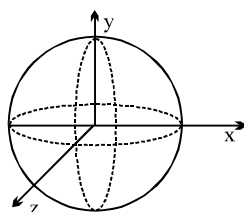


- (A) $\vec{E} = 0$; $\vec{B} = b\hat{j} + c\hat{k}$ (B) $\vec{E} = a\hat{i}$; $\vec{B} = c\hat{k} + a\hat{i}$
 (C) $\vec{E} = 0$; $\vec{B} = c\hat{j} + b\hat{k}$ (D) $\vec{E} = a\hat{i}$; $\vec{B} = c\hat{k} + b\hat{j}$
13. Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A. (NCERT)
- (A) 2×10^{-5} N ; attractive force normal to B towards A.
 (B) 2×10^{-5} N ; attractive force normal to A towards B.
 (C) 5×10^{-5} N ; attractive force normal to A towards B.
 (D) 5×10^{-5} N ; attractive force normal to B towards A.

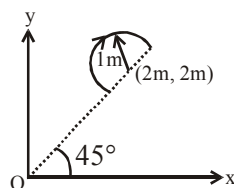
14. A wire of mass 100 g carrying a current of 2A towards increasing x is in the form of $y = x^2$ ($-2\text{m} \leq x \leq +2\text{m}$). This wire is placed in a magnetic field $B = -0.02\hat{k}$ Tesla & gravity free region. The acceleration of the wire (in m/s^2) is :-
 (A) $-1.6\hat{j}$ (B) $-3.2\hat{j}$ (C) $1.6\hat{j}$ (D) $2.4\hat{j}$
15. The square loop ABCD, carrying a current I , is placed in a uniform magnetic field B , as shown. The loop can rotate about the axis XX' . The plane of the loop makes an angle θ ($\theta < 90^\circ$) with the direction of B . Through what angle will the loop rotate by itself before the torque on it becomes zero :-



- (A) θ (B) $90^\circ - \theta$ (C) $90^\circ + \theta$ (D) $180^\circ - \theta$
16. Three rings, each having equal radius R , are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is

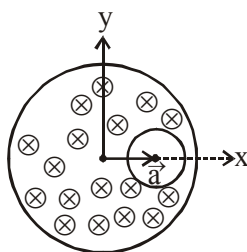


- (A) $\sqrt{3} \frac{\mu_0 I}{2R}$ (B) zero (C) $(\sqrt{2} - 1) \frac{\mu_0 I}{2R}$ (D) $(\sqrt{3} - \sqrt{2}) \frac{\mu_0 I}{2R}$
17. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j} + \hat{k})$ Tesla exist in a region of space. A semicircular wire of radius 1 m carrying a current of 1A having its centre at (2,2,0) m is placed on the X-Y plane as shown. The force on the semicircular wire will be



- (A) $\frac{1}{\sqrt{2}} (\hat{i} - \hat{j} + \hat{k}) \text{ N}$ (B) $\sqrt{2} (\hat{i} - \hat{j} + \hat{k}) \text{ N}$ (C) $\frac{1}{\sqrt{2}} (\hat{i} + \hat{j} - \hat{k}) \text{ N}$ (D) $\sqrt{2} (\hat{i} + \hat{j} - \hat{k}) \text{ N}$

18. A bar magnet of magnetic moment $2A\text{-m}^2$ free to rotate about a vertical axis passing through its centre. The magnet is released from rest from east-west position. Then the kinetic energy of the magnet as it takes North-South position is (horizontal component of the earth's field is $25\mu\text{T}$ and angle of declination is zero:-
 (A) $25\mu\text{J}$ (B) $50\mu\text{J}$ (C) $100\mu\text{J}$ (D) $125\mu\text{J}$
19. A magnet is suspended in the magnetic meridian with an untwisted wire. The upper end of wire is rotated through 180° to deflect the magnet by 30° from magnetic meridian. When the magnet is replaced by another magnet, the upper end of wire is rotated through 270° to deflect the magnet 30° from magnetic meridian. The ratio of magnetic moments of magnets is :-
 (A) 1 : 5 (B) 1 : 8 (C) 5 : 8 (D) 8 : 5
20. If a solution of ferromagnetic material is poured into a U-tube and one arm of this is placed between the poles of a strong magnet with the meniscus in line with the field, then the level of the solution will:
 (A) Rise (B) Fall
 (C) Rise till the liquid comes out (D) Remain unchanged
21. A sample of paramagnetic salt contains 2.0×10^{24} atomic dipoles each of dipole moment $1.5 \times 10^{-23}\text{J T}^{-1}$. The sample is placed under a homogeneous magnetic field of 0.84T , and cooled to a temperature of 4.2K . The degree of magnetic saturation achieved is equal to 15%. What is the total dipole moment (approximate) of the sample for a magnetic field of 0.98T and a temperature of 2.8K ? (Assume Curie's law).
 (A) 7.9J T^{-1} (B) 52.5J T^{-1} (C) 30J T^{-1} (D) 4.6J T^{-1}
22. In a moving coil galvanometer the number of turns $N = 24$, area of the coil $A = 2 \times 10^{-3}\text{m}^2$, and the magnetic field strength $B = 0.2\text{T}$. To increase its current sensitivity by 25% we :-
 (A) increase B to 0.30T (B) decrease A to $1.5 \times 10^{-3}\text{m}^2$
 (C) increase N to 30 (D) none of the above
23. Figure shows cross-section view of a infinite cylindrical wire with a cylindrical cavity, current density is uniform $\vec{J} = -j_0\hat{k}$ as shown in figure.



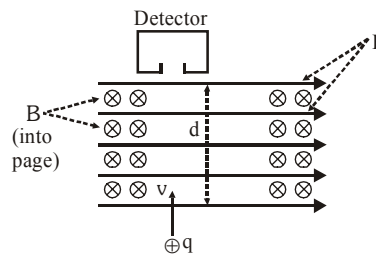
- (A) Field inside cavity is uniform
 (B) Field inside cavity is along $\vec{a}$
 (C) Field inside cavity is perpendicular to $\vec{a}$
 (D) If an electron is projected with velocity $v_0\hat{j}$ it will move undeviated before colliding with cavity wall

24. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a , b and c be the ratio of their time periods, radii and pitches of the helical paths than
- (A) $abc = 1$ (B) $abc > 1$ (C) $abc < 1$ (D) $a = bc$

COMPREHENSION TYPE QUESTIONS

Paragraph for Questions no. 25 to 28

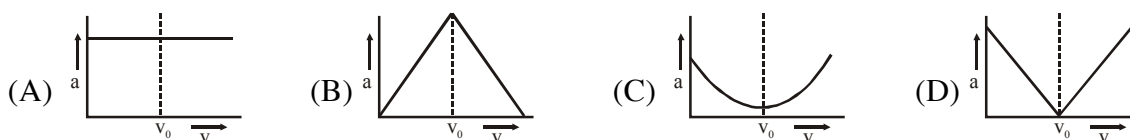
A velocity filter uses the properties of electric and magnetic fields to select charged particles that are moving with a specific velocity. Charged particles with varying speeds are directed into the filter as shown in figure. The filter consists of an electric field E and a magnetic field B , each of constant magnitude, directed perpendicular to each other as shown. The particles that move straight through the filter with their direction unaltered by the fields have the specific filter speed, v_0 . Those with speeds to v_0 may experience sufficiently little deflection that they also enter the detector.



The charged particle will experience a force due to the electric field given by the relationship $\vec{F} = q\vec{E}$, where q is the charge of the particle and $\vec{E}$ is the electric field. The moving particle will also experience a force due to the magnetic field. This force acts to oppose the force due to the electric field. The strength of the force due to the magnetic field is given by the relationship $\vec{F} = q(\vec{v} \times \vec{B})$, where q is the charge of the particle, $\vec{v}$ is the speed of the particle, and $\vec{B}$ is the magnetic field strength. When the forces due to the two fields are equal and opposite, the net force on the particle will be zero, and the particle will pass through the filter with its path unaltered. The electric and magnetic field strengths can be adjusted to choose the specific velocity to be filtered. The effects of gravity can be neglected.

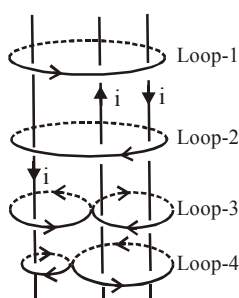
25. The electric and magnetic fields in the filter of figure are adjusted to detect particles with positive charge q of a certain speed, v_0 . Which of the following expressions is equal to this speed ?
- (A) $B/(q^2E)$ (B) $E/(q^2B)$ (C) B/E (D) E/B
26. Which of the following is true about the velocity filter shown in figure ?
- (A) It would not work with negatively charged particles
- (B) The wider the detector entrance, the more narrow the range of speed detected
- (C) The greater the distance d , the more narrow the range of speeds detected
- (D) The detector may not detect a charged particle with the desired filter speed if its charge is too high

27. Which of the following statements is true regarding a charged particle that is moving through the filter at a speed that is less than the filter speed ?
- (A) It experiences a greater force due to the magnetic field than due to the electric field
- (B) It experiences a greater force due to the electric field than due to the magnetic field
- (C) It experiences equal force due to both fields but greater acceleration due to the electric field
- (D) It experiences equal force due to both fields but greater acceleration due to the magnetic field
28. Particles of identical mass and charge are sent through the filter at varying speeds, and the magnitude of acceleration of each particle is recorded as it first begins to be deflected. If the filter is set to detect particles of speed v_0 , which one of the following is correct graph between acceleration and velocity of particle:



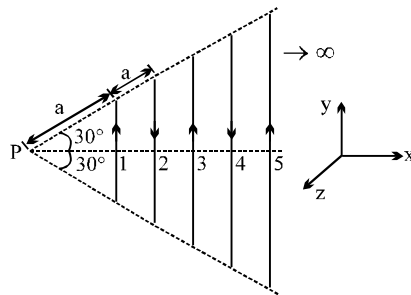
MATRIX MATCH TYPE QUESTION

29. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in figure.



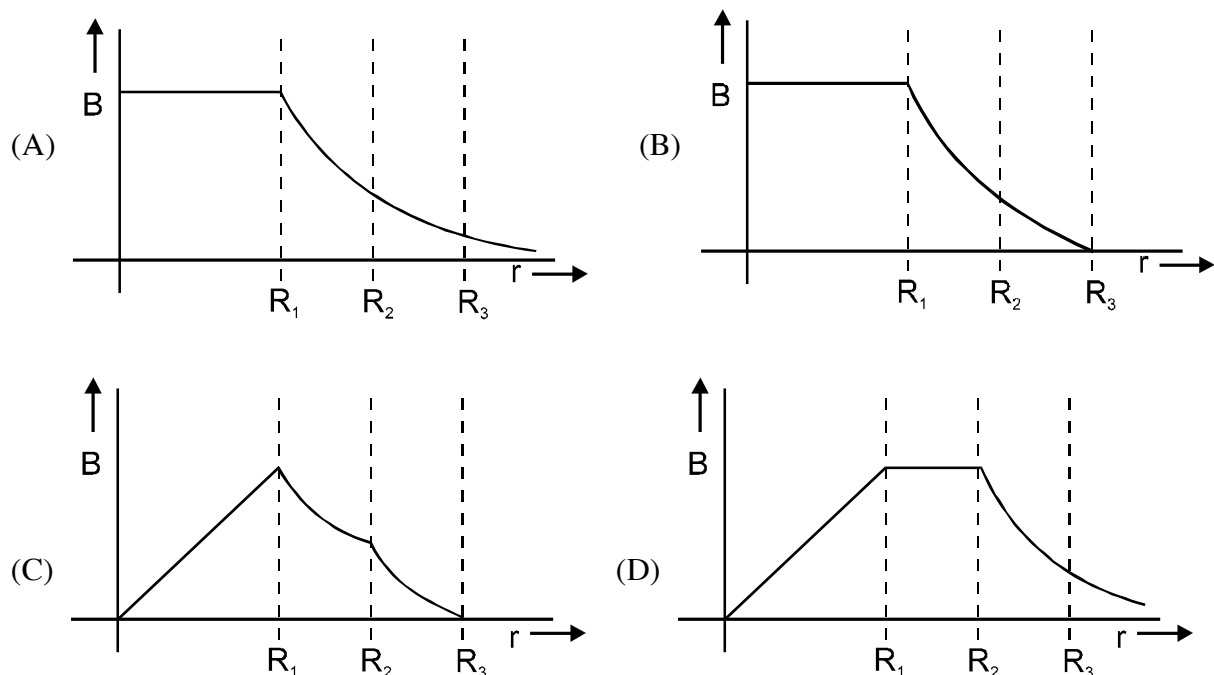
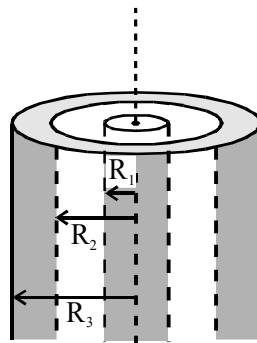
Column-I	Column-II
(A) Along closed loop-1	(p) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
(B) Along closed loop-2	(q) $\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
(C) Along closed loop-3	(r) $\oint \vec{B} \cdot d\vec{\ell} = 0$
(D) Along closed loop-4	(s) Net work done by the magnetic force to move a unit charge along the loop is zero

30. Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is :-

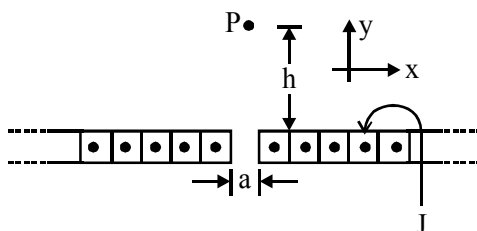


- (A) $\frac{\mu_0 I}{4\pi} \frac{\ln 2}{\sqrt{3} a} \hat{k}$ (B) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3} a} \hat{k}$ (C) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3} a} (-\hat{k})$ (D) Zero

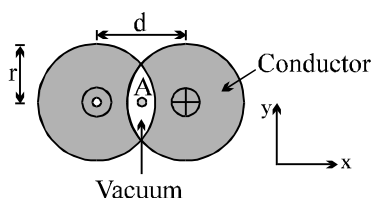
31. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 & the outer conductor is hollow of inner radius R_2 and outer radius R_3 . The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as:



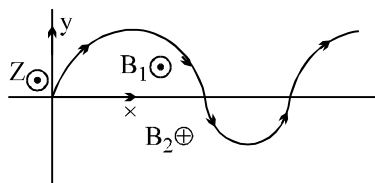
32. A small segment of length a is cut along z -axis from a infinite sheet having a surface current density J (current per unit width). The magnetic field at P is :-



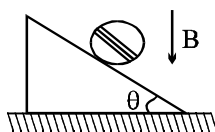
- (A) $2\mu_0 J \left(1 - \frac{h}{a\pi}\right) \hat{i}$ (B) $\frac{\mu_0 J h}{2a\pi} \hat{i}$ (C) $\frac{\mu_0 J}{2} \left(\frac{a}{h\pi} - 1\right) \hat{i}$ (D) $-\frac{\mu_0 J}{2} \left(\frac{h}{a\pi} - 1\right) \hat{i}$
33. Two long conductors are arranged as shown above to form overlapping cylinders, each of radius r , whose centers are separated by a distance d . Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?



- (A) $(\mu_0/2\pi)\pi dJ$, in the $+y$ -direction (B) $(\mu_0/2\pi)d^2/r$, in the $+y$ -direction
 (C) $(\mu_0/2\pi)4d^2J/r$, in the $-y$ -direction (D) $(\mu_0/2\pi)Jr^2/d$, in the $-y$ -direction
34. At $t=0$ a charge q is at the origin and moving in the y -direction with velocity $\vec{v} = v \hat{j}$. The charge moves in a magnetic field that is for $y > 0$ out of page and given by $B_1 \hat{z}$ and for $y < 0$ into the page and given by $-B_2 \hat{z}$. The charge's subsequent trajectory is shown in the sketch. From this information, we can deduce that



- (A) $q > 0$ and $|B_1| < |B_2|$ (B) $q < 0$ and $|B_1| < |B_2|$
 (C) $q > 0$ and $|B_1| > |B_2|$ (D) $q < 0$ and $|B_1| > |B_2|$
35. In the figure shown a coil of single turn is wound on a sphere of radius R and mass m . The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i . The value of B if the sphere is in equilibrium is :-



- (A) $\frac{mg \cos \theta}{\pi i R}$ (B) $\frac{mg}{\pi i R}$ (C) $\frac{mg \tan \theta}{\pi i R}$ (D) $\frac{mg \sin \theta}{\pi i R}$

36. Two protons move parallel to each other, keeping distance r between them, both moving with same velocity $\vec{v}$. Then the ratio of the electric and magnetic force of interaction between them is :-
- (A) c^2/V^2 (B) $2c^2/V^2$ (C) $c^2/2V^2$ (D) None
37. A particle of mass m and charge q moving with velocity $\vec{v}$ enters a region of uniform magnetic field of induction $\vec{B}$. Then
- (A) its path in the region of the field is always circular
(B) its path in the region of the field is circular if $\vec{v} \cdot \vec{B} = 0$
(C) its path in the region of the field is a straight line if $\vec{v} \times \vec{B} = 0$
(D) distance travelled by the particle in time T does not depend on the angle between $\vec{v}$ and $\vec{B}$

MATRIX MATCH TYPE QUESTION

38. A charged particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charged particle are given, while in column II the possible paths of charged particle is mentioned. Match the entries of Column I with the entries of Column II.

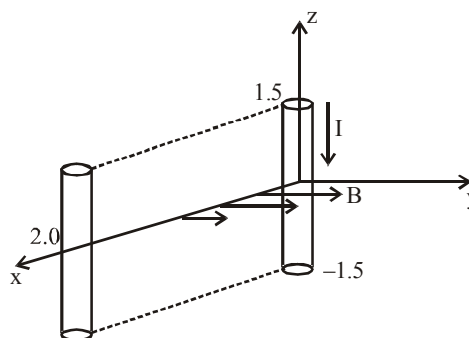
Column-I

Column-II

- | | |
|---|---|
| (A) $\vec{E} = 0, \vec{B} \neq 0$ and initial velocity is at an unknown angle with $\vec{B}$ | (P) Straight line |
| (B) $\vec{E} \neq 0, \vec{B} = 0$ and initial velocity is at an unknown angle with $\vec{E}$ | (Q) Parabola |
| (C) $\vec{E} \neq 0, \vec{B} \neq 0, \vec{E} \parallel \vec{B}$ and initial velocity is perpendicular to $\vec{E}$ | (R) Circular |
| (D) $\vec{E} \neq \vec{B}, \vec{B} \neq 0, \vec{E}$ perpendicular $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$ | (S) Helical path with nonuniform pitch
(T) Helical path with uniform pitch |

39. A conductor lies along the z -axis at $-1.5 \leq z < 1.5$ m and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y$ T, find the power required to move the conductor at constant speed to $x = 2.0$ m, $y = 0$ m in 5×10^{-3} s. Assume parallel motion along the x -axis.

[JEE Mains - 2014]

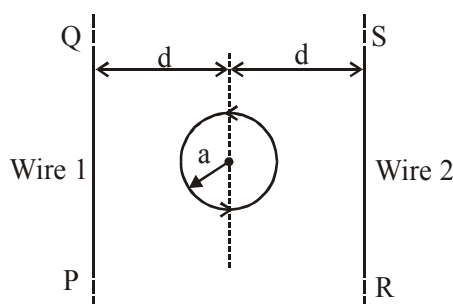


- (1) 14.85 W (2) 29.7 W (3) 1.57 W (4) 2.97 W

Paragraph for Questions 40 & 41

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d . The loop and the wires are carrying the same current I . The current in the loop is in the counterclockwise direction if seen from above.

[JEE-Advanced-2014]



40. When $d \approx a$ but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case
- (A) current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx a$
- (B) current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx a$
- (C) current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx 1.2 a$
- (D) current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx 1.2 a$
41. Consider $d \gg a$, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

[JEE-Advanced-2014]

- (A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3} \mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3} \mu_0 I^2 a^2}{2d}$

ANSWER KEY

EXERCISE (S-2)

1. Ans. $B_1 = \frac{\mu_0 b r_1^2}{3}$, $B_2 = \frac{\mu_0 b R^3}{3 r_2}$

2. Ans. $i_1 = 0.1110 \text{ A}$, $i_2 = 0.096 \text{ A}$

3. Ans. (a) $6.6 \times 10^{-5} \text{ T}$, (b) 0, 0, $2 \times 10^{-5} \ell \ln\left(\frac{4}{3}\right) \text{ N}$

4. Ans. $\frac{\mu_0 i q v}{2 \pi a}$

5. Ans. $\frac{m E I}{B e}$

6. Ans. 8

7. Ans. 4

8. Ans. $z = 0$, $x = \pm \frac{d}{\sqrt{3}}$, (ii) $\frac{I}{2 \pi d} \sqrt{\frac{\mu_0}{\pi \lambda}}$

9. Ans. $F = \alpha a^2 \hat{i} \hat{j}$

10. Ans. 4

11. Ans. (a) $k = NAB$, (b) $C = \frac{2 i_0 NAB}{\pi}$, (c) $Q \times \sqrt{\frac{NAB \pi}{2 i_0}}$

EXERCISE (O-2)

1. Ans. (C)

2. Ans. (C)

3. Ans. (C)

4. Ans. (A)

5. Ans. (C)

6. Ans. (B)

7. Ans. (C)

8. Ans. (C)

9. Ans. (C)

10. Ans. (A)

11. Ans. (C)

12. Ans. (A)

13. Ans. (D)

14. Ans. (A,B,C)

EXERCISE-JA

1. Ans. (A,C) 2. Ans. (A)

3. Ans. (A)→Q, R; (B)→P; (C)→Q, R; (D)→Q or (A)→Q,R; (B)→P; (C)→Q,R; (D)→Q,S

4. Ans. (A,C,D) 5. Ans. (C) 6. Ans. (A)→p,r,s (B)→r, s (C)→p,q,t (D)→r,s

7. Ans. (A)→p, t (B)→q,s,t (C)→p,r,t (D)→q 8. Ans. 7 9. Ans. (C) 10. Ans. (A)

11. Ans. (B) 12. Ans. (BC,BD, BCD) 13. Ans. (A) 14. Ans. (C,D) 15. Ans. 5

16. Ans. (D) 17. Ans. (B) 18. Ans. (A, C) 19. Ans. (A, D) 20. Ans. 3

21. Ans. (A,B,C)

REVIEW TEST

1. Ans. (a) 1.4, (b) 1

2. Ans. $\sqrt{15} \text{ C}$

3. Ans. (6.4 m, 0,0) (6.4m, 0, 2m)

4. Ans. (i) $-\frac{\mu_0 I}{4R} q v_0 \hat{k}$ (ii) $F_1 = 2 I R B$ $F_2 = 2 I R B$, Net force $= F_1 + F_2 = 4 I R B \hat{i}$

5. Ans. (a) $\frac{3mv^2}{4qa}$, (b) $\frac{3mv^3}{4a}$, (c) zero

6. Ans. 5

7. Ans. 020

8. Ans. (a) current in loop PQRS is clockwise from P to QRS., (b) $\vec{F} = BI_0 b (3\hat{k} - 4\hat{i})$, (c) $I = \frac{mg}{6bB_0}$

9. Ans. $\omega = \frac{dT_0}{QR^2B}$

10. Ans. (C)

11. Ans. (B)

12. Ans. (B)

13. Ans. (B)

14. Ans. (C)

15. Ans. (C)

16. Ans. (A)

17. Ans. (B)

18. Ans. (B)

19. Ans. (C)

20. Ans. (A)

21. Ans. (A)

22. Ans. (C)

23. Ans. (A, C, D)

24. Ans. (A, D)

25. Ans. (D)

26. Ans. (C)

27. Ans. (B)

28. Ans. (D)

29. Ans. (A)-q,s; (B)-p,s; (C)-q,s; (D)-p,s

30. Ans. (B)

31. Ans. (C)

32. Ans. (C)

33. Ans. (A)

34. Ans. (A)

35. Ans. (B)

36. Ans. (A)

37. Ans. (B,C,D)

38. Ans. (A) $\rightarrow$ (P,R, T) ; (B) $\rightarrow$ (P,Q) ; (C) $\rightarrow$ (S) ; (D) $\rightarrow$ (P)

39. Ans. (4)

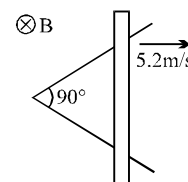
40. Ans. (C)

41. Ans. (B)

ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

EXERCISE (S-2)

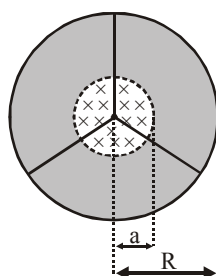
1. Two straight conducting rails form a right angle where their ends are joined. A conducting bar contact with the rails starts at vertex at the time $t = 0$ & moves symmetrically with a constant velocity of 5.2 m/s to the right as shown in figure. A 0.35 T magnetic field points out of the page. Calculate:



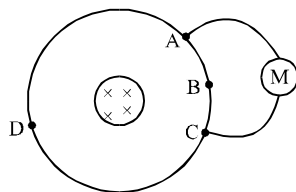
- (i) The flux through the triangle by the rails & bar at $t = 3.0 \text{ s}$.
 - (ii) The emf around the triangle at that time.
 - (iii) In what manner does the emf around the triangle vary with time.
2. A metal rod of resistance 20Ω is fixed along a diameter of a conducting ring of radius 0.1 m and lies on $x-y$ plane. There is a magnetic field $\vec{B} = (50\text{T})\hat{k}$. The ring rotates with an angular velocity $\omega = 20 \text{ rad/sec}$ about its axis. An external resistance of 10Ω is connected across the centre of the ring and rim. Find the current through external resistance.
3. A uniform but time varying magnetic field $B = Kt - C$; ($0 \leq t \leq C/K$), where K and C are constants and t is time, is applied perpendicular to the plane of the circular loop of radius 'a' and resistance R . Find the total charge that will pass around the loop.
4. A charged ring of mass $m = 50 \text{ gm}$, charge 2 coulomb and radius $R = 2\text{m}$ is placed on a smooth horizontal surface. A magnetic field varying with time at a rate of $(0.2 \text{ t}) \text{ Tesla/sec}$ is applied on to the ring in a direction normal to the surface of ring. Find the angular speed attained in a time $t_1 = 10 \text{ sec}$. Assume that the magnetic field is cylindrically symmetric and covering the entire ring.
5. A line charge λ per unit length is lodged uniformly onto the rim of a wheel of mass M and radius R . The wheel has light nonconducting spokes and is free to rotate without friction about its axis see figure. A uniform magnetic field extends over a circular region within the rim. It is given by,
- $$\vec{B} = -B_0\hat{k} \quad (r \leq a; a < R)$$
- $= 0$ (otherwise)

What is the angular velocity of the wheel after the field is suddenly switched off ?

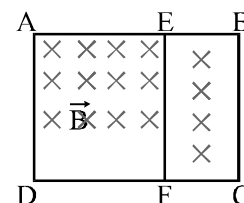
(NCERT)



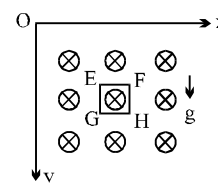
6. A variable magnetic field creates a constant emf E in a conductor ABCDA. The resistances of portion ABC, CDA and AMC are R_1 , R_2 and R_3 respectively. What current will be shown by meter M? The magnetic field is concentrated near the axis of the circular conductor.



7. A rectangular frame ABCD made of a uniform metal wire has a straight connection between E & F made of the same wire as shown in the figure. AEFD is a square of side 1 m & $EB = FC = 0.5$ m. The entire circuit is placed in a steadily increasing uniform magnetic field directed into the plane of the paper & normal to it. The rate of change of the magnetic field is 1 T/s, the resistance per unit length of the wire is $1 \Omega/\text{m}$. Find the current in segments AE, BE & EF.

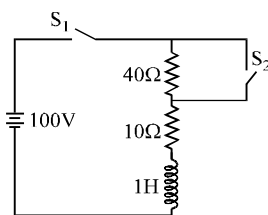


8. A magnetic field $B = (B_0 y / a) \hat{k}$ is into the plane of paper in the $+z$ direction. B_0 and a are positive constants. A square loop EFGH of side a , mass m and resistance R , in x - y plane, starts falling under the influence of gravity. Note the directions of x and y axes in the figure. Find

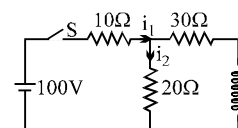


- (a) the induced current in the loop and indicate its direction,
 (b) the total Lorentz force acting on the loop and indicate its direction,
 (c) an expression for the speed of the loop, $v(t)$ and its terminal value.

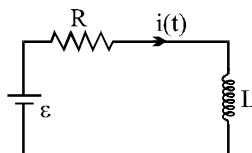
9. In the circuit shown in the figure the switches S_1 and S_2 are closed at time $t = 0$. After time $t = (0.1) \ln 2$ sec, switch S_2 is opened. Find the current in the circuit at time $t = (0.2) \ln 2$ sec.



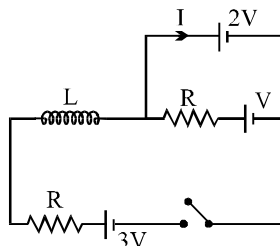
10. Find the values of i_1 and i_2
 (i) immediately after the switch S is closed.
 (ii) long time later, with S closed.
 (iii) immediately after S is open.
 (iv) long time after S is opened.



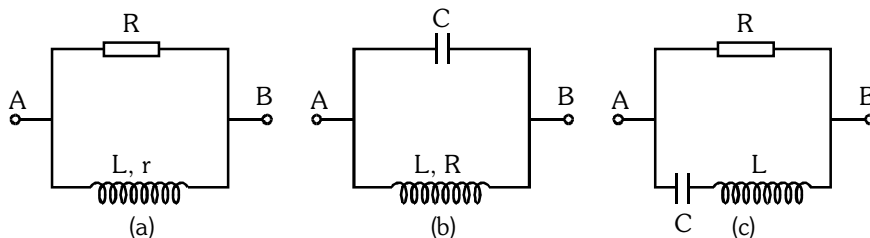
11. Suppose the emf of the battery in the circuit shown varies with time t so the current is given by $i(t) = 3 + 5t$, where i is in amperes & t is in seconds. Take $R = 4\Omega$, $L = 6\text{H}$ & find an expression for the battery emf as function of time.



12. A zero resistance coil of inductance L connects the upper ends of two vertical parallel long conductors. A horizontal sliding conductor, free to slide up and down, always maintaining contact with the vertical conductors, starts falling from rest at $t = 0$, due to its own weight mg . A uniform magnetic field of magnitude B exists in the region horizontally and perpendicular to the plane of the conductors. The distance between the vertical conductors is ' l '. After what time does the conductor come back to its starting position? Also find maximum speed achieved.
13. In the LR circuit shown, what is the variation of the current I as a function of time? The switch is closed at time $t = 0$ sec.



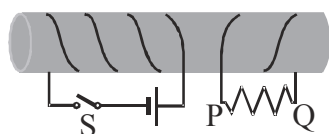
14. Two coils, 1 & 2, have a mutual inductance $= M$ and resistances R each. A current flows in coil 1, which varies with time as: $I_1 = kt^2$, where k is a constant and ' t ' is time. Find the total charge that has flown through coil 2, between $t = 0$ and $t = T$.
15. Draw the approximate vector diagrams of currents in the circuits shown in Fig. The voltage applied across the points A and B is assumed to be sinusoidal; the parameters of each circuit are so chosen that the total current I_0 lags in phase behind the external voltage by an angle ϕ .



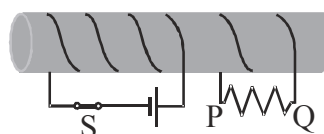
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

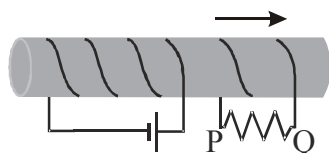
1. For each of the experiments (1, 2, 3, 4) shown in figure. Choose the **CORRECT** option(s) which shows the direction of current flow through the resistor PQ? Note that the wires are not always wrapped around the plastic tube in the same way.



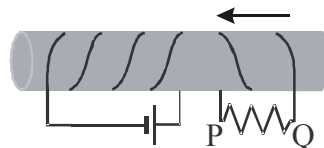
(1) S to be closed



(2) S to be opened



(3) Resistor coil PQ moves to right



(4) Resistor coil PQ

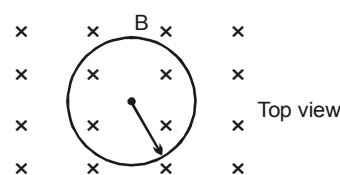
- | | | | |
|------------|--------|-------------------|--------|
| (1) | (2) | (3) moves to left | (4) |
| (A) P to Q | P to Q | P to Q | P to Q |
| (B) P to Q | Q to P | P to Q | Q to P |
| (C) Q to P | Q to P | Q to P | Q to P |
| (D) Q to P | Q to P | P to Q | P to Q |

2. An electron is moving in a circular orbit of radius R with an angular acceleration α . At the centre of the orbit is kept a conducting loop of radius r , ($r \ll R$). The e.m.f induced in the smaller loop due to the motion of the electron is :-

- (A) zero, since charge on electron is constant (B) $\frac{\mu_0 e r^2}{4R} \alpha$
- (C) $\frac{\mu_0 e r^2}{4\pi R} \alpha$ (D) none of these

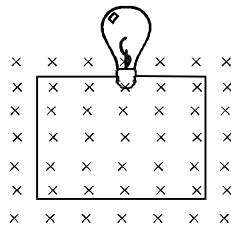
3. A non conducting ring (of mass m , radius r , having charge Q) is placed on a rough horizontal surface (in a region with transverse magnetic field). The field is increasing with time at the rate R and coefficient of friction between the surface and the ring is μ . For ring to remain in equilibrium μ should be greater than :-

- (A) $\frac{QrR}{mg}$ (B) $\frac{QrR}{2mg}$ (C) $\frac{QrR}{3mg}$ (D) $\frac{2QrR}{mg}$



Top view

4. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt . The bulb will shine with full brightness if Δt is equal to :-



- (A) 20 ms (B) 0.02 ms (C) 2 ms (D) 0.2 ms
5. The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m² acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1 μ V, is :-

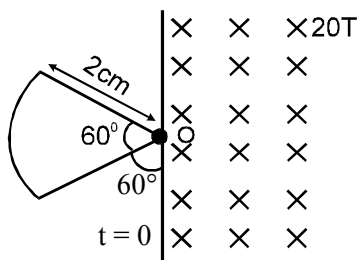
- (A) $\frac{2}{\pi}$ cm (B) $\frac{3}{\pi}$ cm (C) $\frac{4}{\pi}$ cm (D) $\frac{5}{\pi}$ cm

6. A composite rod of length ℓ is one fourth insulator and remaining conductor is made to rotate freely with angular velocity ω , in a space free of any gravitational, electric & magnetic field. Then potential difference across the conducting region will be (rotation is about insulating end).

- (A) $\frac{3m_e \omega^2 \ell^2}{4e}$ (B) $\frac{1}{4} \frac{m_e \omega^2 \ell^2}{e}$ (C) $\frac{1}{16} \frac{m_e \omega^2 \ell^2}{e}$ (D) $\frac{15}{32} \frac{m_e \omega^2 \ell^2}{e}$

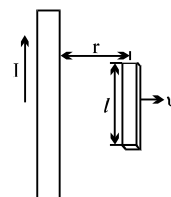
7. A uniform magnetic field 20 T exists on right side of the boundary in a gravity free space as shown in figure. The given circular arc of radius 2 cm made of conducting wire of total resistance 4Ω is rotated around point O at a constant angular speed 2 rad per second. Power required to maintain the constant angular

velocity between time interval $t = \frac{\pi}{6}$ s to $t = \frac{\pi}{3}$ s is :-



- (A) 64 μ W (B) 32 μ W (C) 128 μ W (D) 16 μ W

8. A conducting rod moves with constant velocity v perpendicular to the long, straight wire carrying a current I as shown. Compute the emf generated between the ends of the rod.



- (A) $\frac{\mu_0 v I l}{\pi r}$ (B) $\frac{\mu_0 v I l}{2\pi r}$
 (C) $\frac{2\mu_0 v I l}{\pi r}$ (D) $\frac{\mu_0 v I l}{4\pi r}$
9. A metallic rod of length L and mass M is moving under the action of two unequal forces F_1 and F_2 (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is (e/m) , then the potential difference between the ends of the rod in steady state must be
- (A) $|F_1 - F_2| mL/eM$ (B) $(F_1 + F_2) mL/eM$
 (C) $[mL/eM] \ln [F_1/F_2]$ (D) None
10. Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v . If r is the resistance per unit length of the wire the current in the circuit will be

- (A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$ (C) Bvr (D) Bv

11. The magnetic field in a region is given by $\vec{B} = B_0 \left(1 + \frac{x}{a} \right) \hat{k}$. A square loop of edge-length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{V} = V_0 \hat{i}$. The emf induced in the loop is

- (A) $\frac{V_0 B_0 d^2}{a}$ (B) $\frac{V_0 B_0 d^2}{2a}$ (C) $\frac{V_0 B_0 a^2}{d}$ (D) None

12. A rectangular coil of single turn, having area A , rotates in a uniform magnetic field B with an angular velocity ω about an axis perpendicular to the field. If initially the plane of coil is perpendicular to the field, then the average induced e.m.f. when it has rotated through 90° is :-

- (A) $\frac{\omega BA}{\pi}$ (B) $\frac{\omega BA}{2\pi}$ (C) $\frac{\omega BA}{4\pi}$ (D) $\frac{2\omega BA}{\pi}$

13. In figure (a) a solenoid produces a magnetic field whose strength increases into the plane of the page. An induced emf is established in a conduction loop surrounding the solenoid, and this emf lights bulbs A and B. In figure (b) point P and Q are shorted. After the short is inserted

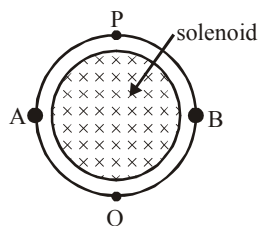


fig. (a)

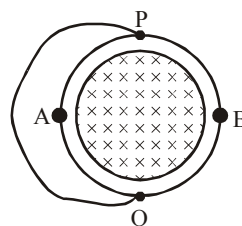
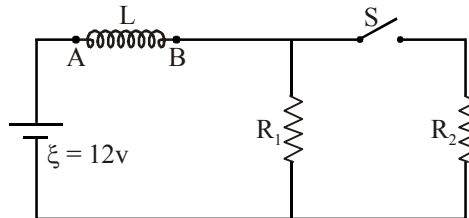


fig. (b)

- (A) Bulb A goes out, bulb B gets brighter
 (B) Bulb B goes out, bulb A gets brighter
 (C) Bulb A goes out, bulb B gets dimmer
 (D) Bulb B goes out, bulb A gets dimmer

14. The circuit shown has been operating for a long time. The instant after the switch in the circuit labeled S is opened, what is the voltage across the inductor V_L and which labeled point (A or B) of the inductor is at a higher potential ? Take $R_1 = 4.0 \Omega$, $R_2 = 8.0 \Omega$, and $L = 2.5 \text{ H}$.

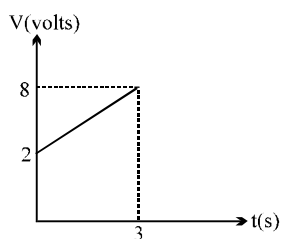


- (A) $V_L = 12 \text{ V}$; Point A is at the higher potential
 (B) $V_L = 12 \text{ V}$; Point B is at the higher potential
 (C) $V_L = 6 \text{ V}$; Point A is at the higher potential
 (D) $V_L = 6 \text{ V}$; Point B is at the higher potential
15. An AC current is given by $I = I_0 + I_1 \sin \omega t$ then its rms value will be
 (A) $\sqrt{I_0^2 + 0.5 I_1^2}$ (B) $\sqrt{I_0^2 + 0.5 I_0^2}$ (C) 0 (D) $I_0 / \sqrt{2}$
16. Power factor of an L-R series circuit is 0.6 and that of a C-R series circuit is 0.5. If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L-R circuit to the resistance in the C-R circuit is
 (A) 6/5 (B) 5/6 (C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

MULTIPLE CORRECT TYPE QUESTIONS

17. Figure shows a conducting rod of negligible resistance that can slide on smooth U-shaped rail made of wire of resistance $1 \Omega/\text{m}$. Position of the conducting rod at $t = 0$ is shown. A time dependent magnetic field $B = 2t$ Tesla is switched on at $t = 0$. After the magnetic field is switched on, the conducting rod is moved to the left perpendicular to the rails at constant speed 5 cm/s by some external agent.
-
- (A) The current in the loop at $t = 0$ due to induced emf is 0.16 A , clockwise
 (B) At $t = 2 \text{ s}$, induced emf has magnitude 0.08 V
 (C) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2 \text{ s}$, is equal to 0.08 N
 (D) The magnitude of the force required to move the conducting rod at constant speed 5 cm/s at $t = 2 \text{ s}$, is equal to 0.16 N

18. A circuit element is placed in a closed box. At time $t = 0$, constant current generator supplying a current of 1 amp, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is :

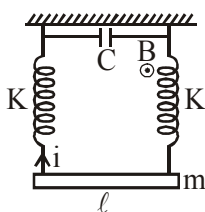


- (A) resistance of 2Ω (B) battery of emf 6V
 (C) inductance of 2H (D) capacitance of 0.5F
19. Two coils A and B have coefficient of mutual inductance $M = 2H$. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then
 (A) the change in current in B in this time interval is 0.5 A
 (B) the change in current in B in this time interval is 2A
 (C) the change in current in B in this time interval is 8A
 (D) a change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 20 and 21.

In the figure shown a uniform conducting rod of mass m and length ℓ is suspended in vertical plane by two conducting springs of spring constant K . Upper end of spring are connected to each other by capacitor of capacitance C . A uniform horizontal magnetic field (B_0) perpendicular to plane of spring exists in space. Initially rod is in equilibrium but if centre of rod is pulled down and released, it performs SHM. Assume that the spring is small and neglect the magnetic force of interaction between circular section of springs & self inductance of rod.



20. Find time period of oscillation of rod :-

(A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{B^2\ell^2C}{K}}$ (C) $\pi\sqrt{\frac{m + B^2\ell^2C}{K}}$ (D) $2\pi\sqrt{\frac{B^2\ell^2C + m}{2K}}$

21. Choose correct options from following :-

- (A) Electrical energy stored in capacitor is maximum when rod is at its lower extreme position
 (B) Electrical energy stored in capacitor is maximum when rod is at its mean position
 (C) Current in rod is maximum at mean position of rod
 (D) If magnetic field is switched off then mean position of rod will change

Paragraph for question nos. 22 to 24.

In a series L-R circuit, connected with a sinusoidal ac source, the maximum potential difference across L and R are respectively 3 volts and 4 volts.

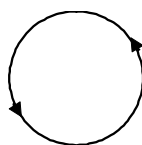
22. At an instant the potential difference across resistor is 2 volts. The potential difference in volt, across the inductor at the same instant will be :
- (A) $3 \cos 30^\circ$ (B) $3 \cos 60^\circ$ (C) $3 \cos 45^\circ$ (D) None of these
23. At the same instant, the magnitude of the potential difference in volt, across the ac source may be :-
- (A) $4 + 3\sqrt{3}$ (B) $\frac{4 + 3\sqrt{3}}{2}$ (C) $1 + \frac{\sqrt{3}}{2}$ (D) $2 + \frac{\sqrt{3}}{2}$
24. If the current at this instant is decreasing the magnitude of potential difference at that instant across the ac source is :-
- (A) Increasing (B) Decreasing (C) Constant (D) Cannot be said

EXERCISE - JA

(Not to be discussed in class)

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

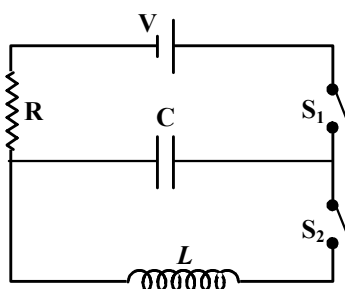
1. In the given diagram, a line of force of a particular force field is shown. Out of the following options, it can never represent [JEE 2006]



- (A) an electrostatic field (B) a magnetostatic field
 (C) a gravitational field of a mass at rest (D) an induced electric field

Comprehension-I

The capacitor of capacitance C can be charged (with the help of a resistance R) by a voltage source V , by closing switch S_1 while keeping switch S_2 open. The capacitor can be connected in series with an inductor ' L ' by closing switch S_2 and opening S_1 .



2. Initially, the capacitor was uncharged. Now, switch S_1 is closed and S_2 is kept open. If time constant of this circuit is τ , then [JEE 2006]
- (A) after time interval τ , charge on the capacitor is $CV/2$
 (B) after time interval 2τ , charge on the capacitor is $CV(1-e^{-2})$
 (C) the work done by the voltage source will be half of the heat dissipated when the capacitor is fully charged.
 (D) after time interval 2τ , charge on the capacitor is $CV(1-e^{-1})$
3. After the capacitor gets fully charged, S_1 is opened and S_2 is closed so that the inductor is connected in series with the capacitor. Then, [JEE 2006]
- (A) at $t = 0$, energy stored in the circuit is purely in the form of magnetic energy
 (B) at any time $t > 0$, current in the circuit is in the same direction
 (C) at $t > 0$, there is no exchange of energy between the inductor and capacitor
 (D) at any time $t > 0$, instantaneous current in the circuit may $V\sqrt{\frac{C}{L}}$

4. If the total charge stored in the LC circuit is Q_0 , then for $t \geq 0$ [JEE 2006]

(A) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} + \frac{t}{\sqrt{LC}}\right)$

(B) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} - \frac{t}{\sqrt{LC}}\right)$

(C) the charge on the capacitor is $Q = -LC \frac{d^2Q}{dt^2}$

(D) the charge on the capacitor is $Q = -\frac{1}{\sqrt{LC}} \frac{d^2Q}{dt^2}$

Paragraph for Question 5 to 7

Modern trains are based on Maglev technology in which trains are magnetically levitated, which runs its EDS Maglev system.

There are coils on both sides of wheels. Due to motion of trains, current induces in the coil of track which levitate it. This is in accordance with Lenz's law. If trains lower down then due to Lenz's law a repulsive force increases due to which train gets uplifted and if it goes much high then there is a net downward force due to gravity. The advantage of Maglev train is that there is no friction between the train and the track, thereby reducing power consumption and enabling the train to attain very high speeds.

Disadvantage of Maglev train is that as it slows down the electromagnetic forces decreases and it becomes difficult to keep it levitated and as it moves forward according to Lenz's law there is an electromagnetic drag force. [JEE 2006]

5. What is the advantage of this system ?
 (A) No friction hence no power consumption (B) No electric power is used
 (C) Gravitation force is zero (D) Electrostatic force draws the train
6. What is the disadvantage of this system ?
 (A) Train experience upward force according to Lenz's law
 (B) Friction force create a drag on the train
 (C) Retardation
 (D) By Lenz's law train experience a drag
7. Which force causes the train to elevate up ?
 (A) Electrostatic force (B) Time varying electric field
 (C) Magnetic force (D) Induced electric field

8. Match the following Columns [JEE 2006]

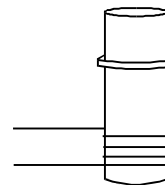
Column-I

- (A) Dielectric ring uniformly charged
 (B) Dielectric ring uniformly charged rotating with angular velocity .
 (C) Constant current i_0 in ring
 (D) Current $i = i_0 \cos \omega t$ in ring

Column-II

- (P) Time independent electrostatic field out of system
 (Q) Magnetic field
 (R) Induced electric field
 (S) Magnetic moment

9. **STATEMENT-1** : A vertical iron rod has a coil of wire wound over it at the bottom end. An alternating current flows in the coil. The rod goes through a conducting ring as shown in the figure. The ring can float at a certain height above the coil



because

STATEMENT-2 : In the above situation, a current is induced in the ring which interacts with the horizontal component of the magnetic field to produce an average force in the upward direction.

[JEE 2007]

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

10. Column I gives certain situations in which a straight metallic wire of resistance R is used and Column II gives some resulting effects. Match the statements in Column I with the statements in Column II. [2007]

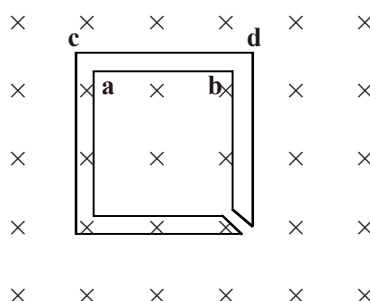
Column I

- (A) A charged capacitor is connected to the ends of the wire
 (B) The wire is moved perpendicular to its length with constant velocity in a uniform magnetic field perpendicular to the plane of motion.
 (C) The wire is placed in a constant electric field that has a direction along the length of the wire
 (D) A battery of constant emf is connected to the ends of the wire

Column II

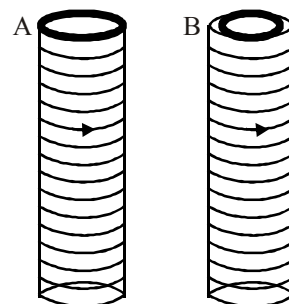
- (p) A constant current flows through the wire
 (q) Thermal energy is generated in the wire
 (r) A constant potential difference develops between the ends of the wire
 (s) Charges of constant magnitude appear at the ends of the wire

11. The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure. The magnitude of the field increases with time. I_1 and I_2 are the currents in the segments **ab** and **cd**. Then, [JEE-2009]



- (A) $I_1 > I_2$
 (B) $I_1 < I_2$
 (C) I_1 is in the direction **ba** and I_2 is in the direction **cd**
 (D) I_1 is in the direction **ab** and I_2 is in the direction **dc**

12. Two metallic rings A and B, identical in shape and size but having different resistivities ρ_A and ρ_B , are kept on top of two identical solenoids as shown in the figure. When current I is switched on in both the solenoids in identical manner, the rings A and B jump to heights h_A and h_B , respectively, with $h_A > h_B$. The possible relation(s) between their resistivities and their masses m_A and m_B is(are)

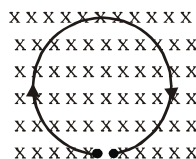


[JEE-2009]

- (A) $\rho_A > \rho_B$ and $m_A = m_B$ (B) $\rho_A < \rho_B$ and $m_A = m_B$
(C) $\rho_A > \rho_B$ and $m_A > m_B$ (D) $\rho_A < \rho_B$ and $m_A < m_B$
13. An AC voltage source of variable angular frequency ω and fixed amplitude V_0 is connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased

[JEE 2010]

- (A) the bulb glows dimmer (B) the bulb glows brighter
(C) total impedance of the circuit is unchanged (D) total impedance of the circuit increases
14. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is :-



- (A) IBL (B) $\frac{IBL}{\pi}$ (C) $\frac{IBL}{2\pi}$ (D) $\frac{IBL}{4\pi}$

15. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **Column II**. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 (indicated in circuits) are related as shown in **Column I**. Match the two

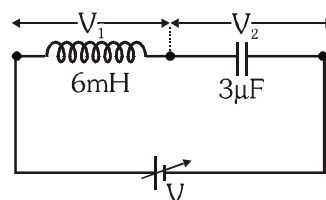
[JEE 2010]

Column I

Column II

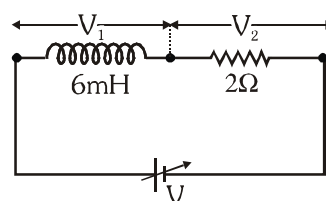
- (A) $I \neq 0$, V_1 is proportional to I

(p)

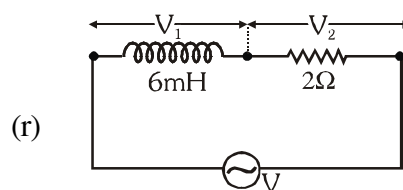


- (B) $I \neq 0$, $V_2 > V_1$

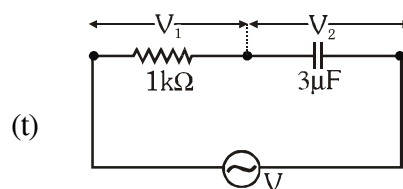
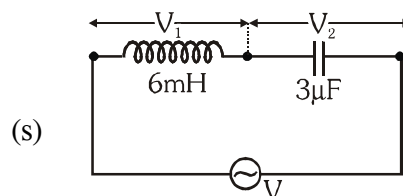
(q)



(C) $V_1=0, V_2=V$

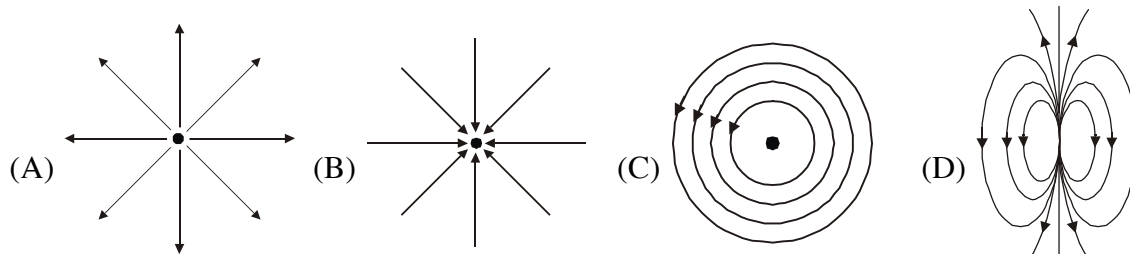


(D) $I \neq 0, V_2$ is proportional to I



16. Which of the field patterns given below is valid for electric field as well as for magnetic field?

[JEE 2011]



17. A series R-C circuit is connected to AC voltage source. Consider two cases ; (A) when C is without a dielectric medium and (B) when C is filled with dielectric of constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in the two cases. Which of the following is/are true?

[JEE 2011]

(A) $I_R^A > I_R^B$

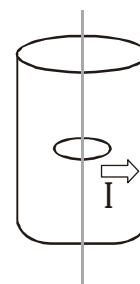
(B) $I_R^A < I_R^B$

(C) $V_C^A > V_C^B$

(D) $V_C^A < V_C^B$

18. A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005 ohm and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0 \cos(300t)$ where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \sin(300t)$, then 'N' is

[JEE 2011]

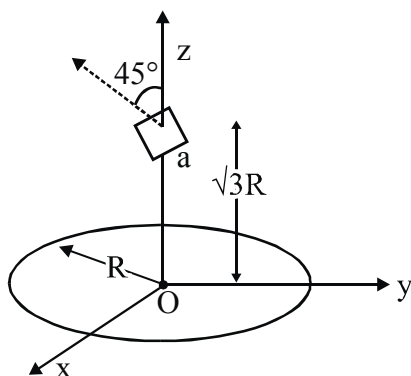


19. A series R-C combination is connected to an AC voltage of angular frequency $\omega = 500$ radian/s. If the impedance of the R-C circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is :-

[JEE 2011]

20. A circular wire loop of radius R is placed in the x-y plane centred at the origin O. A square loop of side a ($a \ll R$) having two turns is placed with its centre at $z = \sqrt{3}R$ along the axis of the circular wire loop, as shown in figure. The plane of the square loop makes an angle of 45° with respect to the z-axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R}$, then the value of p is :-

[JEE 2012]



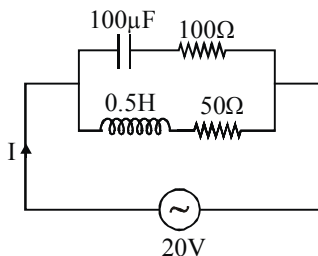
21. A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it. The correct statement(s) is (are)

[JEE 2012]

- (A) The emf induced in the loop is zero if the current is constant.
(B) The emf induced in the loop is infinite if the current is constant.
(C) The emf induced in the loop is zero if the current decreases at a steady rate.
(D) The emf induced in the loop is finite if the current decreases at a steady rate.

22. In the given circuit, the AC source has $\omega = 100$ rad/s. Considering the inductor and capacitor to be ideal, the correct choice (s) is(are)

[JEE 2012]



- (A) The current through the circuit, I is 0.3 A. (B) The current through the circuit, i is $0.3\sqrt{2}$ A.
(C) The voltage across 100Ω resistor = $10\sqrt{2}$ V. (D) The voltage across 50Ω resistor = 10 V.

Paragraph for Questions 23 and 24

A point charge Q is moving in a circular orbit of radius R in the x - y plane with an angular velocity ω .

This can be considered as equivalent to a loop carrying a steady current $\frac{Q\omega}{2\pi}$. A uniform magnetic field along the positive z -axis is now switched on, which increases at a constant rate from 0 to B in one second. Assume that the radius of the orbit remains constant. The application of the magnetic field induces an emf in the orbit. The induced emf is defined as the work done by an induced electric field in moving a unit positive charge around a closed loop. It is known that for an orbiting charge, the magnetic dipole moment is proportional to the angular momentum with a proportionality constant γ .

[JEE Advance-2013]

23. The change in the magnetic dipole moment associated with the orbit, at the end of the time interval of the magnetic field change is :-

(A) $-\gamma BQR^2$ (B) $-\gamma \frac{BQR^2}{2}$ (C) $\gamma \frac{BQR^2}{2}$ (D) γBQR^2

24. The magnitude of the induced electric field in the orbit at any instant of time during the time interval of the magnetic field change is

(A) $\frac{BR}{4}$ (B) $\frac{BR}{2}$ (C) BR (D) $2BR$

Paragraph for Questions 25 and 26

A thermal power plant produces electric power of 600 kW and 4000 V, which is to be transported to a place 20 km away from the power plant for consumers' usage. It can be transported either directly with a cable of large current carrying capacity or by using a combination of step-up and step-down transformers at the two ends. The drawback of the direct transmission is the large energy dissipation. In the method using transformers, the dissipation is much smaller. In this method, a step-up transformer is used at the plant side so that the current is reduced to a smaller value. At the consumers' end, a step-down transformer is used to supply power to the consumers at the specified lower voltage. It is reasonable to assume that the power cable is purely resistive and the transformers are ideal with a power factor unity. All the currents and voltages mentioned are rms values. [JEE Advance-2013]

25. In the method using the transformers, assume that the ratio of the number of turns in the primary to that in the secondary in the step-up transformer is 1 : 10. If the power to the consumers has to be supplied at 200 V, the ratio of the number of turns in the primary to that in the secondary in the step-down transformer is

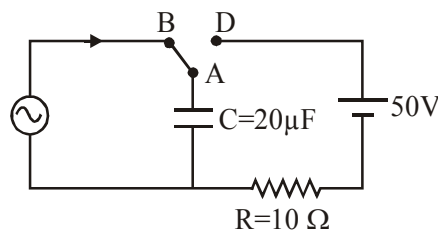
(A) 200 : 1 (B) 150 : 1 (C) 100 : 1 (D) 50 : 1

26. If the direct transmission method with a cable of resistance $0.4 \Omega \text{ km}^{-1}$ is used, the power dissipation (in %) during transmission is :-

(A) 20 (B) 30 (C) 40 (D) 50

27. At time $t = 0$, terminal A in the circuit shown in the figure is connected to B by a key and an alternating current $I(t) = I_0 \cos(\omega t)$, with $I_0 = 1 \text{ A}$ and $\omega = 500 \text{ rad s}^{-1}$ starts flowing in it with the initial direction shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If $C = 20 \mu\text{F}$, $R = 10 \Omega$ and the battery is ideal with emf of 50 V , identify the correct statement (s).

[JEE Advance-2014]



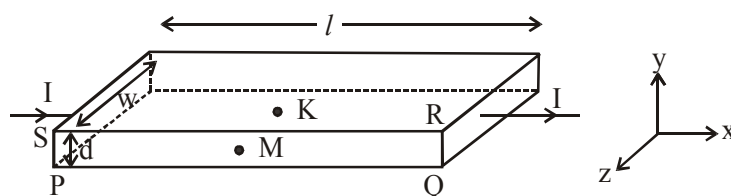
- (A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is $1 \times 10^{-3} \text{ C}$.
- (B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise.
- (C) Immediately after A is connected to D, the current in R is 10 A
- (D) $Q = 2 \times 10^{-3} \text{ C}$

Paragraph for Question No. 28 and 29

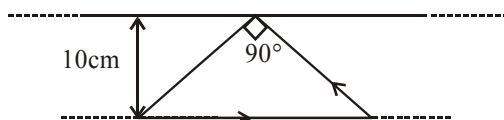
In a thin rectangular metallic strip a constant current I flows along the positive x -direction, as shown in the figure. The length, width the thickness of the strip are l , w and d , respectively.

A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y -direction. Due to this, the charge carriers experience a net deflection along the z -direction. This results in accumulation of charge carriers on the surface PQRS and appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z -direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons.

[JEE Advance-2015]

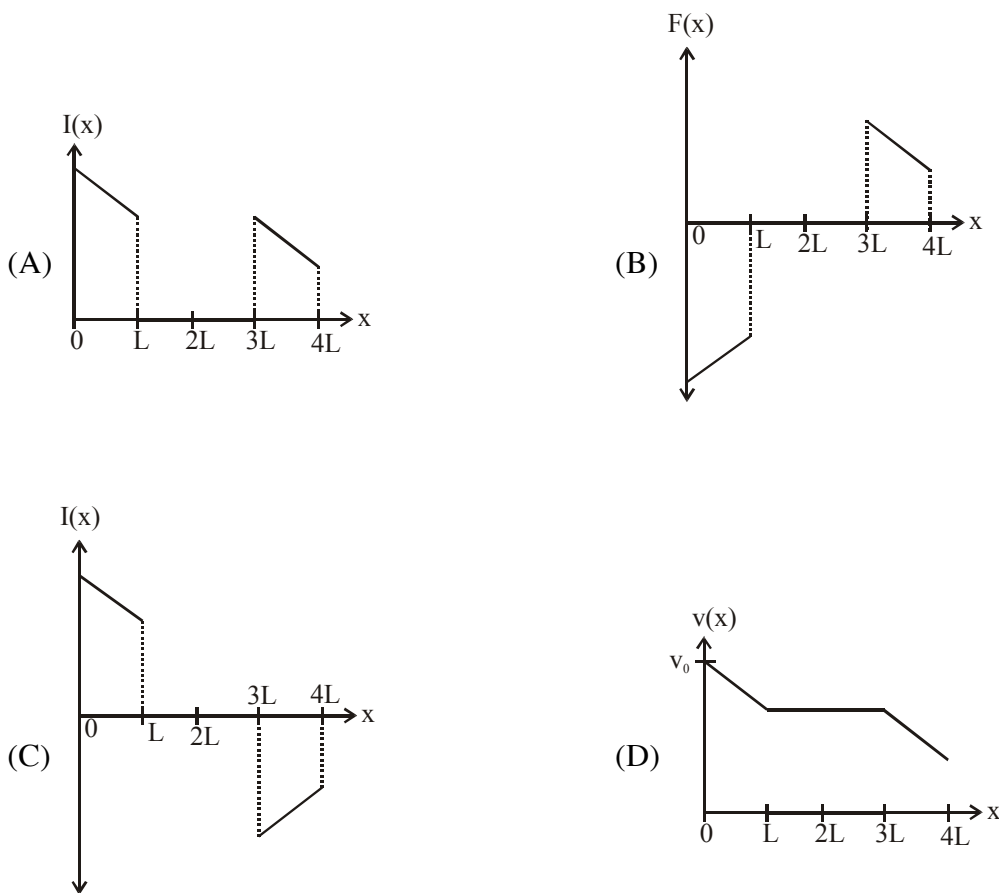
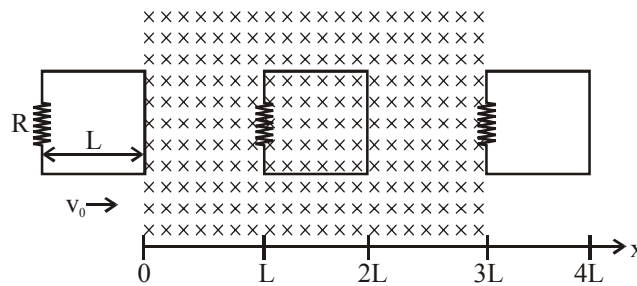


28. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2 , respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B , the correct statement(s) is(are)
- (A) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$
 (B) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$
 (C) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$
 (D) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$
29. Consider two different metallic strips (1 and 2) of same dimensions (length l , width w and thickness d) with carrier densities n_1 and n_2 , respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2 , both along positive y-direction. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)
- (A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$
 (B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$
 (C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$
 (D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$
30. A conducting loop in the shape of right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at constant rate of 10 A s^{-1} . Which of the following statement(s) is(are) true? **[JEE Advance-2016]**



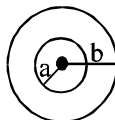
- (A) The induced current in the wire is in opposite direction to the current along the hypotenuse.
 (B) There is a repulsive force between the wire and the loop
 (C) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire
 (D) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volt.
31. Two inductors L_1 (inductance 1 mH, internal resistance 3Ω) and L_2 (inductance 2mH, internal resistance 4Ω), and a resistor R (resistance 12Ω) are all connected in parallel across a 5V battery. The circuit is switched on at time $t = 0$. The ratio of the maximum to the minimum current ($I_{\max}/I_{\min}$) drawn from the battery is. **[JEE Advance-2016]**

32. A rigid wire loop of square shape having side of length L and resistance R is moving along the x -axis with a constant velocity v_0 in the plane of the paper. At $t = 0$, the right edge of the loop enters a region of length $3L$ where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0 , the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let $v(x)$, $I(x)$ and $F(x)$ represent the velocity of the loop, current in the loop, and force on the loop, respectively, as a function of x . Counter-clockwise current is taken as positive. Which of the following schematic plot(s) is(are) correct? (Ignore gravity) [JEE Advance-2016]

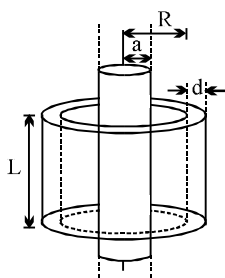
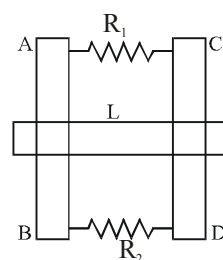


REVIEW TEST

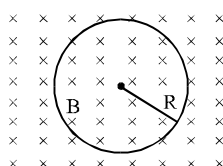
1. Two concentric and coplanar circular coils have radii a and $b (>> a)$ as shown in figure. Resistance of the inner coil is R . Current in the outer coil is increased from 0 to i , then find the total charge circulating the inner coil.



2. An LCR series circuit with 100Ω resistance is connected to an ac source of 200 V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60° . When only the inductance is removed, the current leads the voltage by 60° . Calculate the current and the power dissipated in the LCR circuit.
3. Two parallel vertical metallic rails AB & CD are separated by 1 m. They are connected at the two ends by resistance R_1 & R_2 as shown in the figure. A horizontally metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6 T perpendicular to the plane of the rails, it is observed that when the terminal velocity is attained, the power dissipated in R_1 & R_2 are 0.76 W & 1.2 W respectively. Find the terminal velocity of bar L & value R_1 & R_2 .
4. A box P and a coil Q are connected in series with an ac source of variable frequency. The emf of source at 10 V. Box P contains a capacitance of $1\mu\text{F}$ in series with a resistance of 32Ω while coil Q has a self-inductance 4.9 mH and a resistance of 68Ω series. The frequency is adjusted so that the maximum current flows in P and Q. Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively.
5. A long solenoid of radius a and number of turns per unit length n is enclosed by cylindrical shell of radius R , thickness d ($d \ll R$) and length L . A variable current $i = i_0 \sin \omega t$ flows through the coil. If the resistivity of the material of cylindrical shell is ρ , find the induced current in the shell. **[JEE 2005]**

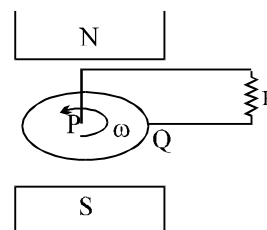


6. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time ' t ', as $R = R_0 + t$. The e.m.f induced in the loop is

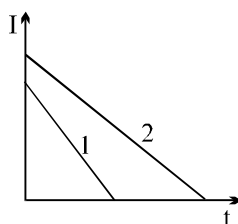


- (A) $2\pi(R_0 + t)B$ clockwise (B) $\pi(R_0 + t)B$ clockwise
 (C) $2\pi(R_0 + t)B$ anticlockwise (D) zero

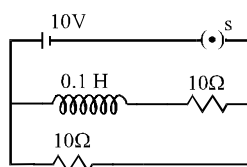
7. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. What current, if any, flows through R?



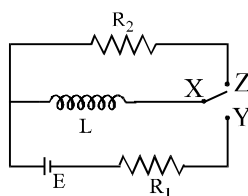
- (A) a current from P to Q
(B) a current from Q to P
(C) no current, because the emf in the disc is opposed by the back emf
(D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
(E) no current, because no radial emf is induced in the disc
8. Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



- (A) 1
(B) 2
(C) same
(D) data are insufficient to decide
9. In the adjoining circuit, initially the switch S is open. The switch 'S' is closed at $t = 0$. The difference between the maximum and minimum current that can flow in the circuit is

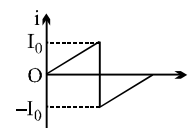
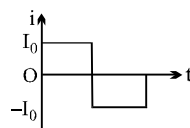
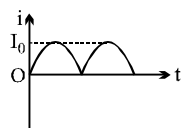
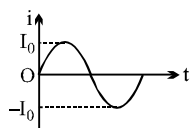


- (A) 2 Amp
(B) 3 Amp
(C) 1 Amp
(D) nothing can be concluded
10. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :

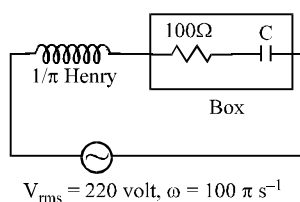


- (A) $\frac{LE^2}{2R_1^2}$
(B) $\frac{LE^2}{2R_2^2}$
(C) $\frac{LE^2}{2R_1R_2}$
(D) $\frac{LE^2R_2}{2R_1^2}$

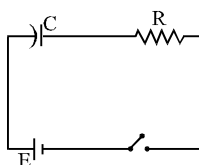
11. If I_1 , I_2 , I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



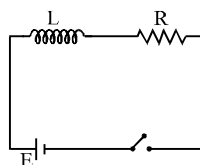
- (A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$
12. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is :-



- (A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$
13. The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10 cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3, the time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbondioxide will be close to ($\ln 5 = 1.601$, $\ln 2 = 0.693$).
- [JEE Main Online 2014]
- (A) 231 s (B) 208 s (C) 142 s (D) 161 s
14. The switches in figures (a) and (b) are closed at $t = 0$



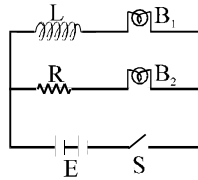
(a)



(b)

- (A) The charge on C just after $t = 0$ is EC .
 (B) The charge on C long after $t = 0$ is EC .
 (C) The current in L just after $t = 0$ is E/R .
 (D) The current in L long after $t = 0$ is E/R .

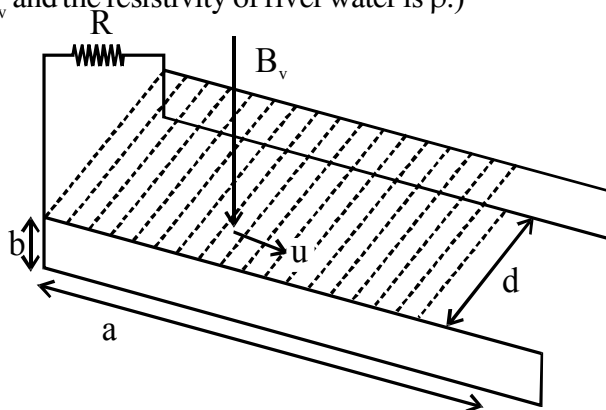
15. An inductor L , a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R . Which of the following statement gives the correct description of the happenings when the switch S is closed?



- (A) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
 (B) B_1 lights up earlier and finally both the bulbs acquire equal brightness.
 (C) B_2 lights up earlier and finally B_1 shines brighter than B_2 .
 (D) B_1 and B_2 light up together with equal brightness all the time.
16. Two different coils have self inductance 8mH and 2mH . The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are I_1 , V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are I_2 , V_2 and W_2 respectively. Then:

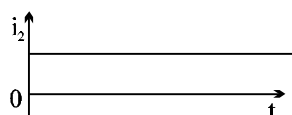
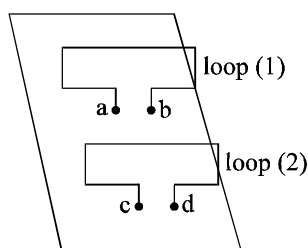
(A) $\frac{I_1}{I_2} = \frac{1}{4}$ (B) $\frac{I_1}{I_2} = 4$ (C) $\frac{W_2}{W_1} = 4$ (D) $\frac{V_2}{V_1} = \frac{1}{4}$

17. The figure shows an apparatus suggested by Faraday to generate electric current from a flowing river. Two identical conducting plates of length a and width b are placed parallel facing one another on opposite sides of the river following with velocity u at a distance d apart. Now both the plates are connected by a load resistance R . Then the current through the load R is :- (Consider vertical component of the magnetic field produced by earth is B_v and the resistivity of river water is ρ .)

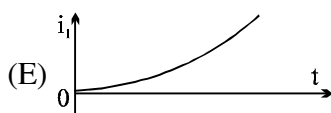
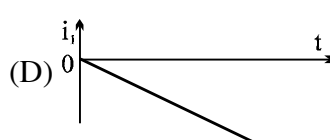
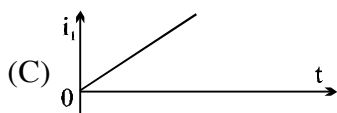
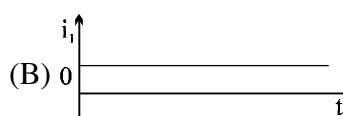
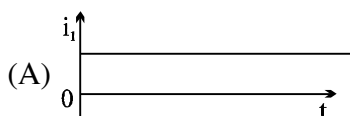


(A) $\frac{B_v ub}{R}$ (B) $\frac{B_v ub}{R + \frac{\rho d}{ab}}$ (C) $\frac{B_v ud}{R + \frac{\rho d}{ab}}$ (D) None of the above

18. An electric current i_1 can flow either direction through loop (A) and induced current i_2 in loop (B). Positive i_1 is when current is from 'a' to 'b' in loop (A) and positive i_2 is when the current is from 'c' to 'd' in loop (B). In an experiment, the graph of i_2 against time 't' is as shown below

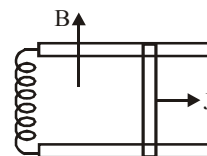


Which one(s) of the following graphs could have caused i_2 to behave as given above.



19. Two parallel resistanceless rails are connected by an inductor of inductance L at

one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the **CORRECT** option (s).



- (A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

$$d = \sqrt{\frac{3J^2 L}{4B^2 \ell^2 m}}$$

- (B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial

$$\text{velocity is } i = \sqrt{\frac{3J^2}{4Lm}}$$

- (C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

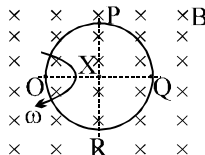
$$d = \sqrt{\frac{3J^2 L}{B^2 \ell^2 m}}$$

- (D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial

$$\text{velocity is } i = \sqrt{\frac{3J^2}{mL}}$$

Paragraph for Question No. 20 to 22

A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The rotational velocity is ω .



20. Choose the correct statement(s) related to the potential of the points P, Q and R

- (A) $V_P - V_O > 0$ and $V_R - V_O < 0$ (B) $V_P = V_R > V_O$
(C) $V_O > V_P = V_Q$ (D) $V_Q - V_P = V_P - V_O$

21. Choose the correct statement(s) related to the magnitude of potential differences

- (A) $V_P - V_O = \frac{1}{2} B\omega a^2$ (B) $V_P - V_Q = \frac{1}{2} B\omega a^2$
(C) $V_Q - V_O = 2B\omega a^2$ (D) $V_P - V_R = 2B\omega a^2$

22. Choose the correct statement(s) related to the induced current in the ring

- (A) Current flows from $Q \longrightarrow P \longrightarrow O \longrightarrow R \longrightarrow Q$
(B) Current flows from $Q \longrightarrow R \longrightarrow O \longrightarrow P \longrightarrow Q$
(C) Current flows from $Q \longrightarrow P \longrightarrow O$ and from $Q \longrightarrow R \longrightarrow O$
(D) No current flows

ANSWER KEY

EXERCISE (S-2)

1. Ans. (i) 85.22 Tm^2 ; (ii) 56.8 V ; (iii) linearly 2. Ans. $\frac{1}{3} A$ 3. Ans. $C \pi a^2 / R$

4. Ans. 200 rad/sec 5. Ans. $-\frac{B \pi a^2 \lambda}{MR} \hat{k}$ 6. Ans. $\frac{E R_1}{R_1 R_2 + R_2 R_3 + R_3 R_1}$

7. Ans. $I_{EA} = \frac{7}{22} A$; $I_{BE} = \frac{3}{11} A$; $I_{FE} = \frac{1}{22} A$

8. Ans. (a) $i = \frac{B_0 a v}{R}$ in anticlockwise direction, $v = \text{velocity at time } t$, (b) $F_{\text{net}} = B_0^2 a^2 v / R$,

(c) $V = \frac{mgR}{B_0^2 a^2} \left(1 - e^{-\frac{B_0^2 a^2 t}{mR}} \right)$

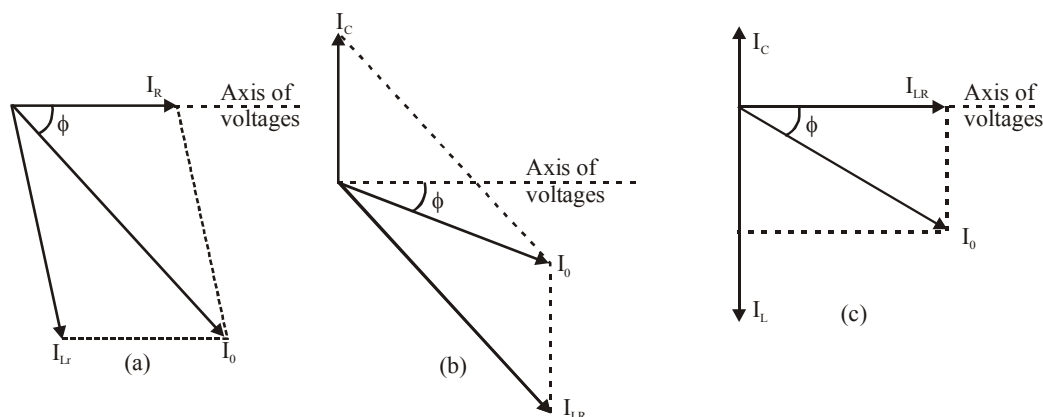
9. Ans. $67/32 A$

10. Ans. (i) $i_1 = i_2 = 10/3 A$, (ii) $i_1 = 50/11 A$; $i_2 = 30/11 A$, (iii) $i_1 = 0$, $i_2 = 20/11 A$, (iv) $i_1 = i_2 = 0$

11. Ans. $42 + 20t \text{ volt}$

12. Ans. $2\pi \frac{\sqrt{mL}}{lB}, g \frac{\sqrt{mL}}{lB}$ 13. Ans. $-\frac{V}{R} e^{-\frac{Rt}{L}}$ 14. Ans. $kMT^2/(R)$

15. Ans.



EXERCISE (O-2)

1. Ans. (A) 2. Ans. (B) 3. Ans. (B) 4. Ans. (A) 5. Ans. (D) 6. Ans. (D)
 7. Ans. (D) 8. Ans. (B) 9. Ans. (A) 10. Ans. (A) 11. Ans. (A) 12. Ans. (D)

13. Ans. (A) 14. Ans. (D) 15. Ans. (A) 16. Ans. (D) 17. Ans. (A, B, C)
 18. Ans. (D) 19. Ans. (B) 20. Ans. (D) 21. Ans. (B) 22. Ans. (A) 23. Ans. (B)
 24. Ans. (A)

EXERCISE - JA

1. Ans. (A,C) 2. Ans. (B) 3. Ans. (D) 4. Ans. (C) 5. Ans. (A)
 6. Ans. (D) 7. Ans. (C) 8. Ans. (A) P; (B) P, Q, S; (C) Q,S ; (D) Q, R, S
 9. Ans. (A) 10. Ans. (A) q (B) r,s (C) s (D) p,q,r 11. Ans. (D) 12. Ans. (B,D)
 13. Ans. (B) 14. Ans. (C) 15. Ans. (A)-R,S,T; (B)-Q,R,S,T; (C)-P,Q; (D)-Q,R,S,T
 16. Ans. (C) 17. Ans. (B,C) 18. Ans. 6 19. Ans. 4 20. Ans. 7
 21. Ans. (A,C) 22. Ans. (C) or (AC) 23. Ans. (B) 24. Ans. (B)
 25. Ans. (A) 26. Ans. (B) 27. Ans. (C,D) 28. Ans. (A,D) 29. Ans. (A,C)
 30. Ans. (B,D) 31. Ans. 8 32. Ans. (C, D)

REVIEW TEST

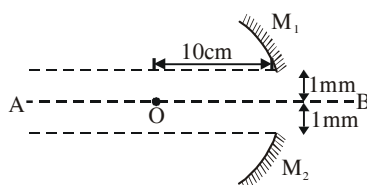
1. Ans. $\frac{\mu_0 i a^2 \pi}{2Rb}$ 2. Ans. 2A, 400W
 3. Ans. $V = 1 \text{ ms}^{-1}$, $R_1 = 0.47 \Omega$, $R_2 = 0.30 \Omega$ 4. Ans. 77Ω , 97.6Ω , 7.7 V , 9.76 V
 5. Ans. $I = \frac{(\mu_0 n i_0 \omega \cos \omega t) \pi a^2 (Ld)}{\rho 2\pi R}$ 6. Ans. (C)
 7. Ans. (A) 8. Ans. (A)
 9. Ans. (C) 10. Ans. (A)
 11. Ans. (B) 12. Ans. (A)
 13. Ans. (D) 14. Ans. (B, D)
 15. Ans. (A) 16. Ans. (A, C, D)
 17. Ans. (C) 18. Ans. (D) 19. Ans. (A, B)
 20. Ans. (B, D) 21. Ans. (C) 22. Ans. (D)

GEOMETRICAL OPTICS

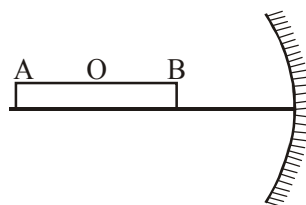
EXERCISE (S-2)

(Not to be discussed in class)

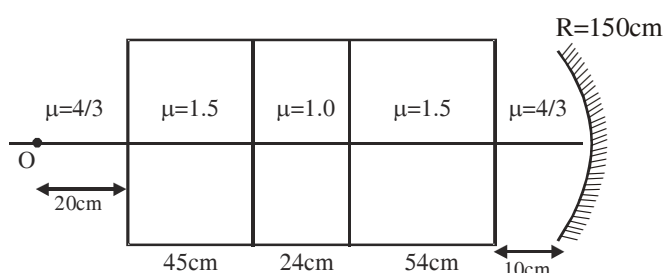
1. A concave mirror of focal length 20 cm is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 mm from the previous principal axis AB. If an object is placed at a distance of 10 cm from the mirror on the line AB then, find the distance between the images formed by the two parts?



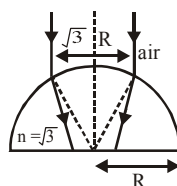
2. A rod of length ℓ is moving with constant speed 1 m/s towards the pole of a concave mirror of focal length ℓ . Find the rate of change in length of the image of the rod (in m/s) at the moment centre of the rod coincides with the centre of curvature of the mirror.



3. A composite slab consisting of different media is placed in front of a concave mirror of radius of curvature 150 cm. The whole arrangement is placed in water. An object O is placed at a distance 20 cm from the slab. The R.I. of different media are given in the diagram. Find the position of the final image formed by the system.



4. A light beam of diameter $\sqrt{3} R$ is incident symmetrically on a glass hemisphere of radius R and of refractive index $n = \sqrt{3}$. Find the radius of the beam at the base of hemisphere.



5. An object is approaching a thin convex lens of focal length 0.3 m with a speed of 0.01 m/s. Find the magnitudes of the rates of change of position and lateral magnification of image when the object is at a distance of 0.4 m from the lens.

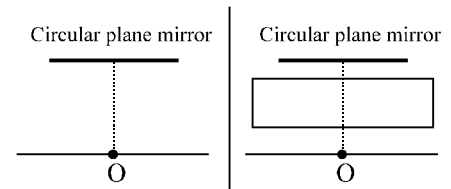
[IIT-JEE 2004]

EXERCISE O-2

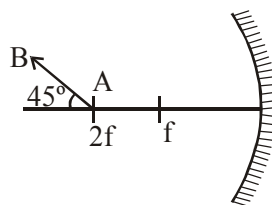
SINGLE CORRECT TYPE QUESTIONS

1. In the diagram shown below, a point source O is placed vertically below the center of a circular plane mirror. The light rays starting from the source are reflected from the mirror such that a circular area A on the ground receives light. Now, a glass slab is placed between the mirror and the source O. What will the magnitude of the new area on the ground receiving light?

- (A) A
(B) Greater than A
(C) Less than A
(D) Cannot tell, as the information given is insufficient



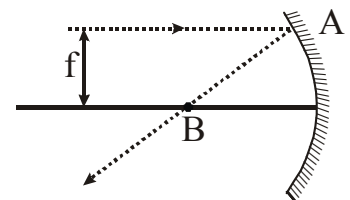
2. The shape of image formed of an object AB due to the concave mirror shown in the figure is best represented by (assume point A is at the centre of curvature of the mirror) :-



- (A) (B) (C) (D)

3. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray) :-

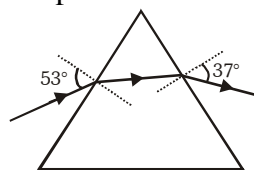
- (A) $\frac{2}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2}$
(C) $\frac{2}{3}$ (D) $\frac{1}{2}$



4. An infinitely long rod lies along the axis of a concave mirror of focal length f. The near end of the rod is at distance $u > f$ from the mirror. Its image will have a length :-

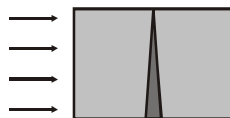
- (A) $\frac{uf}{u-f}$ (B) $\frac{uf}{u+f}$ (C) $\frac{f^2}{u+f}$ (D) $\frac{f^2}{u-f}$

5. A ray incident at an angle 53° on a prism emerges at an angle 37° as shown. If the angle of incidence is made 50° , which of the following is a possible value of the angle of emergence?

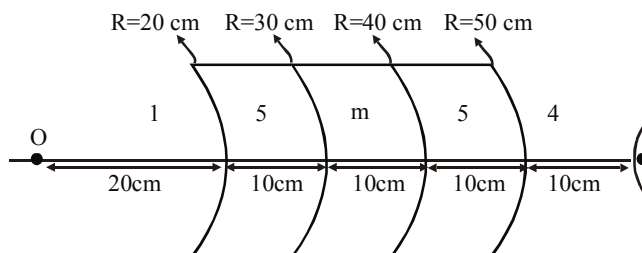


- (A) 35° (B) 42° (C) 40° (D) 38°

6. A thin isosceles prism with angle 4° and refractive index $5/4$ is placed inside a transparent tube with liquid (refractive index = 1.5) as shown. The deviation of light due to prism will be :-



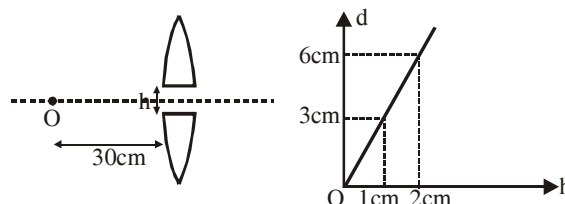
- (A) 0.8° upward (B) 0.8° downward (C) 0.67° upward (D) 0.67° downward
7. In the arrangement shown, separation between observer and object as seen by observer is :-



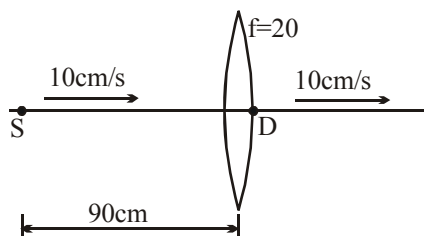
- (A) 60 cm (B) more than 60 cm (C) less than 60 cm (D) depends on μ

MULTIPLE CORRECT TYPE QUESTIONS

8. Figure shows a convex lens cut symmetrically into two equal halves and separated laterally by a distance h . A point object O placed symmetrically at a distance 30 cm, from the lens halves, forms two images separated by a distance d . A plot of d versus h is shown in figure. The focal length of the lens is :-



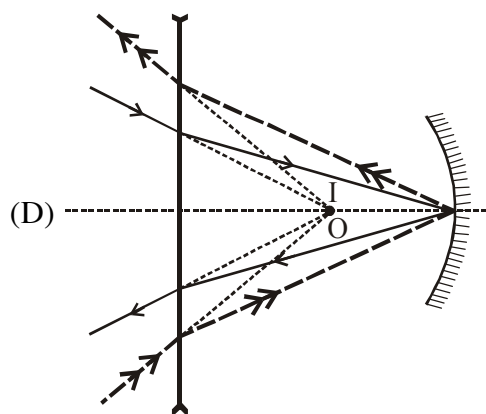
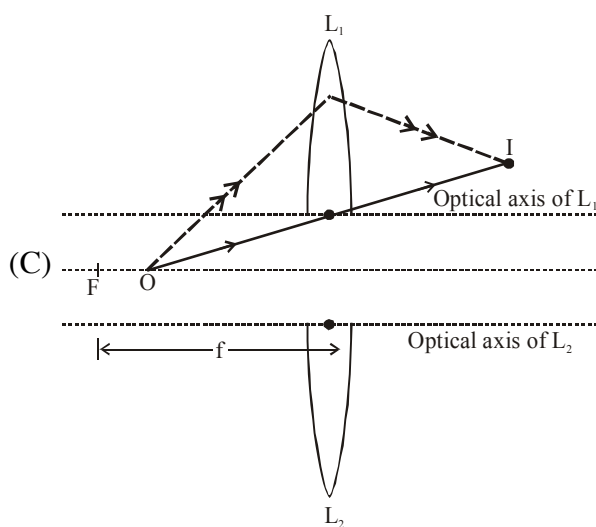
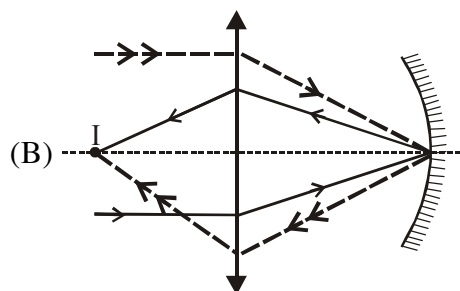
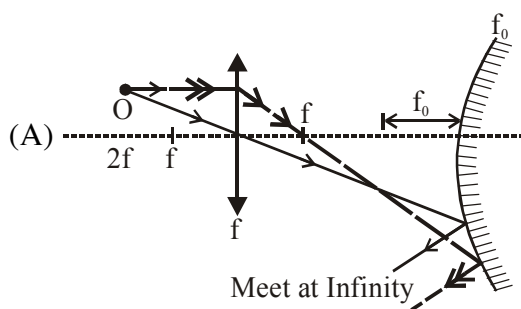
- (A) 22.5 cm (B) 40 cm (C) 45 cm (D) 20 cm
9. The figure shows initial positions of a source of light S and a light detector D. Both S & D move with velocity 10 cm/s, as shown. The converging lens of focal length 20 cm is fixed. Time instant(s) when the detector detects maximum intensity of light is /are :-



- (A) 3 sec (B) 4.5 sec
 (C) 6 sec (D) equal intensity at all time

10. Choose the incorrect ray diagram(s). All the rays shown are paraxial.

[$\uparrow$ denotes converging lens and $\downarrow$ denotes diverging lens]



EXERCISE-(JA)

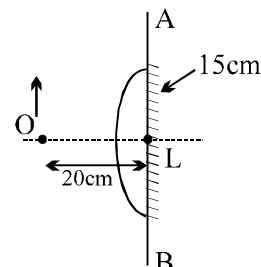
(Not to be discussed in class)

ALL ANSWERS OF THE QUESTIONS **JEE-2007** ONWARDS ARE AS GIVEN ON **IIT-JEE WEBSITE.**

1. A point object is placed at a distance of 20 cm from a thin plano-convex lens of focal length 15 cm, if the plane surface is silvered. The image will form at

[IIT-JEE 2006]

- (A) 60 cm left of AB
 (B) 30 cm left of AB
 (C) 12 cm left of AB
 (D) 60 cm right of AB



2. Parallel rays of light from Sun falls on a biconvex lens of focal length f and the circular image of radius r is formed on the focal plane of the lens. Then which of the following statement is correct?

- (A) Area of image πr^2 directly proportional to f
 (B) Area of image πr^2 directly proportional to f^2
 (C) Intensity of image increases if f is increased.

[IIT-JEE 2006]

- (D) If lower half part is converted by black sheet the area of the image is equal to $\frac{\pi r^2}{2}$

3. A ray of light travelling in water is incident on its surface open to air. The angle of incidence is θ , which is less than the critical angle. Then there will be

[IIT-JEE 2007]

- (A) only a reflected ray and no refracted ray
 (B) only a refracted ray and no reflected ray
 (C) a reflected ray and a refracted ray and the angle between them would be less than $180^\circ - 2\theta$
 (D) a reflected ray and a refracted ray and the angle between them would be greater than $180^\circ - 2\theta$

4. STATEMENT-1

The formula connecting u , v and f for a spherical mirror is valid only for mirrors whose sizes are very small compared to their radii of curvature.

[IIT-JEE 2007]

because

STATEMENT-2

Laws of reflection are strictly valid for plane surfaces, but not for large spherical surfaces.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

5. In an experiment to determine the focal length (f) of a concave mirror by the u - v method, a student places the object pin A on the principal axis at a distance x from the pole P. The student looks at the pin and its inverted image from a distance keeping his/her eye in line with PA. When the student shifts his/her eye towards left, the image appears to the right of the object pin. Then :-

[IIT-JEE 2007]

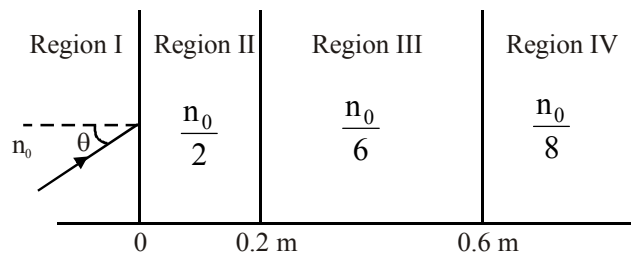
- (A) $x < f$ (B) $f < x < 2f$ (C) $x = 2f$ (D) $x > 2f$

6. Two beams of red and violet colours are made to pass separately through a prism (angle of the prism is 60°). In the position of minimum deviation, the angle of refraction will be : [IIT-JEE 2008]

- (A) 30° for both the colours (B) greater for the violet colour
(C) greater for the red colour (D) equal but not 30° for both the colours

7. A light beam is traveling from Region I to Region IV (Refer Figure). The refractive index in Regions

I, II, III and IV are n_0 , $\frac{n_0}{2}$, $\frac{n_0}{6}$ and $\frac{n_0}{8}$, respectively. The angle of incidence θ for which the beam just misses entering Region IV is figure [IIT-JEE 2008]

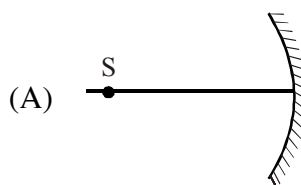


- (A) $\sin^{-1} \left(\frac{3}{4} \right)$ (B) $\sin^{-1} \left(\frac{1}{8} \right)$ (C) $\sin^{-1} \left(\frac{1}{4} \right)$ (D) $\sin^{-1} \left(\frac{1}{3} \right)$

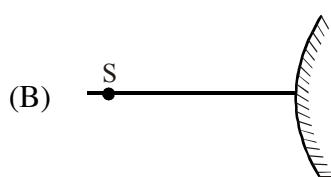
8. An optical component and an object S placed along its optic axis are given in **Column-I**. The distance between the object and the component can be varied. The properties of images are given in **Column-II**. Match all the properties of images from **Column-II** with the appropriate components given in **Column I**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [IIT-JEE 2008]

Column-I

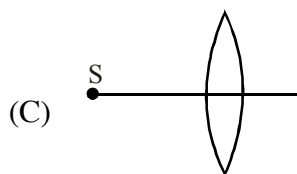
Column-II



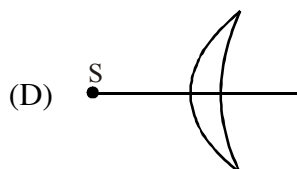
(p) Real image



(q) Virtual image



(r) Magnified image



(s) Image at infinite

9. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is $4/3$. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant, when the ball is 12.8 m above the water surface, the fish sees the speed of ball as [$g = 10 \text{ m/s}^2$.]

[IIT-JEE 2009]

(A) 9 m/s (B) 12 m/s (C) 16 m/s (D) 21.33 m/s

10. A student performed the experiment of determination of focal length of a concave mirror by u-v method using an optical bench of length 1.5 m. The focal length of the mirror used is 24 cm. The maximum error in the location of the image can be 0.2 cm. The 5 sets of (u, v) values recorded by the student (in cm) are : (42, 56), (48, 48), (60, 40), (66, 33), (78, 39). The data set(s) that cannot come from experiment and is (are) incorrectly recorded, is (are) :

[IIT-JEE 2009]

(A) (42, 56) (B) (48, 48) (C) (66, 33) (D) (78, 39)

11. A biconvex lens of focal length 15 cm is in front of a plane mirror. The distance between the lens and the mirror is 10 cm. A small object is kept at a distance of 30 cm from the lens. The final image is

(A) virtual and at a distance of 16 cm from the mirror

[IIT-JEE 2010]

(B) real and at a distance of 16 cm from the mirror

(C) virtual and at a distance of 20 cm from the mirror

(D) real and at a distance of 20 cm from the mirror

12. A ray OP of monochromatic light is incident on the face AB of prism ABCD near vertex B at an incident angle of 60° (see figure). If the refractive index of the material of the prism is $\sqrt{3}$, which of the following is (are) correct?

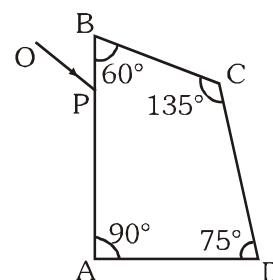
[IIT-JEE 2010]

(A) The ray gets totally internally reflected at face CD

(B) The ray comes out through face AD

(C) The angle between the incident ray and the emergent ray is 90°

(D) The angle between the incident ray and the emergent ray is 120°



13. Two transparent media of refractive indices μ_1 and μ_3 have a solid lens shaped transparent material of refractive index μ_2 between them as shown in figures in **Column II**. A ray traversing these media is also shown in the figures. In **Column I** different relationships between μ_1 , μ_2 and μ_3 are given. Match them to the ray diagrams shown in **Column II**. [IIT-JEE 2010]

Column I

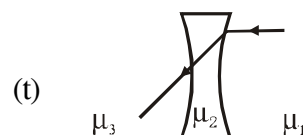
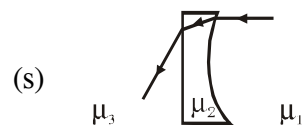
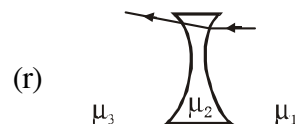
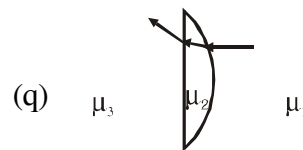
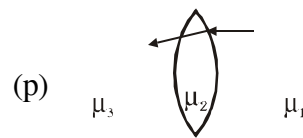
(A) $\mu_1 < \mu_2$

(B) $\mu_1 > \mu_2$

(C) $\mu_2 = \mu_3$

(D) $\mu_2 > \mu_3$

Column II



14. The focal length of a thin biconvex lens is 20 cm. When an object is moved from a distance of 25 cm in front of it to 50 cm, the magnification of its image changes from m_{25} to m_{50} . The ratio $\frac{m_{25}}{m_{50}}$ is

[IIT-JEE 2010]

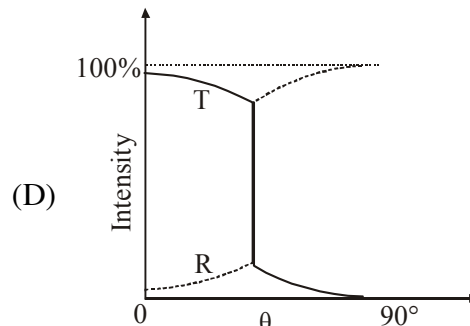
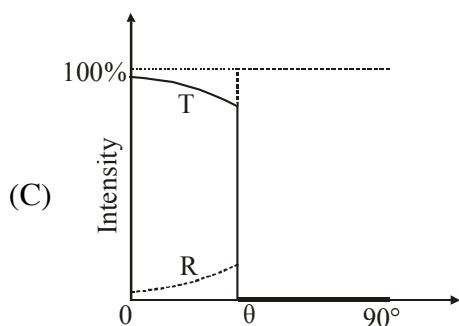
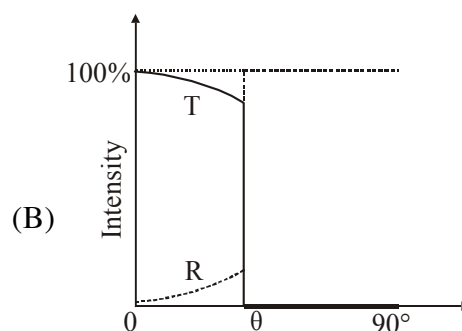
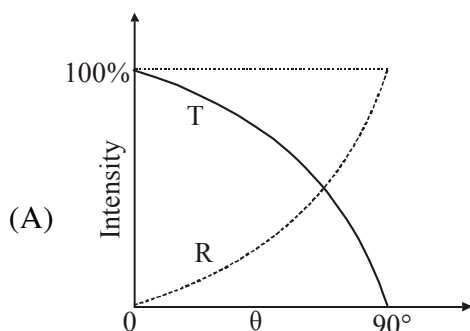
15. Image of an object approaching a convex mirror of radius of curvature 20 m along its optical axis is observed to move from $\frac{25}{3}$ m to $\frac{50}{7}$ m in 30 seconds. What is the speed of the object in km per hour?

[IIT-JEE 2010]

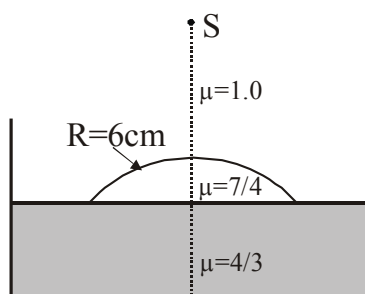
16. A large glass slab $\left(\mu = \frac{5}{3}\right)$ of thickness 8 cm is placed over a point source of light on a plane surface. It is seen that light emerges out of the top surface of the slab from a circular area of radius R cm. What is the value of R?

[IIT-JEE 2010]

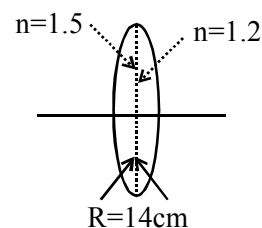
17. A light ray traveling in glass medium is incident on glass-air interface at an angle of incidence θ . The reflected (R) and transmitted (T) intensities, both as function of θ , are plotted. The correct sketch is [IIT-JEE 2011]



18. Water (with refractive index $= \frac{4}{3}$) in a tank is 18 cm deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature 'R = 6 cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at 'x' cm above the bottom of the tank. Then 'x' is [IIT-JEE 2011]



19. A biconvex lens is formed with two thin plano-convex lenses as shown in the figure, Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature $R = 14$ cm. For this biconvex lens, for an object distance of 40 cm, the image distance will be :- [IIT-JEE 2012]



- (A) -280.0 cm
(B) 40.0 cm
(C) 21.5 cm
(D) 13.3 cm

Paragraph for Questions 20 and 21

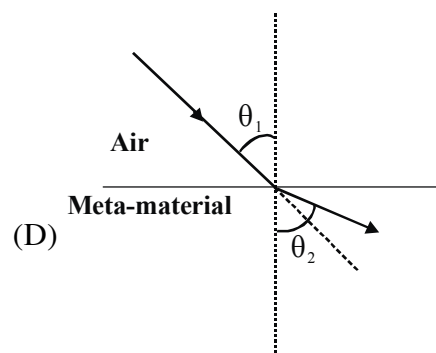
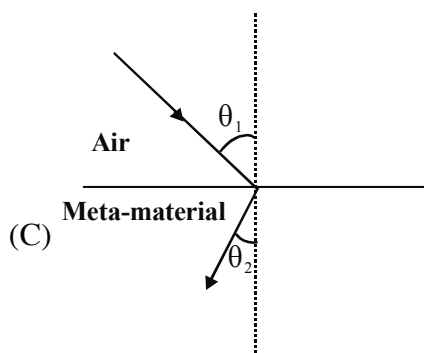
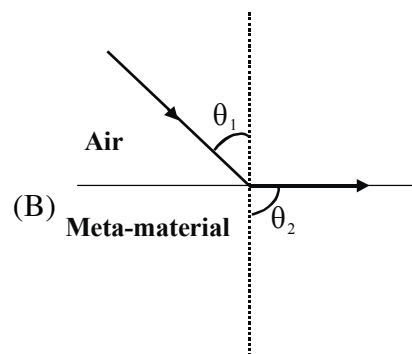
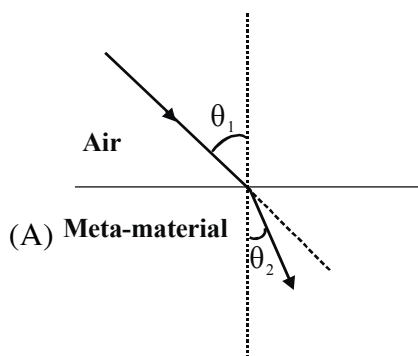
Most materials have the refractive index, $n > 1$. So, when a light ray from air enters a naturally occurring material, then by Snell's law, $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$, it is understood that the refracted ray bends towards the normal. But it never emerges on the same side of the normal as the incident ray. According to

electromagnetism, the refractive index of the medium is given by the relation, $n = \left(\frac{c}{v} \right) = \pm \sqrt{\epsilon_r \mu_r}$,

where c is the speed of electromagnetic waves in vacuum, v its speed in the medium, ϵ_r and μ_r are the relative permittivity and permeability of the medium respectively.

In normal materials, both ϵ_r and μ_r are positive, implying positive n for the medium. When both ϵ_r and μ_r are negative, one must choose the negative root of n . Such negative refractive index materials can now be artificially prepared and are called meta-materials. They exhibit significantly different optical behaviour, without violating any physical laws. Since n is negative, it results in a change in the direction of propagation of the refracted light. However, similar to normal materials, the frequency of light remains unchanged upon refraction even in meta-materials. **[IIT-JEE-2012]**

20. For light incident from air on a meta-material, the appropriate ray diagram is



21. Choose the correct statement.

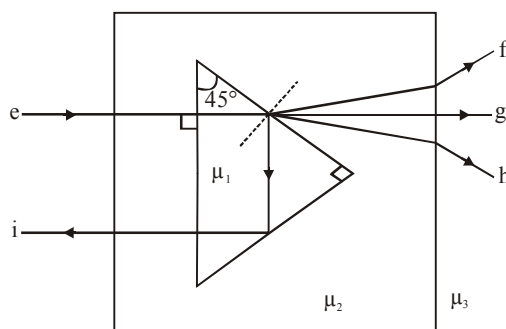
(A) The speed of light in the meta-material is $v = c|n|$

(B) The speed of light in the meta-material is $v = \frac{c}{|n|}$

(C) The speed of light in the meta-material is $v = c$.

(D) The wavelength of the light in the meta-material (λ_m) is given by $\lambda_m = \lambda_{\text{air}}|n|$, where λ_{air} is the wavelength of the light in air.

22. A ray of light travelling in the direction $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$. The angle of incidence is :- **[JEE-Advance-2013]**
- (A) 30° (B) 45° (C) 60° (D) 75°
23. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space. The radius of the curved surface of the lens is :- **[JEE-Advance-2013]**
- (A) 1 m (B) 2 m (C) 3 m (D) 6 m
24. A right angled prism of refractive index μ_1 is placed in a rectangular block of refractive index μ_2 , which is surrounded by a medium of refractive index μ_3 , as shown in the figure. A ray of light 'e' enters the rectangular block at normal incidence. Depending upon the relationships between μ_1 , μ_2 , and μ_3 , it takes one of the four possible paths 'ef', 'eg', 'eh' or 'ei'. **[JEE-Advance-2013]**



Match the paths in List-I with conditions of refractive indices in List-II and select the correct answer using the codes given below the lists :

List-I

- P. $e \rightarrow f$
 Q. $e \rightarrow g$
 R. $e \rightarrow h$
 S. $e \rightarrow i$

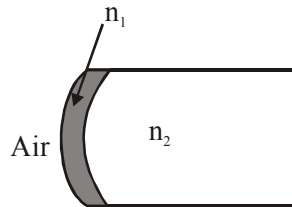
List-II

1. $\mu_1 > \sqrt{2}\mu_2$
2. $\mu_2 > \mu_1$ and $\mu_2 > \mu_3$
3. $\mu_1 = \mu_2$
4. $\mu_2 < \mu_1 < \sqrt{2}\mu_2$ and $\mu_2 > \mu_3$

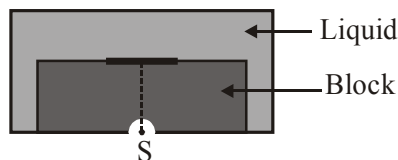
Codes :

	P	Q	R	S
(A)	2	3	1	4
(B)	1	2	4	3
(C)	4	1	2	3
(D)	2	3	4	1

25. A transparent thin film of uniform thickness and refractive index $n_1 = 1.4$ is coated on the convex spherical surface of radius R at one end of a long solid glass cylinder of refractive index $n_2 = 1.5$, as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance f_1 from the film, while rays of light traversing from glass to air get focused at distance f_2 from the film. Then **[JEE-Advance-2014]**

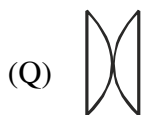
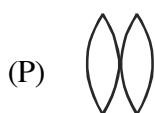


- (A) $|f_1| = 3R$ (B) $|f_1| = 2.8R$ (C) $|f_2| = 2R$ (D) $|f_2| = 1.4R$
26. A point source S is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is :- **[JEE-Advance-2014]**



- (A) 1.21 (B) 1.30 (C) 1.36 (D) 1.42
27. Four combinations of two thin lenses are given in List-I. The radius of curvature of all curved surfaces is r and the refractive index of all the lenses is 1.5. Match lens combinations in List-I with their focal length in List-II and select the correct answer using the code given below the lists. **[JEE-Advance-2014]**

List-I



List-II

(1) $2r$

(2) $\frac{r}{2}$

(3) $-r$

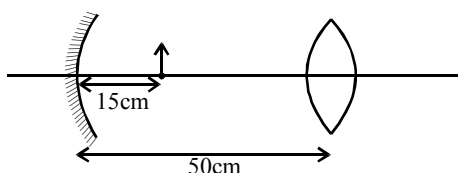
(4) r

Code :

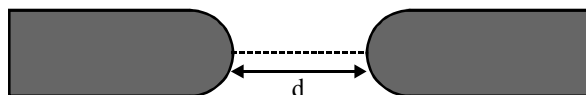
- (A) P-1, Q-2, R-3, S-4
(C) P-4, Q-1, R-2, S-3

- (B) P-2, Q-4, R-3, S-1
(D) P-2, Q-1, R-3, S-4

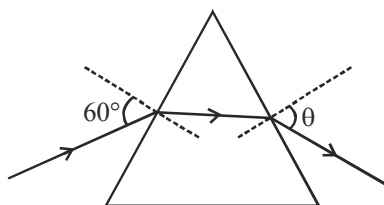
28. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification M_1 . When the set-up is kept in a medium of refractive index $7/6$ the magnification becomes M_2 . The magnitude $\left| \frac{M_2}{M_1} \right|$ is. [JEE-Advance-2015]



29. Two identical glass rods S_1 and S_2 (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance d as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light P is placed inside rod S_1 on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside S_2 . The distance d is : [JEE-Advance-2015]



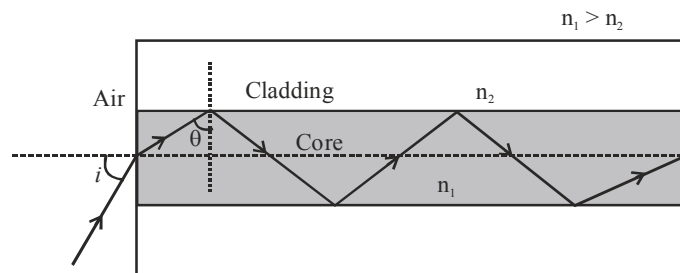
- (A) 60 cm (B) 70 cm
 (C) 80 cm (D) 90 cm
30. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle $\theta(n)$ with the normal (see the figure).
 For $n = \sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of m is [JEE-Advance-2015]



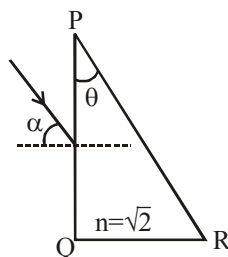
Paragraph for Question No. 31 and 32

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index n_1 surrounded by a medium of lower refractive index n_2 . The light guidance in the structure takes place due to successive total internal reflections at the interface of the media n_1 and n_2 as shown in the figure. All rays with the angle of incidence i less than a particular value of i_m are confined in the medium of refractive index n_1 . The numerical aperture (NA) of the structure is defined as $\sin i_m$.

[JEE-Advance-2015]

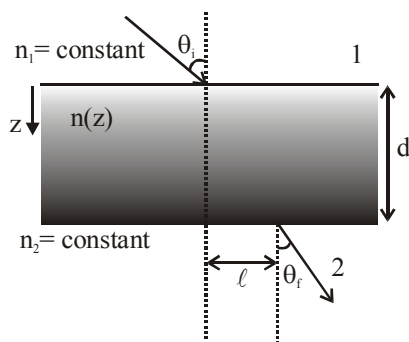


31. For two structures namely S_1 with $n_1 = \sqrt{45}/4$ and $n_2 = 3/2$, and S_2 with $n_1 = 8/5$ and $n_2 = 7/5$ and taking the refractive index of water to be $4/3$ and that of air to be 1, the correct option(s) is (are)
- (A) NA of S_1 immersed in water is the same as that of S_2 immersed in liquid of refractive index $\frac{16}{3\sqrt{15}}$.
- (B) NA of S_1 immersed in liquid of refractive index $\frac{6}{\sqrt{15}}$ is the same as that of S_2 immersed in water.
- (C) NA of S_1 placed in air is the same as that of S_2 immersed in liquid of refractive index $\frac{4}{\sqrt{15}}$.
- (D) NA of S_1 placed in air is the same as that of S_2 placed in water.
32. If two structures of same cross-sectional area, but different numerical apertures NA_1 and NA_2 ($NA_2 < NA_1$) are joined longitudinally, the numerical aperture of the combined structure is
- (A) $\frac{NA_1 NA_2}{NA_1 + NA_2}$ (B) $NA_1 + NA_2$ (C) NA_1 (D) NA_2
33. A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45° . The angle θ of the prism is : [JEE-Advance-2016]

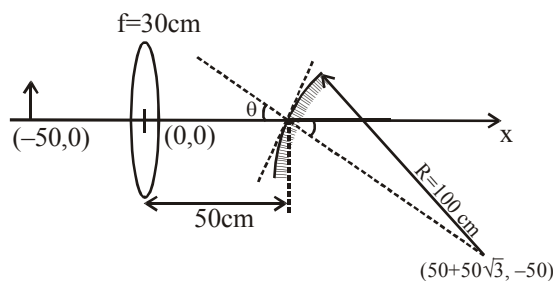


- (A) 15° (B) 22.5° (C) 30° (D) 45°

34. A transparent slab of thickness d has a refractive index $n(z)$ that increases with z . Here z is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices n_1 and n_2 ($> n_1$), as shown in the figure. A ray of light is incident with angle θ_i from medium 1 and emerges in medium 2 with refraction angle θ_f with a lateral displacement ℓ . Which of the following statement(s) is(are) true ? **[JEE-Advance-2016]**



- (A) ℓ is independent of n_2 (B) ℓ is dependent on $n(z)$
 (C) $n_1 \sin \theta_i = (n_2 - n_1) \sin \theta_f$ (D) $n_1 \sin \theta_i = n_2 \sin \theta_f$
35. A plano-convex lens is made of a material of refractive index n . When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true? **[JEE-Advance-2016]**
- (A) The refractive index of the lens is 2.5
 (B) The radius of curvature of the convex surface is 45 cm
 (C) The faint image is erect and real
 (D) The focal length of the lens is 20 cm.
36. A small object is placed 50 cm to the left of thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle $\theta = 30^\circ$ to the axis of the lens, as shown in the figure. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point (x, y) at which the image is formed are : **[JEE-Advance-2016]**

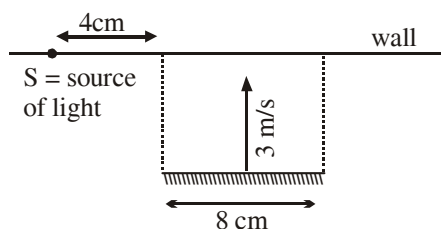


- (A) $(25, 25\sqrt{3})$ (B) $\left(\frac{125}{3}, \frac{25}{\sqrt{3}}\right)$ (C) $(50 - 25\sqrt{3}, 25)$ (D) $(0, 0)$

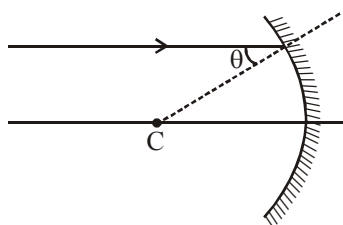
REVIEW TEST

(Not to be discussed in class)

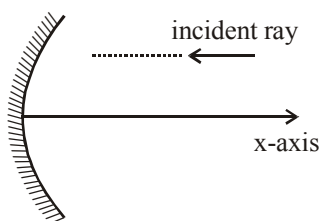
1. A fluorescent lamp of length 1 m is placed horizontally at a depth of 1.2 m below a ceiling. A plane mirror of length 0.6 m is placed below the lamp parallel to and symmetric to the lamp at a distance 2.4m from it. The length of the reflected patch of length on the ceiling _____ .
2. A plane mirror of length 8 cm is moving with speed 3 m/s towards a wall in situation as shown in figure. Size of spot formed on the wall is $32/x$ cm. Find the value of x .



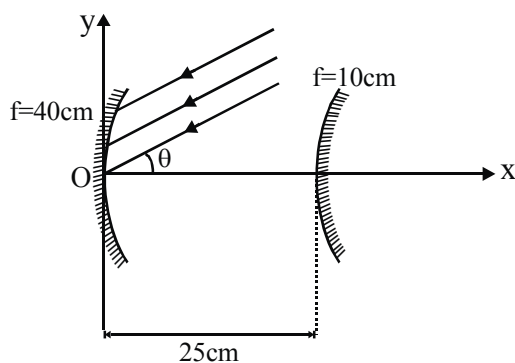
3. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 m while it is 0.30 m in front of the plane mirror. Find the radius of curvature (in cm) of the convex mirror.
4. A ray incident parallel to principal axis on a concave mirror at an angle θ after two reflection, reflected ray becomes parallel to incident ray. If $\theta = \frac{\pi}{N}$ (in radian) then find the value of N .



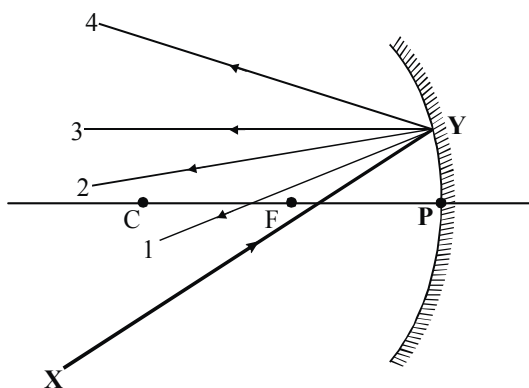
5. A parabolic reflecting surface given by $x = \frac{y^2}{2}$, is placed at origin, as shown. An incident ray moving along $y = 1$ is incident on it. The ray gets reflected by the surface twice. The deviation suffered by the ray is $\frac{\pi}{n}$ radians. Find n .



6. A cylindrical bucket of depth 60 cm is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as seen by the observer is given as α cm. Fill $\left(\frac{\alpha}{5}\right)$ in OMR sheet.
7. There are two thin symmetrical lenses, one is converging, with refractive index $n_1 = 1.70$, and the other is diverging with refractive index $n_2 = 1.50$. Both lenses have the same curvature radius of their surface equal to $R = 10$ cm. The lenses are put close together and submerged into water. The focal length of lens system is found to be $\frac{100}{x}$ cm in water. What is the value of x . (R.I. of water = $4/3$)
8. Consider a concave mirror kept at origin with focal length 40 cm. Parallel rays which subtend an angle $\theta = \frac{1}{40}$ radian are incident on it. A convex mirror is kept at a distance 25 cm from the first mirror as shown. Find the y -coordinate of the image (in cm) formed by the system of mirrors after two reflections.



9. Figure shows a small concave mirror with CP as its principal axis. A ray XY is incident on the mirror. Which of the four rays can be the reflected ray.



(A) 1

(B) 2

(C) 3

(D) 4

10. An object is placed in front of a spherical mirror whose 2 times magnified image is formed on screen.

Then choose **CORRECT** option :-

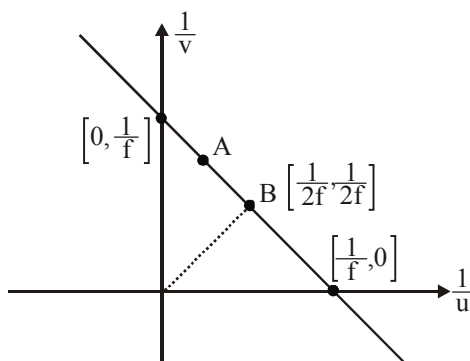
- (A) Mirror is concave $m = +2$ (B) Mirror is concave $m = -2$
(C) Mirror is convex $m = +2$ (D) Mirror is convex $m = -2$

11. The table below lists object and image positions for four objects placed in front of mirrors, using Cartesian sign convention with pole of the mirror as origin and direction of incident rays as positive. In the following cases, the case in which images is formed by a convex spherical mirror, is :-

Object position Image position

- (A) -25.0 cm -16.7 cm
(B) -5.0 cm 10.0 cm
(C) -20.0 cm 5.71 cm
(D) -40.0 cm 80.0 cm

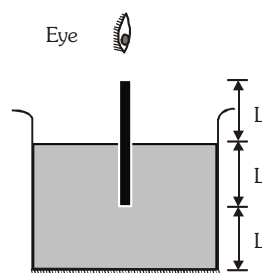
12. Figure shows $\frac{1}{v}$ vs $\frac{1}{u}$ curve for convex mirror. Nature of image at point A is :-



- (A) Real, erect, magnified (B) virtual, inverted, magnified
(C) virtual, inverted, diminished (D) virtual, erect, diminished

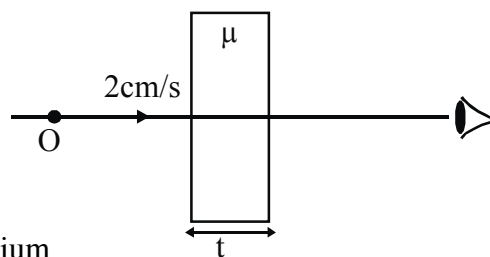
13. What is the length of the image of the rod in mirror, according to the observer in air? (refractive index of the liquid is μ)

- (A) $\mu L + L$ (B) $L + \frac{L}{\mu}$
(C) $L\mu + \frac{L}{\mu}$ (D) None of these

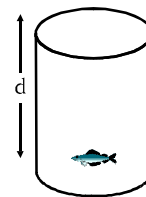


14. A glass slab of width ' t ', refractive index ' μ ' is placed as shown in the figure. If the point object, moves with a speed 2 cm/s towards the slab the speed observed will be :-

- (A) 2 cm/s
(B) less than 2 cm/s
(C) greater than 2 cm/s
(D) dependent on the refractive index of surrounding medium

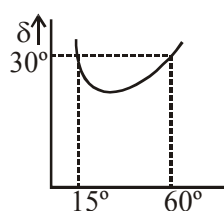


15. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 cm, and the critical angle for liquid is 37° . At what maximum depth below the

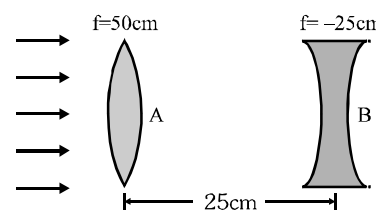


surface of the liquid, d , must the fish be floating? $\left(\sin 37^\circ = \frac{3}{5} \right)$

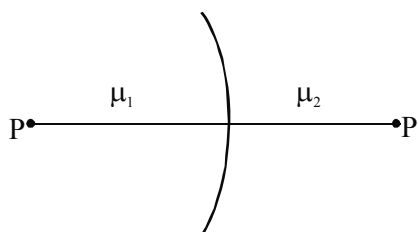
- (A) 16 cm (B) 20 cm (C) 11.25 cm (D) 25 cm
16. Figure shows graph of deviation δ versus angle of incidence for a light ray striking a prism. Angle of prism is :-



- (A) 30° (B) 45° (C) 60° (D) 75°
17. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is :
- (A) Initially parallel and then diverging
 (B) Always diverging
 (C) Initially converging and finally parallel
 (D) Always parallel

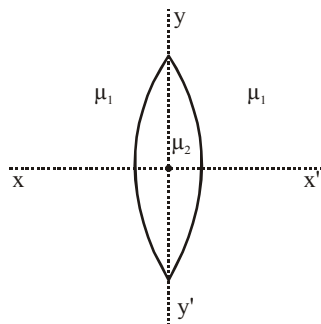


18. Two refracting media are separated by a spherical interface as shown in the figure. PP' is the principal axis, μ_1 and μ_2 are the refractive indices of medium of incidence and medium of refraction respectively, then :-



- (A) if $\mu_2 > \mu_1$, then there cannot be a real image of real object.
 (B) if $\mu_2 > \mu_1$, then there cannot be a real image of virtual object.
 (C) if $\mu_1 > \mu_2$, then there cannot be a virtual image of virtual object.
 (D) if $\mu_1 > \mu_2$, then there cannot be a real image of real object.

19. A lens is formed by a material having refracting index $\frac{3}{2}$ & radii of curvature 20 cm & 10 cm. Then choose the correct option(s) :-
- (A) If lens is concavo-convex & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm
- (B) If lens is concavo-convex & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm
- (C) If lens is convexo-concave & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm
- (D) If lens is convexo-concave & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm
20. An equi-convex lens of refractive index μ_2 and focal length f (in air) is kept in medium of refractive index μ_1 .



Column-A

- (P) If lens is cut in two equal parts by a plane yy' . ($\mu_1 = 1$)
- (Q) If lens is cut in two equal parts by a plane xx' . ($\mu_1 = 1$)
- (R) If $\mu_1 = \mu_2$.
- (S) If $\mu_1 > \mu_2$.

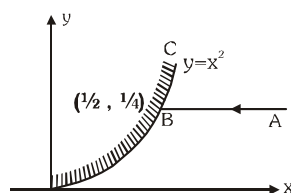
Column-B

- (1) Lens will act as a glass slab
- (2) Lens will be converging and focal length will change.
- (3) Lens will be converging and focal length will remain same
- (4) Lens will be diverging and focal length will change.

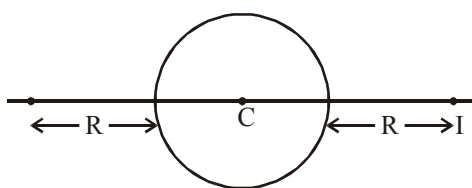
Codes :

	P	Q	R	S
(A)	2	4	1	3
(B)	2	3	1	4
(C)	3	4	2	1
(D)	1	2	3	4

21. In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is :-



- (A) 30° (B) 60° (C) 45° (D) None of these
22. In the given situation, a spherical transparent surface of radius R is placed. The refractive index of sphere is such that image for an object kept at distance R is formed at the same distance of the opposite side. What is refractive index of sphere?



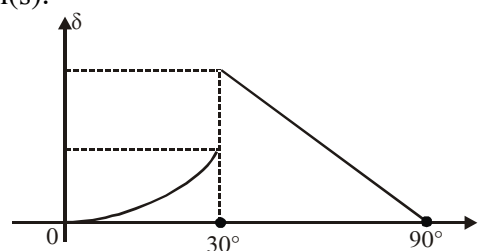
- (A) 1.5 (B) 1.75 (C) 1.25 (D) 2
23. Figure shows graph of angle of deviation δ vs angle of incidence i for a light ray. Incident ray goes from medium 1 (μ_1) to medium 2 (μ_2). Mark the **correct** option(s).

(A) $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

(B) Critical angle is 30°

(C) $\mu_1 > \mu_2$

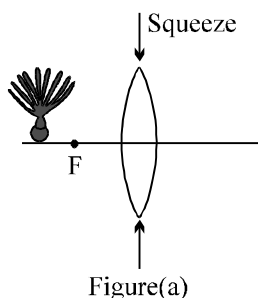
(D) Maximum deviation is 120°



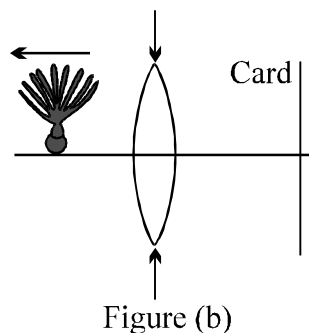
COMPREHENSION TYPE QUESTIONS

Paragraph for Question 24 to 26

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(a)], you can change the curvature of its front and rear sides.



24. When you squeeze the lens, the image.
 (A) moves towards the lens (B) moves away from the lens
 (C) shifts up (D) remains as it is
25. When you squeeze the lens, the lateral height of image.
 (A) increases (B) decreases (C) remains same (D) data insufficient
26. Suppose that a sharp image must be formed on a card which is at a certain distance behind the lens [figure(b)], while you move the turnip away from the lens, then you should



- (A) decrease the squeeze of the lens (B) increase the squeeze of the lens
 (C) keep the card and lens as it is. (D) move the card away from the lens
27. **Statement-1:** When light falls on a sphere made of diamond total internal reflection takes place which makes it shine more than a similar sphere made of common glass.
Statement-2: Refractive index for diamond is more than refractive index of cheap glass.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
28. When the object is at distances u_1 and u_2 the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :
 (A) $\frac{1}{2}\sqrt{u_1 u_2}$ (B) $\frac{u_1 + u_2}{2}$ (C) $\sqrt{u_1 u_2}$ (D) $\sqrt{(u_1 + u_2)}$
29. It is desired to make an achromatic combination of two lenses (L_1 & L_2) made of materials having dispersive powers ω_1 and ω_2 ($<\omega_1$). If the combination of lenses is converging then :-
 (A) L_1 is converging
 (B) L_2 is converging
 (C) Power of L_1 is greater than the power of L_2
 (D) None of these

30. In the figure (a) light is incident at an angle θ which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index n_3 is placed on surface AB in fig. (b). Which of the following statements are correct :-

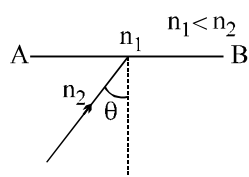


Fig. (a)

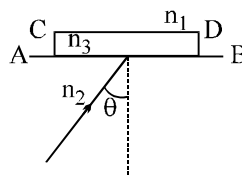
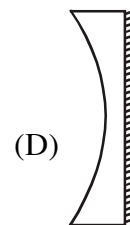
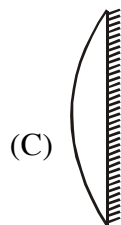
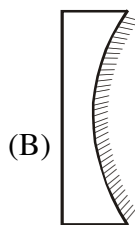
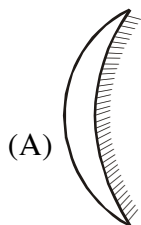


Fig. (b)

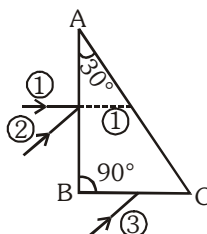
- (A) total internal reflection occurs at AB for $n_3 < n_1$
 (B) total internal reflection occurs at AB for $n_3 > n_1$
 (C) the ray will return back to the same medium for all values of n_3
 (D) total internal reflection occurs at CD for $n_3 < n_1$
31. Which of the following silvered lenses kept in air may form real image of a real object.



32. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.
- (A) the image will look like a white donkey on the photograph.
 (B) the image will look like a Zebra on the photograph.
 (C) the image will be more intense compared to the case in which no such glass is used.
 (D) the image will be less intense compared to the case in which no such glass is used.

Paragraph for Question No. 33 to 35

ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC.



Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) to have the same frequency, answer the following questions.

33. Refractive index of the material of prism is :-
 (A) $\sqrt{2}$ (B) 2 (C) 1.5 (D) 2.2
34. Angle of incidence of ray (2) is :-
 (A) 0° (B) 45° (C) 60° (D) 90°
35. Deviation suffered by ray (3) is :-
 (A) 60° (B) 30° (C) 90° (D) 120°
36. Column-II shows the optical phenomenon that can be associated with optical components given in column-I. Note that column-I may have more than one matching options in column-II.

Column-I

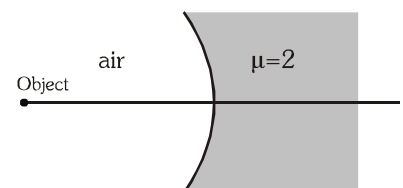
- (A) Convex mirror
 (B) Converging lens
 (C) Thin prism
 (D) Glass slab

Column-II

- (P) Dispersion
 (Q) Deviation
 (R) Real image of real object
 (S) Virtual images of real object

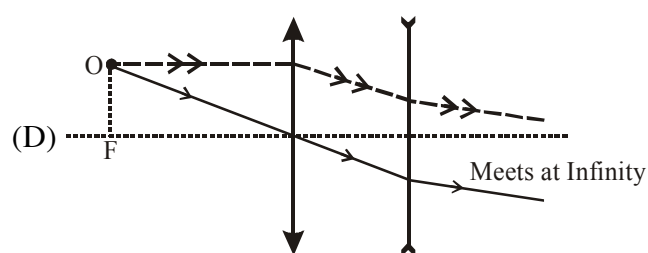
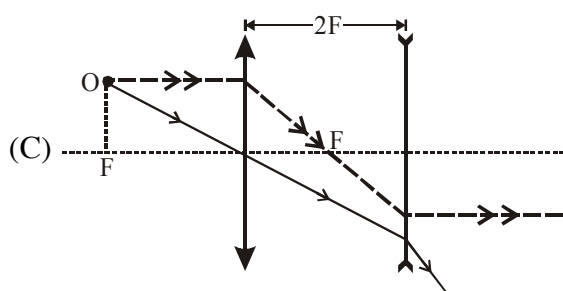
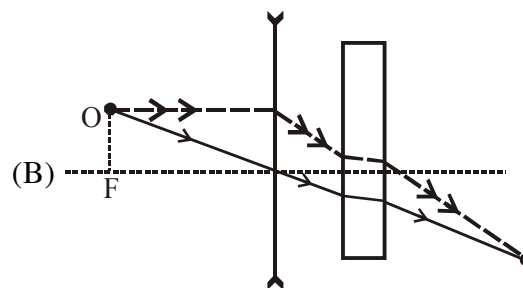
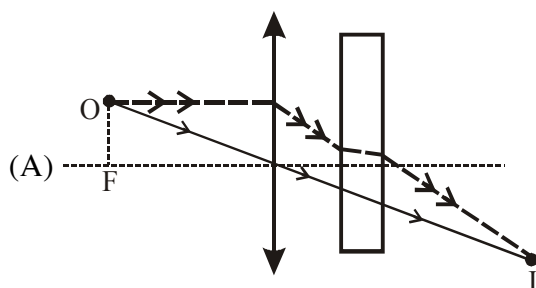
Paragraph for Question 37 to 40

37. If an astronomical telescope has objective and eye-pieces of focal lengths 200 cm and 4 cm respectively, then the magnifying power of the telescope for the normal vision is :-
 (A) 42 (B) 50 (C) 58 (D) 204
38. In the above question the length of the telescope for normal vision, is :-
 (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm
39. In the above question magnifying power of the telescope for distinct vision is :-
 (A) 42 (B) 50 (C) 58 (D) 204
40. In the above question the length of the telescope for distinct vision is :-
 (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm
41. A light ray is incident on a transparent sphere of index $=\sqrt{2}$, at an angle of incidence $= 45^\circ$. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air ?
 (A) 270° (B) 240° (C) 120° (D) 180°
42. A concave spherical surface with radius R separates a medium with index of refraction 2 from air. If an object is moved toward the surface from far away along the central axis, its image :
 (A) Changes from virtual to real when it is $R/2$ from the surface
 (B) Changes from virtual to real when it is $2R$ from the surface
 (C) Changes from real to virtual when it is $R/2$ from the surface
 (D) Remains virtual



43. Choose incorrect ray diagram [$\updownarrow$ denotes converging lens and I denotes diverging lens]

All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is F)



ANSWER KEY EXERCISE (S-2)

1. Ans. 2 mm 2. Ans. $\frac{32}{9}$ 3. Ans. On the object itself
4. Ans. $\frac{R}{\sqrt{3}}$
5. Ans. 0.09 m/s ; Magnitude of the rate of change of lateral magnification is 0.3 s^{-1} .

EXERCISE (O-2)

1. Ans. (A) 2. Ans. (D) 3. Ans. (A) 4. Ans. (D)
5. Ans. (D) 6. Ans. (C) 7. Ans. (A) 8. Ans. (B,D)
9. Ans. (A,C) 10. Ans. (A,B,C)

EXERCISE-(JA)

1. Ans. (C) 2. Ans. (B) 3. Ans. (C) 4. Ans. (C) 5. Ans. (B) 6. Ans. (A)
7. Ans. (B) 8. Ans. (A) p, q, r, s (B) q, (C) p, q, r, s (D) p, q, r, s 9. Ans. (C)
10. Ans. (C, D) 11. Ans. (B) 12. Ans. (A,B,C)
13. Ans. (A)–pr, (B) –qst, (C) –prt, (D) –qs 14. Ans. 6 15. Ans. 3 16. Ans. 6
17. Ans. (C) 18. Ans. 2 19. Ans. (B) 20. Ans. (C) 21. Ans. (B) 22. Ans. (A)
23. Ans. (C) 24. Ans. (D) 25. Ans. (A,C) 26. Ans. (C) 27. Ans. (B) 28. Ans. 7
29. Ans. (B) 30. Ans. 2 31. Ans. (A,C) 32. Ans. (D) 33. Ans. (A) 34. Ans. (A,B,D)
35. Ans. (A,D) 36. Ans. (A)

REVIEW TEST

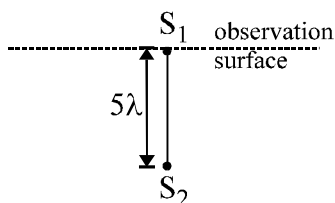
1. Ans. 3m 2. Ans. 2 3. Ans. 25 4. Ans. 4
5. Ans. 1 6. Ans. 8 7. Ans. 3 8. Ans. 2
9. Ans. (D) 10. Ans. (B) 11. Ans. (C) 12. Ans. (C)
13. Ans. (B) 14. Ans. (A) 15. Ans. (B) 16. Ans. (B)
17. Ans. (A) 18. Ans. (A,C) 19. Ans. (A,D) 20. Ans. (B)
21. Ans. (C) 22. Ans. (D) 23. Ans. (B,C,D) 24. Ans. (A)
25. Ans. (B) 26. Ans. (A) 27. Ans. (D) 28. Ans. (B)
29. Ans. (B) 30. Ans. (A,C) 31. Ans. (A, C) 32. Ans. (A, D)
33. Ans. (B) 34. Ans. (D) 35. Ans. (D)
36. Ans. (A)–Q,S; (B)–P,Q,R, S; (C)–P,Q,S (D)–S 37. Ans. (B) 38. Ans. (A)
39. Ans. (C) 40. Ans. (D) 41. Ans. (A) 42. Ans. (D)
43. Ans. (A,B,C,D)

[illegible]

WAVE OPTICS

EXERCISE (S-2)

1. In a Young's double slit experiment, two wavelengths of 500 nm and 700 nm were used. What is the minimum distance from the central maximum where their maximas coincide again? (Take $D/d = 10^3$. Symbols have their usual meanings.) **[IIT-JEE 2004]**
2. The distance between two slits in a YDSE apparatus is 3mm. The distance of the screen from the slits is 1m. Microwaves of wavelength 1 mm are incident on the plane of the slits normally. Find the distance of the first maxima on the screen from the central maxima. Also find the total number of maxima on the screen.
3. A monochromatic light of $\lambda = 5000 \text{ \AA}$ is incident on two slits separated by a distance of $5 \times 10^{-4} \text{ m}$. The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness $1.5 \times 10^{-6} \text{ m}$ & refractive index $\mu = 1.5$ is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is I_0 in the absence of the plate. Also find the lateral shift of the central maximum.
4. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.

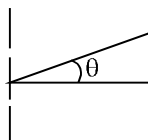


5. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength $5400 \text{ \AA}$. It is found that the point P on the screen where the central maximum ($n = 0$) fell before the glass plates were inserted now has $3/4$ the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of the glass plate. (Absorption of light by glass plate may be neglected). **[IIT-JEE 1997]**
6. One radio transmitter A operating at 60.0 MHz is 10.0 m from another similar transmitter B that is 180° out of phase with transmitter A. How far must an observer move from transmitter A toward transmitter B along the line connecting A and B to reach the nearest point where the two beams are in phase?

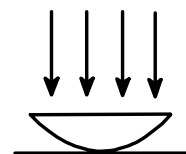
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. Two slits are separated by 0.3 mm. A beam of 500 nm light strikes the slits producing an interference pattern. The number of maxima observed in the angular range $-30^\circ < \theta < 30^\circ$.



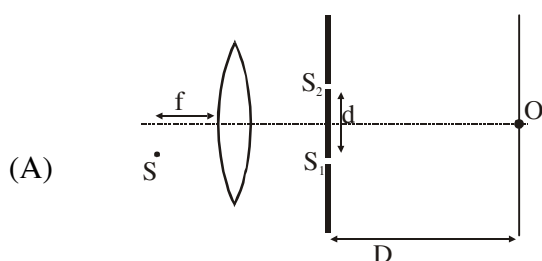
- (A) 300 (B) 150 (C) 599 (D) 601
2. A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be [IIT-JEE '99]
- (A) straight
(B) circular
(C) equally spaced
(D) having fringe spacing which increases as we go outwards.



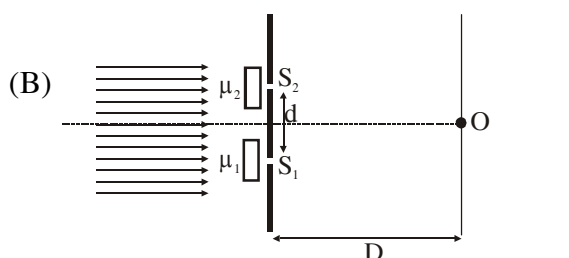
MATRIX MATCH TYPE QUESTIONS

3. Column-I shows some modifications in a standard YDSE setup. Column-II shows the associated characteristics.

Column-I



monochromatic point source
S placed in focal plane.

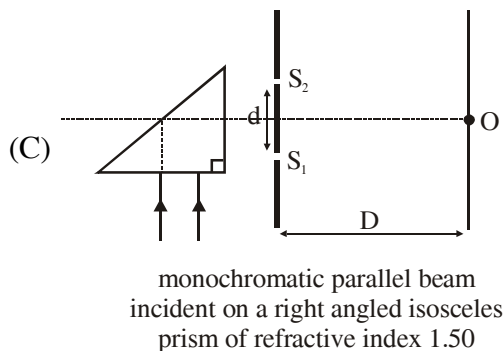


monochromatic parallel beam incident
on S_1, S_2 through transparent
slabs of same thickness but $\mu_1 > \mu_2$

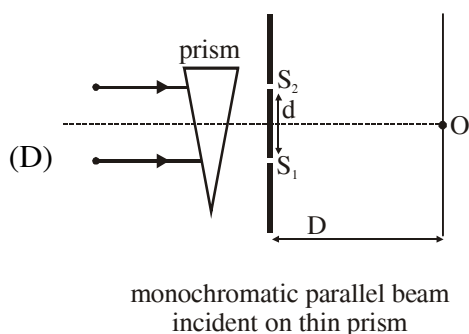
Column-II

(P) Zero order maxima lies above O.

(Q) If a transparent mica sheet is introduced
infront of S_2 central bright fringe can be
obtained at O.



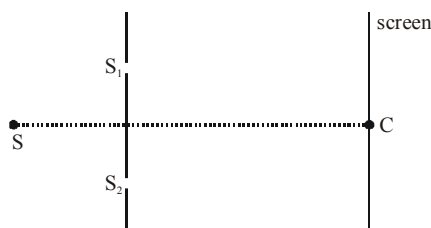
(R) Fringe width $\beta = \frac{\lambda D}{d}$



(S) Point O can be a minima

(T) Point O can be a least order minima.

4. In a YDSE setup, light of wavelength $4000 \text{ \AA}$ is used. Distance of screen from the slits is 2m and distance between slits is 1mm . There are three slabs slab 1 (thickness 2mm , $\mu = 2$), slab 2 (thickness 1mm , $\mu = 3$), slab 3 (thickness 4mm $\mu = \frac{3}{2}$)



- | | |
|---|---|
| (A) If slab 1 is placed in front of slit S_1 | (P) Central maxima at C |
| (B) If the slab 2 is placed in front of slit S_2 along with condition (A) | (Q) Central maxima above C |
| (C) If slab 3 is placed in front of slab 1 along with condition (B) | (R) Fringe width is equal to 0.8 mm |
| | (S) No. of fringes crossing central maxima |

EXERCISE-(J-A)

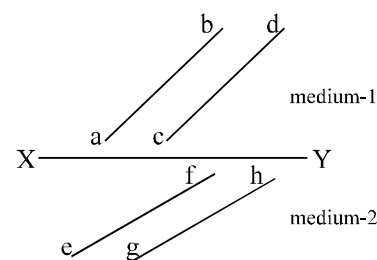
(Not to be discussed in class)

1. Some laws/processes are given in Column-I. Match these with the physical phenomena given in Column-II **[IIT-JEE 2006]**

Column-I	Column-II
(A) Intensity of light received by lens	(p) Radius of aperture (R)
(B) Angular magnification	(q) dispersion of lens
(C) Length of telescope	(r) focal length f_o, f_e
(D) Sharpness of image	(s) spherical aberration

Paragraph for Question Nos. 2 to 4

The figure shows a surface XY separating two transparent media, medium-1 and medium-2. The lines ab and cd represent wavefronts of a light wave traveling in medium-1 and incident on XY . The lines ef and gh represent wavefronts of the light wave in medium-2 after refraction. **[IIT-JEE 2007]**

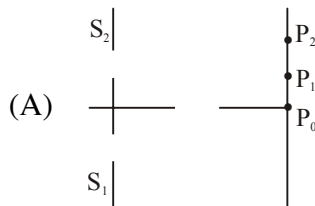


2. Light travels as a :-
- (A) parallel beam in each medium
 (B) convergent beam in each medium
 (C) divergent beam in each medium
 (D) divergent beam in one medium and convergent beam in the other medium
3. The phases of the light wave at c, d, e and f are ϕ_c, ϕ_d, ϕ_e and ϕ_f respectively. It is given that $\phi_c \neq \phi_f$.
- (A) ϕ_c cannot be equal to ϕ_d (B) ϕ_d can be equal to ϕ_e
 (C) $(\phi_d - \phi_f)$ is equal to $(\phi_c - \phi_e)$ (D) $(\phi_d - \phi_c)$ is not equal to $(\phi_f - \phi_e)$
4. Speed of light is
- (A) the same in medium-1 and medium-2 (B) larger in medium-1 than in medium-2
 (C) larger in medium-2 than in medium-1 (D) different at b and d
5. In a Young's double slit experiment, the separation between the two slits is d and the wavelength of the light is λ . The intensity of light falling on slit 1 is four times the intensity of light falling on slit 2. Choose the correct choice(s). **[IIT-JEE 2008]**
- (A) If $d = \lambda$, the screen will contain only one maximum
 (B) If $\lambda < d < 2\lambda$, at least one more maximum (besides the central max.) will be observed on the screen
 (C) If the intensity of light falling on slit 1 is reduced so that it becomes equal to that of slit 2, the intensities of the observed dark and bright fringes will increase
 (D) If the intensity of light falling on slit 2 is increased so that it becomes equal to that of slit 1, the intensities of the observed dark and bright fringes will increase

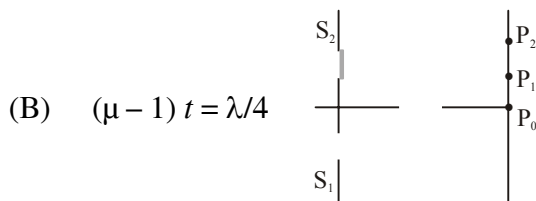
6. **Column I** shows four situations of standard Young's double slit arrangement with the screen placed far away from the slits S_1 and S_2 . In each of these cases $S_1P_0 = S_2P_0$, $S_1P_1 - S_2P_1 = \lambda/4$ and $S_1P_2 - S_2P_2 = \lambda/3$, where λ is the wavelength of the light used. In the cases B, C and D, a transparent sheet of refractive index μ and thickness t is pasted on slit S_2 . The thicknesses of the sheets are different in different cases. The phase difference between the light waves reaching a point P on the screen from the two slits is denoted by $\delta(P)$ and the intensity by $I(P)$. Match each situation given in **Column I** with the statement(s) in **Column II** valid for that situation. [IIT-JEE-2009]

Column-I

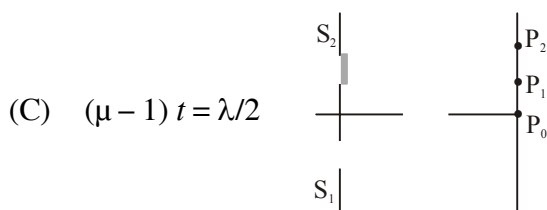
Column-II



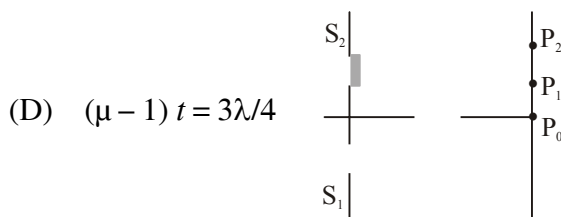
(p) $\delta(P_0) = 0$



(q) $\delta(P_1) = 0$



(r) $I(P_1) = 0$



(s) $I(P_0) > I(P_1)$

(t) $I(P_2) > I(P_1)$

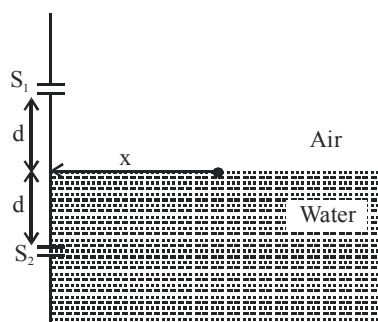
7. Young's double slit experiment is carried out by using green, red and blue light, one color at a time. The fringe widths recorded are β_G , β_R and β_B , respectively. Then [IIT-JEE-2012]

(A) $\beta_G > \beta_B > \beta_R$ (B) $\beta_B > \beta_G > \beta_R$ (C) $\beta_R > \beta_B > \beta_G$ (D) $\beta_R > \beta_G > \beta_B$

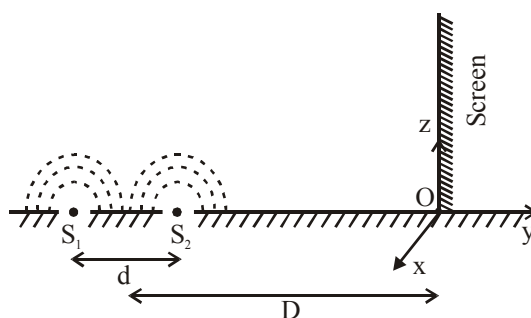
8. In the Young's double slit experiment using a monochromatic light of wavelength λ , the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is :- [JEE Advanced 2013]

(A) $(2n+1)\frac{\lambda}{2}$ (B) $(2n+1)\frac{\lambda}{4}$ (C) $(2n+1)\frac{\lambda}{8}$ (D) $(2n+1)\frac{\lambda}{16}$

9. A light source, which emits two wavelengths $\lambda_1 = 400$ nm and $\lambda_2 = 600$ nm, is used in a Young's double slit experiment. If recorded fringe widths for λ_1 and λ_2 are β_1 and β_2 and the number of fringes for them within a distance y on one side of the central maximum are m_1 and m_2 , respectively, then :-
[JEE Advanced 2014]
- (A) $\beta_2 > \beta_1$
 (B) $m_1 > m_2$
 (C) From the central maximum, 3rd maximum of λ_2 overlaps with 5th minimum of λ_1
 (D) The angular separation of fringes of λ_1 is greater than λ_2
10. A Young's double slit interference arrangement with slits S_1 and S_2 is immersed in water (refractive index = $4/3$) as shown in the figure. The positions of maxima on the surface of water are given by $x^2 = p^2 m^2 \lambda^2 - d^2$, where λ is the wavelength of light in air (refractive index = 1), $2d$ is the separation between the slits and m is an integer. The value of p is.
[JEE Advanced 2015]



11. While conducting the Young's double slit experiment, a student replaced the two slits with a large opaque plate in the x - y plane containing two small holes that act as two coherent point sources (S_1, S_2) emitting light of wavelength 600 nm. The student mistakenly placed the screen parallel to the x - z plane (for $z > 0$) at a distance $D = 3$ m from the mid-point of $S_1 S_2$, as shown schematically in the figure. The distance between the sources $d = 0.6003$ mm. The origin O is at the intersection of the screen and the line joining $S_1 S_2$. Which of the following is (are) true of the intensity pattern on the screen?
[JEE Advanced 2016]



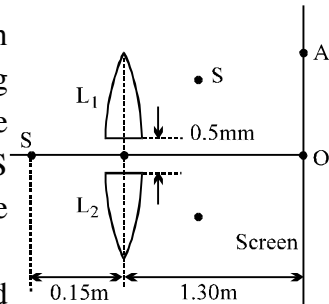
- (A) Hyperbolic bright and dark bands with foci symmetrically placed about O in the x -direction
 (B) Semi circular bright and dark bands centered at point O
 (C) The region very close to the point O will be dark
 (D) Straight bright and dark bands parallel to the x -axis

Review Test

(Not to be discussed in class)

1. A long narrow horizontal slit lies 1mm above a plane mirror as in Lloyd's mirror. The interference pattern produced by the slit and its image is viewed on a screen distant 1m from the slit. The wavelength of light is 600nm. Find the distance of first maximum above the mirror.

2. In the figure shown S is a monochromatic point source emitting light of wavelength $\lambda = 500$ nm. A thin lens of circular shape and focal length 0.10 m is cut into two identical halves L_1 and L_2 by a plane passing through a diameter. The two halves are placed symmetrically about the central axis SO with a gap of 0.5 mm. The distance along the axis from S to L_1 and L_2 is 0.15 m, while that from L_1 & L_2 to O is 1.30 m. The screen at O is normal to SO .



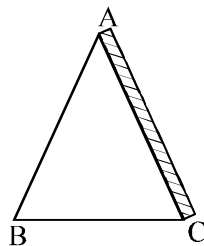
[IIT-JEE 1993]

- (i) If the third intensity maximum occurs at the point A on the screen, find the distance OA .
- (ii) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA increase, decrease or remain the same?

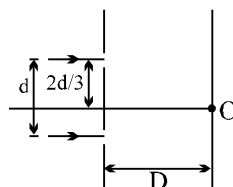
3. A prism ($\mu_p = \sqrt{3}$) has an angle of prism $A = 30^\circ$. A thin film ($\mu_f = 2.2$) is coated on face AC as shown in the figure. Light of wavelength 550 nm is incident on the face AB at 60° angle of incidence. Find

[IIT-JEE 2003]

- (i) the angle of its emergence from the face AC and
- (ii) the minimum thickness (in nm) of the film for which the emerging light is of maximum possible intensity.



4. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the white spot on the screen from O is [Assume $d \ll D$, $\lambda \ll d$]



- (A) 0 (B) $d/2$ (C) $d/3$ (D) $d/6$

5. Light of wavelength λ in air enters a medium of refractive index μ . Two points in this medium, lying along the path of this light, are at a distance x apart. The phase difference between these points is :

- (A) $\frac{2\pi\mu x}{\lambda}$ (B) $\frac{2\pi x}{\mu\lambda}$ (C) $\frac{2\pi(\mu-1)x}{\lambda}$ (D) $\frac{2\pi x}{(\mu-1)\lambda}$

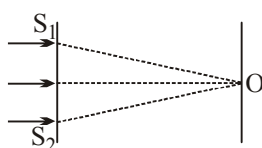
6. In a Young's double-slit experiment, let A and B be the two slits. Films of thicknesses t_A and t_B and refractive indices μ_A and μ_B , are placed in front of A and B respectively. If $\mu_A t_A = \mu_B t_B$, the central maximum will :

(A) not shift (B) shift towards A
 (C) shift towards B (D) option (B), if $t_B > t_A$; option (C) if $t_B < t_A$

7. **Statement-1 :** In YDSE, as shown in figure, central bright fringe is formed at O. If a liquid is filled between plane of slits and screen, the central bright fringe is shifted in upward direction.

and

Statement-2 : If path difference at O increases, y-coordinate of central bright fringe will change.



- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

8. **Statement-1 :** In glass, red light travels faster than blue light.

and

Statement-2 : Red light has a wavelength longer than blue.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

9. In a double slit experiment, the separation between the slits is $d = 0.25$ cm and the distance of the screen $D = 100$ cm from the slits. If the wavelength of light used is $\lambda = 6000 \text{ \AA}$ and I_0 is the intensity of the central bright fringe, the intensity at a distance $x = 4 \times 10^{-5}$ m from the central maxima is :-

(A) I_0 (B) $I_0/2$ (C) $3I_0/4$ (D) $I_0/3$

10. In a Young's double-slit experiment, let A and B be the two slits. A thin film of thickness t and refractive index μ is placed in front of A. Let β = fringe width. The central maximum will shift :

(A) towards A (B) towards B (C) by $t(\mu - 1) \frac{\beta}{\lambda}$ (D) by $\mu t \frac{\beta}{\lambda}$

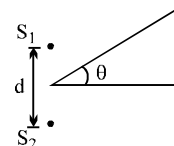
11. In an interference arrangement similar to Young's double-slit experiment, the slits S_1 & S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance $d = 150.0$ m. The intensity $I(\theta)$ is measured as a function of θ at a large distance from S_1 & S_2 , where θ is defined as shown. If I_0 is the maximum intensity then $I(\theta)$ for $0 \leq \theta \leq 90^\circ$ is given by :

(A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30^\circ$

(B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90^\circ$

(C) $I(\theta) = I_0$ for $\theta = 0^\circ$

(D) $I(\theta)$ is constant for all values of θ .



12. In a YDSE with two identical slits, when the upper slit is covered with a thin, perfectly transparent sheet of mica, the intensity at the centre of screen reduces to 75% of the initial value. Second minima is observed to be above this point and third maxima below it. Which of the following can be a possible value of phase difference caused by the mica sheet

(A) $\frac{\pi}{3}$

(B) $\frac{13\pi}{3}$

(C) $\frac{17\pi}{3}$

(D) $\frac{11\pi}{3}$

13. A beam of 2000 eV electrons are incident normally on the surface of crystal whose inter atomic separation is 0.11 nm. The mass of the electron can be taken as 9×10^{-31} kg. At what angle to the normal can we observe a diffraction maxima.

(A) $\sin^{-1}\left(\frac{1}{4}\right)$

(B) $\cos^{-1}\left(\frac{1}{4}\right)$

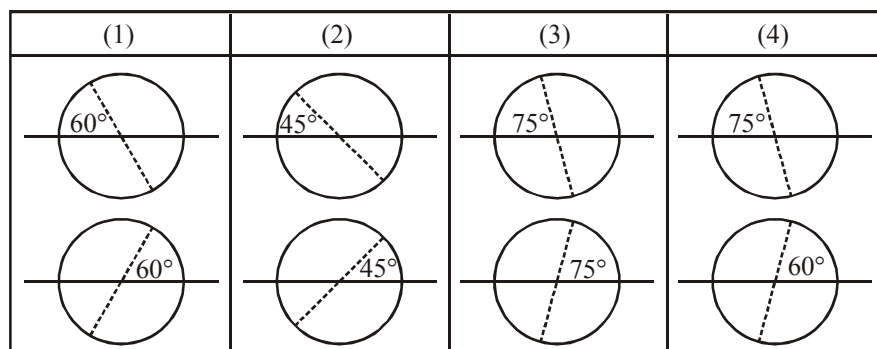
(C) $\sin^{-1}\left(\frac{1}{2}\right)$

(D) $\cos^{-1}\left(\frac{1}{2}\right)$

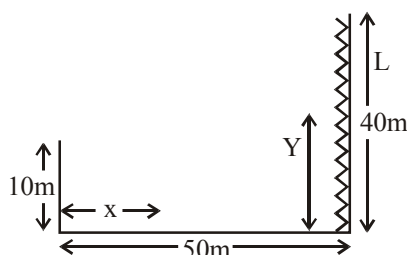
14. A plane electromagnetic wave traveling in the positive direction of x axis in vacuum has components $E_x = E_y = 0$ and $E_z = (2.0 \text{ V/m}) \cos [(\pi \times 10^{15} \text{ S}^{-1})(t - x/c)]$. (a) What is the amplitude of the magnetic field component? (b) Parallel to which axis does the magnetic field oscillate? (c) When the electric field component is in the positive direction of the z axis at a certain point P, what is the direction of the magnetic field component there?

15. A nickel crystal is used as a diffraction grating for x-rays. Then the same crystal is used to diffract electrons. If the two diffraction patterns are identical, and the energy of each x-ray photon is $E = 20.0$ keV. What is the kinetic energy of each electron?

16. The diagrams show four pairs of polarizing sheets, with the polarizing directions indicated by dashed lines. The two 'sheets of each pair are placed one behind the other and the front sheet is illuminated by unpolarized light. The incident intensity is the same for all pairs of sheets. Rank the pairs according to the intensity of the transmitted light, least to greatest.

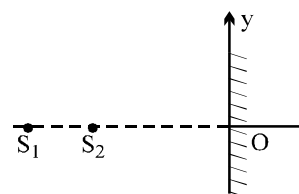


- (A) 1, 2, 3, 4 (B) 4, 2, 1, 3 (C) 2, 4, 3, 1 (D) 2, 1, 4, 3
17. A person lives in a high-rise building on the bank of a river 50m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water $\approx 4/3$)



- (A) 13 m, 27 m (B) 22 m, 13 m (C) 25 m, 10 m (D) 17 m, 29 m
18. Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of a large screen. The sources emit waves in phase with each other. The distance between S_1 and S_2 is d while their distance from the screen is much larger.
- (1) If $d = 7\lambda/2$, O will be a minima
 - (2) If $d = 4.3\lambda$, there will be a total of 8 minima on y -axis.
 - (3) If $d = 7\lambda$, O will be a maxima.
 - (4) If $d = \lambda$, there will be only one maxima on the screen.

Which is the set of correct statement :



- (A) 1, 2 & 3 (B) 2, 3 & 4 (C) 1, 2, 3 & 4 (D) 1, 3 & 4

19. A parallel coherent beam of light falls on Fresnel biprism of refractive index μ and angle α . The fringe width on a screen at a distance D from biprism will be (wavelength = λ)

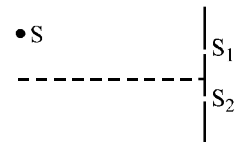
(A) $\frac{\lambda}{2(\mu-1)\alpha}$ (B) $\frac{\lambda D}{2(\mu-1)\alpha}$ (C) $\frac{D}{2(\mu-1)\alpha}$ (D) none

20. In Young's double slit arrangement, water is filled in the space between screen and slits. Then :

- (A) fringe pattern shifts upwards but fringe width remains unchanged.
(B) fringe width decreases and central bright fringe shifts upwards.
(C) fringe width increases and central bright fringe does not shift.
(D) fringe width decreases and central bright fringe does not shift.

21. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then,

- (A) the fringe width will increase and fringe pattern will shift down.
(B) the fringe width will remain same but fringe pattern will shift up.
(C) the fringe width will decrease and fringe pattern will shift down.
(D) the fringe width will remain same but fringe pattern will shift down.



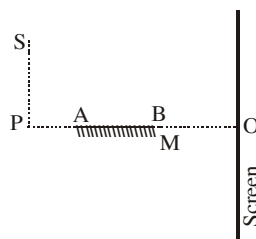
22. In a double slit experiment, when the width of one slit is made twice as wide as the other in compared to normal YDSE having slits of equal width. Then, in the interference pattern [IIT-JEE(Scr.) 2000]

- (A) the intensities of both the maxima and the minima increase.
(B) the intensity of the maxima increases and the minima has zero intensity.
(C) the intensity of the maxima decreases and that of the minima increases.
(D) the intensity of the maxima decreases and the minima has zero intensity.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 23 and 24

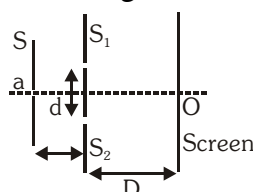
In an experiment on interference due to single mirror, a light wave emitted directly by the source S (narrow slit) interferes with the wave reflected from the mirror M of length 2 mm. Source and screen are separated by distance 90 cm. Source S is at the height of 3 mm, from the point P and the middle point of mirror is at distance of 2mm from point P . Point P and mirror are in the same plane of screen is perpendicular to this plane.



23. If fringe width is 0.1 mm then what is the wavelength of light used?
 (A) 3.3×10^{-7} m (B) 6.7×10^{-7} m (C) 1.0×10^{-7} m (D) 4×10^{-7} m
24. If the mirror is shifted towards left then how does the fringe pattern on screen change?
 (A) Fringe width decreases and the region in which interference is formed shifts downward.
 (B) Fringe width decreases and region in which interference is formed shifts upwards
 (C) Fringe width does not change and region in which interference is formed shifts upwards
 (D) Fringe width does not change and region in which interference is formed shifts downwards

Paragraph for Question No. 25 to 27.

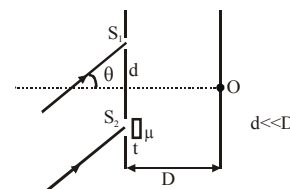
The figure shows a schematic diagram showing the arrangement of Young's Double Slit Experiment



25. Choose the correct statement(s) related to the wavelength of light used
 (A) Larger the wavelength of light larger the fringe width
 (B) The position of central maxima depends on the wavelength of light used
 (C) If white light is used in YDSE, then the violet colour forms its first maxima closest to the central maxima
 (D) The central maxima of all the wavelengths coincide
26. If the distance D is varied, then choose the correct statement(s)
 (A) The angular fringe width does not change
 (B) The fringe width changes in direct proportion
 (C) The change in fringe width is same for all wavelengths
 (D) The position of central maxima remains unchanged
27. If the distance d is varied, then identify the correct statement(s)
 (A) The angular width does not change (B) The fringe width changes in inverse proportion
 (C) The positions of all maxima change (D) The positions of all minima change

Paragraph for Question No. 28 to 30.

A monochromatic beam of light falls on Young's double slit experiment apparatus as shown in figure. A thin sheet of glass is inserted in front of lower slit S_2 .



28. The central bright fringe can be obtained
 (A) at O only (B) at O or below O only
 (C) at O or above O only (D) Anywhere on the screen
29. If central bright fringe is obtained on screen at O :-

(A) $(\mu-1)t = d \sin \theta$ (B) $(\mu-1)t = d \cos \theta$ (C) $(\mu-1)t + d \sin \theta = 0$ (D) $\frac{t}{\mu-1} = \frac{d}{\sin \theta}$

30. The phase difference between the waves interfering at fifth minima is
 (A) 5π (B) 7π (C) 9π (D) 11π

ANSWER KEY

EXERCISE (S-2)

1. Ans. 3.5 mm 2. Ans. 35.35 cm approximate, 5

3. Ans. 0, 1.5 mm 4. Ans. 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.; 

5. Ans. 9.3 μm

6. Ans. 1.25 m

EXERCISE (O-2)

1. Ans. (C) 2. Ans. (A)

3. Ans. (A)-PRST; (B)-QRST; (C)-R; (D)-PRST

4. Ans. (A)-Q,R,S ; (B)-P,R ; (C)-Q,R,S

EXERCISE-(J-A)

1. Ans. (A) (p) ; (B) (r) ; (C) (r) ; (D) (p, q, r,s)

2. Ans. (A)

3. Ans. (C)

4. Ans. (B)

5. Ans. (A,B)

6. Ans. (A) p, s ; (B) q ; (C) t ; (D) r,s,t

7. Ans. (D)

8. Ans. (B)

9. Ans. (A,B,C)

10. Ans. 3

11. Ans. (B, C)

REVIEW TEST

1. Ans. 0.15 mm

2. Ans. (i) 1 mm (ii) increase

3. Ans. 0, 125 nm

4. Ans. (D)

5. Ans. (A)

6. Ans. (D)

7. Ans. (D)

8. Ans. (A)

9. Ans. (C)

10. Ans. (A,C)

11. Ans. (A,C)

12. Ans. (B,C,D)

13. Ans. (A,C)

14. Ans. (a) 6.7 nT; (b) y ; (c) negative direction of y.

15. Ans. 391 eV

16. Ans. (D)

17. Ans. (A)

18. Ans. (C)

19. Ans. (A)

20. Ans. (D)

21. Ans. (D)

22. Ans. (A)

23. Ans. (B)

24. Ans. (C)

25. Ans. (A,C,D)

26. Ans. (A,B,D)

27. Ans. (B,D)

28. Ans. (D)

29. Ans. (A)

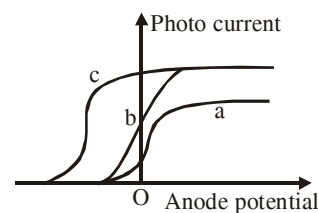
30. Ans. (C)

[illegible]

MODERN PHYSICS-1

EXERCISE (O-2)

- In photoelectric effect, stopping potential depends on
(A) frequency of the incident light
(B) intensity of the incident light by varies source distance
(C) emitter's properties
(D) frequency and intensity of the incident light
- The figure shows the variation of photo current with anode potential for a photosensitive surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a, b and c respectively :
(A) $f_a = f_b$ (B) $I_a < I_b$
(C) $f_c < f_b$ (D) $I_c > I_b$
- Two electrons are moving with the same speed v . One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field, then after sometime if the de-Broglie wavelengths of the two are λ_1 and λ_2 , then:
(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 > \lambda_2$ (C) $\lambda_1 < \lambda_2$ (D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$
- An electron in hydrogen atom first jumps from second excited state to first excited state and then, from first excited state to ground state. Let the ratio of wavelength, momentum and energy of photons in the two cases be x , y and z , then select the wrong answer/(s) :
(A) $z = 1/x$ (B) $x=9/4$ (C) $y=5/27$ (D) $z=5/27$
- An electron is in an excited state in hydrogen-like atom. It has a total energy of -3.4 eV. If the kinetic energy of the electron is E and its de-Broglie wavelength is λ , then
(A) $E = 6.8$ eV, $\lambda = 6.6 \times 10^{-10}$ m (B) $E = 3.4$ eV, $\lambda = 6.6 \times 10^{-10}$ m
(C) $E = 3.4$ eV, $\lambda = 6.6 \times 10^{-11}$ m (D) $E = 6.8$ eV, $\lambda = 6.6 \times 10^{-11}$ m
- The energy levels of a hypothetical one electron atom are shown in the figure



c	_____	0 eV
n = 5	_____	-0.80 eV
n = 4	_____	-1.45 eV
n = 3	_____	-3.08 eV
n = 2	_____	-5.30 eV
n = 1	_____	-15.6 eV

- (A) The ionization potential of this atom is 15.6 V
 (B) The short wavelength limit of the series terminating at $n = 2$ is 2339 Å
 (C) The excitation potential for the state $n = 3$ is 12.52 V
 (D) Wave number of the photon emitted for the transition $n = 3$ to $n = 1$ is $1.009 \times 10^7 \text{ m}^{-1}$
7. If radiation of all wavelengths from ultraviolet to infrared is passed through hydrogen gas at room temperature, absorption lines will be observed in the :
 (A) Lyman series (B) Balmer series (C) both (A) and (B) (D) neither (A) nor (B)
8. In the hydrogen atom, if the reference level of potential energy is assumed to be zero at the ground state level. Choose the incorrect statement.
 (A) The total energy of the shell increases with increase in the value of n
 (B) The total energy of the shell decrease with increase in the value of n .
 (C) The difference in total energy of any two shells remains the same.
 (D) The total energy at the ground state becomes 13.6 eV.
9. X-rays are produced by accelerating electrons across a given potential difference to strike a metal target of high atomic number. If the electrons have same speed when they strike the target, the X-ray spectrum will exhibit.
 (A) a minimum wavelength
 (B) a continuous spectrum
 (C) some discrete comparatively prominent wavelength
 (D) uniform density over the whole spectrum
10. In a Coolidge tube experiment, the minimum wavelength of the continuous X-ray spectrum is equal to 66.3 pm, then
 (A) electrons accelerate through a potential difference of 12.75 kV in the Coolidge tube
 (B) electrons accelerate through a potential difference of 18.75 kV in the Coolidge tube
 (C) de-Broglie wavelength of the electrons reaching the anticathode is of the order of $10 \mu\text{m}$.
 (D) de-Broglie wavelength of the electrons reaching the anticathode is 0.01 Å .
11. The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation:
 (A) the intensity increases (B) the minimum wave length increases
 (C) the intensity decreases (D) the minimum wave length decreases
12. Match the entries of column-I with the entries of column-II :-
- | Column-I | Column-II |
|-----------------------------|---|
| (I) Characteristic X-ray | (P) Inverse process of photoelectric effect |
| (II) Photoelectric effect | (Q) Emission of electrons |
| (III) Thermo-ionic emission | (R) Moseley's law |
| (IV) Continuous X-ray | (S) Emission of radiations |
- Then choose the correct matching.
 (A) (i) $\rightarrow$ (RS) ; (ii) $\rightarrow$ (RS) ; (iii) $\rightarrow$ (S) ; (iv) $\rightarrow$ (PS)
 (B) (i) $\rightarrow$ (RS) ; (ii) $\rightarrow$ (Q) ; (iii) $\rightarrow$ (Q) ; (iv) $\rightarrow$ (PS)
 (C) (i) $\rightarrow$ (RS) ; (ii) $\rightarrow$ (S) ; (iii) $\rightarrow$ (S) ; (iv) $\rightarrow$ (PRS)
 (D) (i) $\rightarrow$ (RS) ; (ii) $\rightarrow$ (Q) ; (iii) $\rightarrow$ (Q) ; (iv) $\rightarrow$ (PRS)

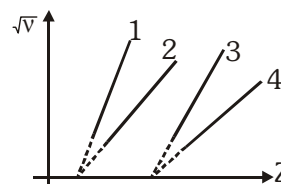
13. $\sqrt{v}$ versus Z graph for characteristic X-rays is as shown in figure. Match the following :

Column I

- (A) Line – 1
- (B) Line – 2
- (C) Line – 3
- (D) Line – 4

Column II

- (P) L_{α}
- (Q) L_{β}
- (R) K_{α}
- (S) K_{β}
- (T) Both K_{α} and L_{β}



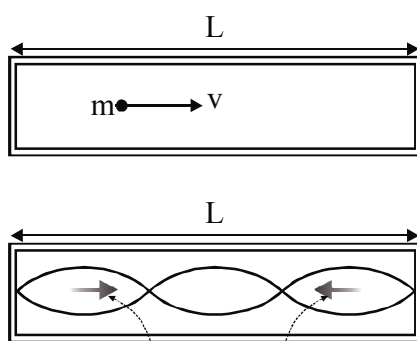
Paragraph for Q. 14 to 16

While conducting his doctoral research in theoretical physics and with no experimental evidence to go on, de Broglie reasoned by analogy with Einstein's equation $E = hf$ and with some of the ideas of his theory of relativity. The details need not concern us, but they led de Broglie to postulate that if a material particle of

momentum $p = mv$ has a wave-like nature, then its wavelength must be given by $\lambda = \frac{h}{p} = \frac{h}{mv}$ where h is

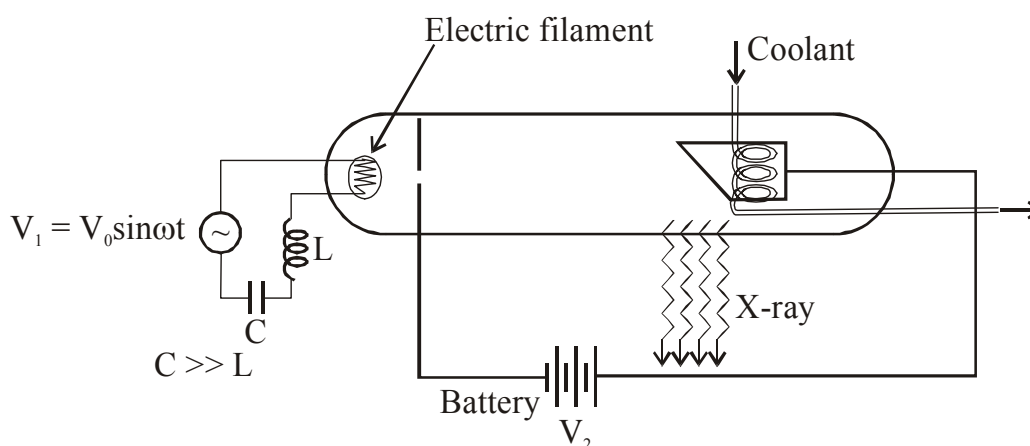
Planck's constant. This is called the de-Broglie wavelength.

de-Broglie considered a matter wave to be a traveling wave. But suppose that a "particle" of matter is confined to a small region of space and cannot travel. How do the wave-like properties manifest themselves? This is the problem of "a particle in a box." Figure shows a particle of mass m moving in one dimension as it bounces back and forth with speed v between the ends of a box of length L . We'll call this a one-dimensional box; its width isn't relevant. A particle in a box creates a standing de Broglie wave as it reflects back and forth.



Matter waves travel in both directions.

14. What should be de-Broglie wavelength of confined particle in the box [here $n \in \mathbb{N}$]
- (A) $\frac{L}{2n}$ (B) $\frac{2L}{n}$ (C) $\frac{L}{n}$ (D) nL
15. Confined particle's energy is given by
- (A) $\frac{n^2 h^2}{2mL^2}$ (B) $\frac{2n^2 h^2}{mL^2}$ (C) $\frac{n^2 h^2}{8mL^2}$ (D) $\frac{n^2 h^2}{4mL^2}$
16. Consider an oil drop from Millikan's oil drop experiment having diameter $1 \mu\text{m}$ confined between the plates separated by $10 \mu\text{m}$. Density of oil is 900 kg/m^3 . What is minimum energy of such an oil drop? [Given : $h = 6.63 \times 10^{-34} \text{ Js}$]
- (A) $2.4 \times 10^{-42} \text{ J}$ (B) $1.2 \times 10^{-42} \text{ J}$ (C) $3.6 \times 10^{-42} \text{ J}$ (D) $4.8 \times 10^{-42} \text{ J}$
17. Figure shows a Coolidge tube being used to produce X-ray. Column-I shows change in various parameters while column-II shows effects occurring due to change in the parameters. Match them correctly

**Column-I**

- (A) On increasing frequency (ω) of AC source
- (B) On increasing pressure of gas inside the tube keeping temperature of gas constant
- (C) On increasing peak voltage (V_0) of AC
- (D) On increasing emf (V_2) of battery.

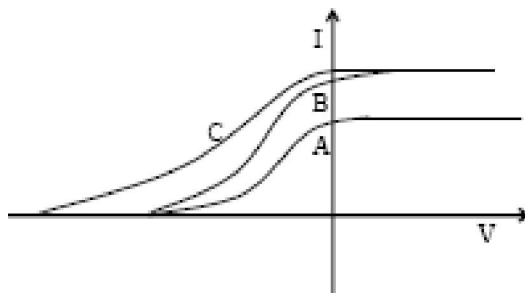
Column-II

- (P) Intensity of X-rays increases.
- (Q) Intensity of X-rays decreases.
- (R) X-ray becomes harder.
- (S) X-ray becomes softer.
- (T) Temperature of coolant at output would be higher.

EXERCISE (JA)

ALL ANSWERS OF THE QUESTIONS **JEE-2007** ONWARDS ARE AS GIVEN ON **IIT-JEE** WEBSITE.

- If the atom ${}_{100}\text{Fm}^{257}$ follows the Bohr model and the radius of ${}_{100}\text{Fm}^{257}$ is n times the Bohr radius, then find n .
(A) 100 (B) 200 (C) 4 (D) $1/4$ [JEE 2003]
- The attractive potential for an atom is given by $v = v_0 \ln(r/r_0)$, v_0 and r_0 are constant and r is the radius of the orbit. The radius r of the n^{th} Bohr's orbit depends upon principal quantum number n as:
[JEE' 2003 (Scr)]
(A) $r \propto n$ (B) $r \propto 1/n^2$ (C) $r \propto n^2$ (D) $r \propto 1/n$
- Frequency of a photon emitted due to transition of electron of a certain element from L to K shell is found to be 4.2×10^{18} Hz. Using Moseley's law, find the atomic number of the element, given that the Rydberg's constant $R = 1.1 \times 10^7 \text{ m}^{-1}$.
[JEE' 2003]
- In a photoelectric experiment set up, photons of energy 5 eV fall on the cathode having work function 3 eV.
(a) If the saturation current is $i_A = 4 \mu\text{A}$ for intensity 10^{-5} W/m^2 , then plot a graph between anode potential and current.
(b) Also draw a graph for intensity of incident radiation of $2 \times 10^{-5} \text{ W/m}^2$. [JEE' 2003]
- In a photoelectric experiment anode potential is plotted against plate current [JEE-2004 (Scr.)]



- A and B will have different intensities while B and C will have different frequencies.
 - B and C will have different intensities while A and C will have different frequencies.
 - A and B will have different intensities while A and C will have equal frequencies.
 - A and B will have equal intensities while B and C will have different frequencies.
- A proton has kinetic energy $E = 100 \text{ keV}$ which is equal to that of a photon. The wavelength of photon is λ_2 and that of proton is λ_1 . The ratio of λ_2 / λ_1 is proportional to
[JEE-2004 (Scr.)]
(A) E^2 (B) $E^{1/2}$ (C) E^{-1} (D) E^{-2}
 - In a photoelectric setup, the radiations from the Balmer series of hydrogen atom are incident on a metal surface of work function 2 eV. The wavelength of incident radiations lies between 450 nm to 700 nm. Find the maximum kinetic energy of photoelectron emitted. (Given $hc/e = 1242 \text{ eV-nm}$).
[JEE-2004]

8. The wavelength of K_{α} X-ray of an element having atomic number $Z = 11$ is λ . The wavelength of K_{α} X-ray of another element of atomic number Z' is 4λ . Then Z' is [JEE' 2005 (Scr)]

(A) 11 (B) 44 (C) 6 (D) 4

9. A photon of 10.2 eV energy collides with a hydrogen atom in ground state inelastically. After few microseconds one more photon of energy 15 eV collides with the same hydrogen atom. Then what can be detected by a suitable detector. [JEE' 2005 (Scr)]

(A) one photon of 10.2 eV and an electron of energy 1.4 eV
 (B) 2 photons of energy 10.2 eV
 (C) 2 photons of energy 3.4 eV
 (D) 1 photon of 3.4 eV and one electron of 1.4 eV

10. In Young's double slit experiment an electron beam is used to form a fringe pattern instead of light. If speed of the electrons is increased then the fringe width will : [JEE' 2005 (Scr)]

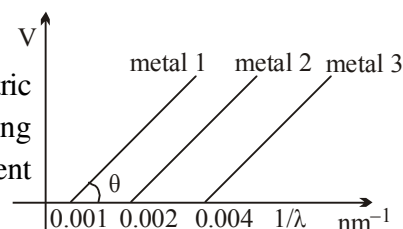
(A) increase (B) decrease
 (C) remain same (D) no fringe pattern will be formed

11. The potential energy of a particle varies as $U(x) = E_0$ for $0 \leq x \leq 1$ and $U(x) = 0$ for $x > 1$. For $0 \leq x \leq 1$, de-Broglie wavelength is λ_1 and for $x > 1$ the de-Broglie wavelength is λ_2 .

Total energy of the particle is $2E_0$. Find $\frac{\lambda_1}{\lambda_2}$.

[JEE 2005]

12. The graph between $1/\lambda$ and stopping potential (V) of three metals having work functions ϕ_1 , ϕ_2 and ϕ_3 in an experiment of photo-electric effect is plotted as shown in the figure. Which of the following statement(s) is/are correct? [Here λ is the wavelength of the incident ray]. [JEE 2006]



(A) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$
 (B) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$
 (C) $\tan \theta$ is directly proportional to hc/e , where h is Planck's constant and c is the speed of light
 (D) The violet colour light can eject photoelectrons from metals 2 and 3.

13. In hydrogen-like atom ($Z = 11$), n^{th} line of Lyman series has wavelength λ equal to the de-Broglie's wavelength of electron in the level from which it originated. What is the value of n ? [Take: Bohr radius (r_0) = $0.53 \text{ \AA}$ and Rydberg constant (R) = $1.1 \times 10^7 \text{ m}^{-1}$] [JEE 2006]

14. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is [JEE 2007]

(A) 802 nm (B) 823 nm (C) 1882 nm (D) 1648 nm

15. **STATEMENT-1** : If the accelerating potential in an X-ray tube is increased, the wavelengths of the characteristic X-rays do not change. [JEE 2007]

because

STATEMENT-2 : When an electron beam strikes the target in an X-ray tube, part of the kinetic energy is converted into X-ray energy

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
16. Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is [JEE 2007]

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$ (C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

17. Which one of the following statements is **WRONG** in the context of X-rays generated from a X-ray tube ? [JEE 2008]
- (A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases
 (B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target
 (C) Intensity of the characteristic X-rays depends on the electrical power given to the X-rays tube
 (D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

Paragraph for Question Nos. 18 to 20

In a mixture of H – He⁺ gas (He⁺ is singly ionized He atom), H atoms and He⁺ ions are excited to their respective first excited states. Subsequently, H atoms transfer their total excitation energy to He⁺ ions (by collisions). Assume that the Bohr model of atom is exactly valid.

18. The quantum number n of the state finally populated in He⁺ ions is [JEE 2008]
 (A) 2 (B) 3 (C) 4 (D) 5
19. The wavelength of light emitted in the visible region by He⁺ ions after collisions with H atoms is
 (A) 6.5×10^{-7} m (B) 5.6×10^{-7} m (C) 4.8×10^{-7} m (D) 4.0×10^{-7} m
20. The ratio of the kinetic energy of the $n = 2$ electron for the H atom to that of He⁺ ion is
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

Paragraph for Question Nos. 21 to 23

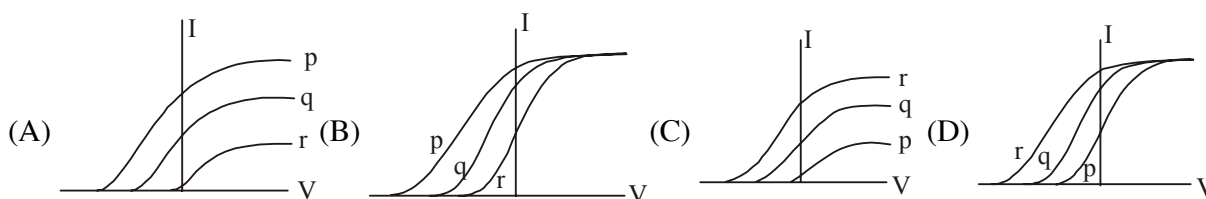
When a particle is restricted to move along x-axis between $x = 0$ and $x = a$, where a is of nanometer dimension, its energy can take only certain specific values. The allowed energies of the particle moving in such a restricted region, correspond to the formation of standing waves with nodes at its ends $x = 0$ and $x = a$. The wavelength of this standing wave is related to the linear momentum p of the particle according to

the de Broglie relation. The energy of the particle of mass m is related to its linear momentum as $E = \frac{p^2}{2m}$.

Thus, the energy of the particle can be denoted by a quantum number ' n ' taking values 1, 2, 3, ... ($n = 1$, called the ground state) corresponding to the number of loops in the standing wave. Use the model described above to answer the following three questions for a particle moving in the line $x = 0$ to $x = a$. Take $h = 6.6 \times 10^{-34}$ Js and $e = 1.6 \times 10^{-19}$ C. [JEE-2009]

21. The allowed energy for the particle for a particular value of n is proportional to
 (A) a^{-2} (B) $a^{-3/2}$ (C) a^{-1} (D) a^2

22. If the mass of the particle is $m = 1.0 \times 10^{-30}$ kg and $a = 6.6$ nm, the energy of the particle in its ground state is closest to :
 (A) 0.8 meV (B) 8 meV (C) 80 meV (D) 800 meV
23. The speed of the particle, that take discrete values, is proportional to
 (A) $n^{-3/2}$ (B) n^{-1} (C) $n^{1/2}$ (D) n
24. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions $\phi_p = 2.0$ eV, $\phi_q = 2.5$ eV and $\phi_r = 3.0$ eV, respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiment is : **[JEE-2009]**



25. An α -particle and a proton are accelerated from rest by a potential difference of 100 V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$, to the nearest integer, is

[JEE 2010]**Paragraph for Question Nos. 26 to 28**

The key feature of Bohr's theory of spectrum of hydrogen atom is the quantization of angular momentum when an electron is revolving around a proton. We will extend this to a general rotational motion to find quantized rotational energy of a diatomic molecule assuming it to be rigid. The rule to be applied is Bohr's quantization condition.

[JEE 2010]

26. A diatomic molecule has moment of inertia I . By Bohr's quantization condition its rotational energy in the n^{th} level ($n=0$ is not allowed) is

(A) $\frac{1}{n^2} \left(\frac{h^2}{8\pi^2 I} \right)$ (B) $\frac{1}{n} \left(\frac{h^2}{8\pi^2 I} \right)$ (C) $n \left(\frac{h^2}{8\pi^2 I} \right)$ (D) $n^2 \left(\frac{h^2}{8\pi^2 I} \right)$

27. It is found that the excitation frequency from ground to the first excited state of rotation for the CO molecule is close to $\frac{4}{\pi} \times 10^{11}$ Hz. Then the moment of inertia of CO molecule about its center of mass is

close to [Take $h = 2\pi \times 10^{-34}$ Js]

(A) 2.76×10^{-46} kg m² (B) 1.87×10^{-46} kg m² (C) 4.67×10^{-47} kg m² (D) 1.17×10^{-47} kg m²

28. In a CO molecule, the distance between C (mass = 12 a.m.u.) and O (mass = 16 a.m.u.),

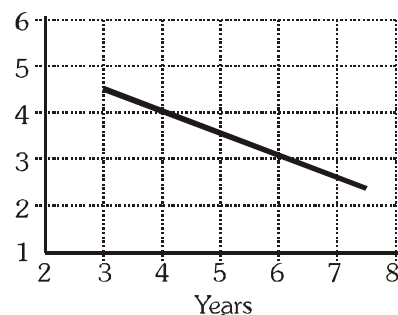
where 1 a.m.u. = $\frac{5}{3} \times 10^{-27}$ kg, is close to

(A) 2.4×10^{-10} m (B) 1.9×10^{-10} m (C) 1.3×10^{-10} m (D) 4.4×10^{-11} m

29. To determine the half life of a radioactive element, a student

plots a graph of $\ln \left| \frac{dN(t)}{dt} \right|$ versus t . Here $\frac{dN(t)}{dt}$ is the rate

of radioactive decay at time t . If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is [JEE 2010]



30. The wavelength of the first spectral line in the Balmer series of hydrogen atom is 6561 Å. The wavelength of the second spectral line in the Balmer series of singly-ionized helium atom is

(A) 1215 Å (B) 1640 Å (C) 2430 Å (D) 4687 Å [JEE 2011]

Paragraph for Question Nos. 31 and 32

A dense collection of equal number of electrons and positive ions is called neutral plasma. Certain solids containing fixed positive ions surrounded by free electrons can be treated as neutral plasma. Let 'N' be the number density of free electrons, each of mass 'm'. When the electrons are subjected to an electric field, they are displaced relatively away from the heavy positive ions. If the electric field becomes zero, the electrons begins to oscillate about the positive ions with a natural angular frequency ' ω_p ', which is called the plasma frequency. To sustain the oscillations, a time varying electric field needs to be applied that has an angular frequency ω , where a part of the energy is absorbed and a part of it is reflected. As ω approaches ω_p , all the free electrons are set to resonance together and all the energy is reflected. This is the explanation of high reflectivity of metals. [JEE 2011]

31. Taking the electronic charge as 'e' and the permittivity as ' ϵ_0 ', use dimensional analysis to determine the correct expression for ω_p .

(A) $\sqrt{\frac{Ne}{m\epsilon_0}}$ (B) $\sqrt{\frac{m\epsilon_0}{Ne}}$ (C) $\sqrt{\frac{Ne^2}{m\epsilon_0}}$ (D) $\sqrt{\frac{m\epsilon_0}{Ne^2}}$

32. Estimate the wavelength at which plasma reflection will occur for a metal having the density of electrons $N \approx 4 \times 10^{27} \text{ m}^{-3}$. Take $\epsilon_0 \approx 10^{-11}$ and $m \approx 10^{-30}$, where these quantities are in proper SI units

(A) 800 nm (B) 600 nm (C) 300 nm (D) 200 nm

33. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s. The mass of an atom of this radioisotope is 10^{-25} kg. The mass (in mg) of the radioactive sample is [JEE 2011]

34. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^Z$ (where $1 < A < 10$). The value of 'Z' is [JEE 2011]

35. A proton is fired from very far away towards a nucleus with charge $Q = 120 e$, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is (Take : The proton mass, $m_p = (5/3) \times 10^{-27}$ kg;

$$h/e = 4.2 \times 10^{-15} \text{ J.s/C}; \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ m/F}; 1 \text{ fm} = 10^{-15} \text{ m} \quad [\text{JEE 2012}]$$

Paragraph for Questions 36 and 37

The β -decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e. $n \rightarrow p + e^- + \bar{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino ($\bar{\nu}_e$) to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is 0.8×10^6 eV. The kinetic energy carried by the proton is only the recoil energy.

36. If the anti-neutrino had a mass of $3 \text{ eV}/c^2$ (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K , of the electron? [JEE 2012]

- (A) $0 \leq K \leq 0.8 \times 10^6 \text{ eV}$ (B) $3.0 \text{ eV} \leq K \leq 0.8 \times 10^6 \text{ eV}$
 (C) $3.0 \text{ eV} \leq K < 0.8 \times 10^6 \text{ eV}$ (D) $0 \leq K < 0.8 \times 10^6 \text{ eV}$

37. What is the maximum energy of the anti-neutrino? [JEE 2012]

- (A) zero (B) much less than $0.8 \times 10^6 \text{ eV}$
 (C) Nearly $0.8 \times 10^6 \text{ eV}$ (D) Much larger than $0.8 \times 10^6 \text{ eV}$

38. The work functions of Silver and sodium are 4.6 and 2.3 eV, repetitively. The ratio of the slope of the stopping potential versus frequency plot for Silver to that of Sodium is. [JEE 2013]

39. The radius of the orbit of an electron in a Hydrogen-like atom is $4.5 a_0$, where a_0 is the Bohr radius. Its orbital angular momentum is $\frac{3h}{2\pi}$. It is given that h is Planck constant and R is Rydberg constant. The possible wavelength (s), when the atom de-excites, is (are) :- [JEE Advanced 2013]

- (A) $\frac{9}{32R}$ (B) $\frac{9}{16R}$ (C) $\frac{9}{5R}$ (D) $\frac{4}{3R}$

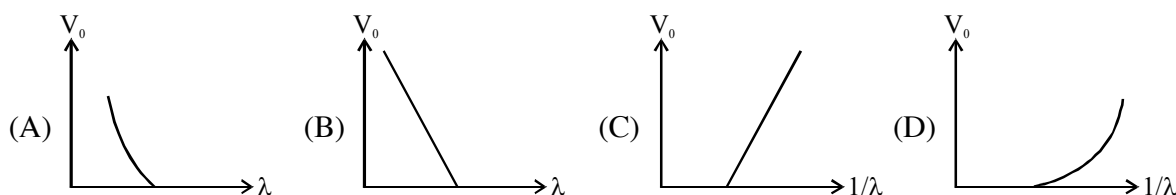
40. If λ_{Cu} is the wavelength of K_α X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength of the K_α X-ray line of molybdenum (atomic number 42), then the ratio $\frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$ is close to [JEE Advanced 2014]

- (A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

41. A metal surface is illuminated by light of two different wavelength 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2 , respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240 \text{ eV nm}$, the work function of the metal is nearly [JEE Advanced 2014]

- (A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

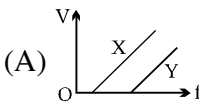
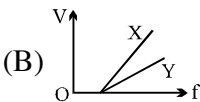
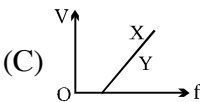
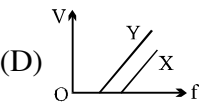
42. Consider a hydrogen atom with its electron in the n^{th} orbital. An electromagnetic radiation of wavelength 90 nm is used to ionize the atom. If the kinetic energy of the ejected electron is 10.4 eV, then the value of n is ($hc = 1242 \text{ eV nm}$). [JEE Advanced-2015]
43. Planck's constant h , speed of light c and gravitational constant G are used to form a unit of length L and a unit of mass M . Then the correct option(s) is(are) :- [JEE Advanced-2015]
- (A) $M \propto \sqrt{c}$ (B) $M \propto \sqrt{G}$ (C) $L \propto \sqrt{h}$ (D) $L \propto \sqrt{G}$
44. For photo-electric effect with incident photon wavelength λ , the stopping potential is V_0 . Identify the correct variation(s) of V_0 with λ and $1/\lambda$. [JEE Advanced-2015]



45. An electron in an excited state of Li^{2+} ion has angular momentum $3h/2\pi$. The de Broglie wavelength of the electron in this state is $p\pi a_0$ (where a_0 is the Bohr radius). The value of p is [JEE Advanced-2015]

REVIEW TEST

- In an experiment on photoelectric effect, the slope of the cut-off voltage versus frequency of incident light is found to be 4.12×10^{-15} V s. Calculate the value of Planck's constant.
- A 100W sodium lamp radiates energy uniformly in all directions. The lamp is located at the centre of a large sphere that absorbs all the sodium light which is incident on it. The wavelength of the sodium light is 589 nm. (a) What is the energy per photon associated with the sodium light ? (b) At what rate are the photons delivered to the sphere?
- Estimating the following two numbers should be interesting. The first number will tell you why radio engineers do not need to worry much about photons! The second number tells you why our eye can never 'count photons', even in barely detectable light.
 - The number of photons emitted per second by a Medium wave transmitter of 10 kW power, emitting radiowaves of wavelength 500 m.
 - The number of photons entering the pupil of our eye per second corresponding to the minimum intensity of white light that we humans can perceive ($\sim 10^{-10}$ W m⁻²). Take the area of the pupil to be about 0.4 cm², and the average frequency of white light to be about 6×10^{14} Hz.
- Light of wavelength 330 nm falling on a piece of metal ejects electrons with sufficient energy which requires voltage V_0 to prevent a electron from reaching collector. In the same setup, light of wavelength 220 nm, ejects electrons which require twice the voltage V_0 to stop them in reaching a collector. Find the numerical value of voltage V_0 .
(Take plank's constant, $h = 6.6 \times 10^{-34}$ Js and $1 \text{ eV} = 1.6 \times 10^{-19}$ J)
- Light of intensity 10^{-5} W m⁻² falls on a sodium photo-cell of surface area 2 cm². Assuming that the top 5 layers of sodium absorb the incident energy, estimate time required for photoelectric emission in the wave-picture of radiation. The work function for the metal is given to be about 2 eV. What is the implication of your answer ?
- When photons of energy 4.25eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_a eV and de Broglie wavelength λ_a . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.7eV is $T_b = (T_a - 1.5)$ eV. If the De Broglie wavelength of these photoelectrons is $\lambda_b = 2\lambda_a$, then find
 - The work function of a
 - The work function of b is
 - T_a and T_b
- A single electron orbits a stationary nucleus of charge Ze where Z is a constant and e is the electronic charge. It requires 47.2eV to excite the electron from the 2nd Bohr orbit to 3rd Bohr orbit. Find
 - the value of Z ,
 - energy required to excite the electron from the third to the fourth orbit
 - the wavelength of radiation required to remove the electron from the first orbit to infinity
 - the kinetic energy, potential energy and angular momentum in the first Bohr orbit
 - the radius of the first Bohr orbit.
- A hydrogen like atom (atomic number Z) is in higher excited state of quantum number n . This excited atom can make a transition to the first excited state by successively emitting two photons of energy 22.95eV and 5.15eV respectively. Alternatively, the atom from the same excited state can make transition to the second excited state by successively emitting two photons of energies 2.4eV and 8.7eV respectively. Find the values of n and Z .

9. A 20 keV energy electron is brought to rest in an X-ray tube, by undergoing two successive bremsstrahlung events, thus emitting two photons. The wavelength of the second photon is 130×10^{-12} m greater than the wavelength of the first emitted photon. Calculate the wavelengths of the two photons.
10. Photoelectrons are emitted when 400 nm radiation is incident on a surface of work function 1.9 eV. These photoelectrons pass through a region containing α -particles. A maximum energy electron combines with an α -particle to form a He^+ ion, emitting a single photon in this process. He^+ ions thus formed are in their fourth excited state. Find the energies in eV of the photons, lying in the 2 to 4 eV range, that are likely to be emitted during and after the combination.
[Take, $h = 4.14 \times 10^{-15}$ eV-s]
11. In a photo-emissive cell, with exciting wavelength λ , the maximum kinetic energy of electron is K. If the exciting wavelength is changed to $\frac{3\lambda}{4}$ the kinetic energy of the fastest emitted electron will be:
(A) $3K/4$ (B) $4K/3$ (C) less than $4K/3$ (D) greater than $4K/3$
12. In a photoelectric experiment, the collector plate is at 2.0 V with respect to the emitter plate made of copper $\phi = 4.5$ eV. The emitter is illuminated by a source of monochromatic light of wavelength 200 nm.
(A) The minimum kinetic energy of the photoelectrons reaching the collector is 0.
(B) The maximum kinetic energy of the photoelectrons reaching the collector is 3.7 eV.
(C) If the polarity of the battery is reversed then answer to part A will be 0.
(D) If the polarity of the battery is reversed then answer to part B will be 1.7 eV.
13. In a photoelectric experiment, electrons are ejected from metals X and Y by light of intensity I and frequency f. The potential difference V required to stop the electrons is measured for various frequencies. If Y has a greater work function than X; which one of the following graphs best illustrates the expected results?
- (A)  (B)  (C)  (D) 
14. An image of the sun is formed by a lens of focal-length of 30 cm on the metal surface of a photo-electric cell and a photo-electric current I is produced. The lens forming the image is then replaced by another of the same diameter but of focal length 15 cm. The photo-electric current in this case is
(A) $I/2$ (B) I (C) 2I (D) 4I
15. Monochromatic light with a frequency well above the cutoff frequency is incident on the emitter in a photoelectric effect apparatus. The frequency of the light is then doubled while the intensity is kept constant. How does this affect the photoelectric current?
(A) The photoelectric current will increase. (B) The photoelectric current will decrease.
(C) The photoelectric current will remain the same. (D) None of these
16. Imagine a Young's double slit interference experiment performed with waves associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease maximum if
(A) the accelerating voltage in the electron gun is decreased
(B) the accelerating voltage is increased and the distance of the screen from the slits is decreased
(C) the distance of the screen from the slits is increased.
(D) the distance between the slits is decreased.

17. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4 eV . Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero.
 (A) -4 (B) -6 (C) -8 (D) -10
18. In hydrogen and hydrogen like atoms, the ratio of difference of energies $E_{4n} - E_{2n}$ and $E_{2n} - E_n$ varies with its atomic number Z and n as:
 (A) Z^2/n^2 (B) Z^4/n^4 (C) Z/n (D) Z^0/n^0
19. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$ whose n_1 and n_2 are the principal quantum numbers of the two states. Assume the Bohr model to be valid. The frequency of orbital motion of the electron in the initial state is $1/27$ of that in the final state. The possible values of n_1 and n_2 are
 (A) $n_1 = 4, n_2 = 2$ (B) $n_1 = 3, n_2 = 1$ (C) $n_1 = 8, n_2 = 1$ (D) $n_1 = 6, n_2 = 3$
20. Electron in a hydrogen atom is replaced by an identically charged particle muon with mass 207 times that of electron. Now the radius of K shell will be
 (A) $2.56 \times 10^{-3} \text{ \AA}$ (B) $109.7 \text{ \AA}$ (C) $1.21 \times 10^{-3} \text{ \AA}$ (D) $22174.4 \text{ \AA}$
21. An electron collides with a fixed hydrogen atom in its ground state. Hydrogen atom gets excited and the colliding electron loses all its kinetic energy. Consequently the hydrogen atom may emit a photon corresponding to the largest wavelength of the Balmer series. The min. K.E. of colliding electron will be
 (A) 10.2 eV (B) 1.9 eV (C) 12.1 eV (D) 13.6 eV
22. The magnitude of angular momentum, orbit radius and frequency of revolution of electron in hydrogen atom corresponding to quantum number n are L, r and f respectively. Then according to Bohr's theory of hydrogen atom,
 (A) fr^2L is constant for all orbits (B) frL is constant for all orbits
 (C) f^2rL is constant for all orbits (D) frL^2 is constant for all orbits
23. A small plate of area 1 cm^2 is placed at a distance of $\frac{1}{\sqrt{\pi}} \text{ m}$ from an isotropic point source emitting light of frequency $\frac{1}{6.63} \times 10^{14} \text{ Hz}$, at a power of 2.00 mW . Assume the plate to be normal to the incident photons. [where $h = 6.63 \times 10^{-34} \text{ J-s}$]. Select **CORRECT** alternative(s)
 (A) Energy possessed by each photon is 10^{-20} J
 (B) Photon emission rate is $2 \times 10^{17} \text{ s}^{-1}$
 (C) The fraction of area intercepted by the plate is $\frac{1}{4} \times 10^{-4}$
 (D) The rate of photons striking the plate is 5×10^{12} per second.
24. A metallic sphere of radius r remote from all other bodies is irradiated with a radiation of wavelength λ which is capable of causing photoelectric effect.
 (A) the maximum potential gained by the sphere will be independent of its radius
 (B) the net positive charge appearing on the sphere after a long time will depend on the radius of the sphere
 (C) the kinetic energy of the most energetic electrons emanating from the sphere will keep on declining with time
 (D) the kinetic energy of the most energetic electrons emanating from the sphere initially will be independent of the radius of the sphere.

25. A particle moves in a closed orbit around the origin, due to a force which is directed towards the origin. The de Broglie wavelength of the particle varies cyclically between two values λ_1, λ_2 with $\lambda_1 > \lambda_2$. Which of the following statements is/are true ?
- (A) The particle could be moving in a circular orbit with origin as centre
 (B) The particle could be moving in an elliptic orbit with origin as its focus.
 (C) When the de Broglie wave length is λ_1 , the particle is nearer the origin than when its value is λ_2 .
 (D) When the de Broglie wavelength is λ_2 , the particle is nearer the origin than when its value is λ_1 .
26. A particular hydrogen like atom has its ground state binding "energy 122.4eV. Its is in ground state. Then:
- (A) Its atomic number is 3
 (B) An electron of 90eV can excite it.
 (C) An electron of kinetic energy nearly 91.8eV can be brought to almost rest by this atom.
 (D) An electron of kinetic energy 2.6eV may emerge from the atom when electron of kinetic energy 125eV collides with this atom.
27. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x-direction. Assume that all photons emitted due to electron transition inside the gas emerge in the y-direction. Let A and B denote the lights emerging from the gas in the x and y directions respectively.
- (A) Some of the incident wavelengths will be absent in A.
 (B) Only those wavelengths will be present in B which are absent in A.
 (C) B will contain some visible light.
 (D) B will contain some infrared light.
28. A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct (Assume that the hydrogen atom and neutron has same mass) :
- (A) If kinetic energy of the neutron is less than 20.4 eV collision must be elastic.
 (B) If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic.
 (C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 eV.
 (D) Perfectly inelastic collision can not take place.
29. A free hydrogen atom in ground state is at rest. A neutron of kinetic energy 'K' collides with the hydrogen atom. After collision hydrogen atom emits two photons in succession one of which has energy 2.55 eV. (Assume that the hydrogen atom and neutron has same mass)
- (A) minimum value of 'K' is 25.5 eV.
 (B) minimum value of 'K' is 12.75 eV
 (C) the other photon has energy 10.2 eV
 (D) the upper energy level is of excitation energy 12.75 eV.

Paragraph for Q. 30 to 32

During the study of cathode rays some invisible rays emerged out from the tube which were not even stopped when tube was wrapped several fold by black paper. They were called as X-rays which travel by speed of light. These rays are produced by bombarding highly energetic electrons on a heavy target having high melting point. When electrons enter the target, they are decelerated by lattice and according to electromagnetic theory they emit radiations which is called X-radiation. But only a fraction (0.5%) of incident electron energy is converted into X-radiation and remaining converts into heat that's why target metal used is of high melting point. Study of X-ray spectra when linked with quantum theory explain that there are two types of spectra 'continuous' and 'characteristic'.

There is existence of definite minimum wavelength in continuous spectrum which indicates complete conversion of energy of electron in photon energy and characteristic spectra is because of knocking out of electrons of the atoms from inner most shells by accelerated electron and subsequent fall back of same electron or of another electron in gap created in inner shells. The tube in which this experiment is carried out is having high vacuum. The X-rays are of order of 1×10^{-10} m wavelength. They travel in straight lines and cannot be deflected by electric magnetic fields. They can also ionize a gas.

30. The potential difference applied to an X-ray tube is increased. Therefore in emitted radiation
 (A) Intensity will increase
 (B) Minimum wavelength will increase
 (C) Intensity remains same and minimum wavelength decreases
 (D) Intensity decreases and minimum wavelength increases
31. An X-ray tube operates at 3×10^4 volts and current through it is 2 mA. The electric power input, no. of electrons striking target in one second and lower wavelength limit of X-rays emitted are respectively
 (A) 60 W, 1.25×10^{16} , $0.41 \text{ \AA}$
 (B) 60 mW, 1.25×10^{16} , $0.82 \text{ \AA}$
 (C) 30 W, 1.6×10^{19} , $0.82 \text{ \AA}$
 (D) 30 mW, 1.6×10^{24} , $0.41 \text{ \AA}$
32. In X-ray tube, the beam coming out will be
 (A) having all wavelength from zero to infinity
 (B) having all wavelengths from some minimum value to infinity
 (C) having all wavelength from some minimum value to maximum value
 (D) having all wavelength from zero to fixed maximum value
33. Some quantities related to the photoelectric effect are mentioned under Column I and Column II. Match each quantity in Column I with the corresponding quantities in Column II on which it depends

Column I

- (A) Saturation current
 (B) Stopping potential
 (C) de-Broglie wavelength of photoelectron
 (D) Force due to radiation falling on the photo-plate.

Column II

- (P) Frequency of light
 (Q) Work function
 (R) Area of photosensitive plate
 (S) Intensity of light (at constant frequency)
 (T) None of these

34. When we write expression for energy of electron in n^{th} orbit of helium ion (He^+) we take zero potential energy for $n = \infty$, but the potential energy depends on reference. If we take total energy of atom for $n = 1$ orbit as zero then

Column-I

- (A) Total energy of electron in $n = 2$
 (B) Ionization energy from ground state
 (C) Energy required to exit electron from $n = 1$ to $n = 2$
 (D) Negative of potential energy of electron in $n = 1$

Column-II

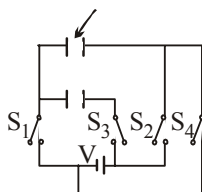
- (P) 54.4 eV
 (Q) 40.8 eV
 (R) depends on reference level
 (S) independent of reference level.
 (T) 70.3 eV

35. In each situation of column-I a physical quantity related to orbiting electron in a hydrogen like atom is given. The terms 'Z' and 'n' given in column-II have usual meaning in Bohr's theory. Match the quantities in column-I with the terms they depend on in column-II :-

Column-I	Column-II
(A) Frequency of orbiting electron	(P) Is directly proportional to Z^2
(B) Angular momentum of orbiting electron	(Q) Is directly proportional to n
(C) Magnetic moment of orbiting electron	(R) Is inversely proportional to n^3
(D) The average current due to orbiting of electron	(S) Is independent of Z
	(T) None of these

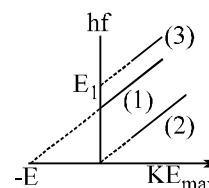
Paragraph for question nos. 36 to 38

The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance C. Light is incident on the left plate of the upper capacitor. When all the switches are open then the hf versus $KE_{\max}$ is shown by the straight line (A). In all the cases, we are measuring the $KE_{\max}$ when the electron reaches the opposite plate.



When only S_1 the switches and S_2 are closed, the graph becomes (B).

When only S_3 and S_4 are closed then the graph becomes (C).



36. What is the work function of the cathode?
 (A) E (B) E_1 (C) $E + E_1$ (D) none of these
37. What is the value of eV?
 (A) E (B) E_1 (C) $E + E_1$ (D) none of these
38. What is the value of E_1 ?
 (A) 3E (B) $3E/2$ (C) $E/2$ (D) none of these

Paragraph for Question Nos. 39 to 41

An electron is orbiting in a circular orbit of radius r under the influence of a constant strong magnetic field of strength B. Assuming that Bohr's postulate regarding the quantisation of angular momentum holds good for this electron, find

39. The allowed values of the radius 'r' of the orbit $\left(\text{here } a_0 = \sqrt{\frac{h}{2\pi eB}} \right)$

- (A) $r = \sqrt{n} a_0$ (B) $r = \frac{1}{2} \sqrt{n} a_0$ (C) $r = 2\sqrt{n} a_0$ (D) None of these

40. The time period of revolution of electron in n^{th} orbit is :-

- (A) $\frac{3\pi m}{eB}$ (B) $\frac{\pi m}{eB}$ (C) $\frac{\pi m}{2eB}$ (D) $\frac{2\pi m}{eB}$

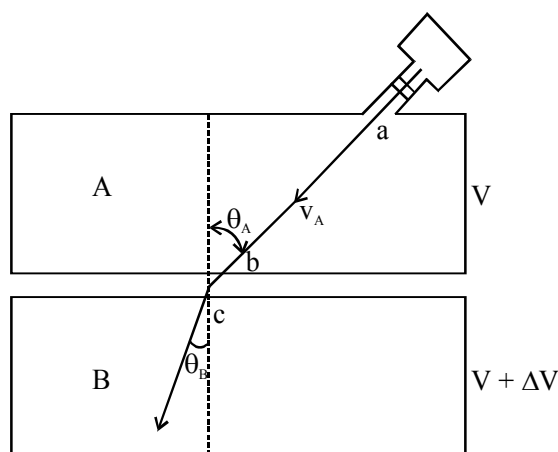
41. The total magnetic flux due to the magnetic field B passing through the n^{th} orbit

- (A) $\phi = \frac{2n h}{e}$ (B) $\phi = \frac{n h}{e}$ (C) $\phi = \frac{n h}{2e}$ (D) $\phi = \frac{4n h}{e}$

Paragraph for Question Nos. 42 to 44

Let a pencil of electrons from a suitable gun G enter through orifice in an enclosed metal box A , which has potential V relative to filament of the gun. Let these electrons emerge from A through orifice b and enter through c to another box B , which is maintained at a potential $V + \Delta V$. The electric field between the two boxes changes the component of velocity of the electrons perpendicular to the adjacent surface, and the electrons enter B with a change in their direction of motion. Let v_A and v_B be the velocities of the electrons in A and B , respectively, and θ_A, θ_B the angles between these directions and the normal to the box faces at b and c . Since the electric field does not change the horizontal component of velocity,

$$v_A \sin \theta_A = v_B \sin \theta_B \quad \frac{\sin \theta_A}{\sin \theta_B} = \frac{v_B}{v_A}$$



The "refraction" of a pencil of electrons

Now if we dealing with light waves undergoing refraction, or any other kind of wave, the relation would be

$$\frac{\sin \theta_A}{\sin \theta_B} = \mu = \frac{u_A}{u_B}$$

where μ is the relative refractive index of the two media and u_A, u_B are the corresponding velocities of light wave. Comparison of the last two equations gives the result $\frac{u_A}{u_B} = \frac{v_B}{v_A}$. We may conclude that if matter waves follow the electron along its path, the wave speed u is inversely proportional to the speed v of the electron, or $u = \frac{h}{mv}$.

42. If we define the frequency of matter waves as $f = \frac{u}{\lambda}$, the
- (A) Frequency of matter waves in medium A is more than that in medium B
 - (B) Frequency of matter waves in medium A is less than that in medium B
 - (C) Frequency of matter waves in medium A is same as that in medium B
 - (D) Cannot be predicted
43. Suppose $V_A = 20$ volt and $V_B = 15$ volt. Choose the **CORRECT** statement :-
- (A) The speed of electrons as well as speed of matter waves inside box B is more
 - (B) The speed of electrons as well as speed of matter waves inside box B is less
 - (C) The speed of electrons inside the box B is more, but speed of matter waves in box B is less
 - (D) The speed of electrons inside the box B is less, but speed of matter waves in box B is more
44. The refractive index for matter waves can be defined as $\frac{c}{u}$ where c is some constant. So refractive index for matter waves
- (A) is inversely proportional to λ
 - (B) is independent of λ
 - (C) is directly proportional of λ
 - (D) is proportional to $\sqrt{\lambda}$

ANSWER KEY

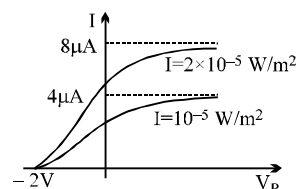
EXERCISE (O-2)

Q.1 AC	Q.2 ABD	Q.3 AD	Q.4 B	Q.5 B
Q.6 ABCD	Q.7 A	Q.8 B	Q.9 ABC	Q.10 B
Q.11 AD	Q.12 B			
Q.13 (A) $\rightarrow$ (S); (B) $\rightarrow$ (R); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (P)				
Q.14 B	Q.15 C	Q.16 B		
Q.17 (A) $\rightarrow$ (Q); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (P, T); (D) $\rightarrow$ (P, R, T)				

EXERCISE (J-A)

1. Ans. (D) 2. Ans. (A) 3. Ans. $Z = 42$

4. Ans.



- | | | | | |
|-----------------|---------------------|-----------------|------------------------|-----------------|
| 5. Ans. (A) | 6. Ans. (B) | 7. Ans. 0.55 eV | 8. Ans. (C) | 9. Ans. (A) |
| 10. Ans. (B) | 11. Ans. $\sqrt{2}$ | 12. Ans. (A, C) | 13. Ans. $n = 24$ | 14. Ans. (B) |
| 15. Ans. (B) | 16. Ans. (A) | 17. Ans. (B) | 18. Ans. (C) | 19. Ans. (C) |
| 20. Ans. (A) | 21. Ans. (A) | 22. Ans. (B) | 23. Ans. (D) | 24. Ans. (A) |
| 25. Ans. 3 | 26. Ans. (D) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. 8 |
| 30. Ans. (A) | 31. Ans. (C) | 32. Ans. (B) | 33. Ans. 1 | 34. Ans. 7 |
| 35. Ans. 7 | 36. Ans. (D) | 37. Ans. (C) | 38. Ans. 1 | 39. Ans. (A, C) |
| 40. Ans. (B) | 41. Ans. (A) | 42. Ans. 2 | 43. Ans. (A), (C), (D) | |
| 44. Ans. (A, C) | 45. Ans. 2 | | | |

REVIEW TEST

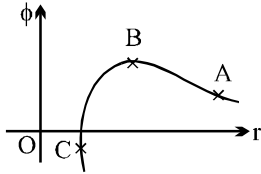
1. 6.59×10^{-34} Js 2. (a) 3.38×10^{-19} J = 2.11 eV (b) 3.0×10^{20} photons/s
3. (a) 3×10^{31} s⁻¹ (b) 10^4 s⁻¹ 4. 15/8 V
5. 1.6×10^7 s 6. (a) 2.25 eV, (b) 4.2 eV, (c) 2.0 eV, 0.5 eV
7. (i) 5, 16.5 eV, 36.4 A, 340 eV, -680 eV, $\frac{h}{2\pi} 1.06 \times 10^{-11}$ m
8. Z = 3, n = 7 9. 62.5×10^{-12} , 192.5×10^{-12}
10. during combination = 3.365 eV; after combination = 3.88 eV (5→3) & 2.63 eV (4→3)
- Q.11** D **Q.12** B **Q.13** A **Q.14** B **Q.15** B **Q.16** B **Q.17** D
- Q.18** D **Q.19** B **Q.20** A **Q.21** C **Q.22** B
- Q.23** ABCD **Q.24** ABCD **Q.25** BD **Q.26** ACD **Q.27** ACD **Q.28** AC
- Q.29** ACD **Q.30** B **Q.31** A **Q.32** B
- Q.33** (A) RS (B) PQ (C) PQ (D) PRS **Q.34** (A) QR, (B) PS, (C) QS, (D) PR
- Q.35** (A)→(P,R) ; (B)→(Q,S) ; (C)→(Q, S) ; (D)→(P,R)
- Q.36** A **Q.37** A **Q.38** B **Q.39** A
- Q.40** D **Q.41** C **Q.42** C **Q.43** D
- Q.44** A

MODERN PHYSICS-2

EXERCISE (O-2)

- When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:
 - both Z and A will decrease, if the process is α decay
 - Z will decrease but A will not change, if the process is β^+ decay
 - Z will decrease but A will not change, if the process is β^- decay
 - Z and A will remain unchanged, if the process is γ decay.
- When the atomic number A of the nucleus increases
 - initially the neutron-proton ratio is constant = 1
 - initially neutron-proton ratio increases and later decreases
 - initially binding energy per nucleon increases and later decreases
 - the binding energy per nucleon increases when the neutron-proton ratio increases.
- Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mass of a ${}^{20}_{10}\text{Ne}$ nucleus and M_2 the mass of a ${}^{40}_{20}\text{Ca}$ nucleus. Then
 - $M_2 = 2M_1$
 - $M_2 > 2M_1$
 - $M_2 < 2M_1$
 - $M_1 < 10(m_n + m_p)$
- The decay constant of a radio active substance is $0.173 \text{ (years)}^{-1}$. Therefore :
 - Nearly 63% of the radioactive substance will decay in $(1/0.173)$ year.
 - half life of the radio active substance is $(1/0.173)$ year.
 - one -forth of the radioactive substance will be left after nearly 8 years.
 - all the above statements are true.
- Which of the following statement(s) is/are correct?
 - The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.
 - The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons.
 - In nuclear fusion, energy is released by fusion of two nuclei of medium mass (approx. 100 amu)
 - In nuclear fusion, energy is released by fragmentation of a very heavy nucleus.
- The graph shown by the side shows the variation of potential energy ϕ of a proton with its distance 'r' from a fixed sodium nucleus, as it approaches the nucleus, placed at origin O. Then the portion.

- AB indicates dominance of nuclear repulsion
 - AB indicates dominance of electrostatic repulsion
 - BC indicates dominance of nuclear attraction
 - BC represents dominance of electrostatic interaction


- A nitrogen nucleus ${}^7_7\text{N}^{14}$ absorbs a neutron and can transform into lithium nucleus ${}^3_3\text{Li}^7$ under suitable conditions, after emitting :
 - 4 protons and 3 neutrons
 - 5 protons and 1 negative beta particle
 - 1 alpha particles and 2 gamma particles
 - 4 protons and 4 neutrons

- 8.** The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be
- (A) a nucleus of excess nucleons is α active
(B) an excited nucleus of excess protons is β^- active
(C) an excited nucleus of excess protons is β^+ active
(D) a nucleus of excess neutrons is β^- active
- 9.** In β -decay, the Q-value of the process is E. Then
- (A) K.E. of a β -particle cannot exceed E.
(B) K.E. of anti neutrino emitted lies between Zero and E.
(C) N/Z ratio of the nucleus is altered.
(D) Mass number (A) of the nucleus is altered.
- 10.** Consider the following nuclear reactions and select the correct statements from the options that follow.
- Reaction I :** $n \rightarrow p + e^- + \bar{\nu}$ **Reaction II :** $p \rightarrow n + e^+ + \nu$
- (A) Free neutron is unstable, therefore reaction I is possible
(B) Free proton is stable, therefore reaction II is not possible
(C) Inside a nucleus, both decays (reaction I and II) are possible
(D) Inside a nucleus, reaction I is not possible but reaction II is possible.
- 11.** When the nucleus of an electrically neutral atom undergoes a radioactive decay process, it will remain neutral after the decay if the process is :
- (A) α decay (B) β^- decay (C) γ decay (D) K-capture
- 12.** In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

Column I		Column II	
(i)	Energy of thermal neutrons	(A)	0.025 eV
(ii)	Energy of X-rays	(B)	0.5 eV
(iii)	Binding energy per nucleon	(C)	3 eV
(iv)	Photoelectric threshold of metal	(D)	20 eV
		(E)	10 keV
		(F)	8 MeV

EXERCISE - JA

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

1. The volume and mass of a nucleus are related as [JEE 2003 (Scr)]
(A) $v \propto m$ (B) $v \propto 1/m$
(C) $v \propto m^2$ (D) $v \propto 1/m^2$
2. The nucleus of element X ($A = 220$) undergoes α -decay. If Q-value of the reaction is 5.5 MeV, then the kinetic energy of α -particle is : [JEE 2003 (Scr)]
(A) 5.4 MeV (B) 10.8 MeV
(C) 2.7 MeV (D) None
3. A radioactive sample emits n β -particles in 2 sec. In next 2 sec it emits 0.75 n β -particles, what is the mean life of the sample? [JEE 2003]
4. A 280 days old radioactive substance shows an activity of 6000 dps, 140 days later it's activity becomes 3000dps. What was its initial activity. [JEE 2004 (Scr)]
(A) 20000 dps (B) 24000 dps
(C) 12000 dps (D) 6000 dps
5. The age of a rock containing lead and uranium is equal to 1.5×10^9 yrs. The uranium is decaying into lead with half life equal to 4.5×10^9 yrs. Find the ratio of lead to uranium present in the rock, assuming initially no lead was present in the rock. (Given $2^{1/3} = 1.259$). [JEE 2004]
6. If a star converts all of its Helium into oxygen nucleus, find the amount of energy released per nucleus of oxygen. $O = 15.9994$ amu and $He = 4.0026$ amu [JEE' 2005 (Scr)]
(A) 7.26 MeV (B) 7 MeV
(C) 10.24 MeV (D) 5.12 MeV
7. Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of helium nucleus is $(14)^{1/3}$. Find [JEE 2005]
(a) atomic number of the nucleus
(b) the frequency of K_{α} line of the X-ray produced. ($R = 1.1 \times 10^7 \text{ m}^{-1}$ and $c = 3 \times 10^8 \text{ m/s}$)
8. Given a sample of Radium-226 having half-life of 4 days. Find the probability, a nucleus disintegrates within 2 half lives. [JEE 2006]
(A) 1 (B) 1/2 (C) 3/4 (D) 1/4

9. Match the following Columns

[JEE 2006]

Column I

- (A) Nuclear fusion
- (B) Nuclear fission
- (C) β -decay
- (D) Exothermic nuclear reaction

Column II

- (P) Converts some matter into energy
- (Q) Generally occurs for nuclei with low atomic number
- (R) Generally occurs for nuclei with higher atomic number
- (S) Essentially proceeds by weak nuclear forces

10. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is

[JEE 2007]

- (A) $E(^{236}_{92}\text{U}) > E(^{137}_{53}\text{I}) + E(^{97}_{39}\text{Y}) + 2E(n)$ (B) $E(^{236}_{92}\text{U}) < E(^{137}_{53}\text{I}) + E(^{97}_{39}\text{Y}) + 2E(n)$
- (C) $E(^{236}_{92}\text{U}) < E(^{140}_{56}\text{Ba}) + E(^{94}_{36}\text{Kr}) + 2E(n)$ (D) $E(^{236}_{92}\text{U}) = E(^{140}_{56}\text{Ba}) + E(^{94}_{36}\text{Kr}) + 2E(n)$

11. Some laws/processes are given in **Column I**. Match these with the physical phenomena given in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS.

[JEE 2007]

Column I

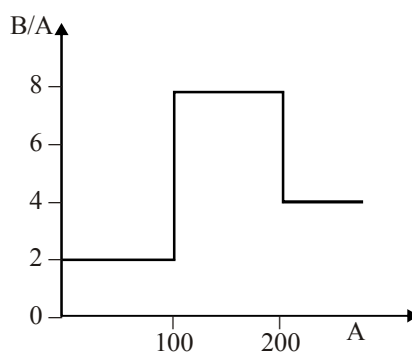
- (A) Transition between two atomic energy levels
- (B) Electron emission from a material
- (C) Mosley's law
- (D) Change of photon energy into kinetic energy of electrons

Column II

- (P) Characteristic X-rays
- (Q) Photoelectric effect
- (R) Hydrogen spectrum
- (S) β -decay

12. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below :

[JEE 2008]



- (A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy
- (B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy
- (C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments
- (D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

13. A radioactive sample S1 having an activity of $5\mu\text{Ci}$ has twice the number of nuclei as another sample S2 which has an activity of $10\mu\text{Ci}$. The half lives of S1 and S2 can be : **[JEE 2008]**
- (A) 20 years and 5 years, respectively
 (B) 20 years and 10 years, respectively
 (C) 10 years each
 (D) 5 years each

Paragraph for Question Nos. 14 to 16

Scientists are working hard to develop nuclear fusion reactor. Nuclei of heavy hydrogen, ${}^2_1\text{H}$, known as deuteron and denoted by D, can be thought of as a candidate for fusion reactor. The D-D reaction is ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + n + \text{energy}$. In the core of fusion reactor, a gas of heavy hydrogen is fully ionized into deuteron nuclei and electrons. This collection of ${}^2_1\text{H}$ nuclei and electrons is known as plasma. The nuclei move randomly in the reactor core and occasionally come close enough for nuclear fusion to take place. Usually, the temperatures in the reactor core are too high and no material wall can be used to confine the plasma. Special techniques are used which confine the plasma for a time t_0 before the particles fly away from the core. If n is the density (number/volume) of deuterons, the product nt_0 is called Lawson number. In one of the criteria, a reactor is termed successful if Lawson number is greater than $5 \times 10^{14} \text{ s/cm}^3$. It may be helpful to use the following :

$$\text{Boltzmann constant } k = 8.6 \times 10^{-5} \text{ eV/K} ; \frac{e^2}{4\pi\epsilon_0} = 1.44 \times 10^{-9} \text{ eVm.}$$

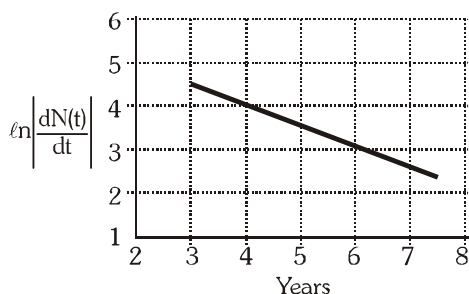
14. In the core of nuclear fusion reactor, the gas becomes plasma because of **[JEE-2009]**
- (A) strong nuclear force acting between the deuterons
 (B) Coulomb force acting between the deuterons
 (C) Coulomb force acting between deuteron-electron pairs
 (D) the high temperature maintained inside the reactor core
15. Assume that two deuteron nuclei in the core of fusion reactor at temperature T are moving towards each other, each with kinetic energy 1.5 kT , when the separation between them is large enough to neglect Coulomb potential energy. Also neglect any interaction from other particles in the core. The minimum temperature T required for them to reach a separation of $4 \times 10^{-15} \text{ m}$ is in the range **[JEE-2009]**
- (A) $1.0 \times 10^9 \text{ K} < T < 2.0 \times 10^9 \text{ K}$ (B) $2.0 \times 10^9 \text{ K} < T < 3.0 \times 10^9 \text{ K}$
 (C) $3.0 \times 10^9 \text{ K} < T < 4.0 \times 10^9 \text{ K}$ (D) $4.0 \times 10^9 \text{ K} < T < 5.0 \times 10^9 \text{ K}$

16. Results of calculations for four different designs of a fusion reactor using D-D reaction are given below. Which of these is most promising based on Lawson criterion? [JEE-2009]

- (A) deuteron density = $2.0 \times 10^{12} \text{ cm}^{-3}$, confinement time = $5.0 \times 10^{-3} \text{ s}$
 (B) deuteron density = $8.0 \times 10^{14} \text{ cm}^{-3}$, confinement time = $9.0 \times 10^{-1} \text{ s}$
 (C) deuteron density = $4.0 \times 10^{23} \text{ cm}^{-3}$, confinement time = $1.0 \times 10^{-11} \text{ s}$
 (D) deuteron density = $1.0 \times 10^{24} \text{ cm}^{-3}$, confinement time = $4.0 \times 10^{-12} \text{ s}$

17. To determine the half life of a radioactive element, a student plots a graph of $\ln \left| \frac{dN(t)}{dt} \right|$ versus t.

Here $\frac{dN(t)}{dt}$ is the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is [JEE-2010]



18. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s . The mass of an atom of this radioisotope is 10^{-25} kg . The mass (in mg) of the radioactive sample is [JEE-2011]
19. A freshly prepared sample of a radioisotope of half-life 1386 s has activity 10^3 disintegrations per second. Given that $\ln 2 = 0.693$, the fraction of the initial number of nuclei (expressed in nearest integer percentage) that will decay in the first 80 s after preparation of the sample is. [JEE-2013]

Paragraph for Questions 20 and 21

The mass of a nucleus ${}^A_Z\text{X}$ is less than the sum of the masses of $(A - Z)$ number of neutrons and Z number of protons in the nucleus. The energy equivalent to the corresponding mass difference is known as the binding energy of the nucleus. A heavy nucleus of mass M can break into two light nuclei of masses m_1 and m_2 only if $(m_1 + m_2) < M$. Also two light nuclei of masses m_3 and m_4 can undergo complete fusion and form a heavy nucleus of mass M' only if $(m_3 + m_4) > M'$. The masses of some neutral atoms are given in the table below :- [JEE Advance-2013]

${}^1_1\text{H}$	1.007825 u	${}^2_1\text{H}$	2.014102 u	${}^3_1\text{H}$	3.016050 u	${}^4_2\text{He}$	4.002603 u
${}^6_3\text{Li}$	6.015123 u	${}^7_3\text{Li}$	7.016004 u	${}^{70}_{30}\text{Zn}$	69.925325 u	${}^{82}_{34}\text{Se}$	81.916709 u
${}^{152}_{64}\text{Gd}$	151.919803 u	${}^{206}_{82}\text{Pb}$	205.974455 u	${}^{209}_{83}\text{Bi}$	208.980388 u	${}^{210}_{84}\text{Po}$	209.982876 u

$$(1\text{u} = 932 \text{ MeV}/c^2)$$

20. The kinetic energy (in keV) of the alpha particle, when the nucleus ${}^{210}_{84}\text{Po}$ at rest undergoes alpha decay, is :-
 (A) 5319 (B) 5422 (C) 5707 (D) 5818

21. The correct statement is :-

- (A) The nucleus ${}^6_3\text{Li}$ can emit an alpha particle
 (B) The nucleus ${}^{210}_{84}\text{Po}$ can emit a proton
 (C) Deuteron and alpha particle can undergo complete fusion
 (D) The nuclei ${}^{70}_{30}\text{Zn}$ and ${}^{82}_{34}\text{Se}$ can undergo complete fusion

22. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists:

List I**List II****[JEE Advance-2013]**

- | | |
|--------------------|--|
| P. Alpha decay | 1. ${}^{15}_8\text{O} \rightarrow {}^{15}_7\text{N} + \dots$ |
| Q. β^+ decay | 2. ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$ |
| R. Fission | 3. ${}^{185}_{83}\text{Bi} \rightarrow {}^{184}_{82}\text{Pb} + \dots$ |
| S. Proton emission | 4. ${}^{239}_{94}\text{Pu} \rightarrow {}^{140}_{57}\text{La} + \dots$ |

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 4 | 2 | 1 | 3 |
| (B) | 1 | 3 | 2 | 4 |
| (C) | 2 | 1 | 4 | 3 |
| (D) | 4 | 3 | 2 | 1 |

23. If λ_{Cu} is the wavelength of K_α X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength of the K_α X-ray line of molybdenum (atomic number 42), then the ratio $\frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$ is close to :-

[JEE Advance-2014]

- (A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

24. A metal surface is illuminated by light of two different wavelength 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2 , respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240 \text{ eV nm}$, the work function of the metal is nearly :-

[JEE Advance-2014]

- (A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

25. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5% of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of nT years, then the value of n is.

[JEE Advance-2015]

26. Match the nuclear processes given in column-I with the appropriate option(s) in column-II.

[JEE Advance-2015]

Column-I

Column-II

- | | |
|----------------------------------|--|
| (A) Nuclear fusion | (P) Absorption of thermal neutrons by ${}_{92}^{235}\text{U}$ |
| (B) Fission in a nuclear reactor | (Q) ${}_{27}^{60}\text{Co}$ nucleus |
| (C) β -decay | (R) Energy production in stars via hydrogen conversion to helium |
| (D) γ -ray emission | (S) Heavy water |
| | (T) Neutrino emission |

27. For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t . Two radioactive sources P (mean life τ) and Q (mean life 2τ) have the same activity at $t = 0$. Their rates of change of activities at $t = 2\tau$ are R_P and R_Q , respectively. If $\frac{R_P}{R_Q} = \frac{n}{e}$, then the value of n is : [JEE Advance-2015]

28. A fission reaction is given by ${}_{92}^{236}\text{U} \rightarrow {}_{54}^{140}\text{Xe} + {}_{38}^{94}\text{Sr} + x + y$, where x and y are two particles. Considering ${}_{92}^{236}\text{U}$ to be at rest, the kinetic energies of the products are denoted by $K_{\text{Xe}}, K_{\text{Sr}}, K_x$ (2 MeV) and K_y (2 MeV), respectively. Let the binding energies per nucleon of ${}_{92}^{236}\text{U}$, ${}_{54}^{140}\text{Xe}$ and ${}_{38}^{94}\text{Sr}$ be 7.5 MeV, 8.5 MeV and 8.5 MeV, respectively. Considering different conservation laws, the correct option(s) is (are) :- [JEE Advance-2015]

- (A) $x = n, y = n, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (B) $x = p, y = e^-, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (C) $x = p, y = n, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (D) $x = n, y = n, K_{\text{Sr}} = 86 \text{ MeV}, K_{\text{Xe}} = 129 \text{ MeV}$

ANSWER KEY

EXERCISE (O-2)

- | | | | | |
|--------|----------------------------------|--------|--------|---------|
| 1. ABD | 2. AC | 3. CD | 4. AC | 5. AD |
| 6. BC | 7. CD | 8. ACD | 9. ABC | 10. ABC |
| 11. CD | 12. (i) A, (ii) E (iii) F (iv) C | | | |

EXERCISE (J-A)

- | | | |
|---|--|---|
| 1. Ans. (A) | 2. Ans. (A) | 3. Ans. $1.75n = N_0(1 - e^{-4\lambda}), 6.95 \text{ sec}, \frac{2}{\ln\left(\frac{4}{3}\right)}$ |
| 4. Ans. (B) | 5. Ans. 0.259 | 6. Ans. (C) |
| 7. Ans. $v = 1.546 \times 10^{18} \text{ Hz}$ | 8. Ans. (C) | |
| 9. Ans. (A) P, Q; (B) P, R; (C) S, P; (D) P, Q, R | | 10. Ans. (A) |
| 11. Ans. (A) R, P; (B) Q, S; (C) P; (D) Q | | 12. Ans. (B, D) |
| 13. Ans. (A) | 14. Ans. (D) | 15. Ans. (A) 16. Ans. (B) |
| 17. Ans. (8) | 18. Ans. (1) | 19. Ans. 4 20. Ans. (A) |
| 21. Ans. (C) | 22. Ans. (C) | 23. Ans. (B) 24. Ans. (A) |
| 25. Ans. 3 | 26. Ans. (A)-R or R,T; (B)-P & S; (C)-Q & T; (D)-R | |
| 27. Ans. 2 | 28. Ans. (A) | |

IMPORTANT NOTES

[illegible]

[illegible]